Sultanate of Oman

Nama Water Services Company SAOC

Consultancy Services for Design and Supervision for STP, Sewer and TE Networks Systems in Ibri

Tender No. T/2719110/2025

Tutorial T04

From the Meter to the Pipe — Population, Flows, Sewer and Plant Design

Revision 00 · September 2026

A teaching reference for the whole concept design method. Every step, from the electricity meter on a plot to a sized, self-cleansing sewer and a sized treatment plant, is given with its rule, its source page, a worked Ibri number and the script that does it.

It replaces Tutorials T01, T02 and T03 Revision 01 as the teaching reference. Those three stay unchanged as the record.

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Project 2621 · Revision 00 · September 2026

Contents

How to use this tutorial

This tutorial teaches the concept design method of the Ibri sewerage scheme from end to end. It starts at the electricity meter on a plot and ends at a sized, self-cleansing gravity sewer, a sized treatment plant and the appraisal that chooses between options.

It is written for the project's hydraulic engineer, and for anyone who has to check the work. The reader is assumed to know sewer design. What the tutorial adds is this project: every step made explicit, with its authority, a real Ibri number worked through it, and the place in the scripts where it is done.

It is a reference, not a report. It is not addressed to the client. A number that goes to the client is taken from the current Concept Design Report, where the same method is stated in the client's terms.

What it replaces

Tutorial T01 taught the sewage flow and load calculation, T02 the hydraulic design of a gravity sewer, and T03 Revision 01 the concept design methodology. T04 carries all three. The older tutorials stay unchanged as the record of what was taught at the time. Where one of them disagrees with T04, T04 governs, and the chapter concerned says in one sentence what changed. The changes a reader of the older tutorials will meet most often are these.

Table 1 What changed from the earlier tutorials

Topic	T01 to T03 taught	T04 teaches
Occupancy rate	One rate for the whole study area: 5 persons per property (T01, T02) and 5.32 (Concept Design Report R1; T03 Revision 01 gives the method but prints no value)	A rate for each settlement: its 2024 population divided by its counted properties, never below 4.0 (project rule)
Load of a plot	A flat load per property	The plot's own meters: homes at occupancy × 164 l/d per person; shops and government at a unit rate per meter, from the settlement's 22 % and 14 % shares (G201-p60)
The street pipe	"Lateral"	"Secondary main sewer" (G203-p17, p21). "Lateral" means only the tertiary pipe from the house connection to the main sewer (G203-p22 Tab 6)
Connection ratio	Applied along the projected series	Applied only to the 2030 self-cleansing case. Pipes and plant are sized at 100 % connection (G201-p73)

Reading paths

If you want to	Read
See the whole chain in ten minutes	The chain at a glance, which follows this section
Carry out one step correctly	The chapter for that step. Each is self-contained
Check a number before it goes to the client	The step's "where it lives" part, then the field in the layer, then the guideline page
Know where the guidelines are silent, wrong or departed from	Chapter 24, and the caution boxes throughout
Rerun the chain after the data changes	Appendix A

The five parts of every step

Every step is written in the same five parts, in the same order, so the tutorial can be read by its shape.

1. **What it is for.** One or two sentences on why the step exists and what it feeds.

2. **The rule.** The equation, numbered and followed by a table of every symbol with its meaning and unit; or the decision rule, stated exactly.
3. **The source.** The guideline and page; or "project rule" with who decided it and when; or "outside assumption" with where it comes from and its tag.
4. **A worked Ibri number.** A real number from the live data, carried through the rule so the arithmetic can be followed by hand.
5. **Where it lives.** The script that does the step and the field or sheet that holds the result, so the number can be found in the layer and checked.

Each chapter ends with a short list headed "Check it yourself": what to open, what to sum or filter, and what the answer should be.

Where the numbers come from

Every Ibri number in this tutorial is read from the outputs of the load chain at the moment the document is built: the plot layer, the settlement layer, the growth workbook and the design-flow file. None is typed in. When the data changes and the chain is rerun, the tutorial is rebuilt and its numbers change with it. The method does not.

This copy read the data on 2026-09-11. On that reading the study area holds 119,893 people in 2024, the sum of the population by year sheet of the growth workbook. A later copy may print a different figure for the same sentence.

Two consequences follow. Quote the method from this tutorial, but quote a number only from the current report or from the layer itself. And a number that does not come from the data, such as a guideline constant or a value read from a table, carries its page beside it every time it appears.

Numbers are rounded half up and printed with a thousands separator, so a total prints the same wherever it appears.

How guideline pages are cited

A guideline value is cited by document and page, for example G203-p26. Where a table or section number helps to find it, that follows: G203-p27 Tab 10. The page is the one printed at the foot of the guideline, "Page 26 of 201". In G201 and G203 the printed page and the PDF page number are the same, so the page box of the viewer goes straight to it.

Table 2 Guideline page references

Written here	Older project files	Document
G201-p##	G1-p##	PAM-GUD-201 General Design Guidelines v1.0, 152 pages
G202-p##	G2-p##	PAM-GUD-202 Water and TSE Design Guidelines v1.0, 177 pages
G203-p##	p## (no prefix)	PAM-GUD-203 Wastewater Design Guidelines v1.0, 201 pages

The older shorthand survives in the scripts and the design criteria register. It points to the same pages.

> **Nothing is quoted from memory.** Every guideline value in this tutorial was read from the page of the guideline. Equations were read from the page image, because the text layer of these PDFs mangles fraction bars, roots and exponents, and some equations exist only as pictures. The tractive-force equation of G203-p27 is one of them.

Three kinds of value, and how to tell them apart

Every value in the design is one of three kinds. The kind decides who may change it and what happens if it is wrong, so the source line of every step says which kind it is.

Table 3 The three kinds of value

Kind	What it is	How it is marked	Ibri example	Who can change it
Guideline value	A number or rule printed in G201, G202 or G203	Document and page	0.75 m/s minimum velocity at peak flow (G203-p26)	NWS. Any departure is declared and needs NWS agreement
Project rule	A decision made for this project where the guideline is silent or leaves a choice	"Project rule", with who decided it and the date	Size on saturation; check self-cleansing on 2030 × 0.61 (engineer, 2026-09-11)	The engineer, recorded in the design criteria register; shown to the client
Outside assumption	A value from outside the three guidelines, used because a guideline requires a method and gives no value for it	"Outside assumption", its origin and its gap number	Tractive tension τ = 1 Pa for the G203-p27 test (Mara; GAP-9). The 1.5 l/s low-flow threshold (Mara; not in G203)	Carried as a parameter until NWS confirms it

The three fail in different ways. A wrong guideline value is a compliance error, and a reviewer will find it. A project rule is a decision the client is entitled to challenge, so it must be visible. An outside assumption is the dangerous one: once its tag is lost it reads as a guideline value, and nobody asks for it to be confirmed.

A fourth label, pending data, marks something held out of the calculation until data arrives. It is not a value. Tanker deliveries are the main example: no filling-station records are held, so no flow in this tutorial contains them.

Where a formula is not NWS's, it says so. Several formulae an engineer expects to find are not written anywhere in the three guidelines: net present value, life cycle cost, total dynamic head, pump power, and the friction equations in written form. Each is attributed in this tutorial to its real source. Citing NWS for them would be a false reference.

Where the guideline is wrong, it is printed and flagged. The guidelines contain printing and arithmetic errors. This tutorial prints what the guideline prints, then says plainly what is wrong with it. It does not silently correct the client's own standard. Chapter 24 lists them together.

Where each step lives

The table maps every chapter to the script that does the work and to the field or sheet where the result can be read. Scripts named without a folder are in W14/py. Where no script exists yet, the method lives in the section of the Concept Design Report named in the table.

Table 4 Where each step lives: the script and the field or sheet that holds the result

Ch	Step	Script	Main output: field or sheet
1	The data	plots_meters_load.py (run in QGIS); inventory _BRAIN/03_DATA_INVENTORY.md	W14/shp/PLOTS_load.shp: Name, Moh_Classi, Buiding_St, AREA_M2; meters ELE_meters_on_plots.shp
2	From meter to plot, and the use of every plot	plots_meters_load.py, ndvi_plots.py, plot_class_v2_apply.py	meter PLACE; plot N_ACC, G_DOM to G_AGR, DERIVED, WHYC, GREEN_M2
3	Properties per plot and the occupancy rate	plot_class_v2_apply.py	G_DOM, PPP_S, OR_S; W14/analysis/occupancy_by_settlement.csv
4	The empty land and its capacity	plot_class_v2_apply.py	HOMESHAPE, HOMESH_S, FUT_CAP, SPREAD_W; settlement CAP_POP
5	Growth, overflow and saturation	growth_by_settlement.py	workbook sheets Population by year, Overflow routes, Settlements; SAT_YEAR, POP_ULT
6	Water demand	plot_class_v2_apply.py	U_NDOM, U_GOV, W_DOM, W_NDOM, W_GOV, W_SPEC, W_TOT

Ch	Step	Script	Main output: field or sheet
7	The sewage of every plot	plot_class_v2_apply.py	S_DOM, S_NDOM, S_GOV, S_SPEC, QADF
8	Flow through time and the two design cases	growth_by_settlement.py, make_design_flows.py	Q_2030, Q_2055, Q_ULT; sheet Qadf by year m3d; W14/analysis/design_flows.json
9	Peak flow and infiltration	W13/py/sewnet/criteria.py, hydra.py	pipe N_PROPS, QADF_M3D, PF, PF_PELT, QPEAK_LS
10	The network and its tiers	W13/py/sewnet/skeleton.py, pipeline.py	pipe TIER, ON_TRUNK, IS_JOIN
11	Gravity sewer hydraulics	W13/py/sewnet/hydra.py, stages/hydraulic.py	pipe DN_MM, SLOPE_PCT, VEL_MS, DOD; chamber DEPTH
12	Self-cleansing and the early years	W13/tmp3/py/sewnet/hydra.py (smin_tractive), stages/audit.py	audit findings; the four classes are a rule, not yet a field
13	Lifting stations and force mains	W13/py/sewnet/pipeline.py, catchments.py; limit MAX_DEPTH in criteria.py	chamber IS_PUMP, LIFT_M; pipe RISE_MAIN, QDUTY_LS
14	Septicity and odour	no script; method in report R2 §30 (W14/report/rpt_gh.py)	none yet
15	Utilities and crossings	no script; report R2 §33 (W14/report/rpt_gh.py)	pipe IS_XING (crosses a dual carriageway)
16	Hydraulic modelling	W13/py/sewnet/export_gems.py	CONDUITS.shp, MANHOLES.shp, LOADS.xlsx, REFEREE_pipes.csv
17	The existing network	W13/py/export_existing.py; report R2 §18 (W14/report/rpt_ef.py)	NAMA KMZ records as shapefiles, indicative only; filter on OP_STATUE (built or proposed) before quoting a length
18	Treatment plant flows and loads	make_design_flows.py; report R2 §25 (W14/report/rpt_ef.py)	sheet Qadf by year m3d; stp_ultimate_with_margin_m3d in design_flows.json
19	Treated effluent	no script; report R2 §17 and §24 (rpt_cd.py, rpt_ef.py)	none yet
20	Sludge	no script; report R2 §27 (W14/report/rpt_ef.py)	none yet
21	Cost estimation	no script; report R2 §35 (W14/report/rpt_gh.py)	none yet
22	Financial appraisal	report R2 §35.4 (rpt_gh.py); W9/analysis/W9_PIAD_financial_review.md	none yet
23	Options and the recommendation	report R2 §21, §22, §38 (rpt_ef.py, rpt_gh.py)	none yet
24	Where the guidelines cannot be relied on	_BRAIN/05_GAPS.md, _BRAIN/02_DESIGN_CRITERIA.md; departures report R2 §10.1 (rpt_cd.py)	the gap register
A	Equations, fields and the rerun order	make_design_flows.py; the rerun order in Appendix A	the fields listed in design_flows.json

The pipe-laying engine does not read the plot flows yet. Chapters 9 to 13 point to the network engine, which still loads every plot with one flat figure per property. Before any design run it has to read each plot's own saturation flow Q_ULT and opening-year flow Q_2030 from the plot layer. Its layout results on the test boundary must still hold after that change. Its flow results will be re-baselined.

Abbreviations and symbols

Term	Meaning
Qadf	Average daily dry-weather flow, m^3/d (Ml/d in Merrimack, l/s in Peltier)
QADF, Q_2030, Q_2055, Q_ULT	The plot's average sewage flow in 2024, 2030, 2055 and at saturation, m^3/d
Qpdf, PF	Peak flow, and the peak factor that turns an average flow into it
OR	Occupancy rate, persons per domestic property
N_dom, N_nd, N_gov	Domestic, non-domestic and governmental meters on a plot
U_nd, U_gov	Unit rate per non-domestic and per governmental meter, l/d
d/D	Proportional depth: depth of flow over pipe diameter
DN	Nominal diameter, mm
τ	Tractive tension on the pipe wall, Pa
SLS	Sewage lifting station
TE, TSE	Treated effluent (see the note below)
NWS	Nama Water Services, the client
NCSI	National Centre for Statistics and Information
MoHUP	Ministry of Housing and Urban Planning
G201, G202, G203	PAM-GUD-201 General Design, -202 Water and TSE, -203 Wastewater

One term to be careful with. In G201 and G203, TE means trade effluent and TSE means treated sewage effluent. This project, its terms of reference and the tender use TE for treated effluent. Wherever this tutorial says TSE it means the treated product of the plant. Say so explicitly in anything sent to NWS, or a reviewer who knows the guidelines will read a trade effluent network into the design.

The chain at a glance

The whole method on one page. Chapters 1 to 8 are one calculation, so an error in the count of properties or the occupancy rate reaches every flow downstream. Chapters 9 to 17 take the flow into the network, 18 to 20 to the plant, 21 to 23 decide what is built, and 24 lists where the guidelines give no value.

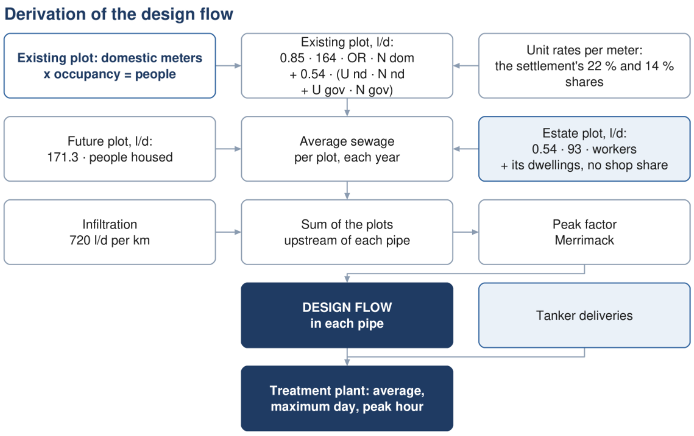

Figure 1   Derivation of the design flow, from the meters on a plot to the flow in each pipe and at the plant

1. **The data.** What was supplied, what each dataset is trusted for, and what the chain reads.
2. **From meter to plot, and the use of every plot.** Meters go onto plots; their mix sets the use.
3. **Properties per plot and the occupancy rate.** A home meter is a property; a rate per settlement.
4. **The empty land and its capacity.** The people the home-shaped empty plots can still house.
5. **Growth, overflow and saturation.** Own growth until full, then overflow to neighbours.
6. **Water demand.** Litres per person in the homes; shop and government shares landed on their own meters.
7. **The sewage of every plot.** The return to sewer turns each plot's water into its average dry-weather flow.
8. **Flow through time and the two design cases.** Size on saturation; test self-cleansing on the low case.
9. **Peak flow and infiltration.** Merrimack or Peltier by properties upstream; infiltration per km.
10. **The network and its tiers.** Primary, secondary and tertiary pipes, and the roads each may follow.
11. **Gravity sewer hydraulics.** Diameter, gradient, depth of flow and velocity, each against its limit.
12. **Self-cleansing and the early years.** Velocity and tractive tests; four classes of pipe.
13. **Lifting stations and force mains.** Where gravity would go too deep, and how the pumped main is sized.
14. **Septicity and odour.** Retention, turbulence and the measures that keep hydrogen sulphide down.
15. **Utilities and crossings.** Clearances, corridors, dual carriageways and wadis.
16. **Hydraulic modelling.** The SewerGEMS model as the independent check on the design calculation.
17. **The existing network.** What the records show, what they cannot show, and how the network is assessed.
18. **Treatment plant flows and loads.** Average, maximum-day and peak flows, margin and loads.
19. **Treated effluent.** How much the plant produces, how much reaches customers, and who takes it.
20. **Sludge.** How much is produced and where it goes.
21. **Cost estimation.** Capital and operating cost, on one basis for every option.
22. **Financial appraisal.** Net present value and life cycle cost decide; payback is reported, not used.
23. **Options and the recommendation.** Three options per system, scored on the same criteria.
24. **Where the guidelines cannot be relied on.** Missing formulae and values, defects, departures.

Appendix A   Equations, fields and the rerun order. Every equation, every field, and what to rerun first.

The headline numbers

The table gives the numbers the rest of the tutorial keeps returning to. They are average dry-weather flows summed over the study area, read from the growth workbook and the design-flow file on this build.

Table 5 Headline population and average flow, study area (concept stage)

Case	Year	People	Qadf, m³/d	Used for
Today	2024	119,893	20,852	The base year of the meters and the census
Opening year	2030	139,535	24,216	The first model year
Opening year, low case	2030	85,117 connected	14,772	Self-cleansing only: 2030 flow × 0.61 connected (project rule)
Model year	2055	244,914	42,266	Model year; plant phasing
Saturation	**2070**	**349,029**	**60,099**	**Pipe size and capacity**, at 100 % connection (G201-p73)
Plant at saturation	2070		66,109	Saturation plus the 10 % margin for new plants (G201-p73)
Capacity of the empty land		229,136		The people the empty plots can still house

Saturation is today's population plus the capacity of the empty land: 119,893 + 229,136 = 349,029. The growth rates decide the year the land is full. They do not decide how many people it finally holds; the land does.

The domestic properties number 22,559 in 2024, 26,255 in 2030 and 70,328 at saturation. Their count upstream of a pipe decides which peak-factor formula applies to it. The sizing case counts the properties standing at saturation. The low case counts only the connected ones, the 2030 properties × 0.61, which is 16,016 over the study area. That low-case count is recommended in the design-flow handoff, to be confirmed by the engineer. Ibri, the largest settlement, is full in 2056; the last settlement fills in 2070, and that year is saturation.

> **These are concept-stage figures.** They are average dry-weather flows. They contain no peak and no infiltration, which are added pipe by pipe in the design, and no plant margin except in the plant row. They contain no tanker deliveries and no private wells, which are pending data. 126 meters lie more than 15 m from any plot and carry no load in the plot table. Every figure here changes when the data changes. Quote the current report, not this page.

1 The data

The load chain rests on five datasets. None of them was made for sewer design, and each answers only part of the question. The electricity meters say how many premises there are and what kind, but not how large they are. The cadastre says where the plots are and whether they are built, but its land-use field is unreliable. The settlement series says how many people live in each settlement and how fast that number grows, but not where inside the settlement they live. The public record finds the few large users that none of the three shows. This chapter describes what each dataset carries, what it lacks, and the decisions taken to make it usable. Chapter 2 then puts the meters on the plots and reads the use of every plot from them.

Table 6 The five datasets and what each can and cannot tell

Dataset	Carries	Does not carry	Used for
Electricity meters, 2024	33,971 points, a tariff name each	consumption, address, land use, floor area	properties; non-domestic and governmental premises
Large-consumer register	499 accounts on the Cost Reflective Tariff, placed by public data	any use in the source data; the use was found by search	the category of the largest consumers
Cadastre, September 2026	77,265 plots, geometry, a built flag, a class	a government class; a reliable use	the unit the load lands on; built or empty
Settlements and population series	25 settlement outlines; population per settlement, 2023 to 2100	where inside a settlement people live	occupancy in 2024; the growth rate
Identified projects	estates, camps, planned schemes, tanker sources found in public records	workforce, occupancy, discharge	loads kept outside the population ratios

1.1 The electricity meters

What it is for. The meters are the only complete count of premises in the study area. A domestic meter is a property, and a shop or government meter is a non-domestic or governmental premises. Every population and every load placement in this tutorial starts from them.

The dataset holds 33,971 electricity accounts for 2024, the base year of the design. Each is one point with one attribute, the tariff name. There is no consumption figure, no address and no land use. The points were shifted onto the cadastre before use, so that a meter drawn on a house falls inside that house's plot. The dataset therefore says what kind of customer sits at a point and how many there are. It can never say how large the customer is, and that single limit is why the non-domestic demand is set by the guideline ratios and not by the unit rates of G201 Table 12, which are priced in pupils, beds, employees and floor area.

The rule. Each of the thirteen tariffs is folded into one of the guideline's demand categories. A domestic tariff is one property, whatever it is called. The Additional Account Tariff is a second dwelling on the same plot, so it is a property in its own right. Shop, government, farm and large-consumer meters carry no people. An agricultural meter drives an irrigation pump whose water goes onto a field, so it carries no sewage. The Cost Reflective Tariff (CRT) is a consumption class, not a use, and each of its accounts takes the category of the use found for it in Section 1.2.

Table 7 The thirteen tariffs and the category each takes

Tariff, as received	Meters	Category	People
Primary Account Tariff	10,973	Domestic	one property
Primary Account Tariff (with National Subsidy)	5,272	Domestic	one property
Additional Account Tariff	6,344	Domestic	one property, a second dwelling on the plot
Commercial	9,385	Non-domestic	none
Fisheries	6	Non-domestic	none
Tourism	1	Non-domestic	none
Government	966	Governmental	none
MOD	1	Governmental	none
Agricultural	523	Agricultural	none, and no sewage
Industrial	1	Special	none
CRT Seasonal	298	by the use found (1.2)	none
CRT Time of Use	192	by the use found (1.2)	none
CRT Fixed Rate	9	by the use found (1.2)	none
Total	**33,971**		

The number of domestic properties is the sum of the three domestic tariffs.

$$N_{\text{dom}} = N_{\text{P}} + N_{\text{PS}} + N_{\text{A}} \tag{1}$$

Symbol	Meaning	Unit
N dom	domestic properties in the study area	—
N P	meters on the Primary Account Tariff	—
N PS	meters on the Primary Account Tariff with National Subsidy	—
N A	meters on the Additional Account Tariff	—

Ibri number. N_dom = 10,973 + 5,272 + 6,344 = 22,589 domestic properties. The additional accounts are 28.1 per cent of them: counting the primary tariffs alone would lose more than a quarter of the dwellings.

Once the large-consumer accounts are resolved (Section 1.2), every meter sits in one of five categories. The table shows where each total comes from, so the two halves can be checked against each other.

Table 8 Meters by category: from the tariff, from the large-consumer accounts, and in total

Category	From the tariff	From the large-consumer accounts	Meters in the category
Domestic	22,589	0	22,589
Non-domestic	9,392	321	9,713
Governmental	967	135	1,102
Agricultural	523	28	551
Special	1	15	16
Total	**33,472**	**499**	**33,971**

Source. *G201-p59 §7.3 names the components of demand (domestic, non-domestic, tanker, and special categories such as industrial or institutional uses); §7.3.1 domestic G201-p59, §7.3.2 non-domestic G201-p60, §7.3.3 governmental G201-p61, §7.3.4 special G201-p61. The crosswalk from tariff to category is a project inference, not a guideline table (project rule, engineer, 2026-09-09). Table 12 is not used: its drivers were not supplied (G201-p61; locked load basis, engineer, 2026-08-30).*

Where it lives. W14/py/plots_meters_load.py, dictionaries GROUP (tariff to group) and GUD (group to category), run inside QGIS on the shifted meter layer. Output W14/shp/ELE_meters_on_plots.shp, one point per meter, fields TARIFF, GROUP, GUD, USE, PROPERTY (1 for a domestic meter). Counts: facts_w14.meter_counts().

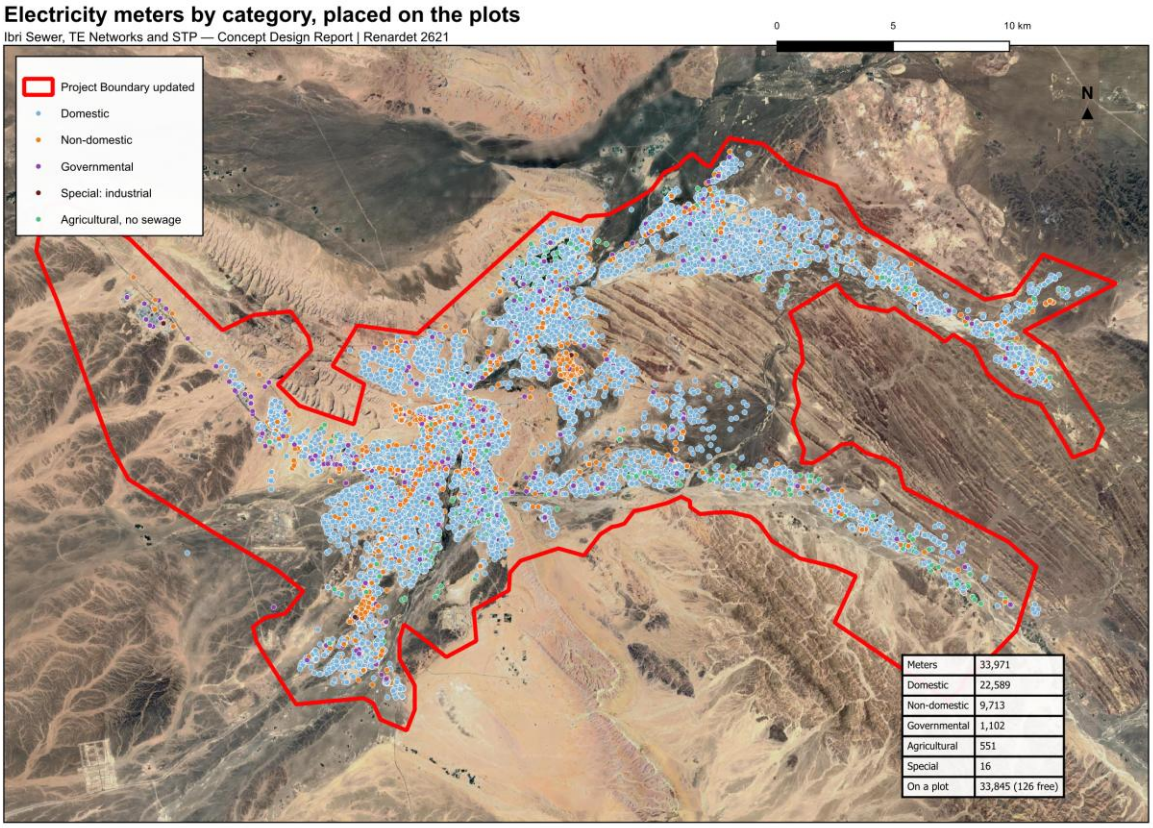

Figure 2 The electricity meters by category after the large consumers are resolved, placed on the plots. The data box gives the count in each category and the meters left free.

1.2 The large-consumer accounts

What it is for. The 499 accounts on the Cost Reflective Tariff are the largest consumers in the study area, and the tariff says nothing about what they are. The tariff applies above a consumption threshold, so a shopping centre, a hospital, a mosque and a factory all fall in it. Each account must be given a category before its water can be placed.

The rule. Each account was placed by public data, in four passes, and takes the category of the use found:

1. the accounts are clustered at 120 m, so a shopping centre with twenty meters is one site and not twenty;
2. each cluster and each single account is matched against the named features of OpenStreetMap within 80 m;
3. what remains is reverse geocoded (Nominatim), and the largest clusters are inspected on the aerial imagery;
4. the identified-projects search (Section 1.5) adds the industrial estates and the institutions named in the press.

An account with no named feature within 80 m is recorded as unresolved and carried as non-domestic, the largest class among the resolved accounts. Every account carries a confidence: certain where a named feature lies within 20 m or the site was inspected; likely where one lies within 80 m or the reverse geocoding names it; guessing where only the surroundings speak.

Table 9 Large-consumer accounts by the use found, the category adopted and the confidence

Use found	Accounts	Category	Certain	Likely	Guessing	Typical sites
Commercial	205	Non-domestic	107	68	30	Bawadi shopping centre, the Al Murtafa souq, the Araqi strip, hypermarkets, banks, fuel
Government	67	Governmental	4	42	21	police headquarters, post office, directorates
Education	37	Governmental	0	32	5	the university campus, the college, schools
Health	15	Governmental	0	15	0	Ibri Hospital, clinics
Religious	16	Governmental	0	15	1	mosques
Industrial	15	Special	8	7	0	mostly the Tanam estate
Agricultural	28	Agricultural	0	9	19	farm pumps at Al Aynayn
Telecommunications, utility	6	Non-domestic	0	6	0	exchanges, water works, masts
Unresolved	110	Non-domestic	0	0	110	no named feature within 80 m
Total	**499**		119	194	186	

Ibri number. 342 of the 499 accounts sit in 52 clusters; the two largest hold 20 and 19 accounts (the Bawadi shopping centre and the Al Murtafa souq). 205 accounts are commercial and 135 governmental, educational, medical or religious. 110 are unresolved and carried as non-domestic.

> **What a wrong category costs.** The non-domestic and governmental water of a settlement is a pool set by its population, not by its meters (Chapter 2 and the load chapters). A large-consumer account in the wrong category therefore moves water from one plot to another inside the settlement; it does not change the total. The exception is a farm pump, which leaves the pool, and an estate meter, which the estate's workforce replaces.

Older project documents left the large-consumer accounts unresolved, pending a site check. They are now resolved account by account, with the evidence recorded.

Source. *Project rule (engineer, 2026-09-09); G201-p59 §7.3.1 keeps specific identified non-domestic projects outside the ratios, which is why the industrial accounts are separated.*

Where it lives. W14/analysis/CRT_accounts_identified.csv, one row per account (500 rows: the 499 large-consumer accounts and the single Industrial-tariff account), fields USE, GUD_CAT, CONF, EVIDENCE and cl (the cluster; -1 for a single account). The register writes CRT_review for an unresolved account; the meter layer carries it as non_domestic. Working files in W14/analysis/crt_identify/ (classify_crt.py, osm_tiles.py, nominatim.csv). The category reaches each meter through the dictionary GUD_OF_USE in W14/py/plots_meters_load.py, field USE on ELE_meters_on_plots.shp. Counts: facts_w14.crt_summary().

1.3 The cadastre

What it is for. The plot is the unit the load lands on. The network passes plots, and a pipe's flow is the sum of the plots upstream of it, so the plot layer is the load table of the design.

The cadastral layer received in September 2026 holds 77,265 plots. Each has a geometry, a class code and name, and a built flag. 16,756 plots are marked built and 60,509 future. In the file as used, the flag and the meters agree exactly: 0 built plots are without a meter and 0 future plots carry one. The built flag can therefore be read as "has a meter", and it is not an independent check on the meters.

The rule. Trust the geometry and the built flag. Use the class for two things only: the Tourism class, which is the old quarter of Ibri, and the Industrial class inside the two estates, which carries the estate workforce. Derive the use of every other plot from its meters (Chapter 2).

The class field cannot be trusted for use, for three reasons, each visible in the data:

- **No government class.** The classes are Residential, Residential-Commercial, Commercial, Agricultural, Industrial, Tourism and Proposed. There is no government class, yet 409 built plots carry a government majority of meters;
- **Placeholder codes.** 18,053 plots carry the codes 0 and 99 under the name Residential. The codes look like placeholders for an unclassified plot, not a survey of use;
- **Disagreement with the meters.** On the 16,756 built plots the class matches the use derived from the meters on 12,769 (76.2 per cent). The largest disagreements are 1,645 plots called Residential that the satellite shows as groves and 874 called Residential whose meters are two thirds or more shops.

Table 10 Class in the cadastre against the use derived from the meters, built plots

Cadastre class	Built plots	Derived use the same	Per cent	Largest other derived use
Agricultural	268	37	14	Government (77)
Commercial	160	90	56	Residential-Commercial (34)
Industrial	246	207	84	Commercial (35)
Proposed	10	0	0	Commercial (5)
Residential	15,573	12,309	79	Agricultural (1,645)
Residential-Commercial	499	126	25	Commercial (245)

Ibri number. 76.2 per cent agreement on the built plots, 18,053 plots on placeholder codes, and no government class at all.

Source. *G201-p58 §7.2.2 names the Ministry of Housing and Urban Planning as the official source of plot information and asks for plots "classified by clear typologies". The cadastre supplies the plots; the typology is derived (project rule, engineer, 2026-09-09).*

Where it lives. W14/shp/PLOTS_load.shp, fields Name, Moh_Classi, Classes and Buiding_St as received, AREA_M2 added. The comparison of class and derived use per plot is the field AGREE in W14/analysis/plot_class_audit.csv (same, differs, new, unmetered).

1.4 The settlements and the population series

What it is for. Every rate in the load chain is a property of a settlement: the occupancy, the properties per plot, the share of empty land that becomes housing, the growth rate and the year the land is full. The series supplies two things and nothing else: each settlement's population in 2024, which the occupancy is measured against, and its growth rate.

The Inception Report supplied 25 settlement outlines and a population series for each settlement, year by year from 2023 to 2100. The outlines are drawn around the built cores. They do not touch one another and they leave the land between them unassigned. The project boundary, 531.4 km², holds plots far outside every outline.

The rule. The series is built in three parts. To 2040 it is the forecast of the National Centre for Statistics and Information for the Wilayat of Ibri, Omani and expatriate population separately. From 2041 to 2050 it is an extrapolation of that forecast, documented in the technical note on population issued with the Inception Report. From 2051 the rate rises from 2.21 per cent to 2.40 per cent a year, reached in 2058, and holds at 2.40 to 2100, the planning horizon instructed by Nama Water Services. The wilayat total is then split between the settlements in fixed shares taken from the census.

$$P_s(y) = k_s \times P_W(y),\ k_s = \frac{P_{s,\text{census}}}{P_{W,\text{census}}} \tag{2}$$

Symbol	Meaning	Unit
P s(y)	population of settlement s in year y	persons
P W(y)	population of the Wilayat of Ibri in year y	persons
k s	settlement's share of the wilayat, fixed at the census	—

Because every share is fixed, every settlement grows at the same annual rate as the wilayat. The series says nothing about any one settlement's own momentum; it is used for the rate only.

Table 11 Mean annual growth rate of the series, the 25 settlements

Period	Mean rate, per cent a year	Basis of the series
2024 to 2030	2.56	national forecast
2030 to 2040	2.51	national forecast
2040 to 2050	2.05	extrapolation of the forecast
2050 to 2060	2.33	ramp from 2.21 in 2051 to 2.40 by 2058
2060 to 2100	2.40	constant rate instructed by Nama Water Services

Ibri number. The wilayat holds 183,564 people in 2024. The 25 settlements in the project boundary hold 116,452, which is 63.4 per cent of it. Ibri itself holds 67,106, a share of k = 0.366. The series' own total in 2100 is not a saturation figure and is not used as one.

The series beyond 2050 is not data. G201-p58 does not recommend extrapolating more than ten years beyond the available forecast. The national forecast ends in 2040, so the series is inside that limit to 2050 and outside it after. The design therefore defends its saturation population on the capacity of the land, counted plot by plot, and uses the series only for the pace at which that land fills.

Each plot is assigned to the settlement whose outline contains its centroid or, where none does, to the nearest outline. The outlines are then redrawn as a partition of the whole boundary, so that every plot lies in exactly one settlement and the settlements meet without gaps. The two outlines of Al Aynayn are treated as one settlement.

Table 12 The 25 settlements: census population 2024, domestic properties and plots

Settlement	Census population 2024	Share of the 25, %	Domestic properties on plots	Built plots	Empty plots	Under 1,000 people
Ibri	67,106	57.6	11,052	7,664	14,203	
Ad Dariz	11,850	10.2	2,439	2,063	4,888	
Al Araqi	10,696	9.2	2,466	1,819	3,360	
Al Aynayn	4,554	3.9	889	735	1,780	
Al Wahrah	3,351	2.9	716	578	2,000	
At Tayyib	3,348	2.9	833	859	10,012	
Al Jibayyah	2,686	2.3	524	422	2,784	
Bat	2,557	2.2	418	355	879	
Ad Dibayshi	2,411	2.1	681	579	2,430	
Tanam	2,116	1.8	403	350	2,045	
Suwayda al Ma	1,559	1.3	301	223	1,593	
Hijar	1,033	0.9	352	223	375	
Al Ghubayrah	631	0.5	161	140	1,448	yes
Al Qurayn	549	0.5	147	142	2,051	yes
Sayh al Masarrat	466	0.4	457	174	1,745	yes
Al Jahli	415	0.4	344	156	1,235	yes

Settlement	Census population 2024	Share of the 25, %	Domestic properties on plots	Built plots	Empty plots	Under 1,000 people
Satwah	263	0.2	27	25	181	yes
Al Akheedar	198	0.2	120	42	120	yes
Al Qali	191	0.2	56	34	220	yes
Shalashil	130	0.1	38	32	3,686	yes
Al Makhtibyah	122	0.1	91	97	1,523	yes
Usaybuq	91	0.1	21	18	60	yes
Wadi al Mankas	52	0.0	8	9	1,580	yes
Ash Shiab	47	0.0	11	11	65	yes
Miayrid	34	0.0	4	6	246	yes
Total	**116,452**	**100.0**	**22,559**	**16,756**	**60,509**	**13**

The 13 settlements of fewer than a thousand people have too few meters to measure a rate on. They are flagged here and take fixed rates in the occupancy chapter.

Source. *G201-p58 §7.2.1: the NCSI is the official source; a wilayat forecast is distributed to settlements in proportion to the latest census; extrapolation beyond ten years of the forecast is not recommended. The 2.40 per cent rate from 2051 is an instruction of Nama Water Services recorded in the Inception Report. Plot-to-settlement assignment and the partition: project rule (engineer, 2026-09-09 and 2026-09-10).*

Where it lives. _CLIENT/Ibri Sewer Demand R0 2026 08 03.xlsx, sheets Pop_Wilayat (the wilayat series) and Project Pop Settlements (the 25 settlements, columns Pop 2023 to Pop 2100). Plot to settlement: W14/py/plots_meters_load.py, field SETTLE on PLOTS_load.shp. Partition: W14/py/settlements_partition.py (Voronoi of the plot centroids, dissolved by settlement, clipped to the boundary, shared edges simplified at 25 m), output W14/shp/Settlements_merged.shp. Rates: facts_w14.growth_rates(); the settlement rows: facts_w14.settlement_table().

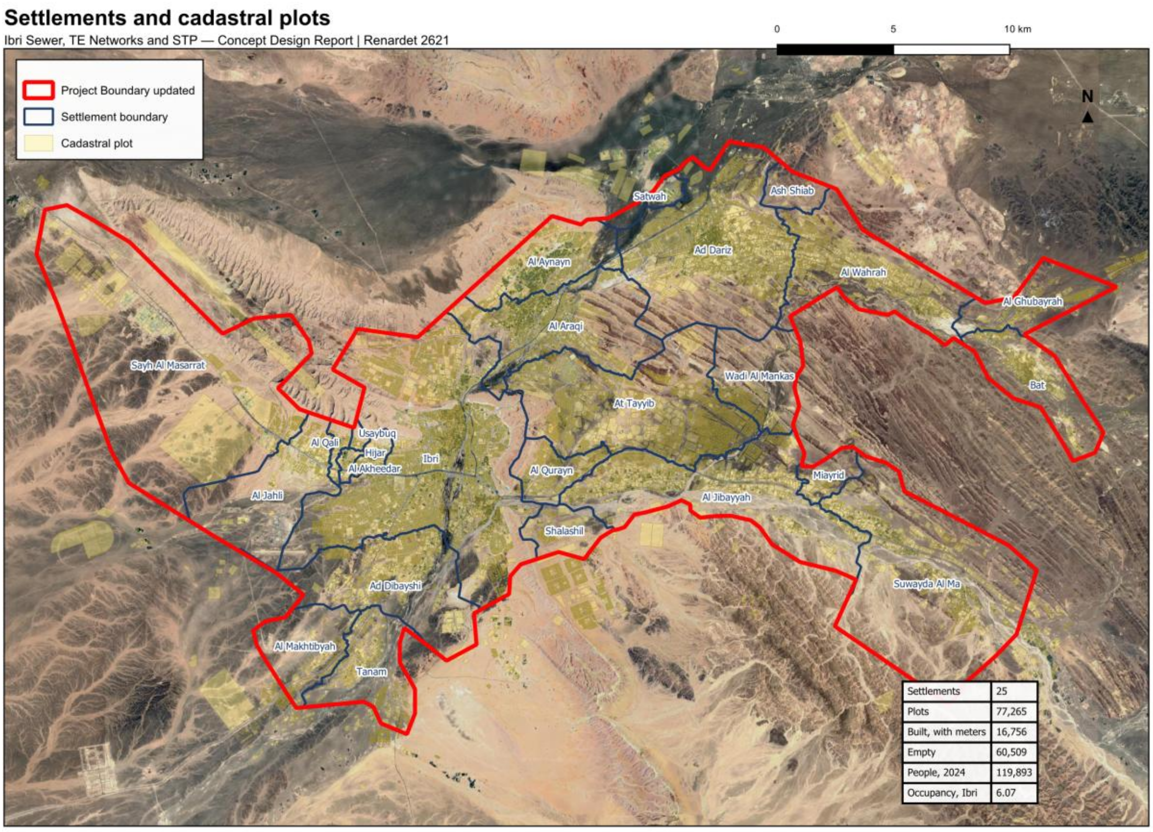

Figure 3 The 25 settlements redrawn as a partition of the project boundary, over the plots.

1.5 The identified projects

What it is for. The guideline's population ratios cover the ordinary shops, offices and public buildings of a town. They exclude economic zones and special consumers such as labour camps and high-water industry, which must be found and added separately. None of these is visible as such in the meters, so the public record was searched for them.

The rule. Each site found is placed in one of three treatments: an identified project inside the boundary whose load is added to its plots; a site recorded but given no load until its data exists; or a source outside the boundary that can only reach the plant by tanker.

Table 13 Identified projects and special consumers

Site	What the data shows	Treatment
Al Tayyeb industrial area	93.8 ha footprint; 280 plots inside it, 202 of them Industrial in the cadastre; 1,035 meters on the Commercial tariff, 2 large-consumer	special: 4,500 workers at 93 l/d
Tanam industrial area	61.1 ha footprint; 185 plots, 105 Industrial; 409 Commercial-tariff meters, 13 large-consumer	special: 1,800 workers at 93 l/d
Army camp, Northern Frontier Regiment	296 ha military land use, no meter in the dataset	recorded; not connected; occupancy to come from the Ministry of Defence
Ibri View resort, As Sulayf	announced, about 2 km², not in the cadastre	recorded; no load until its masterplan is issued
Madayn Ibri Industrial City	outside the boundary, 319 ha footprint, first phase on a collection tank	possible tanker source to the plant
Power station and solar plants	outside the boundary, with a construction camp	possible tanker source to the plant

The two estates were hidden under the Commercial tariff, which is why the tariff alone never showed an industrial area. Their workforce is an assumption made from the meter count, since no employment figure was supplied. It is spread over the estate's Industrial plots by area, and the meters inside the estates are taken out of the non-domestic pool so the estate water is not counted twice.

$$W_{\text{spec,i}} = 93 \times N_{\text{w}} \times \frac{A_i}{\sum_j A_j} \tag{3}$$

Symbol	Meaning	Unit
W spec,i	special water on Industrial plot i of an estate	l/d
93	dry industry, per employee (G201-p61 Table 12)	l/d
N w	assumed workforce of the estate	persons
A i, A j	area of plot i; areas of all the estate's Industrial plots	m²

Ibri number. 4,500 + 1,800 = 6,300 workers × 93 l/d = 585.9 m³/d of water, and × 0.54 returned = 316.4 m³/d of sewage. The plot layer sums to 316.4 m³/d. The dwellings on estate plots (62 and 67 domestic meters) still carry their people at the domestic rate.

Special meters also stand on 2 plots outside both estates, one in Ibri (fid 29186) and one in Sayh al Masarrat (fid 70911). They make their plots industrial in Chapter 2, but they carry no water of their own: they are not in the non-domestic pool and they have no workforce.

Source. *G201-p59 §7.3.1: the ratios do not apply to specific identified non-domestic projects such as economic zones, which are determined case by case. G201-p61 §7.3.4: special consumption is not covered by the population forecast. G201-p61 Table 12: dry industry 93 l/d per employee; army camps 185 l/d per occupant. G201-p71 Table 19: non-domestic return 54 per cent. Workforce 4,500 and 1,800: outside assumption, tagged (engineer, 2026-09-09). Army camp, resort and tanker sources: recorded only (engineer, 2026-09-09).*

Where it lives. W14/analysis/IDENTIFIED_PROJECTS.md (every site, its source and confidence); footprints in W14/shp/Identified_projects_OSM.shp. On the plots: fields ESTATE, WORKERS, W_SPEC and S_SPEC of PLOTS_load.shp, the workforce written by W14/py/plots_meters_load.py and rescaled to the stated totals in W14/py/plot_class_v2_apply.py. Constants facts_w14.WORKERS and facts_w14.L_IND.

1.6 Check it yourself

1. Open ELE_meters_on_plots.shp and count the points by TARIFF; the thirteen counts must match the tariff table and sum to the total.
2. Count the same layer by GUD and compare with the category table: the tariff side plus the large-consumer side must give each category total.
3. Filter CRT_accounts_identified.csv on USE = Unresolved and confirm every row has CONF = Guessing and no feature within 80 m in EVIDENCE.
4. In PLOTS_load.shp, cross-tabulate Classes against DERIVED on the built plots; the diagonal is the agreement quoted in Section 1.3.
5. Sum Pop 2024 over the 25 settlements in the demand workbook and divide by the wilayat's 2024 value in Pop_Wilayat; the share must match Section 1.4.
6. Sum WORKERS over the plots with ESTATE = AL TAYYEB and = TANAM; each must return the stated workforce, and S_SPEC must sum to the estate sewage in Section 1.5.

2 From meter to plot, and the use of every plot

This chapter does two things. It puts every meter on a plot, so that each plot carries a count of its meters in every category. It then reads the main use of every plot from those counts, in a fixed order of rules, with the satellite deciding where the plot is a grove. The first step sets the load of every built plot. The second does not, and it is worth being clear about that before the rules.

2.1 Meter to plot

What it is for. A meter's load reaches the network only through a plot. Every meter must therefore sit on exactly one plot, or be recorded as carrying nothing.

The rule. Each meter is tested in this order:

1. a meter inside a plot belongs to that plot;
2. a meter inside no plot, on a road or in a gap, is moved to the nearest plot within 15 m (the five nearest plots are tested and the closest is taken);
3. a meter further than 15 m from any plot stays free. It is listed with its nearest settlement and carries no load in the plot table.

Each plot then counts its meters twice: by tariff group, as received, and by category, with the large consumers placed by their use. The domestic count is the property count of the plot.

$$G_{\mathrm{DOM}} = N_{\mathrm{DOM}} + N_{\mathrm{DOMADD}} \tag{4}$$

Symbol	Meaning	Unit
G DOM	domestic meters on the plot: its properties	—
N DOM	meters on the primary and subsidised primary tariffs	—
N DOMADD	meters on the Additional Account Tariff	—

Table 14 The meter counts every plot carries

Field	Counts	Includes
N_DOM, N_DOMADD	tariff groups, domestic	primary and subsidised; additional
N_COM, N_GOV	tariff groups	Commercial with Fisheries and Tourism; Government with MOD
N_AGR, N_CRT, N_IND	tariff groups	Agricultural; the three CRT tariffs; Industrial
G_DOM	category, domestic	N_DOM + N_DOMADD
G_NDOM	category, non-domestic	N_COM + large consumers found commercial, utility or unresolved
G_GOV	category, governmental	N_GOV + large consumers found government, education, health, religious
G_SPEC	category, special	N_IND + large consumers found industrial
G_AGR	category, agricultural	N_AGR + large consumers found agricultural

Ibri number. Of the 33,971 meters, 30,931 fall inside a plot, 2,914 were moved to a plot within 15 m and 126 stay free: 30 domestic, 44 non-domestic, 47 governmental and 5 agricultural. The plots therefore hold 22,589 − 30 = 22,559 domestic properties.

The free dwellings are a stated gap, not a rounding. The 30 free domestic meters are real dwellings whose people are not in the plot table. They are too few to move a pipe size, but whether to snap them regardless of distance is an open decision, and the total is stated rather than hidden.

Older project documents left meters outside a plot unassigned until the corrected cadastre arrived, so that a moved meter would not shift load between streets. The corrected cadastre has arrived, and the 15 m rule now applies.

Source. *Project rule (engineer, 2026-09-09). G201-p58 §7.2.1 itself allows population to be distributed within a settlement "on a pro-rata basis using the number of electricity accounts"; the method goes one step finer, to the plot.*

Where it lives. W14/py/plots_meters_load.py (run inside QGIS), constant SNAP_M = 15.0 m; field PLACE (inside, snapped, free) and PLOT_FID on W14/shp/ELE_meters_on_plots.shp; the counts N_* and G_* on W14/shp/PLOTS_load.shp. The layer also has a field SNAP_M for the distance moved, but it reads 0 on all 2,914 moved meters (0 non-zero): the distance was not written, and the 15 m limit is enforced only by the search radius. Counts: facts_w14.meter_counts().

2.2 What the use of a plot does, and what it does not

What it is for. The use class answers questions the meters alone cannot: which built plots are plain homes, so that properties per plot and the home share can be measured on them; and which empty plots can ever become homes or receive growth. It also draws the land-use map.

The use class does not set the load of a built plot. The people come from the domestic meters, the non-domestic water from the shop meters and the governmental water from the government meters, whatever the plot is called. A house in a palm grove is classed as a farm, and its dwellings still carry their people. The only exception is the estate, which is decided by the footprint and not by the class: an estate plot carries its workforce and its dwellings, and its shop meters leave the pool.

Ibri number. The plots classed as farms carry 13,673 people and 2,075 m^3/d of sewage in 2024, because 1,454 of them hold dwelling meters. Reclassifying all of them would not change today's flow by one litre; it would change only the samples and the exclusions described above.

2.3 The land-use rules, in their fixed order

What it is for. One class per plot, set the same way everywhere, so that the samples used later (the pure home plots) and the exclusions (groves, estates, heritage) are reproducible.

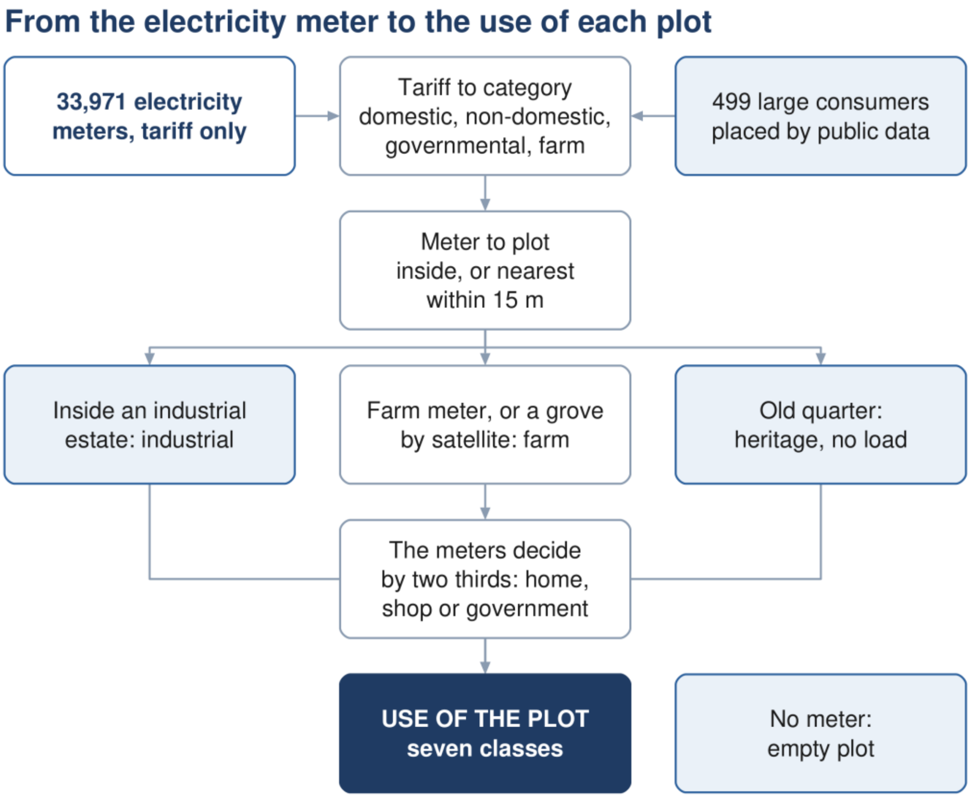

Figure 4   How the use of each plot is derived from the electricity meters, the satellite and the plot itself.

The proportion rules work on the meters that carry people or water. Farm and special meters are kept out of the denominator, so a farm pump does not dilute the home share of a house.

$$N_t = G_{DOM} + G_{NDOM} + G_{GOV}, \quad f_k = \frac{G_k}{N_t} \tag{5}$$

Symbol	Meaning	Unit
N t	meters on the plot that carry people or water: domestic, non-domestic, governmental	—
f k	share of those meters in category k (dom, nd, gov)	—
G k	meters of category k on the plot	—
G SPEC, G AGR	special and agricultural meters; they enter only rules 3 and 6	—

The rule. The first rule that holds sets the class. The order matters: a rule lower down never sees a plot a rule above has taken.

Table 15 The land-use rules in the order they are applied, and the plots each one sets

Rule	Test	Class	WHYC	All plots	Built plots
1	centroid inside the Al Tayyeb or Tanam footprint	Industrial	EST	465	261
2	cadastre class Tourism (the old quarter of Ibri)	Heritage	HER	177	0
3	any agricultural meter, G_AGR > 0	Agricultural	AGR	489	489
4	a grove by satellite (Section 2.4), unless f_nd ≥ 2/3 or f_gov > 1/2	Agricultural	GRN	2,910	1,196
5	no domestic, non-domestic, governmental or special meter	Unmetered (empty)	UNM	58,414	0
6	G_SPEC > 0 and G_SPEC ≥ G_DOM	Industrial	IND	2	2
7	N_t = 0 (a guard; cannot hold after rules 5 and 6)	Unresolved	UNR	0	0
8	f_dom > 2/3	Residential (home)	RES	12,464	12,464
9	f_gov ≥ 2/3	Government	GOV	319	319
10	f_nd ≥ 2/3	Commercial (shop)	COM	1,321	1,321
11a	none of 8 to 10, and G_NDOM ≥ G_GOV	Residential-Commercial (home and shop)	RC	614	614
11b	none of 8 to 10, and G_NDOM < G_GOV	Government	RG	90	90
	Total			**77,265**	**16,756**

Four features of the order carry the method:

- **Estate first.** The estate footprint comes first, so a house or a shop inside an estate is industrial land. Its people still count; its shop meters leave the pool (Section 1.5).
- **Farm by meter, then by satellite.** A farm meter beats every meter mix, because the pump proves the land is irrigated. The satellite grove is weaker evidence and gives way to a clear shop majority (two thirds) or a simple government majority, since a school or a clinic often stands among palms.
- **The strict inequality.** A plot is a home only if more than two thirds of its meters are domestic. Government and shop take two thirds or more. A plot with exactly two thirds dwellings is therefore mixed, not a home, and the home sample stays pure.
- **Mixed plots.** Between the thresholds the shop and government meters decide: a tie goes to home and shop.

Ibri number. Three meter mixes that occur in the data, outside the estates and without farm or special meters:

Table 16 Three meter mixes and the class each takes

Meters on the plot	N t	Shares	Rule	Class	Built plots with this mix
2 domestic, 1 shop	3	f_dom = 0.667, f_nd = 0.333	11a	Residential-Commercial	42 RC, 7 GRN
3 domestic, 1 shop	4	f_dom = 0.750	8	Residential	20 RES, 1 GRN
2 domestic, 1 government	3	f_dom = 0.667, f_gov = 0.333	11b	Government	16 RG, 8 GRN

The plots of the same mix that carry the code GRN are groves: rule 4 took them before the proportion was read. The third row shows what rule 11b does to a small house with a government meter: the plot is classed Government, while its two dwellings keep their people.

Rule 6 fires on 2 plots, the ones with special meters outside the estates: each has no dwelling and one special meter, so G_SPEC ≥ G_DOM holds.

Source. *Project rules: farm first and the two-thirds proportions (engineer, 2026-09-09); the government majority over the satellite grove, the heritage quarter and the satellite test (engineer, 2026-09-10). The guideline asks for plots "classified by clear typologies" (G201-p58 §7.2.2) and gives no rule for deriving them; the thresholds are the project's.*

Where it lives. W14/py/plot_class_v2_apply.py, function classify(r); fields DERIVED (the class) and WHYC (the rule code) on W14/shp/PLOTS_load.shp. The first, naive class and the dropped tests are kept for audit in W14/analysis/plot_class_audit.csv (DERIVED1, AGREE, GREEN, GREEN_IMG, VEGFRAC, HIGH).

2.4 The satellite grove test

What it is for. The farm meter alone under-counts farms. Of the 489 built plots with a farm meter, 186 (38 per cent) show no crop: they are pump sites, and the grove the pump waters lies on a neighbouring plot that carries only a house meter, or none. The satellite finds those groves.

The test uses one cloud-free Sentinel-2 scene of 9 September 2026, red and near-infrared bands at 10 m, from the public archive. One scene is enough because the date palm is evergreen: a grove reads green in any month, and a dry-season scene keeps seasonal crops from passing for groves.

The rule. The vegetation index is computed per pixel from the surface reflectance of the two bands.

$$\mathrm{NDVI} \;=\; \frac{\rho_{\mathrm{NIR}} - \rho_{\mathrm{red}}}{\rho_{\mathrm{NIR}} + \rho_{\mathrm{red}}}, \qquad \rho \;=\; \max\left(0, \frac{\mathrm{DN} - 1000}{10000}\right) \tag{6}$$

Symbol	Meaning	Unit
ρ NIR, ρ red	surface reflectance, band B08 (near-infrared) and band B04 (red)	—
DN	stored value of the Level-2A product, which carries an offset of 1,000	—
NDVI	normalised difference vegetation index	—

The offset matters and the scale does not: the 10,000 cancels in the ratio, but without subtracting the 1,000 a dark grove reads an index above one. Each plot then gets three numbers over the pixels that touch it: the mean index, the share of pixels at 0.30 or more, and the green area.

$$A_{\mathrm{g}} \;=\; s \,\times\, \min(A, 100n) \tag{7}$$

Symbol	Meaning	Unit
A g	green area of the plot (GREEN_M2)	m²
s	share of the plot's pixels with NDVI ≥ 0.30 (NDVI_SHARE)	—
A	plot area (AREA_M2)	m²
n	pixels touching the plot; each covers 100 m²	—

A plot is a grove where either condition holds:

- **Large grove.** A green area of at least 1,000 m² at a mean index of at least 0.20; this catches a large orchard even when part of the plot is bare or built;
- **Small grove.** A plot of at least 800 m² that is at least 60 per cent green at a mean index of at least 0.40; this catches a small, dense garden that never reaches 1,000 m² of green.

The thresholds were set on two groups of built plots: the plots with a farm meter as the positives, and the plots whose meters are more than two thirds domestic and carry no farm meter as the negatives. Mean index and green area were swept against both groups, then the result was checked by eye on contact sheets of aerial thumbnails. An earlier test on the colour of the aerial imagery was tried and dropped, because it put farms on bare ground.

Table 17 What the index reads on the ground, and how the test separates the groups

Group of plots	Plots	Median mean NDVI	Called a grove by the test
Empty plots that fail the test (bare desert)		0.072	—
Built plots classed as homes	12,464	0.075 (90th percentile 0.206)	none, by construction
Built plots with a farm meter (positives)	489	0.374	303 (62 %)
Built plots, more than two thirds domestic, no farm meter	13,603		1,139 (8.4 %)
Plots classed as groves by rule 4	2,910	0.588	all

Ibri number. Of the 2,910 groves, 1,613 pass both conditions, 937 only the large-grove one and 360 only the small-grove one. Two of them, each the median-sized plot of its kind:

Table 18 Two groves worked through the test

	Large grove only	Small grove only
Plot (fid), settlement	21669, Ibri	1336, Al Qurayn
Area A, m²	6,959	999
Share s at NDVI ≥ 0.30	0.302	0.800
A g = s × A, m²	2,102 (stored 2,102)	800 (stored 800)
Mean NDVI	0.271	0.761
Large grove: A g ≥ 1,000 and mean ≥ 0.20	yes	no
Small grove: A ≥ 800, s ≥ 0.60, mean ≥ 0.40	no	yes
Dwelling meters; people carried	1; 6.1	1; 4.0

1,196 of the groves are built plots and 1,714 are empty. The empty ones matter for the future: a grove is never counted as a future home and never receives growth. The override kept 44 green built plots out of the farm class: 27 with a shop majority and 17 with a government majority. 1,139 home-majority plots became groves; they leave the home sample, and their people stay in the load.

Source. *Project calibration, not a guideline value (engineer, 2026-09-10). Sentinel-2 Level-2A, tile 40QDL, scene S2B_40QDL_20260909, bands B04 and B08 at 10 m, read from the public cloud archive.*

Where it lives. W14/py/ndvi_plots.py (the index, the per-plot statistics and the calibration sweeps it prints); the index raster is saved under Hydraulic/Imagery, outside the repository, and never pushed. Per plot: W14/analysis/ndvi_plots.csv, and fields NDVI_MEAN, NDVI_SHARE and GREEN_M2 on PLOTS_load.shp. The test is applied as the flag GREEN in W14/py/plot_class_v2_apply.py (kept in plot_class_audit.csv); the contact sheets come from W14/py/ndvi_contact_sheet.py. facts_w14.farm_bare_share() repeats the test.

2.5 The use of every plot

What it is for. The class table is the input to the later chapters: the home plots give the properties per plot and the home share, and the exclusions decide which empty land can fill.

Table 19 Use of every plot, built and empty

Use	Plots	Per cent of built	Rule (code, built plots)
Home	12,464	74.4	8 (RES)
Farm	1,685	10.1	3 (AGR 489), 4 (GRN 1,196)
Shop	1,321	7.9	10 (COM)
Home and shop	614	3.7	11a (RC)
Government	409	2.4	9 (GOV 319), 11b (RG 90)
Industrial	263	1.6	1 (EST 261), 6 (IND 2)
Built plots with meters	**16,756**	**100.0**	
Heritage, the old quarter	177		2 (HER); no meter, no load
Empty, no meter	58,414		5 (UNM)
Empty grove	1,714		4 (GRN); excluded from future homes
Empty, inside an estate	204		1 (EST); excluded from future homes

Ibri number. 77,265 plots: 16,756 built with meters, of which 74.4 per cent homes and 10.1 per cent farms. 12,443 of the homes have fewer than 15 dwelling meters and form the pure home sample used for properties per plot.

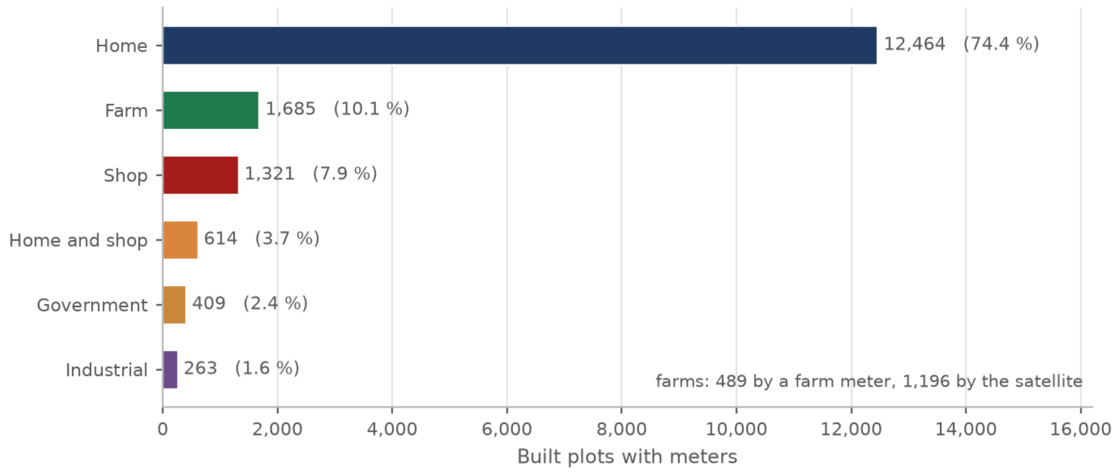

Figure 5 Use of the built plots that carry meters, as derived. Farms are found by the meter or by the satellite.

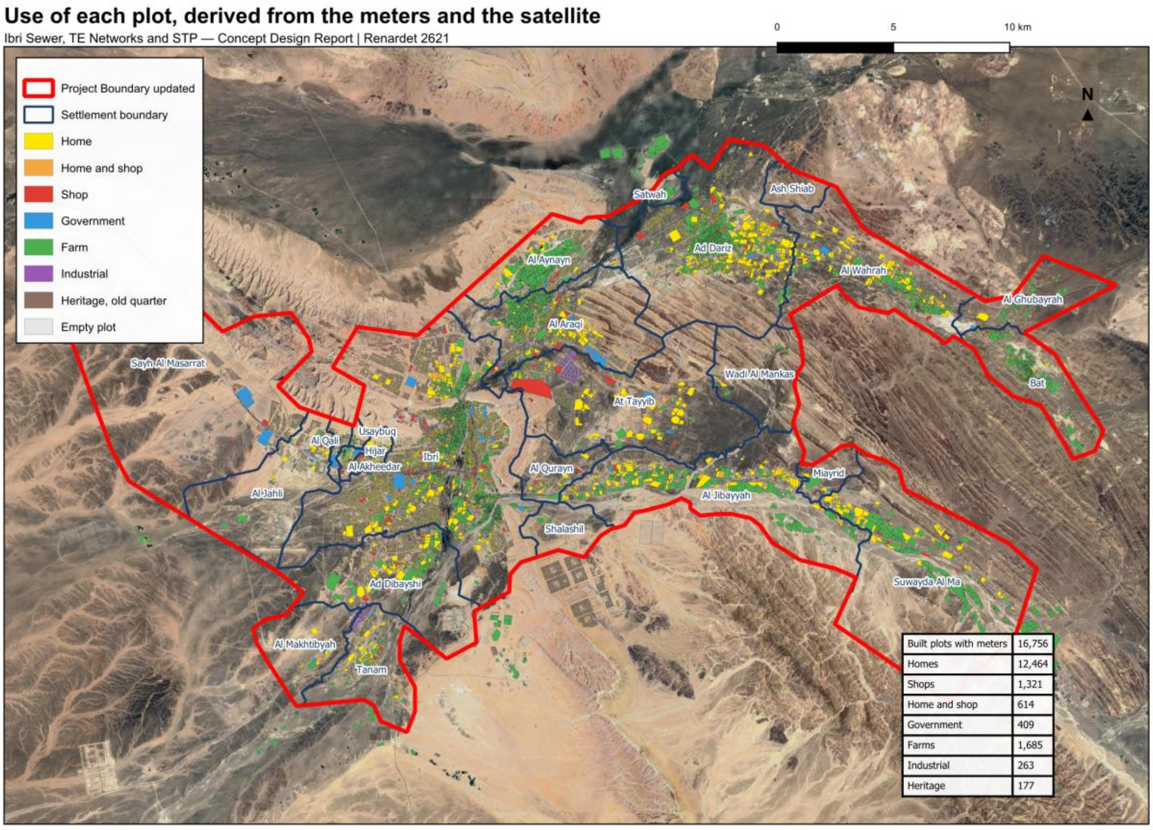

Figure 6 Use of each plot derived from the meters and the satellite. Empty plots are shown in grey.

Source. *The rules of Section 2.3 and the test of Section 2.4.*

Where it lives. field DERIVED on W14/shp/PLOTS_load.shp; counts facts_w14.plot_summary() (keys classes, farms_by, heritage, metered, pure_home_plots). The map is the QGIS layout behind report map M07; the chart is built by W14/report/charts_w14.py.

2.6 Check it yourself

1. Count ELE_meters_on_plots.shp by PLACE; inside, snapped and free must sum to the total. Select the free meters and confirm none lies within 15 m of a plot.
2. Sum G_DOM over PLOTS_load.shp; it must equal the domestic meters less the free domestic ones.
3. Pick any built plot, read G_DOM, G_NDOM, G_GOV, G_SPEC, G_AGR, NDVI_MEAN, NDVI_SHARE, GREEN_M2 and AREA_M2, walk the rules of Section 2.3 in order, and compare the class you reach with DERIVED and the rule with WHYC.
4. Filter WHYC = GRN and G_DOM > 0: these are houses in groves. Their POP is not zero, which confirms that the class does not remove people.
5. Filter G_AGR > 0 on built plots and count those that fail both grove conditions; the share must match the pump-site share in Section 2.4.
6. Count DERIVED on the built plots; the counts must match the class table and the data box of the land-use map.

3 Properties per plot and the occupancy rate

This chapter turns the domestic electricity meters into people. It needs two numbers per settlement. The occupancy rate converts a property into persons, and it sets the population of 2024 on every built plot. The number of properties on a home plot is measured here too, although it is not used for today's population: the meters already count today's properties directly. It is used in Chapter 4, to say how many properties a future home plot will carry.

Both numbers are set per settlement. T03 Revision 01 and Revision 1 of the concept report used one occupancy rate, 5.32, for the whole study area; the rate per settlement described here replaces it, and every population and flow in this tutorial follows from it.

3.1 One domestic meter is one property

What it is for. A property is the unit the occupancy rate multiplies. Counting it from the meters gives the number of dwellings on every plot without assuming a plot typology.

The rule. Every meter on a domestic tariff is one property: the primary account, the primary account with the national subsidy, and the additional account. An additional account is a second dwelling on the same plot, so it counts as a property of its own. Shop, government, farm, industrial and large-consumer meters carry no people.

Source. The guideline accepts electricity accounts as the basis for distributing population below settlement scale (G201-p58, §7.2.1). Counting the domestic accounts as properties is the project's substitute for NCSI housing units, declared as a departure in Section 3.3.

Worked for Ibri. Ibri holds 11,052 domestic properties. Over the study area 22,589 meters are domestic; 22,559 of them lie on a plot or within 15 m of one and are carried in the plot table. The other 30 are among the 126 meters that lie further from any plot and carry no load.

Where it lives. W14/py/plots_meters_load.py assigns each meter to a plot; field G_DOM on W14/shp/PLOTS_load.shp is the count of domestic meters, and so of properties, on the plot. Column PROPS of W14/analysis/settlements_today.csv is its sum per settlement.

3.2 Properties per home plot

What it is for. A future home plot has no meter yet. The number of properties it will carry is taken from the built home plots of the same settlement, so this ratio must be measured on plots that look like what a future home plot will become.

The rule. The ratio is measured per settlement on the pure home plots only: built plots whose use is Residential by the meter rule of Chapter 2 (more than two thirds of the meters domestic) and that carry between 1 and 14 domestic meters.

$$p_s = \frac{\Sigma_{\mathrm{pure},s} N_{\mathrm{dom}}}{n_{\mathrm{pure},s}} \tag{8}$$

Symbol	Meaning	Unit
p s	properties per home plot in settlement s	properties per plot
N dom	domestic meters on one pure home plot	count
n pure, s	number of pure home plots in settlement s	count

Why the two exclusions. A plot with 15 or more dwelling meters is a block of flats or workers' housing, not a home plot. There are 37 such plots with 901 dwellings between them, enough to pull the ratio up in the settlement that holds them. A plot that also carries shops, offices or a farm is counted for its people but is not a template for a future home plot. Counting every built plot with a dwelling was tried first and rejected: it gives Ibri 1.60 (11,052 dwellings on 6,898 plots) against the 1.33 of the pure home plots.

Source. G201-p58, §7.2.2: population from plots uses the average number of properties per plot, taken from plots classified by clear typologies. The pure-home-plot definition and the 15-meter limit are a project rule (engineer, 2026-09-09).

Worked for Ibri. 7,782 domestic meters on 5,853 pure home plots: 7,782 ÷ 5,853 = 1.33 properties per home plot. Over the whole study area, 16,190 meters on 12,443 pure home plots give 1.30.

A settlement with fewer than ten pure home plots has no sample to speak of. Its derived ratio is the study-area value, 1.30, and its derived home share (Chapter 4) the study-area value, 0.87. Every settlement that falls back in this way is under 1,000 people, so the rule of Section 3.5 replaces both values anyway.

Table 20 Properties per home plot by settlement, derived and adopted

Settlement	Domestic properties	Pure home plots measured	Derived	Adopted
Ibri	11,052	5,853	1.33	1.33
Ad Dariz	2,439	1,567	1.20	1.20
Al Araqi	2,466	1,282	1.32	1.32
Al Aynayn	889	560	1.14	1.14
Al Wahrah	716	420	1.22	1.22
At Tayyib	833	623	1.09	1.09
Al Jibayyah	524	291	1.26	1.26
Bat	418	239	1.17	1.17
Ad Dibayshi	681	457	1.16	1.16
Tanam	403	196	1.24	1.24
Suwayda al Ma	301	119	1.60	1.60
Hijar	352	193	1.69	1.69
Al Ghubayrah	161	102	1.12	**1.00**
Al Qurayn	147	119	1.14	**1.00**
Sayh al Masarrat	457	133	2.52	**1.00**
Al Jahli	344	112	2.16	**1.00**
Satwah	27	13	1.23	**1.00**
Al Akheedar	120	25	2.12	**1.00**
Al Qali	56	20	2.20	**1.00**
Shalashil	38	27	1.11	**1.00**
Al Makhtibyah	91	62	1.15	**1.00**
Usaybuq	21	12	1.25	**1.00**
Wadi al Mankas	8	6	1.30	**1.00**
Ash Shiab	11	10	1.10	**1.00**
Miayrid	4	2	1.30	**1.00**

Adopted values in bold are set by the rule for settlements under 1,000 people (Section 3.5).

Where it lives. W14/py/plot_class_v2_apply.py, mask pure and series ppp (constant HIGH = 15, MIN_SAMPLE = 10). Columns PPP_RAW (derived), PROPS_PER_BUILT_PLOT (adopted) and DOM_PLOTS_BUILT (plots measured) in W14/analysis/settlements_today.csv; field PPP_S on every plot of PLOTS_load.shp.

3.3 The occupancy rate

What it is for. The occupancy rate converts a property into persons. It is the one number that ties the meter count to the official population.

The guideline's rule. The guideline defines the rate as population over housing units, both from NCSI, from the most recent data and at the geographic scale of the project.

$$\mathrm{OR} = \frac{\text{Population}}{\text{Housing units}} \tag{9}$$

Source. G201-p58, §7.2.2, for the equation; G201-p59 for the requirement that both terms come from NCSI at the scale of the project area.

> **A declared departure.** NCSI publishes housing units at governorate and wilayat level only (G201-p59). A wilayat rate would put the same household size on Ibri town and on a village. The project therefore divides the population of each settlement by the domestic properties counted in it. The departure is listed in the concept report, Revision 2, Section 10.1, and needs NWS concurrence.

The rule used. For each settlement s, the 2024 population of the Inception Report series is divided by the domestic properties counted in the same settlement. 2024 is the year of the electricity accounts, so the two halves of the fraction describe the same year and the same ground.

$$\mathrm{OR}_{\mathrm{der,s}} = \frac{P_{2024,\mathrm{s}}}{N_{\mathrm{dom,s}}} \tag{10}$$

Symbol	Meaning	Unit
OR der, s	occupancy rate derived for settlement s	persons per property
P 2024, s	population of settlement s in 2024, Inception Report series	persons
N dom, s	domestic properties counted in settlement s (Section 3.1)	properties

Worked for Ibri. 67,106 ÷ 11,052 = 6.07 persons per property. Ibri needs neither the floor nor the cap of Section 3.4, so the adopted rate is the derived one.

Where it lives. W14/py/plot_class_v2_apply.py, series or_raw: the workbook sheet Project Pop Settlements, column Pop 2024, over G_DOM summed per settlement. Columns WB_2024, PROPS and OR_RAW in settlements_today.csv; OR_raw in W14/analysis/occupancy_by_settlement.csv.

3.4 The floor and the cap

What it is for. A derived rate can be wrong for reasons that have nothing to do with household size. The floor and the cap keep those errors out of the design.

The rule. The derived rate is bounded below by 4.0 and above by the highest rate derived in any settlement of 2,000 people or more.

$$\mathrm{OR}_{\mathrm{s}} = \min\left(\max\left(\mathrm{OR}_{\mathrm{der,s}}, 4.0\right), \mathrm{OR}_{\mathrm{cap}}\right) \tag{11}$$

$$\mathrm{OR}_{\mathrm{cap}} = \max_{\mathrm{s:\,P} \geq 2{,}000} \mathrm{OR}_{\mathrm{der,s}} \tag{12}$$

Symbol	Meaning	Unit
OR s	occupancy rate adopted for settlement s	persons per property
OR cap	highest derived rate among the settlements of 2,000 people or more	persons per property

Why the floor. In some settlements the domestic meters include the housing of the police headquarters, the college and similar institutions. Their occupants are not census residents, so the census population is small against the meters and the derived rate falls to between one and four. Those dwellings discharge to the sewer whatever the census says. The floor keeps them in the design. Its value, 4.0, is At Tayyib's measured 4.02 at 3,348 people, the lowest rate measured in any settlement whose derived rate is used without a bound.

Why the cap. A rate above the highest one measured on a large sample says that too few meters were found for the census count, not that households are larger. Satwah is the extreme case: 263 people on 27 meters give 9.73; Miayrid's 34 people on 4 meters give 8.49. The cap is Bat's 6.12.

What they change. The floor lifts Ad Dibayshi (3.54) and Hijar (2.93) to 4.0. The settlements above the cap, Satwah (9.73), Wadi al Mankas (6.49), Miayrid (8.49), are all under 1,000 people and take the rule of Section 3.5 instead. No settlement of 1,000 people or more is cut by the cap.

Source. Project rule (engineer, 2026-09-10). The guideline gives no bound on the rate.

Where it lives. plot_class_v2_apply.py: OR_FLOOR = 4.0, OR_CEIL computed from or_raw over the settlements with a 2024 population of 2,000 or more, or_s = or_raw.clip(OR_FLOOR, OR_CEIL). Column rule in occupancy_by_settlement.csv names the bound that applied.

3.5 Settlements under 1,000 people

What it is for. A small settlement has too few meters to measure any of its three rates on. One rule sets all three for it.

The rule. A settlement with fewer than 1,000 people in 2024 takes an occupancy of 4.0, one property per home plot and a home share of 0.9, whatever its meters return.

Why. There are two reasons. The meters return rates on a handful of accounts that swing from 1.02 to 9.73. And a village of that size does not attract second dwellings on a plot, so one property per plot describes its future homes better than a ratio measured on a dozen plots. The occupancy of 4.0 is the floor of Section 3.4. The home share of 0.9 lies within the range measured in the settlements of 1,000 people or more (Chapter 4).

Which settlements. The rule applies to 13 settlements: Al Ghubayrah, Al Qurayn, Sayh al Masarrat, Al Jahli, Satwah, Al Akheedar, Al Qali, Shalashil, Al Makhtibyah, Usaybuq, Wadi al Mankas, Ash Shiab, Miayrid. Sayh al Masarrat and Al Jahli are among them although their meters say 457 and 344 properties: the census counts 466 and 415 residents there, because the police headquarters housing and the college housing are not census residents. Their derived rates, 1.02 and 1.21, would have been floored to 4.0 in any case.

Source. Project rule (engineer, 2026-09-10).

Where it lives. plot_class_v2_apply.py, SMALL_POP = 1000 and the series small; column SMALL in settlements_today.csv; the derived values stay beside the adopted ones in the columns OR_RAW, PPP_RAW and HOME_SHARE_RAW.

3.6 The population in 2024

The rule. The people on a built plot are its domestic properties times the adopted occupancy of its settlement. The settlement population is the sum over its plots.

$$P_{\text{plot}} = N_{\text{dom}} \times \text{OR}_{\text{s}} \quad (13)$$

Worked for Ibri. A plot with four domestic meters in Ibri carries 4 × 6.07 = 24.3 people. Ibri as a whole carries 11,052 × 6.0718 = 67,106 people, the census figure, because no bound applies to it.

Table 21 Occupancy rate by settlement: derived, adopted, and the people it gives in 2024

Settlement	Domestic properties	Census 2024	Rate derived	Rate adopted	Rule	People 2024	Difference
Ibri	11,052	67,106	6.07	6.07	derived	67,106	0
Ad Dariz	2,439	11,850	4.86	4.86	derived	11,850	0
Al Araqi	2,466	10,696	4.34	4.34	derived	10,696	0
Al Aynayn	889	4,554	5.12	5.12	derived	4,554	0
Al Wahrah	716	3,351	4.68	4.68	derived	3,351	0

Settlement	Domestic properties	Census 2024	Rate derived	Rate adopted	Rule	People 2024	Difference
At Tayyib	833	3,348	4.02	4.02	derived	3,348	0
Al Jibayyah	524	2,686	5.13	5.13	derived	2,686	0
Bat	418	2,557	6.12	6.12	derived	2,557	0
Ad Dibayshi	681	2,411	3.54	4.00	floor	2,724	+313
Tanam	403	2,116	5.25	5.25	derived	2,116	0
Suwayda al Ma	301	1,559	5.18	5.18	derived	1,559	0
Hijar	352	1,033	2.93	4.00	floor	1,408	+375
Al Ghubayrah	161	631	3.92	4.00	under 1,000	644	+13
Al Qurayn	147	549	3.73	4.00	under 1,000	588	+39
Sayh al Masarrat	457	466	1.02	4.00	under 1,000	1,828	+1,362
Al Jahli	344	415	1.21	4.00	under 1,000	1,376	+961
Satwah	27	263	9.73	4.00	under 1,000	108	−155
Al Akheedar	120	198	1.65	4.00	under 1,000	480	+282
Al Qali	56	191	3.41	4.00	under 1,000	224	+33
Shalashil	38	130	3.42	4.00	under 1,000	152	+22
Al Makhtibyah	91	122	1.34	4.00	under 1,000	364	+242
Usaybuq	21	91	4.33	4.00	under 1,000	84	−7
Wadi al Mankas	8	52	6.49	4.00	under 1,000	32	−20
Ash Shiab	11	47	4.27	4.00	under 1,000	44	−3
Miayrid	4	34	8.49	4.00	under 1,000	16	−18
Total	**22,559**	**116,452**	**5.16**	**5.31**		**119,893**	**+3,441**

In the total row the derived rate is the census population over all properties and the adopted rate is the design population over all properties.

What the bounds cost. The floor and the small-settlement rule put 3,643 people above the census in 10 settlements and 203 below it in 5 of the smallest, which the rule brings down to 4.0. The net is 3,441 above the 116,452 of the census series, giving 119,893 people in 2024. The census over the properties of the study area gives 5.16 persons per property; the design population gives 5.31. The meters flush whether the census counts their occupants or not, so the design keeps the higher figure.

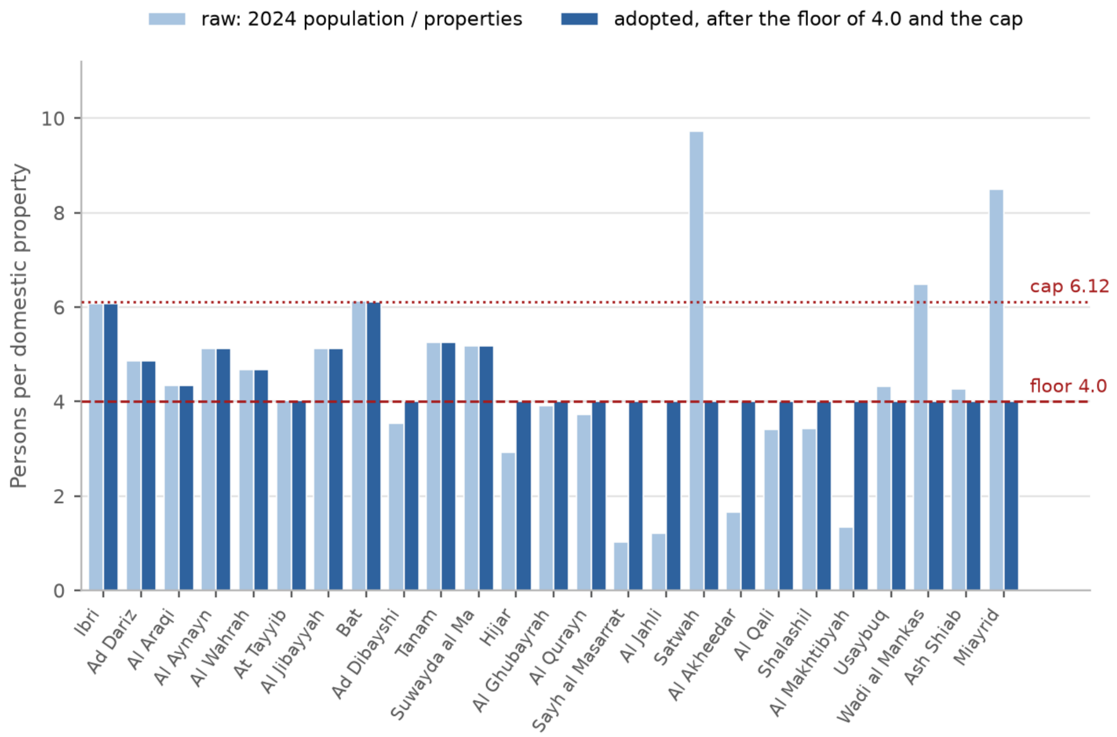

Figure 7   Occupancy by settlement: the rate derived from the 2024 population and the domestic properties, and the rate adopted after the floor of 4.0, the cap and the rule for settlements under 1,000 people.

Where it lives. Field OR_S (the adopted rate, four decimals) and field POP (people in 2024, = G_DOM × OR_S) on PLOTS_load.shp; columns OR_S and POP_TODAY in settlements_today.csv; the full table with the rule in occupancy_by_settlement.csv, read by facts_w14.settlement_table() and facts_w14.occupancy().

3.7 Check it yourself

1. In settlements_today.csv, divide WB_2024 by PROPS for Ibri. The answer, 6.0718, must equal OR_S.
2. In QGIS, select on PLOTS_load.shp: Buiding_St = 'EXisting' and DERIVED = 'Residential' and G_DOM between 1 and 14 and SETTLE = 'IBRI'. The selection holds 5,853 plots and sums to 7,782 domestic meters.
3. Sum POP over the plots of any settlement; it must equal POP_TODAY. The sum over all plots is 119,893.
4. In occupancy_by_settlement.csv, every row whose rule is not 'derived' must be a settlement under 1,000 people or one whose OR_raw is below 4.0.
5. Recompute OR_CEIL: the largest OR_RAW among the rows of settlements_today.csv with WB_2024 of 2,000 or more. It is Bat's 6.1167.

4 The empty land and its capacity

Of the 77,265 plots in the cadastre, 60,509 carry no meter. This chapter sets how many people that empty land can hold when it is fully built. The answer is the capacity of each settlement, and it is the ceiling that Chapter 5 fills with growth. It rests on three measurements taken from the built plots of the same settlement: what a home plot looks like, what share of such plots are homes, and how many properties a home plot carries (Section 3.2). The occupancy of Section 3.4 turns properties into people.

4.1 Which empty plots become homes

What it is for. Not every empty plot will carry a house. A sliver, a strip along a road, a grove or a plot the size of a district will not. The test picks the plots that look like the built home plots.

The rule. An empty plot is counted as a future home plot when all of the following hold: its area is between 200 and 1,000 m²; it is compact, meaning its area is at least 0.6 of the area of its smallest enclosing rotated rectangle; it is not a strip, meaning that rectangle's long side is no more than three times its short side; and its use by the rules of Chapter 2 is not grove, industrial or heritage, and it lies outside the two industrial estates.

Why these bounds. The pure home plots of Section 3.2 have a median area of 717 m², a median compactness of 0.98 and a median aspect of 1.31. The test is stricter than the built stock: only 76 per cent of today's home plots would pass it. The capacity it gives therefore errs towards fewer homes, not more. A plot above 2,000 m² is a future district drawn as one parcel and takes nothing until it is subdivided.

Table 22 From empty plots to counted future home plots, the study area

Test	Plots removed	Plots left
Empty plots in the cadastre		60,509
Smaller than 200 m² (slivers)	5,469	55,040
Larger than 1,000 m² (2,815 up to 2,000 m², 5,235 above)	8,050	46,990
Not compact (below 0.6)	344	46,646
A strip (aspect above 3)	233	46,413
Grove, industrial, heritage or in an estate	304	**46,109**

Source. Project rule (engineer, 2026-09-10). The guideline asks for population from plots classified by clear typologies and warns that plot subdivision must be reflected in the housing units created (G201-p58, §7.2.2, and the NOTE at G201-p59). The shape test is the typology; the exclusion of plots above 2,000 m² is the subdivision caution.

Worked for Ibri. Ibri has 14,203 empty plots, of which 10,469 pass the test. Across the study area the median counted plot is 647 m².

Where it lives. W14/py/plot_class_v2_apply.py: fields COMPACT (area over the area of minimum_rotated_rectangle), ASPECT (long over short side of that rectangle), HOMESHAPE (1 = size and shape pass) and FUT_CAP (1 = counted: empty, home-shaped and not excluded) on PLOTS_load.shp; column FUT_CAP_PLOTS in settlements_today.csv.

4.2 The home share

What it is for. A district needs mosques, shops and offices as well as houses, so not every home-shaped plot becomes a home. The home share is the fraction that does, measured where the district already exists.

The rule. For each settlement, among its built plots that carry a meter and pass the size and shape test of Section 4.1, the home share is the number that are pure home plots (Section 3.2) divided by all of them.

$$h_s = \frac{n_{\text{pure,home-shaped,s}}}{n_{\text{built,metered,home-shaped,s}}} \quad (14)$$

Symbol	Meaning	Unit
h s	home share of settlement s	—
n pure, home-shaped, s	built, metered, home-shaped plots that are pure home plots	count
n built, metered, home-shaped, s	all built, metered, home-shaped plots	count

Worked for Ibri. 4,838 ÷ 5,579 = 0.867. Among the settlements of 1,000 people or more the measured share runs from 0.67 in Tanam, with its workshops, to 0.94 in Al Aynayn. The settlements under 1,000 people take 0.9 (Section 3.5).

Source. Project rule (engineer, 2026-09-10). It is the coverage by land-use typology that the NOTE at G201-p59 asks for.

Where it lives. plot_class_v2_apply.py, masks bh and pure and the series home_share; columns HOME_SHARE_RAW (derived) and HOME_SHARE (adopted) in settlements_today.csv; field HOMESH_S on every plot.

4.3 The capacity equation

What it is for. The capacity is the number of people the empty land of a settlement can hold when every counted plot is built. It is the ceiling of that settlement's growth.

The rule. The counted plots, times the home share, give the future home plots. Times the properties per home plot, they give the future properties. Times the occupancy, they give the people.

$$C_s = N_{\text{hs,s}} \times h_s \times p_s \times \text{OR}_s \quad (15)$$

Symbol	Meaning	Unit
C s	capacity of the empty land of settlement s	persons
N hs, s	empty plots counted as future home plots (Section 4.1)	count
h s	home share (Section 4.2), adopted value	—
p s	properties per home plot (Section 3.2), adopted value	properties per plot
OR s	occupancy rate (Section 3.4), adopted value	persons per property

Source. It is the guideline's equation, population equals number of plots times average properties per plot times occupancy rate (G201-p58, §7.2.2), with the home share as the typology factor. It gives the build-out ceiling only. The NOTE at G201-p59 requires development to be phased, not assumed simultaneous; Chapter 5 does that.

Worked for Ibri. 10,469 counted plots × 0.867 × 1.33 = 12,072 future properties. 12,072 × 6.0718 = 73,299 people. The occupancy is carried to four decimals in the calculation; it prints as 6.07. The properties are rounded to whole numbers before the occupancy is applied, as the script does.

Where it lives. plot_class_v2_apply.py: FUT_CAP_PROPS = round(FUT_CAP_PLOTS × PROPS_PER_BUILT_PLOT × HOME_SHARE) and FUT_CAP_POP = round(FUT_CAP_PROPS × OR_S) in settlements_today.csv; FUT_PROPS on each counted plot is its expected properties, p × h.

4.4 Capacity by settlement

Table 23 Capacity of the empty land by settlement

Settlement	Empty plots	Counted	Home share derived	Home share adopted	Props per plot	Occu-pancy	Capacity, people	People 2024	Full, people
Ibri	14,203	10,469	0.87	0.87	1.33	6.07	73,299	67,106	140,405
Ad Dariz	4,888	3,219	0.90	0.90	1.20	4.86	16,854	11,850	28,704
Al Araqi	3,360	2,463	0.84	0.84	1.32	4.34	11,832	10,696	22,528
Al Aynayn	1,780	1,362	0.94	0.94	1.14	5.12	7,459	4,554	12,013
Al Wahrah	2,000	1,373	0.91	0.91	1.22	4.68	7,109	3,351	10,460
At Tayyib	10,012	8,640	0.92	0.92	1.09	4.02	34,675	3,348	38,023
Al Jibayyah	2,784	1,966	0.89	0.89	1.26	5.13	11,306	2,686	13,992
Bat	879	355	0.88	0.88	1.17	6.12	2,226	2,557	4,783
Ad Dibayshi	2,430	1,887	0.90	0.90	1.16	4.00	7,920	2,724	10,644
Tanam	2,045	1,556	0.67	0.67	1.24	5.25	6,779	2,116	8,895
Suwayda al Ma	1,593	937	0.82	0.82	1.60	5.18	6,369	1,559	7,928
Hijar	375	211	0.90	0.90	1.69	4.00	1,288	1,408	2,696
Al Ghubayrah	1,448	1,094	0.84	0.90	1.00	4.00	3,940	644	4,584
Al Qurayn	2,051	1,830	0.93	0.90	1.00	4.00	6,588	588	7,176
Sayh al Masarrat	1,745	1,384	0.92	0.90	1.00	4.00	4,984	1,828	6,812
Al Jahli	1,235	1,023	0.78	0.90	1.00	4.00	3,684	1,376	5,060
Satwah	181	79	0.87	0.90	1.00	4.00	284	108	392
Al Akheedar	120	45	0.53	0.90	1.00	4.00	160	480	640
Al Qali	220	157	0.92	0.90	1.00	4.00	564	224	788
Shalashil	3,686	3,250	0.91	0.90	1.00	4.00	11,700	152	11,852
Al Makhtibyah	1,523	1,192	0.90	0.90	1.00	4.00	4,292	364	4,656
Usaybuq	60	39	0.87	0.90	1.00	4.00	140	84	224
Wadi al Mankas	1,580	1,394	0.87	0.90	1.00	4.00	5,020	32	5,052
Ash Shiab	65	13	0.87	0.90	1.00	4.00	48	44	92
Miayrid	246	171	0.87	0.90	1.00	4.00	616	16	632
Total	**60,509**	**46,109**					**229,136**	**119,893**	**349,029**

Full, people = people in 2024 + capacity: the population of the settlement in its saturation year.

The empty land of the study area holds 229,136 people, against 119,893 living in it in 2024. When every counted plot is built the study area holds 349,029 people. That is the saturation population of Chapter 5, and it follows from the land alone. The growth series only says when it is reached.

The table also shows why the next chapter needs an overflow rule. At Tayyib has room for 34,675 people against 3,348 living there, far more than its own growth will ever fill. Ibri's capacity of 73,299 is about 1.1 times its 2024 population. It fills in 2056, and from then on the growth of the largest settlement in the study area has to be housed on its neighbours' land.

4.5 Check it yourself

1. Select on PLOTS_load.shp: Buiding_St = 'Future' and FUT_CAP = 1 and SETTLE = 'IBRI'. The selection holds 10,469 plots.
2. Select Buiding_St = 'EXisting' and HOMESHAPE = 1 and N_ACC > 0 and SETTLE = 'IBRI': 5,579 plots. Of these, DERIVED = 'Residential' and G_DOM between 1 and 14: 4,838. The ratio is Ibri's home share.
3. Open any counted plot and check its COMPACT against the area of its oriented minimum bounding box (QGIS: Processing, Oriented minimum bounding box).
4. In settlements_today.csv recompute FUT_CAP_POP from FUT_CAP_PLOTS, PROPS_PER_BUILT_PLOT, HOME_SHARE and OR_S for three settlements; the column sums to 229,136.
5. The sum of FUT_PROPS over a settlement's plots equals FUT_CAP_PROPS within rounding.

5 Growth, overflow and saturation

Chapter 3 put people on the built plots of 2024 and Chapter 4 set how many more the empty land of each settlement can hold. This chapter fills that land year by year. It answers three questions: how fast each settlement grows, where its growth goes once its own land is full, and in which year the study area is full, which is the saturation the design is sized for. The last step places the people housed in each year on the individual empty plots, so that the network sees a load on every plot it passes.

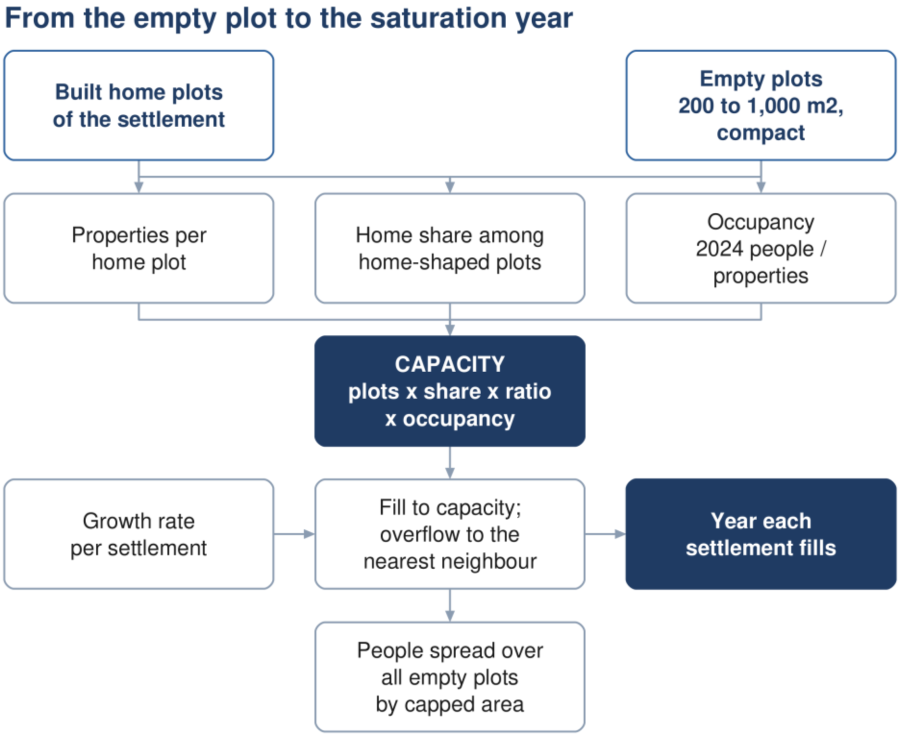

Figure 8 From the empty plot to the saturation year: the three rates of Chapters 3 and 4 set the capacity, the growth rate fills it, the overflow moves what a full settlement cannot house, and the housed people are spread over the plots.

5.1 The growth series

What it is for. The growth series says how fast each settlement's population grows. It is used for its rate only; the starting population is the counted one of Chapter 3.

The series. The population series of the Inception Report is built in three parts. The wilayat total is then split between the settlements in fixed shares from the census, so every settlement carries the same annual rate in every year.

Table 24 The three parts of the growth series and the mean annual rate

Part	Years	Basis	Mean rate, % a year
1	2024 to 2040	NCSI forecast for the Wilayat of Ibri, Omani and expatriate population separately	2.56 (2024 to 2030); 2.51 (2030s)
2	2041 to 2050	Extrapolation of the NCSI forecast, documented in the technical note on population issued with the Inception Report	2.05 (2040s)
3	2051 to 2100	Rising from 2.21 in 2051 to 2.40 by 2058, then constant to 2100, the planning horizon instructed by NWS	2.33 (2050s); 2.40 (2060 on)

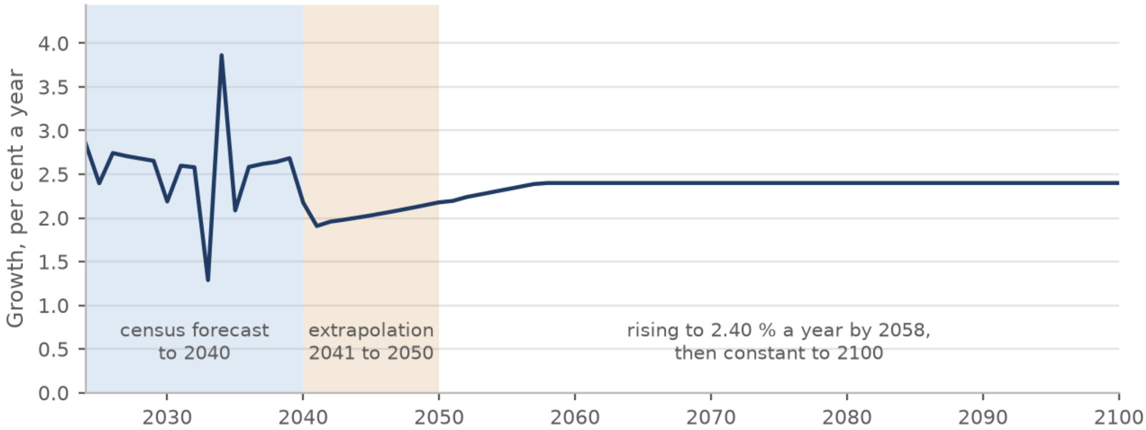

Figure 9 The annual growth rate of the population series by year: the census forecast to 2040, the extrapolation to 2050, and the rise to a constant 2.40 per cent.

The rule. Each settlement's growth factor is its workbook population in year y over its workbook population in 2024. The new people who want a home in the settlement by year y are its counted 2024 population times that factor, less the 2024 population.

$$g(y) = \frac{P_{\mathrm{wb,s}}(y)}{P_{\mathrm{wb,s}}(2024)} \tag{16}$$

$$D_s(y) = P_{2024,s} \times (g(y) - 1) \tag{17}$$

Symbol	Meaning	Unit
g(y)	growth factor from 2024 to year y, the same for every settlement	—
P wb, s (y)	population of settlement s in year y, Inception Report series	persons
P 2024, s	counted population of settlement s in 2024 (Section 3.6)	persons
D s (y)	new people wanting a home in settlement s by year y	persons

Worked for Ibri. The factor to 2030 is 1.1638; for the study-area total it is 1.1638, the same, because the shares are fixed. Ibri's growth to 2030 is 67,106 × (1.1638 − 1) = 10,994 people.

Source. Inception Report Revision 0, Section 6, and its demand workbook, sheet Project Pop Settlements. The split in census shares is the guideline's rule for a forecast held only at wilayat level (G201-p58, §7.2.1).

Where it lives. W14/py/growth_by_settlement.py: P (the workbook sheet), growth = P / P[2024], demand = pop0 × (growth − 1). The rates are printed by facts_w14.growth_rates().

5.2 The ten-year limit, and why saturation rests on the land

The rule. NCSI forecasts cover 20 to 25 years. Beyond them the guideline permits a regression, but states that it is "not recommended to extrapolate more than ten years beyond the available forecast period".

Source. G201-p58, §7.2.1, time scale.

What it means here. The NCSI forecast ends in 2040, so the limit falls in 2050, which is exactly where part 2 of the series stops. Everything after 2050 is outside it. By 2050 the growth has housed 99,033 people, 43 per cent of the capacity; 24 of the 25 settlements fill after 2050, and the study area fills in 2070.

> **The curve dates saturation; it does not size it.** The population at saturation is the 2024 population plus the capacity: 119,893 + 229,136 = 349,029. That sum does not depend on the growth curve. A faster curve brings 2070 forward and a slower one pushes it back; neither changes the 349,029 people or the 60,099 m³/d. Left to run, the series itself would reach 691,264 people in the 25 settlements by 2100, which the land cannot hold. It is not a target and not a saturation figure.

This is why the two flow cases of the network design rest on firm ground. Pipes are sized on the saturation flow, which comes from the land. Self-cleansing is checked on the 2030 flow, which lies inside the NCSI forecast. The 2055 model year, 244,914 people, lies on the extrapolated part of the curve: quote it as a dated estimate, not as data. The concept report records the use of the series beyond 2050 as a departure (Revision 2, Section 10.1).

Where it lives. The two flow cases are the project rule of 2026-09-11 (engineer), taught in Chapters 11 and 12. The saturation population is column SAT_POP of W14/shp/Settlements_merged.shp and field POP_ULT on every plot.

5.3 Filling the land in proportion

What it is for. A settlement's new people have to be housed somewhere on its land. The rule says how, and it sets the year the settlement is full.

The rule. A settlement's growth fills all its counted plots together and in proportion, not plot by plot. Its fill fraction in year y is the people it has housed since 2024 over its capacity. The settlement is full, and that year is its saturation year, when the housed people reach the capacity. Growth it cannot house from then on is overflow (Section 5.4).

$$f_s(y) = \frac{H_s(y)}{C_s} \leq 1, \;\; P_s(y) = P_{2024,s} + H_s(y) \qquad (18)$$

Symbol	Meaning	Unit
f s (y)	fill fraction of settlement s in year y	—
H s (y)	new people housed in s since 2024, its own growth plus what it received	persons
C s	capacity of settlement s (Section 4.3)	persons
P s (y)	population of settlement s in year y	persons

Worked for Ibri. In 2030 Ibri has housed 10,994 people, 15.0 per cent of its capacity of 73,299. In 2055 it has housed 70,093, 95.6 per cent. It is full in 2056, with 140,405 people; from then on its population stays there and its growth goes elsewhere.

Source. Project rule (engineer, 2026-09-10). The gradual fill is the phasing that the NOTE at G201-p59 requires in place of a simultaneous build-out.

Where it lives. growth_by_settlement.py, the year loop: own_c (housed from own growth), in_c (received), sat_year; sheets Population by year, Housed own growth, Inflow from overflow and Fill fraction of W14/analysis/W14_growth_by_settlement.xlsx.

5.4 The overflow rules

What it is for. A full settlement's growth does not stop; it moves to land that is still empty. The rules say where it goes, in which order.

The rule. Each year, the largest settlements are served first, and each settlement's growth is housed in this order.

1. In its own counted plots, while it has room.
2. Ibri's overflow goes in parallel to Al Araqi, Al Qurayn and Shalashil, in shares of 70, 20 and 10 per cent. When one of the three is full, its share is offered again to the others of the same group.
3. When those three are full, Ibri's overflow goes to Ad Dariz.
4. At Tayyib's overflow goes first to Miayrid, the small settlement beside it.
5. After its named receivers, and for every other settlement from the start, the overflow goes to the nearest settlement with room, by distance between the settlement centroids, then the next nearest.

6. Every settlement receives, whatever its size. A receiver's own growth continues alongside what it receives, and both count against its capacity.

Why. The parallel routes out of Ibri reflect local knowledge of where Ibri's growth is going. The nearest-with-room rule stands for everything else. Letting every settlement receive is what lets all 25 fill; with a minimum receiver size the small settlements with large empty land would never fill (Section 5.7).

Source. Project rule (engineer, 2026-09-10), from a person who knows the area.

Where it lives. growth_by_settlement.py: SPILL = {'IBRI': [[AL ARAQI 0.7, AL QURAYN 0.2, SHALASHIL 0.1], [AD DARIZ 1.0]], 'AT TAYYIB': [[MIAYRID 1.0]]}, RECEIVER_MIN_POP = 0, the receivers sorted by centroid distance in order[s]; sheet Overflow routes of the growth workbook.

5.5 What the rules produce

The rule sets the order; the land sets the result. Al Araqi fills in 2057, a year after Ibri, so it takes only 163 of Ibri's people. Its share passes to the other two of the group: Shalashil takes 11,471 and Al Qurayn 5,856 before they fill in 2061 and 2060. Ad Dariz, next in line, is filling with its own growth and takes 440. The rest goes by distance, and the largest single route is Ibri to At Tayyib, 22,078 people. Ibri sends out 55,261 people in all, to 14 settlements; 8 take a thousand or more.

Table 25 The main overflow routes: every route carrying 1,000 people or more at saturation

From	To	People at saturation	Donor full	Receiver starts	Receiver full
Ibri	At Tayyib	22,078	2056	2060	2068
Ibri	Shalashil	11,471	2056	2056	2061
Ibri	Al Qurayn	5,856	2056	2056	2060
Ad Dariz	Wadi al Mankas	4,965	2062	2062	2068
Al Araqi	At Tayyib	4,873	2057	2060	2068
Ibri	Ad Dibayshi	4,078	2056	2062	2063
Ibri	Al Jibayyah	3,927	2056	2068	2070
Bat	Al Ghubayrah	2,493	2052	2052	2070
Ibri	Al Makhtibyah	2,441	2056	2063	2069
Ibri	Tanam	2,010	2056	2063	2069
Al Araqi	Al Aynayn	1,790	2057	2057	2060
Al Aynayn	At Tayyib	1,639	2060	2060	2068
Al Araqi	Al Jibayyah	1,410	2057	2068	2070
Ibri	Sayh al Masarrat	1,225	2056	2063	2069
Ad Dibayshi	Tanam	1,004	2063	2063	2069

Receiver starts is the first year the receiver takes overflow from any donor, which can be before this donor is full.

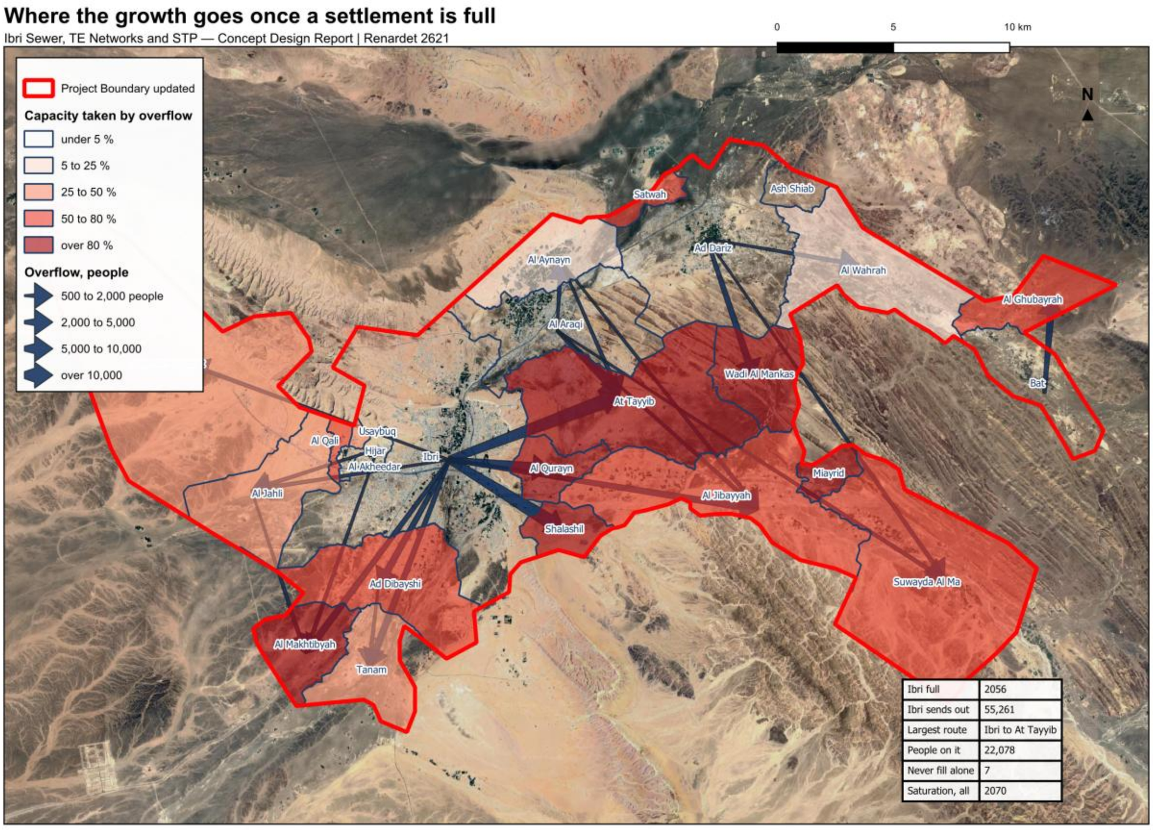

Figure 10 Where the growth goes once a settlement is full. Arrows show the routes carrying 500 people or more at saturation; the shading shows how much of each settlement's capacity is taken by its neighbours' overflow.

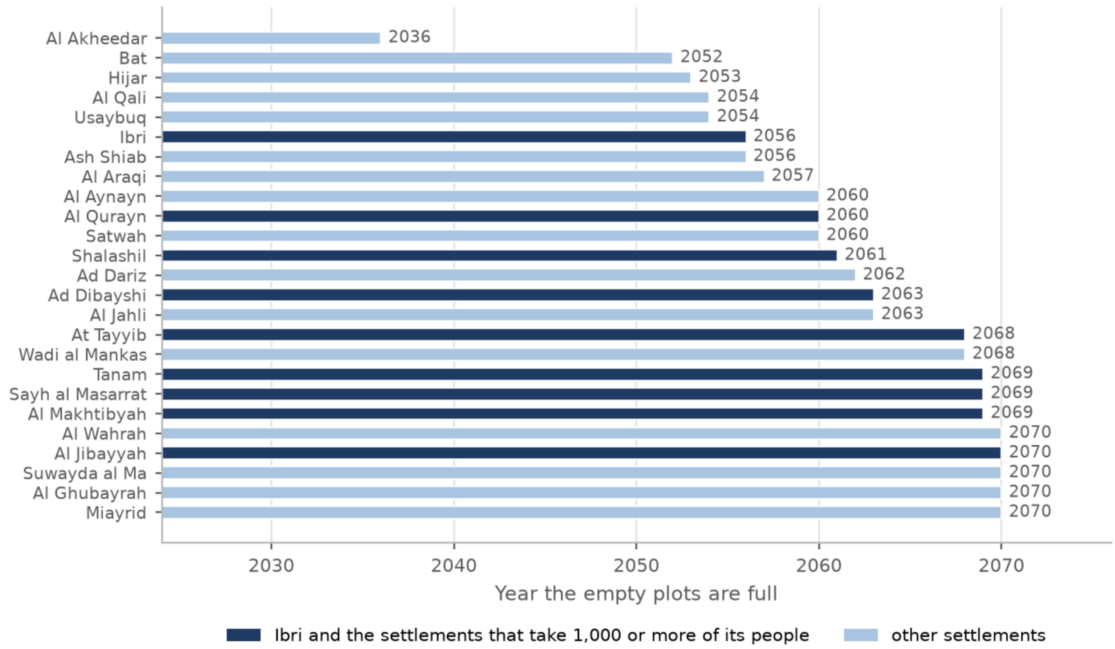

Figure 11 The year each settlement's empty plots are full. Ibri and the settlements that take 1,000 or more of its people are marked.

5.6 The saturation year

The rule. The study area is saturated in the year its last settlement fills. Because every settlement receives overflow, that is also the first year in which growth finds no empty plot anywhere. Growth after it is reported as unhoused and is not placed.

The result. The first settlement to fill is Al Akheedar, in 2036; there is no overflow before then, so the 2030 figures are each settlement's own growth. The last 5 settlements fill in 2070, which is the saturation year, with 349,029 people and 60,099 m³/d of average sewage flow. The table below gives every settlement at five-year steps; a cell is blank once the settlement is full, and the last two columns give its saturation year and its population then.

Source. Project rule (engineer, 2026-09-10). The Terms of Reference ask for the design to completion plus 25 years or to the ultimate saturated condition, with five-year intervals (scope, pp 3 and 14–15).

Table 26 Population at five-year steps to saturation, every settlement

Settlement	2024	2025	2030	2035	2040	2045	2050	2055	2060	2065	2070	Sat. year	Sat. population
Ibri	67,106	68,713	78,099	88,272	100,066	110,349	122,535	137,198				2056	140,405
Ad Dariz	11,850	12,134	13,791	15,587	17,670	19,486	21,638	24,206	27,350			2062	28,704
Al Araqi	10,696	10,952	12,448	14,069	15,949	17,588	19,531	21,849				2057	22,528
Al Aynayn	4,554	4,663	5,300	5,990	6,791	7,489	8,316	9,303	12,013			2060	12,013
Al Wahrah	3,351	3,431	3,900	4,408	4,997	5,511	6,119	6,845	7,703	8,684	10,460	2070	10,460
At Tayyib	3,348	3,428	3,897	4,404	4,993	5,506	6,114	6,839	8,130	23,972		2068	38,023
Ad Dibayshi	2,724	2,789	3,170	3,583	4,062	4,479	4,974	5,564	6,262			2063	10,644
Al Jibayyah	2,686	2,750	3,126	3,533	4,005	4,416	4,904	5,486	6,174	6,951	13,992	2070	13,992
Bat	2,557	2,618	2,976	3,363	3,813	4,204	4,669					2052	4,783
Tanam	2,116	2,167	2,463	2,784	3,156	3,480	3,864	4,323	4,865	5,961		2069	8,895
Sayh al Masarrat	1,828	1,872	2,127	2,405	2,726	3,006	3,338	3,734	4,202	4,786		2069	6,812
Suwayda al Ma	1,559	1,596	1,814	2,050	2,324	2,563	2,846	3,184	3,583	4,034	7,928	2070	7,928
Hijar	1,408	1,442	1,639	1,852	2,100	2,315	2,571					2053	2,696
Al Jahli	1,376	1,409	1,601	1,810	2,052	2,263	2,513	2,831	3,746			2063	5,060
Al Ghubayrah	644	659	750	847	960	1,059	1,176	1,756	2,575	3,501	4,584	2070	4,584
Al Qurayn	588	602	684	773	877	967	1,074	1,201	7,176			2060	7,176
Al Akheedar	480	492	559	631								2036	640
Al Makhtibyah	364	373	424	479	543	599	665	744	837	1,522		2069	4,656
Al Qali	224	229	261	295	410	518	646					2054	788
Shalashil	152	156	177	200	227	250	278	311	8,335			2061	11,852
Satwah	108	111	126	142	161	178	197	221	392			2060	392
Usaybuq	84	86	98	110	125	138	153					2054	224
Ash Shiab	44	45	51	58	66	72	80	90				2056	92
Wadi al Mankas	32	33	37	42	48	53	58	65	74	2,859		2068	5,052
Miayrid	16	16	19	21	24	26	29	33	37	41	632	2070	632
Total	**119,893**	**122,766**	**139,535**	**157,710**	**178,781**	**197,153**	**218,926**	**244,914**	**275,608**	**310,307**	**349,029**	**2070**	**349,029**

Totals are rounded from the unrounded values and may differ by one from the sum of the rows.

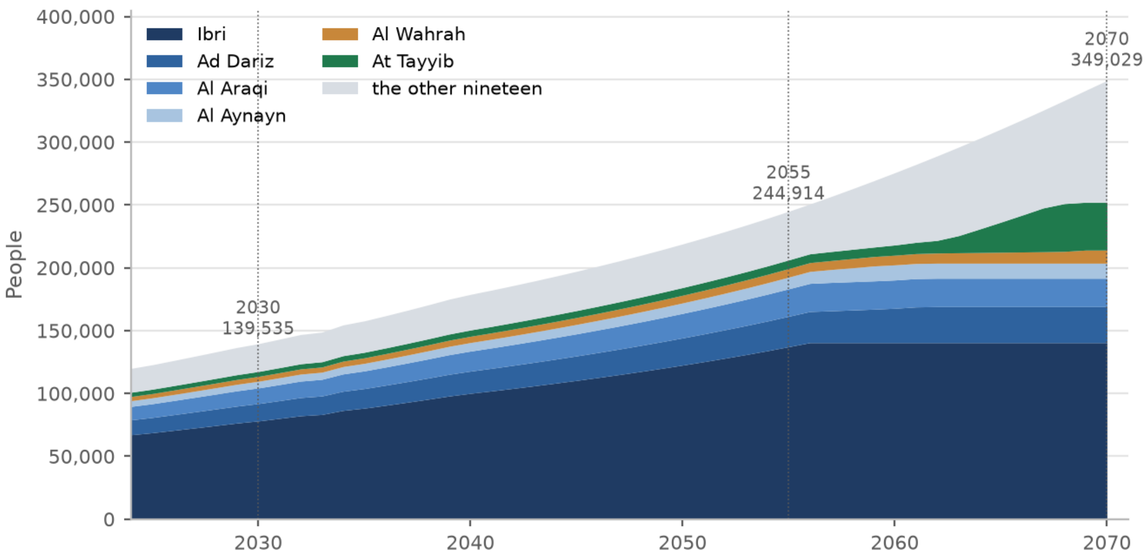

Figure 12 People by year to saturation, the six largest settlements shown separately. The flat top of each band is the year that settlement fills.

Where it lives. growth_by_settlement.py: first_unhoused and ult; sheets Five-year to saturation and Unhoused of the growth workbook; field ULT_YEAR on every plot and column SAT_YEAR on every plot and settlement. facts_w14.five_year_rows('pop') prints the table.

5.7 Saturation with and without the overflow

On its own growth alone, 7 of the 25 settlements would not fill before 2100: At Tayyib, Al Ghubayrah, Al Qurayn, Al Makhtibyah, Shalashil, Wadi al Mankas, Miayrid. With the overflow every settlement fills by 2070. Their land is large against their own population, and it is Ibri's growth and its neighbours' that occupies it. Ibri itself fills a year earlier with the overflow (2056 against 2057), because it receives 117 people from small neighbours that fill before it.

Table 27 The year each settlement is full, on its own growth and with the overflow

Settlement	Capacity, people	Full on own growth	Full with overflow	People received
Ibri	73,299	2057	2056	117
Ad Dariz	16,854	2063	2062	810
Al Araqi	11,832	2057	2057	163
Al Aynayn	7,459	2066	2060	1,790
Al Wahrah	7,109	2073	2070	924
At Tayyib	34,675	not before 2100	2068	28,937
Ad Dibayshi	7,920	2083	2063	4,078
Al Jibayyah	11,306	2095	2070	6,349
Bat	2,226	2052	2052	0
Tanam	6,779	2086	2069	3,014
Sayh al Masarrat	4,984	2081	2069	1,662
Suwayda al Ma	6,369	2094	2070	3,386
Hijar	1,288	2053	2053	0
Al Jahli	3,684	2080	2063	1,743
Al Ghubayrah	3,940	not before 2100	2070	2,751
Al Qurayn	6,588	not before 2100	2060	5,856
Al Akheedar	160	2036	2036	0
Al Makhtibyah	4,292	not before 2100	2069	3,644
Al Qali	564	2078	2054	351
Shalashil	11,700	not before 2100	2061	11,503
Satwah	284	2080	2060	150
Usaybuq	140	2067	2054	60
Ash Shiab	48	2056	2056	0
Wadi al Mankas	5,020	not before 2100	2068	4,965
Miayrid	616	not before 2100	2070	586

Why it matters. The design puts every settlement at its capacity at saturation either way. The overflow decides when each settlement's land fills and who fills it. That sets the population of each settlement in the intermediate years, the loads on its plots in 2055, and the order in which the network and the plant are needed.

Where it lives. growth_by_settlement.py: sat_own (the year demand alone reaches the capacity) and sat_year; columns SAT_OWN (0 = not before 2100), SAT_YEAR, IN_PEOPLE, OUT_PEOPLE, MAIN_TO and INSHARE of Settlements_merged.shp; facts_w14.own_growth_saturation().

5.8 Placing the housed people on the plots

What it is for. The capacity count of Chapter 4 decides how many people a settlement can hold. The network needs to know on which plots they live, and a sewer that serves a street serves every plot on it, whether or not the plot passed the home-shape test.

The rule. The people a settlement has housed by year y are spread over all its empty plots up to 2,000 m² (not grove, industrial, heritage or in an estate), in proportion to plot area capped at 1,000 m². A sliver takes a sliver's share and a large plot no more than a full home plot. The sum over the plots equals the settlement's housed people by construction, so the network total and the plant total are the same figure.

$$n_{\mathrm{i}}(y) = H_{\mathrm{s}}(y) \times \frac{w_{\mathrm{i}}}{\sum_{\mathrm{j} \in \mathrm{s}} w_{\mathrm{j}}}, \quad w_{\mathrm{i}} = \min(A_{\mathrm{i}}, 1{,}000) \qquad (19)$$

Symbol	Meaning	Unit
n i (y)	new people placed on empty plot i in year y	persons
H s (y)	new people housed in settlement s by year y (Section 5.3)	persons
w i	spread weight of plot i: its area capped at 1,000 m²; zero for a plot above 2,000 m² or an excluded use	m²
A i	area of plot i	m²

Worked for Ibri. Ibri has 12,901 plots that receive the spread, with a total weight of 7,907,074 m². A 600 m² plot takes 600 ÷ 7,907,074 of Ibri's housed people: 0.83 people in 2030 and 5.56 at saturation. A 1,500 m² plot takes the capped weight of 1,000 m²: 9.27 people at saturation. Over the study area 54,163 empty plots receive people, and the slivers under 200 m² carry 0.9 per cent of the weight. Chapter 7 turns these people into flow.

Source. Project rule (engineer, 2026-09-10).

Where it lives. plot_class_v2_apply.py writes SPREAD_W (the weight); growth_by_settlement.py writes POP_2030, POP_2055 and POP_ULT = POP + H × SPREAD_W / SPREAD_W_SUM, with Q_2030, Q_2055 and Q_ULT beside them, on PLOTS_load.shp. SPREAD_PLOTS and SPREAD_W_SUM per settlement are in settlements_today.csv.

5.9 Check it yourself

1. In the growth workbook, sheet Population by year, divide Ibri's 2030 value by its 2024 value, then do the same for the total. Both give 1.1638.
2. For every settlement in Settlements_merged.shp, SAT_POP equals POP_2024 plus CAP_POP to within one person. Their sum is 349,029.
3. Sum POP_ULT over all plots of PLOTS_load.shp: 349,029. Sum it over one settlement: its saturation population.
4. On an empty Ibri plot, POP_2030 − POP must equal Ibri's housed people in 2030 × SPREAD_W / 7,907,074.
5. In the sheet Overflow routes, the column for the saturation year summed by receiver equals IN_PEOPLE; summed by donor it equals OUT_PEOPLE.
6. A settlement whose SAT_OWN is 0 must appear in the list of those that never fill on their own growth; there are 7.

6 Water demand

Sewage is calculated from the water people use, so the chain starts with water demand even though this is a sewerage scheme. The guideline divides demand into five components. Three of them are rates applied to the population: domestic, non-domestic and governmental. Two sit outside those rates and are added separately: special consumption and water delivered by tanker. On the wastewater side the guideline adds one more duty, an assessment of private wells and other water that never passes through the network. This chapter takes each in turn, explains the choice the guideline offers between two methods, and names the items that are still waiting for data.

Table 28 The components of water demand and how each is treated here

Component	Guideline basis	Clause	Treated here as
Domestic	population × 164 l/c/d	G201-p59–60, Tab 11	applied to every dwelling
Non-domestic	22 % of domestic, or G201 Table 12 unit rates	G201-p60, §7.3.2	the ratio, placed on the shop meters
Governmental	14 % of domestic, or calculated for the project	G201-p61, §7.3.3	the ratio, placed on the government meters
Special	provided by the developer	G201-p61, §7.3.4	the two industrial estates, workers × 93 l/d
Blue tankers	assessed from filling-station records	G201-p61–62, §7.3.5	pending: no records held
Other sources	private wells and other abstraction assessed	G201-p70, §7.4	pending: no records held

6.1 Domestic demand

What it is for. Domestic demand is the base of the whole chain. The non-domestic and governmental streams are fractions of it, and it carries most of the sewage.

The rule. Domestic water is the population multiplied by the governorate's unit consumption.

$$Q_{\text{dom}} = P \times \text{LPCD} \tag{20}$$

Symbol	Meaning	Unit
Q dom	domestic water demand	l/d
P	population served	persons
LPCD	domestic unit consumption, 164 for Adh Dhahirah	l/c/d

The figure is worth understanding before it is used. The guideline derives it from the water the network delivers to domestic customers:

$$\text{LPCD} = \frac{W_{\text{dom,acc}}}{\text{OR} \times N_{\text{acc}}} \tag{21}$$

Symbol	Meaning	Unit
W dom,acc	total domestic water accounted for by the network	l/d
OR	occupancy rate	persons per property
N acc	active domestic accounts	—

Two things follow. First, the rate measures network water only. Water bought from a tanker or drawn from a private well is not in it, so those sources are added separately and adding them is not double counting (Section 6.5). Second, occupancy sits in the denominator: a higher occupancy gives a lower rate per person for the same metered volume, so the two cannot be chosen independently. The guideline calls the table values indicative figures from the 2024 Integrated Master Plan, to be used in the absence of updated figures and to be validated by Nama Water Services before design. The Inception Report workbook computes 163.5 l/c/d from actual consumption for 2021 to 2024, which agrees for every practical purpose.

Source. *G201-p59–60, §7.3.1 and Table 11, row Adh Dhahirah (164 l/c/d). Inception Report R0 workbook, Water Demand Criteria (163.5 l/c/d), as recorded in _BRAIN/02_DESIGN_CRITERIA.md §12c.*

Worked for Ibri. The study area houses 119,893 people in 2024. At 164 l/c/d their domestic water is 19,663 m³/d. Ibri alone, 67,106 people, uses 11,005 m³/d.

Where it lives. W14/py/plot_class_v2_apply.py. Field POP = G_DOM × OR_S on every plot; field W_DOM = POP × 164 / 1000, in m³/d. The constant is F.LPCD in W14/report/facts_w14.py.

6.2 Non-domestic and governmental demand: the two tiers

What it is for. The guideline offers two methods for the water used by shops, offices, schools and government buildings. Which one applies is set by the data available, not by preference, so the choice has to be made and stated.

The rule. Tier A is the planning method: governorate ratios on domestic demand and published return rates. Tier B is the project-specific method: unit rates from G201 Table 12 and design flows to a stated international standard. The guideline's wording is mandatory for Tier B wherever detailed land use exists.

Table 29 The two tiers of the flow chain

	Tier A: planning ratios	Tier B: project-specific design
Applies to	planning and forecasting across broad service areas	a project where detailed land use information is available
Non-domestic	22 % of domestic (G201 Table 11)	G201 Table 12 unit rates per pupil, bed, employee or m² of floor area
Governmental	14 % of domestic (G201 Table 11)	calculated specifically for the project
Sewage	return rates 85 % and 54 % (G201 Table 19)	design flows to a stated standard such as BS EN 752, with site evidence
Needs	the population	counts of pupils, beds, staff and floor areas
Used here	**yes**, declared as a departure	no: the quantities are not held

The guideline's own words set the order. For non-domestic water, where the project has detailed land use, it "shall be calculated using the reference values presented in the Table 12". Governmental water is to be calculated for the project, "not as a ratio of domestic consumption". On the wastewater side, design flows for such a project follow a standard such as BS EN 752, and "Calculations shall clearly state which standard has been used." The Tier A return rates are described as a framework for "planning and forecasting purposes across broader service areas". Above both tiers, a developer's calculation made on an approved method takes precedence over the whole chain.

Table 30 G201 Table 12: the unit rates Tier B needs (extract)

Category	Quantity it is priced on	Rate
Educational	pupils and staff	130 l/d each
Hospitals	beds and staff	650 l/d each
Commercial shopping	floor area	12.2 l/d per m²
Office	employees	93 l/d each
Mosques	floor area	185 l/d per m²
Dry industry	employees	93 l/d each
Wet industry	none	no rate: supplied by the developer

Every rate in G201 Table 12, extracted above, is priced on a quantity that describes the building in use: its floor area, its pupils, its beds, its staff. None is priced on the plot. Replacing floor area with cadastral plot area is wrong by an order of magnitude, because a mosque or a school stands on a fraction of its plot. Deriving the quantities from plot area, cover and storeys only disguises the same guess.

Departure: Tier A is used. Detailed land use is known plot by plot, so the condition for Tier B is met in form. It is not met in substance. No dataset held records pupils, beds, employees or floor areas, and the electricity file gives the tariff of each meter, not its consumption or the building behind it. The published ratios of G201 Table 11 are therefore used for the volume, the return rates of G201 Table 19 for the sewage, and no design-flow standard is yet named. G201 Table 12 will be adopted for non-domestic and governmental demand the day the quantities arrive. It is never applied together with the ratios: unit rates on the shops plus the 22 % and 14 % uplifts count the same water twice.

Source. *G201-p59 §7.3 (precedence), G201-p60 §7.3.2, G201-p61 §7.3.3 and Table 12, G201-p70–71 §7.4.1 and Table 19; quotations as recorded in _BRAIN/02_DESIGN_CRITERIA.md §11.0. Project rule: the load basis, locked 2026-08-30. Declared in report R2 §10.1, rows "Non-domestic and governmental demand" and "Design flow standard".*

Worked for Ibri. The study area has 9,669 shop meters and 1,055 government meters on its plots. Not one of them carries a floor area, a head count or a metered volume, which is why G201 Table 12 cannot be applied to them.

Where it lives. _BRAIN/02_DESIGN_CRITERIA.md §11.0 (the two tiers) and §11.1 (the locked load basis); the departures table in W14/report/rpt_cd.py, Section 10.1.

6.3 The distributed ratios, and why they are never compounded

What it is for. The two ratios set how much non-domestic and governmental water a settlement uses. They set the volume only. Where that volume is placed is a separate step (Chapter 7).

The rule. Each ratio is a fraction of domestic demand alone. The three streams are added, never multiplied.

$$Q_{\mathrm{nd}} = 0.22 \times Q_{\mathrm{dom}} \tag{22}$$

$$Q_{\mathrm{gov}} = 0.14 \times Q_{\mathrm{dom}} \tag{23}$$

$$Q_{\mathrm{water}} = Q_{\mathrm{dom}} + Q_{\mathrm{nd}} + Q_{\mathrm{gov}} = (1 + 0.22 + 0.14) \times Q_{\mathrm{dom}} \tag{24}$$

Symbol	Meaning	Unit
Q nd	non-domestic water demand	l/d
Q gov	governmental water demand	l/d
Q water	water demand of the three population-based streams	l/d

Per person the streams are 164 l/d domestic, 0.22 × 164 = 36.08 l/d non-domestic and 0.14 × 164 = 22.96 l/d governmental, 223.04 l/d in all.

The column headings of G201 Table 11 read "Distributed Non-Domestic Ratio (% LPCD)" and "Distributed Governmental Ratio (% LPCD)". The word distributed matters. These are not demands a person makes. They are the governorate's recorded non-domestic and governmental volumes from the water balance, expressed against its domestic consumption. Both are measured against the same domestic base, which is why neither may be applied on top of the other.

Never 164 × 1.22 × 1.14. Compounding gives 228.09 l/c/d instead of 223.04. The extra 5.05 l/c/d is 14 % of the non-domestic water, a volume that appears nowhere in the governorate balance the ratios came from. It is 2.3 % too much water, and it cannot be traced back to any line of the guideline.

The guideline also describes the ratios as spatially distributed consumption to be added to the domestic consumption, which reads as spreading them evenly across the population. This design keeps the volume and moves the placement: each settlement's non-domestic and governmental water is placed on the meters that generate it. A street of houses therefore runs at 139.4 l/d of sewage per person, not 171.3, and a commercial street carries the difference. The settlement total is the same either way. This is the second half of the departure, and it is declared as such.

Source. *G201-p59 §7.3.1 ("spatially distributed"), G201-p60 Table 11, row Adh Dhahirah (164 l/c/d, 22 %, 14 %). Project rule: the load basis, locked 2026-08-30 (volume from the ratios, placement from land use). Declared in report R2 §10.1, row "Allocation of non-domestic demand".*

Worked for Ibri. In 2024 the study area's non-domestic water is 4,326 m³/d and its governmental water 2,753 m³/d, against 19,663 m³/d domestic: ratios of 0.22 and 0.14, exactly the published ones. For Ibri the same arithmetic gives 2,421 and 1,541 m³/d on 11,005 m³/d domestic.

Where it lives. W14/py/plot_class_v2_apply.py, the pools pool_nd_s and pool_gv_s (settlement people × 0.22 × 164 and × 0.14 × 164); W14/analysis/settlements_today.csv, columns W_NDOM and W_GOV. Constants F.R_ND and F.R_GOV.

6.4 Special consumption: the two industrial estates

What it is for. Some water is used by identified projects that the population ratios never covered. It has to be added on its own line, or it is missing from the flow.

The rule. An industrial-estate plot carries its workers at the dry-industry rate of 93 litres per employee per day. The estates' meters take no share of the settlement's non-domestic or governmental water, and any dwelling meter on an estate carries the domestic rate like any other.

$$Q_{\text{spec}} = 93 \times N_{\text{w}} \tag{25}$$

Symbol	Meaning	Unit
Q spec	special consumption of an estate plot	l/d
N w	workers assigned to the plot	—
93	G201 Table 12 rate for dry industry	l/d per employee

Two industrial estates lie inside the study area, at Al Tayyeb in the north-east of the town and at Tanam beside the treatment plant. Their meters are on the commercial tariff, which hid them in the electricity data. The guideline states that the ratios do not apply to identified non-domestic projects such as economic zones, which are determined case by case. It also states that special consumption, such as labour camps and industry with high water use, is provided by the developer and is not covered by population forecasts. The two estates are exactly that case.

Using the G201 Table 12 rate here does not mix the tiers. The estates sit outside the ratios by the guideline's own words, so the rate is applied to a stream the ratios never contained and nothing is counted twice. For the same reason the estates' shop meters are kept out of the settlement's non-domestic pool: a share of the pool on an estate plot would load the estate's water a second time.

Outside assumption, tagged. No workforce or discharge record has been supplied. The workforces, 4,500 at Al Tayyeb and 1,800 at Tanam, are estimated from the number of workshop meters and are spread over each estate's industrial plots in proportion to area. They are to be replaced by the estates' own records.

Source. *G201-p59 §7.3.1 (identified projects), G201-p61 §7.3.4 (special consumption) and Table 12, dry industry. Workforce: outside assumption (engineer, 2026-09-09), declared in report R2 §10.1, row "Industrial estates".*

Worked for Ibri. 4,500 × 93 = 418.5 m³/d at Al Tayyeb and 1,800 × 93 = 167.4 m³/d at Tanam, 585.9 m³/d of water in all. The Al Tayyeb estate lies in At Tayyib. The Tanam estate's plots fall in more than one settlement of the partition, so its workforce is split: Ad Dibayshi 286, Tanam 1,130 and Al Makhtibyah 385 workers. The estates hold 1,445 shop meters and 2 government meters that take no share of any pool.

Where it lives. W14/py/plot_class_v2_apply.py: field ESTATE marks the estate plots, WORKERS carries the assigned workers, W_SPEC = WORKERS × 93 / 1000 in m³/d. The workforces are F.WORKERS.

6.5 Blue tankers and private wells: pending

What it is for. Water that reaches a home by truck or from a well still becomes sewage. Because the domestic rate measures network water only, these volumes are outside it and must be assessed on their own.

The rule. Where a tanker filling station lies within or near the project area, tanker consumption must be assessed explicitly from Nama Water Services station records (historical use, the area served and its change over the horizon), validated, and added to demand. Separately, the designer shall assess private wells, private water providers and other non-network abstraction so that their contribution to wastewater is accounted for. Tanker water returns to the sewer at the domestic rate of 85 %.

Neither record is held. The position adopted is stated plainly. A household on tanker supply is metered for electricity like any other and carries the domestic rate, so its people are in the flow; the tanker volume itself is not, and no tanker or well term appears in any flow in this design. Both are open data requests. When the filling-station records arrive, the tanker volume joins the domestic stream at 85 %.

G201 Table 13 cannot be used as printed. G201 Table 13 gives Adh Dhahirah a tanker volume of 5,145 m³/d, averaged over 2021 to 2023, and a ratio of 333 % to network domestic consumption in 2023, by far the highest in Oman. The two figures together imply a governorate network domestic consumption of about 1,545 m³/d. This study area alone uses 19,663 m³/d of domestic water at the guideline's own rate. The volume column is usable; the ratio is not, and it should be raised with Nama Water Services rather than applied.

Source. *G201-p61–62 §7.3.5 and Table 13 (Adh Dhahirah 5,145 m³/d, 333 %); G201-p70 §7.4 and §7.4.1 (other water sources, binding); G201-p71 Table 19 (domestic and tanker 85 %). Declared in report R2 §10.1, rows "Water supplied by tanker" and "Other water sources".*

Worked for Ibri. There is no Ibri number yet, and that is the point of the step: every flow in this tutorial is on the network-accounted basis, and a reader quoting one should say so.

Where it lives. Nowhere in the plot table, by design. W14/docs/DESIGN_FLOWS_FOR_NETWORK.md §8 lists both as not in the flows and not to be assumed; the data request register carries them.

6.6 Check it yourself

1. Open W14/shp/PLOTS_load.shp. Sum W_DOM and POP. W_DOM × 1000 / POP must be 164.0 l/c/d.
2. In W14/analysis/settlements_today.csv divide W_NDOM and W_GOV by W_DOM for any settlement. The answers must be 0.22 and 0.14.
3. Sum W_SPEC over the plots. It must equal the 6,300 workers × 93 l/d, 585.9 m³/d.
4. Filter the plots on ESTATE. Every estate plot must show W_NDOM = 0 and W_GOV = 0.
5. Open PAM-GUD-201 page 60, Table 11, row Adh Dhahirah, and page 61, Table 12, row Dry Industry, and confirm 164 l/c/d, 22 %, 14 % and 93 l/d per employee.
6. Look for a tanker or a well term anywhere in the plot table. There must be none until the records arrive.

7 The sewage of every plot

This chapter turns the water of Chapter 6 into sewage and places it on the ground, plot by plot. The cadastre holds 77,265 plots: 16,756 built plots with electricity meters and 60,509 empty ones. Every built plot gets its own average dry-weather flow for 2024 from its meters. Every empty plot gets a flow from the people it will house in each later year. The network then sums the plots upstream of each pipe.

An earlier tutorial loaded every property at a flat five people and 171 l/c/d. That is replaced here: dwellings carry the domestic rate at their settlement's own occupancy, and the non-domestic and governmental water sits on the meters that use it.

7.1 Return to the sewer

What it is for. Not all water supplied reaches the sewer. The return rate converts each water stream into the flow the pipe actually receives.

The rule. Domestic and tanker water return 85 %. Non-domestic water, which the guideline defines as government and commercial together, returns 54 %.

$$Q_{\mathrm{ww}} = 0.85(Q_{\mathrm{dom}} + Q_{\mathrm{tank}}) + 0.54(Q_{\mathrm{nd}} + Q_{\mathrm{gov}} + Q_{\mathrm{spec}}) \qquad (26)$$

Symbol	Meaning	Unit
Q ww	average dry-weather sewage, before infiltration	l/d or m³/d
Q dom	domestic water	l/d or m³/d
Q tank	tanker water: zero until the filling-station records arrive	l/d or m³/d
Q nd, Q gov	non-domestic and governmental water	l/d or m³/d
Q spec	special consumption of the industrial estates	l/d or m³/d

Per person, the three population streams give 0.85 × 164 = 139.4, 0.54 × 36.08 = 19.48 and 0.54 × 22.96 = 12.40 litres a day, 171.3 l/d of sewage per person in all. Two rules of this design sit beside the guideline's. An agricultural meter supplies an irrigation pump and returns nothing; the dwelling on a farm is metered separately and carries the domestic rate. The estates' dry-industry water returns 54 %, as other non-domestic water does.

Source. *G201-p70–71 §7.4.1 and Table 19 (Domestic & Tanker 85 %; Non-Domestic, government and commercial, 54 %). Project rules: agricultural meters carry no load (engineer, 2026-08-30); the estates return 54 % (engineer, 2026-09-10).*

Table 31 Water and sewage of the study area by stream, 2024

Stream	Water, m³/d	Return	Sewage, m³/d
Domestic	19,663	0.85	16,713
Non-domestic	4,326	0.54	2,336
Governmental	2,753	0.54	1,486
Industrial estates	586	0.54	316
Agricultural meters (546)	not counted	0	0
Total	**27,327**		**20,852**

Worked for Ibri. Ibri's 11,005 m³/d of domestic water returns 9,355 m³/d; its 2,421 and 1,541 m³/d of non-domestic and governmental water return 1,307 and 832 m³/d. Over the whole study area 76.3 % of the water becomes sewage.

Where it lives. W14/py/plot_class_v2_apply.py: S_DOM = W_DOM × 0.85; S_NDOM, S_GOV and S_SPEC = W_NDOM, W_GOV and W_SPEC × 0.54; QADF = their sum, m³/d. Constants F.RET_DOM and F.RET_ND.

7.2 The flow of an existing plot

What it is for. Every built plot gets its own average dry-weather sewage flow for the base year, built from the meters that stand on it and the rates of its settlement.

The rule. The flow of an existing plot, in litres a day, is

$$Q_{\text{plot}} = 0.85 \times 164 \times OR \times N_{\text{dom}} + 0.54 \times \left(U_{\text{nd}} N_{\text{nd}} + U_{\text{gov}} N_{\text{gov}} + 93 N_{\text{w}}\right) \qquad (27)$$

Symbol	Meaning	Unit
Q plot	average dry-weather sewage of the plot, 2024	l/d
OR	occupancy adopted for the plot's settlement	persons per property
N dom	domestic meters on the plot, one property each	—
U nd	non-domestic water per shop meter in the settlement (Section 7.3)	l/d per meter
N nd	shop meters on the plot; none counted on an estate plot	—
U gov	governmental water per government meter in the settlement (Section 7.3)	l/d per meter
N gov	government meters on the plot; none counted on an estate plot	—
N w	workers on an industrial-estate plot (Section 6.4)	—

Each term is one stream of Chapter 6 at its return rate. The first is the domestic stream: properties times occupancy gives people, times 164 gives water, times 0.85 gives sewage. The bracket holds the three streams that return 54 %. A plot of houses has only the first term. A shop, a school or a ministry building has only the bracket. Most plots in a town centre have both. An agricultural meter appears nowhere, because it returns nothing.

Source. *Project rule (engineer, 2026-09-10), built on G201-p60 Table 11, G201-p61 Table 12 (dry industry) and G201-p71 Table 19. Report R2 §15.4.*

Worked for Ibri. Section 7.4 works one Ibri plot through the equation, term by term.

Where it lives. Field QADF on every plot of W14/shp/PLOTS_load.shp, m³/d, written by W14/py/plot_class_v2_apply.py. Its inputs are on the same plot: G_DOM, OR_S, G_NDOM, U_NDOM, G_GOV, U_GOV, WORKERS.

7.3 The unit rates per meter

What it is for. The unit rates turn a settlement's non-domestic and governmental volume into a load on the particular plots that generate it.

The rule. Each settlement's non-domestic and governmental water is a pool, sized by its people. The pool is divided equally among the settlement's shop meters, and among its government meters, outside the industrial estates.

$$U_{\text{nd}} = \frac{0.22 \times 164 \times P_s}{N_{\text{nd,s}}} \qquad (28)$$

$$U_{\text{gov}} = \frac{0.14 \times 164 \times P_s}{N_{\text{gov,s}}} \qquad (29)$$

Symbol	Meaning	Unit
P s	people of the settlement, 2024	persons
N nd,s	shop meters of the settlement outside the industrial estates	—
N gov,s	government meters of the settlement outside the industrial estates	—

The share per meter is equal because the electricity file records no consumption. A large shop and a small one are the same meter to it. The settlement's total is right; how it splits between two shops in the same street is not known.

The estates' meters stay out of the pool for two reasons. Their water is already carried by their workers (Section 6.4), so a pool share would load it twice. And their 1,445 shop meters are 15 % of all the shop meters in the study area: counted in, they would draw that share of the towns' shop water onto two industrial areas and thin the load on every town street.

A settlement with no shop meter or no government meter outside the estates keeps that share on its dwellings, in proportion to their people. Usaybuq, Wadi al Mankas and Ash Shiab have no shop meter outside the estates; Satwah and Wadi al Mankas have no government meter. Their shares ride on the houses, which is where the guideline's own reading of the ratios would have put them in any case.

Where a settlement has few meters the rate per meter swings widely. The government rate runs from 184 l/d per meter in Miayrid to 8,082 in Hijar, which has 4 government meters. The settlement total is right in both, and the volumes behind the extremes are small.

Source. *Project rule (engineer, 2026-09-10): unit rates per meter, the estates' meters out of the pool, the fallback to dwellings. The placement is the declared departure of Section 6.3 (G201-p59).*

Worked for Ibri. Ibri houses 67,106 people and has 4,747 shop meters and 461 government meters outside the estates. U nd = 0.22 × 164 × 67,106 / 4,747 = 510 l/d per shop meter. U gov = 0.14 × 164 × 67,106 / 461 = 3,342 l/d per government meter. The government rate is 6.6 times the shop rate: the governmental pool is 64 % of the shop pool, but it is shared among one-tenth as many meters.

Where it lives. Fields U_NDOM and U_GOV on every plot, l/d per meter; columns NDOM_POOL, GOV_POOL, U_NDOM and U_GOV in W14/analysis/settlements_today.csv. Computed in W14/py/plot_class_v2_apply.py (u_nd_s, u_gv_s), with the fallback to dwellings in the same block.

7.4 One plot worked through

Take an illustrative plot in Ibri with four domestic meters, two shop meters and one government meter. The settlement's occupancy is 6.07 and its unit rates are 510 and 3,342 l/d per meter.

Table 32 The flow of one plot, worked from its meters

Stream and meters	Water, l/d	Return	Sewage, l/d
Domestic: 4 meters × 6.07 persons × 164 l/d	3,983	0.85	3,386
Non-domestic: 2 meters × 510 l/d	1,020	0.54	551
Governmental: 1 meter × 3,342 l/d	3,342	0.54	1,805
Plot	8,345		5,741 = 5.74 m³/d

The plot houses 24.3 people and sends 5,741 l/d to the sewer, 236 l/d per resident. That is well above 171.3 because the plot also carries shop and government water. The same four houses without the shops and the office would send 3,386 l/d, 139.4 per person. This is the placement at work: the flow goes where the meters are.

7.5 Shares and unit rates by settlement

The table sets out, for every settlement, the three water shares, the meters that carry them, the unit rates, the estates' water and the resulting sewage. The plots sum to the settlement in every column. "On dwellings" marks the fallback of Section 7.3.

Table 33 Water shares and unit rates by settlement, 2024

Settlement	People	Domestic m³/d	Non-domestic share m³/d	Governmental share m³/d	Shop meters	Government meters	l/d per shop meter	l/d per government meter	Estates m³/d	Sewage m³/d
Ibri	67,106	11,005	2,421	1,541	4,747	461	510	3,342	–	11,494
Ad Dariz	11,850	1,943	428	272	820	82	521	3,318	–	2,030
Al Araqi	10,696	1,754	386	246	1,018	96	379	2,558	–	1,832

Settlement	People	Domestic m³/d	Non-domestic share m³/d	Governmental share m³/d	Shop meters	Government meters	l/d per shop meter	l/d per government meter	Estates m³/d	Sewage m³/d
Al Aynayn	4,554	747	164	105	204	26	805	4,022	–	780
Al Wahrah	3,351	550	121	77	132	23	916	3,345	–	574
At Tayyib	3,348	549	121	77	179	19	675	4,046	419	799
Al Jibayyah	2,686	440	97	62	96	20	1,009	3,083	–	460
Bat	2,557	419	92	59	51	26	1,809	2,258	–	438
Ad Dibayshi	2,724	447	98	63	236	17	416	3,679	27	481
Tanam	2,116	347	76	49	124	32	616	1,518	105	419
Suwayda al Ma	1,559	256	56	36	34	13	1,654	2,753	–	267
Hijar	1,408	231	51	32	63	4	806	8,082	–	241
Al Ghubayrah	644	106	23	15	105	4	221	3,697	–	110
Al Qurayn	588	96	21	14	14	7	1,515	1,929	–	101
Sayh al Masarrat	1,828	300	66	42	128	99	515	424	–	313
Al Jahli	1,376	226	50	32	143	87	347	363	–	236
Satwah	108	18	4	2	5	0	779	on dwellings	–	19
Al Akheedar	480	79	17	11	69	6	251	1,837	–	82
Al Qali	224	37	8	5	5	22	1,616	234	–	38
Shalashil	152	25	5	3	2	1	2,742	3,490	–	26
Al Makhtibyah	364	60	13	8	45	4	292	2,089	36	82
Usaybuq	84	14	3	2	0	1	on dwellings	1,929	–	14
Wadi al Mankas	32	5	1	1	0	0	on dwellings	on dwellings	–	5
Ash Shiab	44	7	2	1	0	1	on dwellings	1,010	–	8
Miayrid	16	3	1	0	4	2	144	184	–	3
Total	119,893	19,663	4,326	2,753	8,224	1,053	526	2,614	586	20,852

The Tanam estate's plots lie across more than one settlement, which is why estate water appears in several rows. Totals are rounded from unrounded values and may differ by one from the sum of the rows. In 2024 the study area generates 20,852 m³/d: 16,713 domestic, 2,336 non-domestic, 1,486 governmental and 316 from the two estates.

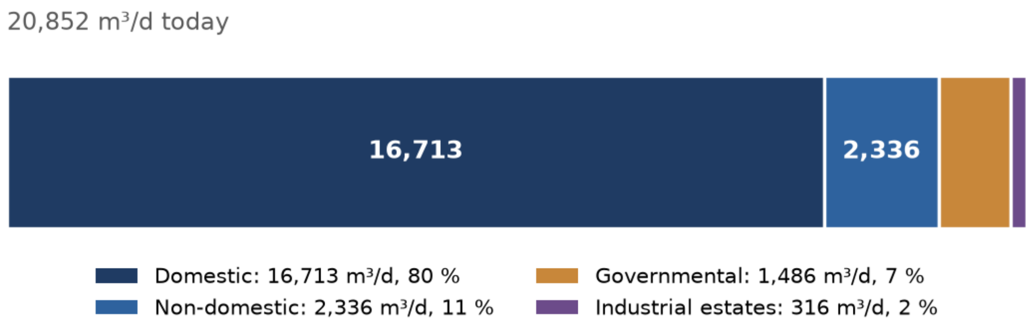

Figure 13   Average dry-weather sewage of the study area in 2024, by stream.

Worked for Ibri. Ibri's row: 11,494 m³/d from 67,106 people is 171.3 l/d per person, the per-person rate of Section 7.1. It has to be: the pools are sized on people, so a settlement with no estate always averages 171.3 l/d per person, whatever the placement inside it.

Where it lives. F.unit_rate_rows() in W14/report/facts_w14.py reads W14/analysis/settlements_today.csv; chart C11 is drawn by W14/report/charts_w14.py.

7.6 The future plot

What it is for. An empty plot has no meters, so its flow is built from the people it houses in each later year.

The rule. A future plot carries its housed people at 171.3 l/d each, all three streams together.

$$q_{\mathrm{fut}} = 164 \times (0.85 + 0.22 \times 0.54 + 0.14 \times 0.54) = 171.3 \quad (30)$$

$$Q_{\mathrm{fut}} = q_{\mathrm{fut}} \times P_{\mathrm{new}} \quad (31)$$

Symbol	Meaning	Unit
q fut	sewage per person on a future plot	l/d
Q fut	sewage of the plot in the year	l/d
P new	people housed on the plot in the year	persons

Where the shops, schools and offices of a district that does not yet exist will stand is not known. Putting each person's share of all three streams on the plot where the person lives keeps the settlement's total exact, and it lands the load on the street the sewer will run in. It is the same total per person as an existing settlement's, as Section 7.5 showed; only the placement differs. The housed people of a settlement are spread over its empty plots of up to 2,000 m², in proportion to plot area capped at 1,000 m², so every plot the network passes carries a share.

An existing plot keeps its base-year load in every later year. All growth, including the growth of a settlement's own people, is housed on empty plots. That is why a residential branch that is already built stays at 139.4 l/d per person, while a new district runs at 171.3.

Source. *G201-p60 Table 11 and G201-p71 Table 19 for the rates. Project rule (engineer, 2026-09-10): all three streams on the future plot; growth spread over empty plots of 2,000 m² or less by area capped at 1,000 m².*

Worked for Ibri. Between 2024 and 2030 the study area houses 19,642 more people. At 171.3 l/d they add 3,364 m³/d, which takes the total from 20,852 to 24,216 m³/d.

Where it lives. W14/py/growth_by_settlement.py: Q_PER_CAP = 0.1713 m³/d per person; the plot fields POP_2030, Q_2030, POP_2055, Q_2055, POP_ULT and Q_ULT, where Q_year = QADF + new people × Q_PER_CAP.

7.7 The fields on the plot

Every term of this chapter is stored on the plot, so any plot's flow can be rebuilt by hand. The plot layer is W14/shp/PLOTS_load.shp.

Table 34 The load fields of the plot layer

Field	Meaning	Unit
SETTLE	settlement the plot belongs to	—
G_DOM, G_NDOM, G_GOV	domestic, shop and government meters on the plot	—
ESTATE, WORKERS	industrial-estate flag and assigned workers	—
OR_S	occupancy adopted for the settlement	persons per property
POP	people, 2024	persons
U_NDOM, U_GOV	the settlement's unit rates per shop and per government meter	l/d per meter
W_DOM, W_NDOM, W_GOV, W_SPEC, W_TOT	water by stream and in total	m³/d
S_DOM, S_NDOM, S_GOV, S_SPEC	sewage by stream	m³/d
QADF	average dry-weather sewage, 2024	m³/d
POP_2030, POP_2055, POP_ULT	people in 2030, 2055 and at saturation	persons
Q_2030, Q_2055, Q_ULT	average dry-weather sewage in 2030, 2055 and at saturation	m³/d

The plot carries average flow only. It carries no peak, no infiltration, no tanker water and no private-well water. Peaking belongs to the accumulated flow in a pipe, infiltration to the pipe's length, and the plant margin to the works. They are added in the design, never stored on the plot.

7.8 Check it yourself

1. Pick any built plot. Rebuild QADF × 1000 from G_DOM, OR_S, G_NDOM, U_NDOM, G_GOV, U_GOV and WORKERS with the equation of Section 7.2. It must match to the litre.
2. Sum QADF by SETTLE and compare with the sewage column of the settlement table. They must agree.
3. For any settlement without an estate, divide its sewage by its people. The answer must be 171.3 l/d per person.
4. Filter the plots on ESTATE: W_NDOM and W_GOV must be zero, and W_SPEC must equal WORKERS × 0.093.
5. For any empty plot, (Q_2030 − QADF) / (POP_2030 − POP) must be 0.1713 m^3/d per person.
6. Pick a plot with only an agricultural meter. Its QADF must be zero.

8 Flow through time and the two design cases

A single saturation flow sizes the pipes. It says nothing about when the treatment plant must be built, and nothing about how little the network will carry in its first years. Both answers need the flow year by year. This chapter builds that series, then sets out the two flows every pipe is checked against: the saturation flow for size, and the opening-year flow at the connection ratio for self-cleansing.

8.1 The model years and the reporting interval

What it is for. The model years fix the flows the deliverables report and the hydraulic models run.

The rule. The Terms of Reference name three design years and the ultimate: a start year to be agreed, 2030 as the middle year, 2055 as the end year, and the ultimate, saturated flow. Population and flow are projected at five-year intervals. The design horizon is the year of completion plus 25 years, or the ultimate saturated flow. The series is calculated every year and reported every five.

Source. *Terms of Reference (Data/scope.pdf) p3 (design horizon), p14 (design years for the models), p15 item 12 (five-year interval). Base year 2024: the year of the electricity accounts.*

Table 35 The model years

Year	Role	People	Qadf, m³/d
2024	base year: today's plots, from the meters	119,893	20,852
2030	middle design year; the opening-year case	139,535	24,216
2055	end design year	244,914	42,266
2070	saturation: the last settlement is full; the sizing case	349,029	60,099

Worked for Ibri. The study area goes from 20,852 m³/d in 2024 to 24,216 in 2030, 42,266 in 2055 and 60,099 at saturation in 2070. Ibri itself is full in 2056.

Where it lives. W14/analysis/W14_growth_by_settlement.xlsx, sheets "Qadf by year m3d" (every settlement, every year 2024 to 2100) and "Five-year Qadf"; F.totals() in W14/report/facts_w14.py.

8.2 How the series is built

What it is for. The series turns today's plots and the growth of each settlement into a flow for every year.

The rule. Existing plots keep their 2024 flow. Each settlement grows at its own rate from the Inception Report, and the new people are housed on its empty plots until they are full; growth a full settlement cannot house overflows to its neighbours. Every person housed adds 171.3 l/d.

$$Q_s(y) = Q_s(2024) + q_{\text{fut}} \times H_s(y) \tag{32}$$

Symbol	Meaning	Unit
Q s(y)	average dry-weather sewage of settlement s in year y	m³/d
Q s(2024)	the sum of its existing plots' QADF	m³/d
q fut	sewage per housed person, 0.1713	m³/d per person
H s(y)	people housed on its empty plots by year y, its own growth and overflow received	persons

Source. *Project rules (engineer, 2026-09-09 and 2026-09-10): growth rates from the Inception Report workbook, sheet Project Pop Settlements; capacity and overflow as the population chapter sets out.*

Worked for Ibri. Ibri grows from 67,106 people in 2024 to 78,099 in 2030, so 10,993 people are housed on its empty plots. At 0.1713 m³/d each they add 1,883 m³/d, which takes Ibri from 11,494 to 13,377 m³/d.

Where it lives. W14/py/growth_by_settlement.py: q = housed × Q_PER_CAP + q0, where q0 is the sum of QADF by settlement; sheet "Qadf by year m3d" of the workbook.

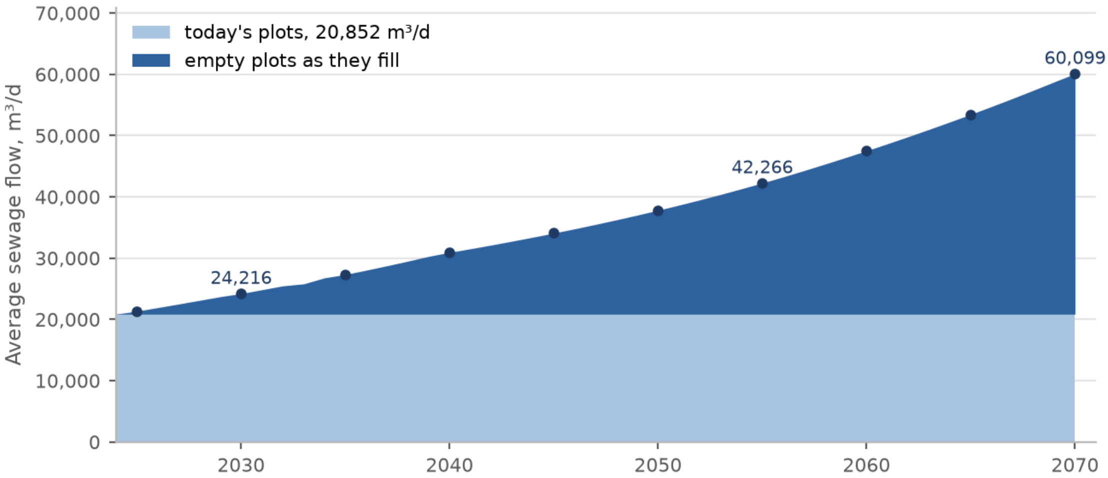

Figure 14 Average dry-weather sewage of the study area by year: today's plots at their base-year load, and the empty plots as they fill. The points mark the five-year reporting intervals.

8.3 The flow at five-year intervals

The table gives every settlement's average flow at five-year steps. A settlement's row is blank once it is full; its last two columns give the year it fills and the flow it then holds, which is the flow its pipes are sized on. Columns past the last saturation year are not printed.

Table 36 Average dry-weather sewage at five-year intervals to saturation, m³/d

Settlement	2024	2025	2030	2035	2040	2045	2050	2055	2060	2065	2070	Saturation year	Saturation flow
Ibri	11,494	11,769	13,377	15,119	17,140	18,901	20,988	23,500				2056	24,049
Ad Dariz	2,030	2,078	2,362	2,670	3,027	3,338	3,706	4,146	4,685			2062	4,917
Al Araqi	1,832	1,876	2,132	2,410	2,732	3,013	3,345	3,742				2057	3,859
At Tayyib	800	813	893	980	1,081	1,169	1,273	1,398	1,618	4,332		2068	6,739
Al Aynayn	780	799	908	1,026	1,163	1,283	1,424	1,593	2,058			2060	2,058
Al Wahrah	574	588	668	755	856	944	1,048	1,173	1,320	1,487	1,792	2070	1,792
Ad Dibayshi	481	492	557	628	710	782	866	967	1,087			2063	1,838
Al Jibayyah	460	471	535	605	686	756	840	940	1,057	1,191	2,397	2070	2,397
Bat	438	448	510	576	653	720	800					2052	819
Tanam	419	428	479	534	597	653	719	797	890	1,078		2069	1,580
Sayh al Masarrat	313	321	364	412	467	515	572	640	720	820		2069	1,167
Suwayda al Ma	267	273	311	351	398	439	488	545	614	691	1,358	2070	1,358
Hijar	241	247	281	317	360	397	440					2053	462
Al Jahli	236	241	274	310	351	388	430	485	642			2063	867
Al Ghubayrah	110	113	128	145	165	181	201	301	441	600	785	2070	785
Al Qurayn	101	103	117	133	150	166	184	206	1,229			2060	1,229
Al Akheedar	82	84	96	108								2036	110
Al Makhtibyah	82	83	92	101	112	122	133	147	163	280		2069	817
Al Qali	38	39	45	51	70	89	111					2054	135
Shalashil	26	27	30	34	39	43	48	53	1,428			2061	2,030
Satwah	19	19	22	24	28	30	34	38	67			2060	67
Usaybuq	14	15	17	19	22	24	26					2054	38
Ash Shiab	8	8	9	10	11	12	14	15				2056	16
Wadi al Mankas	6	6	6	7	8	9	10	11	13	490		2068	865
Miayrid	3	3	3	4	4	5	5	6	6	7	108	2070	108
Total	20,852	21,344	24,216	27,329	30,938	34,085	37,815	42,266	47,523	53,466	60,099	2070	60,099

Worked for Ibri. Ibri fills in 2056 at 24,049 m³/d. The first settlement to fill is Al Akheedar, in 2036; the last are full in 2070, when the study area carries 60,099 m³/d.

Where it lives. F.five_year_rows("q") reads sheet "Five-year Qadf" of W14/analysis/W14_growth_by_settlement.xlsx.

8.4 The opening year and the connection ratio

What it is for. Self-cleansing is decided in the first years, when the pipes carry least. That needs a year and a share of the people actually connected in that year.

The rule. The opening year is the construction year, or 2030, the first design year the Terms of Reference fix; 2030 is used until a construction year is agreed. In that year the connected share is 0.61, the connection ratio of the Inception Report.

The Inception Report carries the connected share of the population as a series. It is 0.51 in 2024, 0.61 in 2028 and 0.67 in 2030. The rule takes 0.61, the value reached in 2028, for the 2030 flow. The Report's own 2030 value is higher, so 0.61 gives less flow than the Report would. For this check less flow is the safe side (Section 8.7).

The guideline assumes that collection coverage reaches 100 % by the end of the planning period, as customers connect to whatever collection system becomes available, and it asks the designer to secure self-cleansing velocities at the initial stage of operation. Full connection belongs to the sizing case; the ratio belongs to the opening years.

Source. *Terms of Reference p14 (start year to be agreed, 2030 middle year). Inception Report R0 workbook, sheet Pop_Wilayat, row W_Pop_Connected. G201-p73 §7.4.4 (coverage 100 % by the end of the period; self-cleansing at the initial stage). Project rule (engineer, 2026-09-11).*

Worked for Ibri. Ibri's 2030 flow is 13,377 m³/d; at 0.61 connected, 8,160 m³/d. The whole study area: 24,216 × 0.61 = 14,772 m³/d.

Where it lives. W14/analysis/design_flows.json: rules.opening_year, rules.connection_ratio_2030, low_case_2030_m3d and settlements[].q_2030_low.

8.5 The two design cases

What it is for. Every pipe is checked twice, on two different flows, because the two checks fail in opposite directions.

The rule. Size and capacity on the saturation flow, fully connected. Self-cleansing on the 2030 flow at the connection ratio. Each is summed over the plots upstream of the pipe, then peaked for that pipe.

$$Q_{\text{size}} = \textstyle\sum_i Q_{\text{ULT,i}} \tag{33}$$

$$Q_{\text{low}} = 0.61 \times \textstyle\sum_i Q_{\text{2030,i}} \tag{34}$$

Symbol	Meaning	Unit
Q size	average flow for sizing: the sum over the upstream plots	m³/d
Q low	average flow for the self-cleansing check	m³/d
Q ULT,i, Q 2030,i	fields Q_ULT and Q_2030 of upstream plot i	m³/d
0.61	connection ratio in the opening year	—

Both averages are then peaked per pipe. Over 100 properties upstream, the Merrimack formula gives the peak daily flow, Qpdf = 2.65 Qadf^0.879, both in Ml/d. At 100 properties or fewer, the Peltier alternative gives the peak factor, PF = 1.5 + 1/√Qm, with the average in l/s. The properties upstream in a year are counted as G_DOM + (POP_year − POP) / OR_S per plot. The sizing case counts the properties at saturation. The low case counts the connected properties, those standing in 2030 × 0.61, so that the count choosing the formula describes the same customers as the flow it peaks. A pipe with more than 100 properties standing can therefore still take Peltier in the low case; the lower count gives the lower peak at small flows, the safe direction for self-cleansing. This connected count is recommended in the design-flow handoff, to be confirmed by the engineer. The sizing case then takes infiltration at 720 l/d per kilometre of the pipe's own length, added along the run.

Table 37 Which flow for which check

Check	Flow	Why this flow
Pipe size and capacity: d/D and velocity limits	Q_ULT upstream, peaked, plus infiltration	saturation and full connection; an under-estimate surcharges the pipe
Self-cleansing: velocity and tractive slope	Q_2030 × 0.61 upstream, peaked	the opening years; an over-estimate declares a silting pipe self-cleansing
Plant capacity and phasing	settlement totals by year, plus infiltration, plus 10 %	the growth curve decides when each stage is needed

The self-cleansing check applies both approaches the guideline requires, velocity and tractive force, and the steeper gradient governs. At the head of a system, where the velocity cannot be reached, tractive force alone sets the gradient. Each pipe falls in one of four classes on the low case:

Table 38 The self-cleansing classes on the low case

Class	Test	Action
Velocity pass	at least 0.75 m/s at the low-case peak	none
Tractive pass	gradient at or above the tractive-force minimum at the low-case peak	none
Early cleansing	flow too small for either test	on the flushing list; not upsized
Fails both: regrade	real flow, but too flat for either test	a design fault: steepen the pipe

Two values the guideline does not give. The guideline sets no numeric tractive tension, so it is carried as a parameter and the class counts are reported against it until Nama Water Services confirms a value. Nor does it give a low-flow threshold for the early-cleansing class: the 1.5 l/s often quoted comes from the literature (Mara), not from PAM-GUD-203, and is used only as a tagged outside assumption.

Source. *Project rule, the two flow cases (engineer, 2026-09-11). G201-p71 §7.4.2 (Merrimack, over 100 properties), G201-p72 (Peltier alternative; hourly peak factor not to exceed 5.0, a recommendation), G201-p72 §7.4.3 (infiltration 720 l/d per km), G201-p73 §7.4.5 (plant +10 %). G203 §4.2.2.1 pp 25–27 (two approaches, the steeper governs; tractive force alone at the head); G203 §4.2.6 p28 (no threshold). Tractive tension: GAP-9. Low-case property count: the connected count, recommended in the design-flow handoff (W14/docs/DESIGN_FLOWS_FOR_NETWORK.md §8), to be confirmed by the engineer.*

Worked for Ibri. For the whole study area the sizing average is 60,099 m³/d and the low-case average 14,772 m³/d, before peaking. Properties number 22,559 in 2024, 26,255 in 2030 and 70,328 at saturation. The low case counts 26,255 × 0.61 = 16,016 of them as connected. The plant, before infiltration, takes 66,109 m³/d at saturation with its margin.

Where it lives. W14/analysis/design_flows.json (rules, totals, properties, low_case_2030_m3d, stp_ultimate_with_margin_m3d; the low-case count under rules.open_rulings_recommended.low_case_property_count); the handoff note W14/docs/DESIGN_FLOWS_FOR_NETWORK.md §2 to §4. The network engine must read Q_ULT and Q_2030 from W14/shp/PLOTS_load.shp.

8.6 Why never the saturation flow times the ratio

What it is for. A shortcut is tempting: take the saturation flow already summed for sizing and multiply it by the connection ratio. This section shows why that is wrong.

The rule. The low case is always Q_2030 × 0.61, never Q_ULT × 0.61.

The shortcut mixes the people of 2070 with the connection share of 2028. It assumes that every district carries 61 % of its saturation flow in the opening year. A built-up district is close to that; a new district carries almost nothing in 2030, because its plots are still empty. The error therefore lands on exactly the pipes where silting happens, and it lands on the unsafe side.

Table 39 The low case against the shortcut, m³/d

Settlement	Q 2030	Q 2030 × 0.61 (right)	Q saturation	Q saturation × 0.61 (wrong)	Overstated by, times
Ibri	13,377	8,160	24,049	14,670	1.8
Ad Dariz	2,362	1,441	4,917	2,999	2.1
At Tayyib	893	545	6,739	4,111	7.5
Al Qurayn	117	72	1,229	750	10.5
Shalashil	30	19	2,030	1,238	66.9
Wadi al Mankas	6	4	865	528	135.3
Study area	**24,216**	**14,772**	**60,099**	**36,660**	**2.5**

Worked for Ibri. Over the study area the shortcut gives 60,099 × 0.61 = 36,660 m³/d against 14,772 m³/d really expected in 2030, 2.5 times too much. In Ibri, already built up, the shortcut is 1.8 times too much. In Shalashil, which is mostly empty plots today, it is 67 times too much: the pipes of its new district would be declared self-cleansing on a flow that will not arrive for decades.

Where it lives. W14/analysis/design_flows.json, settlements[] (q_2030, q_2030_low, q_ult); W14/docs/DESIGN_FLOWS_FOR_NETWORK.md §2.

8.7 Accuracy is directional

What it is for. This is the reason the design carries two cases rather than one best estimate.

The rule. One population cannot serve every use, because the safe direction reverses between them. Size capacity on the high case; set the self-cleansing check and the early washing schedule on the low case; carry the connection ratio explicitly in the low case.

Table 40 The three uses of the flow and their dangerous direction

Use	Driven by	Dangerous direction	Case used
Pipe sizing	peak flow and its distribution	under-estimate: the pipe surcharges	Q_ULT, 2070, fully connected
Plant capacity and staging	total average flow and its growth curve	under-estimate: the plant is too small or a stage comes late	settlement totals by year, fully connected
Self-cleansing and the early washing schedule	minimum flow in the opening years	over-estimate: pipes declared self-cleansing while they silt; washing stopped too soon	Q_2030 × 0.61

An earlier tutorial applied a connection ratio to the flow of every year. That is replaced here: the sizing case and the plant are fully connected, and only the low case carries the ratio.

Source. *Project doctrine, accuracy is directional (_BRAIN/07_PROJECT_STATE.md §2 item 1d; _BRAIN/02_DESIGN_CRITERIA.md §11.1), with G201-p73 §7.4.4 and the Inception Report connection ratio.*

Worked for Ibri. Sized on 60,099 m³/d, checked for self-cleansing on 14,772 m³/d: the same network, two flows 4.1 times apart, each on its safe side.

Where it lives. The two flows are fields Q_ULT and Q_2030 on every plot of W14/shp/PLOTS_load.shp; the ratio is rules.connection_ratio_2030 in W14/analysis/design_flows.json.

8.8 Check it yourself

1. Open sheet "Qadf by year m3d" of W14/analysis/W14_growth_by_settlement.xlsx. The 2030 column must sum to 24,216 m³/d and the 2070 column to 60,099.
2. Sum Q_2030 over the plots and multiply by 0.61. The answer must be 14,772 m³/d, the low_case_2030_m3d of design_flows.json.
3. For one settlement, divide (Q in any year − Q in 2024) by (people in that year − people in 2024). The answer must be 0.1713 m³/d per person.
4. Open the Inception Report workbook, sheet Pop_Wilayat, row W_Pop_Connected, and read the values for 2024, 2028 and 2030.
5. Pick a settlement that is mostly empty today and compare Q_ULT × 0.61 with Q_2030 × 0.61. The gap is the error the shortcut would put into its pipes.
6. Read the Terms of Reference, page 14, for the design years and page 15, item 12, for the five-year interval.

9 Peak flow and infiltration

The chapters before this one end with an average daily flow on every plot, for 2024, 2030, 2055 and saturation. A sewer is not sized on an average. It must carry the peak of the flow that reaches it, plus the groundwater that leaks in through its joints, and it must also run fast enough to clean itself while the flow is still small. This chapter takes the plot averages to those two design flows, pipe by pipe, and then to the flows at the treatment plant.

9.1 The plot carries average flow only

What it is for. To keep each quantity where it belongs. The plot knows how much sewage it makes on an average day. It does not know how that flow combines with its neighbours', so it cannot know its own peak.

The rule. Every plot stores four average dry-weather flows in m³/d and nothing else: 2024, the opening year, 2055 and saturation. Peak factor, infiltration and the plant margin are applied later, to flows that have been summed along the network. Tanker deliveries and private wells are in no flow until their records arrive.

The reason is attenuation. Each house discharges in short bursts. A pipe serving a few houses sees those bursts almost one at a time, so its peak is many times its average. A pipe serving thousands sees them overlap and smooth out, so its peak is a small multiple of its average. The peak factor therefore falls as the flow accumulates, and peaks do not add. Summing the plot peaks, or the peaks of the pipes arriving at a junction, always overstates the peak downstream.

Ibri number. At saturation 70,766 plots carry a flow, 60,099 m³/d in all. Peaked plot by plot with the small-catchment formula of Section 9.5 and added, they would ask the plant for 661,128 m³/d; even with every plot held to a factor of 5.0 it is 297,746 m³/d. The Merrimack formula applied once to the area's accumulated flow gives 97,022 m³/d. The 5.51 km² test area shows the same effect at a smaller scale: the 19 pipes that join the main pipe carry 3,764 m³/d between them at saturation; their own peaks add to 122.0 l/s, while the peak of their combined flow is 98.3 l/s.

Source. PAM-GUD-201 §7.4.2, G201-p71: the peak factor is defined for an area, a catchment or sub-catchment, not for a property. G201-p72: infiltration is a rate per length of sewer. Project rule (engineer, 2026-09-11): the plot stores average flow only, W14/docs/DESIGN_FLOWS_FOR_NETWORK.md §1 and §8.

Where it lives. W14/shp/PLOTS_load.shp, fields QADF (2024), Q_2030, Q_2055, Q_ULT, all m³/d. The rules in machine form: W14/analysis/design_flows.json, key "rules".

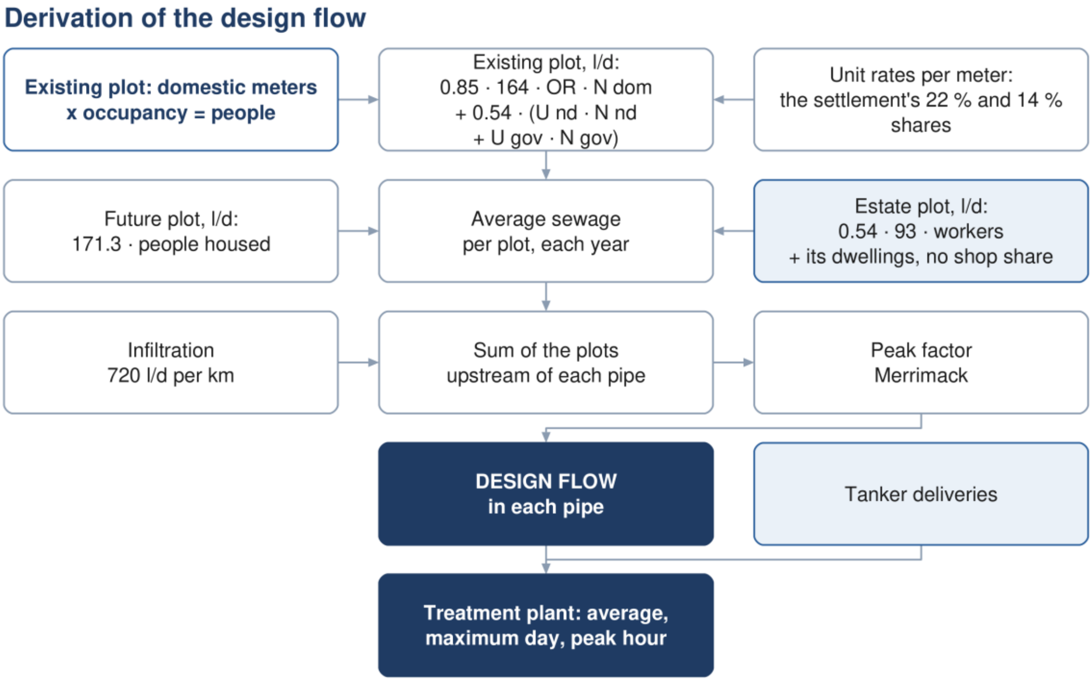

Figure 15 From the plot's average flow to the design flow in each pipe and at the plant.

9.2 Summing upstream and counting properties

What it is for. To give each pipe the two numbers the peak formulas need: the average flow of everything upstream, and the number of properties that make it.

The rule. The average flow of a pipe is the sum of the flows of every plot that drains through it, in the year of the case being checked.

$$Q_{\text{pipe,y}} = \sum_{i \in \text{up}} q_{i,y} \tag{35}$$

The number of properties upstream is counted per plot and summed the same way. For an existing plot it is the domestic properties counted from the electricity meters. Growth adds properties at the settlement's occupancy, so a future plot, or a built plot that gains people, adds its extra people divided by the occupancy.

$$N_{\text{y}} = \sum_{i \in \text{up}} \left(G_{\text{dom,i}} + \frac{P_{\text{y,i}} - P_{\text{2024,i}}}{\text{OR}_{\text{s}}} \right) \tag{36}$$

Symbol	Meaning	Unit
Q pipe,y	average dry-weather flow in the pipe in year y	m³/d
q i,y	average flow of plot i in year y (QADF, Q_2030, Q_2055 or Q_ULT)	m³/d
N y	properties upstream in year y	—
G dom,i	domestic properties counted on plot i in 2024 (G_DOM)	—
P y,i ; P 2024,i	people on plot i in year y (POP_2030, POP_ULT) and in 2024 (POP)	persons
OR s	occupancy of the plot's settlement (OR_S)	persons per property
up	the set of plots draining through the pipe	—

The count is fractional, because future people are spread across plots by capacity rather than by whole houses. The threshold in Section 9.3 compares the sum with 100; the total is not rounded before the comparison.

Ibri number. Over the whole study area the count gives 22,559 properties in 2024, 26,255 in 2030 and 70,328 at saturation. Ibri's occupancy is 6.07, so every 6.07 people added to an Ibri plot count as one more property.

Source. Project rule (engineer, 2026-09-11), W14/analysis/design_flows.json, rules.properties_per_plot_in_year. Occupancy by settlement: report R2 §14.4.

Where it lives. Per plot: G_DOM, POP, POP_2030, POP_ULT, OR_S in W14/shp/PLOTS_load.shp. The pipe summation is the design engine's job; the engine does not read these fields yet (DESIGN_FLOWS_FOR_NETWORK.md §6). The worked pipe below is summed by this chapter's build script, TUTORIALS/T04/ch04_peak_tiers.py, function _net.

9.3 Which peak formula

What it is for. To pick, for each pipe and each flow case, the formula that turns its average into its peak.

The rule. More than 100 properties upstream: the Merrimack formula. 100 properties or fewer: the Peltier formula. The choice is made again for each case, because the same pipe can have more than 100 properties at saturation and fewer in the opening year.

The guideline makes Merrimack obligatory for an area of over 100 properties and gives no method for smaller ones. It then offers Peltier as an alternative, the method of NWS's 2024 Integrated Master Plan. Using Peltier for the small heads is the project's reading of that gap. The two formulas do not meet at 100 properties; Section 9.5 shows the jump.

Source. G201-p71 §7.4.2 (Merrimack, over 100 properties); G201-p72 (Peltier, introduced as the alternative). Project rule for the split (engineer, 2026-09-11), design_flows.json rules.peak_over_100_properties and rules.peak_100_properties_or_fewer.

Ibri number. Of the 1,261 loaded street pipes of the test-area design, 138 carry more than 100 properties at saturation and take Merrimack; 1,123 take Peltier. In the 2030 low case, counting the connected properties (Section 9.9), 78 take Merrimack and 1,183 Peltier; counted on the properties standing it would be 109 and 1,152. Most pipes of a street network are small heads, so Peltier is the formula most pipes use.

Where it lives. design_flows.json rules. The choice per pipe will be a field of the engine's pipe output; the test-area design stored both factors, PF and PF_PELT, side by side.

9.4 Merrimack

What it is for. The peak of a pipe or area serving more than 100 properties.

The rule. The formula gives the peak flow directly; the peak factor is the ratio.

$$Q_{\mathrm{pdf}} = 2.65 \times Q_{\mathrm{adf}}{}^{0.879} \tag{37}$$

$$\mathrm{Pf} = \frac{Q_{\mathrm{pdf}}}{Q_{\mathrm{adf}}} \tag{38}$$

Symbol	Meaning	Unit
Q pdf	peak flow (the guideline calls it the peak daily flow)	**Ml/d**
Q adf	average daily flow	**Ml/d**
Pf	peak factor	—

Both flows are in megalitres per day. The exponent is 0.879, not 1, so the formula is not unit-free. Feeding it m³/d gives a different peak factor, not a scaled one. Divide m³/d by 1,000 before using it, or use the m³/d form in Section 9.7, whose constant is 6.113, not 2.65.

Source. G201-p71 §7.4.2: the formula, its units and the definition Pf = Qpdf / Qadf.

Ibri number. The factor falls steadily as the area grows. The table applies Merrimack to real accumulated flows from the plot layer.

Table 41 Merrimack peak factor at four scales of Ibri flow

Area	Average, m³/d	Average, Ml/d	Peak, m³/d	Pf
The 5.51 km² test area at saturation	3,764.3	3.764	8,497	2.26
Ibri town at saturation	24,048.7	24.049	43,374	1.80
The study area in 2030	24,216.4	24.216	43,639	1.80
The study area at saturation, 2070	60,099.0	60.099	97,022	1.61

Where it lives. Totals: facts_w14.totals() and the plot layer (Ibri: SETTLE = IBRI; test area: plots inside W8/shp/W8_boundary.shp). Formula: function merrimack in this chapter's script.

9.5 Peltier

What it is for. The peak of a pipe serving 100 properties or fewer, which is most of the pipes in a street network.

The rule. The formula gives a peak factor from the average flow in litres per second; the peak is the average times the factor.

$$\mathrm{Pf_{ww}} = 1.5 + \frac{1}{\sqrt{Q_\mathrm{m}}} \tag{39}$$

$$Q_\mathrm{pdf} = Q_\mathrm{adf} \times \mathrm{Pf_{ww}} \tag{40}$$

Symbol	Meaning	Unit
Pf ww	peak factor for wastewater flows	—
Q m	average daily flow	**l/s**
Q adf ; Q pdf	average and peak flow in the second line	m³/d

The guideline prints 1 in the numerator. The Peltier relation as usually published carries 2.5 over the root. The guideline's page prints 1; its text layer reads 1, and the page was also checked as a high-resolution render and a glyph-level dump when the previous tutorial was written. This tutorial uses the formula as printed. The printed form gives lower peaks than the published one: on the self-cleansing side that is the cautious direction, and on these small pipes the minimum diameter, not the peak, sets the size. Whether NWS meant 2.5 is a question for NWS.

Ibri number. An Ibri property at saturation makes 1.040 m³/d on average (24,049 m³/d over 23,124 properties, shops and offices included). A catchment of exactly 100 such properties makes 104.0 m³/d, which is 1.204 l/s. Peltier gives 1.5 + 1/√1.204 = 2.41. One more property moves the pipe to Merrimack, which gives 3.48: the peak jumps by 45 % at the threshold. The jump belongs to the two formulas, not to the calculation, and it is why the formula is recorded with every peak.

Source. G201-p72 §7.4.2: the formula, the note that Qm is in litres per second, and its IMP2024 origin.

Where it lives. Function peltier_pf in this chapter's script; the test-area design stored the Peltier factor as PF_PELT in W8/shp/W8_pipes.shp, on its earlier flat loads.

9.6 The hourly peak factor of 5.0

What it is for. To limit the factor at the very top of the network, where Peltier grows without bound as the flow goes to zero.

The rule. The guideline recommends that the hourly peak factor should not exceed 5.0. It is a recommendation, not a limit. The design computes the factor uncapped, and where it applies 5.0 instead it says so for that pipe; it never truncates silently. With the printed Peltier, the factor passes 5.0 below an average of 0.0816 l/s, which is 7.05 m^3/d.

Source. G201-p72, the second note under the Peltier formula. The factor it limits is the hourly one, which is the sense in which the network uses the peak (Section 9.11).

Ibri number. On the test area, 392 loaded street pipes have an uncapped factor above 5.0 at saturation and 648 in the 2030 low case. They are the first pipes below a head. A single Ibri plot at saturation makes about one cubic metre a day, and Peltier gives it a factor of 10.8. The capacity test on these pipes is not affected, because the minimum diameter governs them; the self-cleansing audit places them in the early-cleansing class.

Where it lives. Peak factor per pipe: function peak in this chapter's script, uncapped. The class list: W14/docs/DESIGN_FLOWS_FOR_NETWORK.md §4.

9.7 Units

What it is for. The two formulas sit on facing pages and want different units. Every mistake in this chapter that is not arithmetic is a unit mistake.

The rule. Store plot flows in m^3/d. Convert at the formula, and convert back.

Table 42 Conversions used in the peak and infiltration steps

From	To	Operation
m^3/d	Ml/d (Merrimack)	÷ 1,000
m^3/d	l/s (Peltier, pipe hydraulics)	÷ 86.4
l/d per km, length in m	m^3/d	× length ÷ 1,000 ÷ 1,000
Merrimack in Ml/d	Merrimack in m^3/d	Qpdf = 6.113 × Qadf^0.879, both in m^3/d

The m^3/d form of Merrimack follows from the first: 1,000 × 2.65 × (Q/1,000)^0.879 equals 2.65 × 1,000^0.121 × Q^0.879, and 1,000^0.121 is 2.3067.

$$Q_{\text{pdf}} = 6.113 \times Q_{\text{adf}}^{0.879} \quad (\text{both in m}^3/\text{d}) \tag{41}$$

Source. G201-p71 (Ml/d), G201-p72 (l/s). The constant is derived here, it is not a guideline value.

Ibri number. The saturation flow of the study area, 60,099 m^3/d, is 60.099 Ml/d or 696 l/s. Merrimack gives 97,022 m^3/d by either form.

Where it lives. Constants LS and MK_M3D in this chapter's script.

9.8 Infiltration

What it is for. Groundwater enters a sewer through joints, cracks, connections and chambers. It takes capacity, adds pumping and treatment, and has to be designed for.

The rule. A new sewer carries 720 litres a day per kilometre of its own length. The allowance belongs to the pipe, not to the plot: each pipe adds it for its own length and passes it downstream, so a pipe carries the allowance of all the sewer above it. It is added after the peak and is not peaked itself, because a steady leak does not follow the daily cycle. Storm water is not considered. Flow collected by tanker or vacuum truck carries no infiltration.

$$Q_{\text{inf}} = \frac{i \times L}{10^6} \tag{42}$$

$$Q_{\text{design}} = Q_{\text{pdf}} + Q_{\text{inf}} \tag{43}$$

Symbol	Meaning	Unit
Q inf	infiltration carried by the pipe	m³/d
i	infiltration rate for a new sewer, 720	l/d per km
L	length of sewer upstream of the pipe's lower end, the pipe included	m
Q design	flow the pipe is sized on	m³/d, then l/s

Table 43 Infiltration allowances in the guideline

Network	Allowance
Newly designed network	**720 l/d per km of sewer**
Existing network, inland or outside groundwater influence	10 % of the wastewater flow
Existing network in a groundwater zone or coastal area	up to 40 % of the wastewater flow
Tanker or vacuum collection	none

Two of the allowances are shares of flow and one is a rate per length, so they are not interchangeable. The 10 % and 40 % apply only where an existing network is kept in service, and choosing between them needs groundwater evidence for that network. The guideline does not say whether infiltration is added before or after peaking; adding it after is the project's practice.

Ibri number. The test-area design has 71.64 km of sewer. At 720 l/d per km that is 51.6 m³/d, against 3,764 m³/d of saturation sewage from the same plots: 1.4 %. Infiltration is small in a new, inland network. It still matters at the plant, where it arrives every hour of the year.

Source. G201-p72 §7.4.3 (the three allowances; the volume must be accounted for; storm water not considered); G201-p73 (tanker or vacuum collection needs none). Added after peaking: project practice, the guideline is silent on the order.

Where it lives. design_flows.json rules.infiltration_l_per_day_per_km and rules.infiltration_applies; pipe lengths LEN_M in W8/shp/W8_pipes.shp; total length net_km in W8/run/summary.json.

9.9 Two flow cases in every pipe

What it is for. One flow cannot serve two tests. The capacity test fails on an under-estimate of flow. The self-cleansing test fails on an over-estimate, because a pipe credited with flow it does not get is declared self-cleansing while it silts.

The rule. Size on the saturation flow, Q_ULT, fully connected. Check self-cleansing on the opening-year flow, Q_2030, times the connection ratio 0.61. Sum each case upstream, count its properties (in the low case the connected ones, see the box below), and peak it by its own formula. The low-case peak feeds both the velocity test and the tractive-slope test. Never use saturation times the ratio.

$$Q_{\text{low}} = 0.61 \times \textstyle\sum_{i \,\in\, \text{up}} q_{2030,\text{i}} \tag{44}$$

Symbol	Meaning	Unit
Q low	low-case average flow for the self-cleansing test	m³/d
q 2030,i	average flow of plot i in 2030 (Q_2030)	m³/d
0.61	share of properties connected, reached by 2028 in the Inception Report	—

Ibri number. For the whole study area the low case is 24,216 × 0.61 = 14,772 m³/d. Saturation times the ratio would give 36,660 m³/d, 2.5 times as much. The error concentrates in the new districts: a pipe in an empty district would be credited with 61 % of its saturation flow in 2030, when it carries almost nothing.

Source. Project rule (engineer, 2026-09-11), _BRAIN/02_DESIGN_CRITERIA.md §1, the two-flow-case row. Coverage 100 % by the end of the planning period: G201-p73 §7.4.4. The opening year is the construction year or 2030 (Terms of Reference). The self-cleansing tests: G203 §4.2.2.1, pp 25–27; early phases: G203 §4.2.6, p28. Property count in the low case: recommended in the design-flow handoff, to be confirmed by the engineer (W14/docs/DESIGN_FLOWS_FOR_NETWORK.md §8).

A point to confirm: the property count in the low case. The low case counts the connected properties: those standing in 2030 times 0.61. This is recommended in the design-flow handoff, to be confirmed by the engineer. Near the threshold the count decides the formula. A pipe with more than 100 and up to about 163.9 properties standing (101 to 163 in whole properties) has 100 or fewer connected, so it takes Peltier instead of Merrimack. On the test area 31 pipes sit in that band and every one of them gets a lower low-case peak, which is the cautious direction for this test. Counted on the properties standing, 109 pipes would take Merrimack in the low case instead of 78. The ruling should be made before the self-cleansing audit runs.

Where it lives. W14/shp/PLOTS_load.shp fields Q_ULT and Q_2030; design_flows.json keys rules.size_on, rules.self_cleansing_on, rules.connection_ratio_2030, low_case_2030_m3d and rules.open_rulings_recommended.low_case_property_count.

9.10 A worked pipe in Ibri

What it is for. To run Sections 9.2 to 9.9 once, end to end, on a real street sewer with real plots.

The pipe is P-0749 of the test-area design, a secondary main sewer (Chapter 10) of DN200 laid at 0.50 %, from chamber MH-0462 to MH-0491. It is the street main sewer that carries the most properties at saturation without itself joining the main pipe. Above its lower end are 79 plots on 1,071 m of sewer: 49 built and 30 empty, with 89 domestic properties counted today. Each plot of the test area is given to its nearest street pipe, and the sums run down the design's own tree.

Table 44 Design flows of P-0749, a secondary main sewer in Ibri

Step	Sizing case, saturation	Low case, 2030
Plots upstream	79	79
Average flow of the year, Σ plots, m³/d	105.85 (Q_ULT)	94.97 (Q_2030)
Connection ratio	1.00	0.61
Average flow of the case, m³/d	**105.85**	**57.93**
in Ml/d	0.1058	0.0579
in l/s	1.225	0.670
Properties upstream, Σ (G_DOM + ΔP / OR_S)	101.3	90.8 standing
Properties counted for the formula	101.3	55.4 connected (× 0.61)
Formula	Merrimack (over 100)	Peltier (100 or fewer)
Peak factor	3.48	2.72
Peak flow, l/s	4.26	1.82
Infiltration, 1,071 m at 720 l/d per km, l/s	0.0089	not added
Design flow, l/s	**4.27**	**1.82**
Used for	capacity: d/D and maximum velocity	velocity and tractive-slope tests

Read the table step by step. At saturation the 79 plots make 105.85 m³/d, which is 0.1058 Ml/d. They hold 101.3 properties, so Merrimack applies: $2.65 \times 0.1058^{0.879} = 0.3681$ Ml/d, a factor of 3.48 and a peak of 4.26 l/s. Infiltration on 1,071 m is 771 l/d, or 0.0089 l/s, which brings the design flow to 4.27 l/s. In the low case the same plots make 94.97 m³/d in 2030; 0.61 of that is 57.93 m³/d, or 0.670 l/s. In 2030 90.8 properties stand and 55.4 of them are connected, so Peltier applies: $1.5 + 1/\sqrt{0.670} = 2.72$, and the peak is 1.82 l/s.

This pipe sits on the threshold, which is why it makes a good example. With the other formula the saturation factor would be 2.40 instead of 3.48, and the low-case factor 3.74 instead of 2.72. The low-case figure is the one that matters: the self-cleansing tests on this pipe are judged on 1.82 l/s, and the other formula would have credited it with 2.51 l/s.

The low case is taken without infiltration: recommended in the design-flow handoff, to be confirmed by the engineer. Here the allowance would add 0.0089 l/s to 1.82 l/s and cannot change the result; on a long pipe with very little flow it could, and adding flow is the unsafe direction for this test.

For scale, the same steps on P-0741, the header at the largest join of the test area (DN315), which collects 1,004 plots from 19.49 km of sewer:

Table 45 Design flows of P-0741, a secondary header at its join to the main pipe

Case	Average, m³/d	Properties	Formula	Pf	Peak, l/s	Infiltration, l/s	Design, l/s
Saturation	1,190.5	1,150	Merrimack	2.59	35.75	0.162	**35.92**
Low, 2030 × 0.61	551.4	534 connected	Merrimack	2.85	18.17	not added	**18.17**

The header's factor is 2.59 against 3.48 for the street sewer, on 11 times the flow. Its infiltration, 0.162 l/s, is 0.5 % of its peak.

Source. Formulas G201-p71–72; infiltration G201-p72; cases and threshold as in Sections 9.3 and 9.9.

Where it lives. Pipes P-0749 and P-0741 in W8/shp/W8_pipes.shp (fields LABEL, ND_UP, ND_DN, DN_MM, SLOPE_PCT, LEN_M, TIER, IS_JOIN); plots in W14/shp/PLOTS_load.shp; the allocation and summation in _net() of this chapter's script. The test-area design's own flow fields (N_PROPS, QADF_M3D, PF, QPEAK_LS) were built on the earlier flat load per property and do not match this table.

9.11 The flows at the treatment plant

What it is for. The plant is sized on three flows, not one. Structures the flow passes through are sized on the peak hour; biological treatment is sized on the annual average with the inlet load.

The rule. The incoming flow is the average sewage flow plus the infiltration of the network, increased by 10 % as an operational safety allowance. The allowance sits on top of any duty and standby redundancy and is never netted against it. The peak hour is the Merrimack peak of the accumulated flow at the plant: the guideline equates the design peak hourly flow with Qpdf, the quantity Merrimack returns.

$$Q_{\mathrm{AAF}} = (1 + m) \times (Q_{\mathrm{adf}} + Q_{\mathrm{inf}}) \tag{45}$$

$$Q_{\mathrm{PHF}} = (1 + m) \times \left(Q_{\mathrm{pdf}} + Q_{\mathrm{inf}}\right) \tag{46}$$

Symbol	Meaning	Unit
Q AAF	average annual flow, the design average flow	m³/d
Q PHF	peak hourly flow	m³/d
Q adf ; Q pdf	average and Merrimack peak of the sewage reaching the plant	m³/d
Q inf	infiltration of the whole network upstream, 720 l/d per km	m³/d
m	design margin, 0.10	—

The maximum day flow, the largest 24-hour volume, is defined by the guideline, but no factor for it is given anywhere in PAM-GUD-201 or PAM-GUD-203. It stays open until NWS supplies inflow records for its plants or confirms the standard to be used for design flows (the guideline names BS EN 752 and the British Water Code of Practice for Flows and Loads). The margin is carried on the peak hour as well as on the average; that is a reading of Table 29, which applies it to the incoming flows.

Table 46   Flows at the plant if the whole study area drains to one works, before infiltration and tankers

Flow	2030	2055	Saturation, 2070
Average sewage from the plots, m³/d	24,216	42,266	60,099
Merrimack peak factor	1.80	1.68	1.61
Peak hour before margin, m³/d	43,639	71,202	97,022
in l/s	505	824	1,123
Average annual flow with 10 %, m³/d	**26,638**	**46,492**	**66,109**
Peak hourly flow with 10 %, m³/d	**48,003**	**78,322**	**106,724**
Infiltration with margin, add to both	+ 0.792 m³/d per km of network		
Maximum day flow	no factor in the guidelines		

Ibri number. At saturation the plant average before infiltration is 60,099 × 1.10 = 66,109 m³/d, the figure in the design-flow handoff. The peak hour is 106,724 m³/d, a factor of 1.61 on the average. The full-area network has not been laid, so its infiltration cannot be totalled yet; each 100 km of new sewer will add 79.2 m³/d with the margin. Tanker deliveries are added when their records arrive. The table assumes one works for the whole area; the plant options decide how the flow is split.

Source. G203-p65 Table 29 (incoming flow; AAF, MDF and PHF defined, PHF = QPDF), G203-p66 (pass-through structures on PHF, biology on AAF and load), G201-p73 §7.4.5 (10 % margin for new STPs, over and above redundancy), G201-p71 (BS EN 752 and British Water named for project-specific design flows).

Where it lives. Totals by year: W14/analysis/W14_growth_by_settlement.xlsx, sheet Qadf by year m3d, through facts_w14.totals(); design_flows.json stp_ultimate_with_margin_m3d.

9.12 Check it yourself

1. In QGIS, open W14/shp/PLOTS_load.shp, filter SETTLE = 'IBRI' and sum Q_ULT. You should get 24,048.7 m³/d, which matches the Ibri row of design_flows.json, 24,048.8, within its rounding.
2. In the same layer, compute "G_DOM" + ("POP_ULT" - "POP") / "OR_S" with the field calculator and sum it over the whole layer. It should be close to 70,328 properties.
3. In a spreadsheet, enter =1000*2.65*(Q/1000)^0.879 with Q the study-area saturation flow, 60,099. You should get 97,022 m³/d. Then drop the two factors of 1,000 and see how far the answer moves.
4. Select pipe P-0749 in W8/shp/W8_pipes.shp and every pipe upstream of chamber MH-0462, then the plots of the test area whose nearest street pipe is in that set. You should find 79 plots and 105.85 m³/d of Q_ULT.
5. Check that the low case of the study area is 24,216 × 0.61 = 14,772 m³/d, and that no calculation uses 36,660 m³/d instead.
6. Re-read G201 pages 71 to 73 and G203 pages 65 and 66 against Sections 9.3 to 9.11. Every constant here is on those pages, except the connection ratio, which comes from the Inception Report.

10 The network and its tiers

A sewer network is a hierarchy. Houses drain into short connecting pipes, those into the street sewers, the street sewers into a few collecting mains, and only the collecting mains reach the trunk to the plant. This chapter sets out the guideline's names for those tiers, the names this project uses from now on, what the built network teaches about how the tiers connect, and the fixed inputs every layout starts from.

10.1 What the tiers are for

What it is for. The tier of a pipe decides its minimum diameter, the gradient table it follows, what it may connect to, and what it is called in the drawings and the model. A layout without tiers is a set of branches; a layout with them can be built.

The rule. Every pipe carries one tier, named in the guideline's words.

Source. G203 §3.2, p17; G203 §4, p21; G203 §5, p35.

Ibri number. Every one of the 1,414 pipes of the test-area design carries a tier value; Section 10.4 shows how those values translate.

Where it lives. Field TIER of the pipe layer of every design run.

10.2 The guideline's three tiers

What it is for. To use the same words as NWS, so that a pipe called a lateral in our drawings is what NWS means by a lateral.

The rule. The guideline divides the gravity network into three tiers.

Table 47 The tiers of a gravity sewer network in PAM-GUD-203

Tier	Pipes	What it does	Size	Gradient	Page
Primary	Trunk mains	Carries the wastewater of the catchments to the main pumping station or the plant. NWS applies the term to pipes above 800 mm, over 1,000 without connections (the page prints mm), upstream of the plant or main station.	Material table from 600 mm	G203 Table 11	G203-p35
Secondary	Headers and main sewers	The headers or main sewers laid under the streets, serving the watersheds and taking the tertiary network's flow to the trunk mains.	Main sewer OD200 minimum; typically 200 to 400 mm, the 400 not mandatory	G203 Table 11: DN200 5.00 mm/m, flatter for larger pipes	G203-p21, p22, p23, p29
Tertiary	Rider sewers and lateral sewers, with the property connection sewers often counted in	Collects from the house connection chambers and discharges into a main sewer.	Rider OD160, lateral OD200 minimum; lateral at most 45 m long	G203 Table 5: 1 % to 10 %	G203-p17, p18, p22

Two things in this table catch people out. First, the guideline's lateral is a short pipe, at most 45 m, from a house connection chamber or a rider to the main sewer. It is not a street pipe. Second, the tertiary gradients are separate from G203 Table 11 and much steeper: a lateral needs at least 1 %, while G203 Table 11 lets a DN200 main sewer run at 0.5 %. Laying a lateral at the main sewer's minimum is a design error.

Source. G203-p17 §3.2 (the tiers named, the 45 m lateral), G203-p18 Table 5 (tertiary slopes), G203-p21 §4 (secondary and tertiary defined), G203-p22 Table 6 (minimum sizes), G203-p23 (secondary range), G203-p29 Table 11 (via _BRAIN/02_DESIGN_CRITERIA.md §2), G203-p35 §5 (trunk mains).

Ibri number. On the test area, 1,167 street pipes, 56.8 km, are secondary main sewers. Not one of them is a guideline lateral: the tertiary layer has not been drawn yet (Section 10.9).

Where it lives. _BRAIN/02_DESIGN_CRITERIA.md §2 (tier names read from the PDF on 2026-09-11) and §3 (minimum sizes, trunk definition).

10.3 From the house to the main sewer

What it is for. The tertiary layer is where the network meets the plot. Its rules decide how deep the street sewer must be to take a house.

The rule. The wastewater of a property passes, in order, through the property connection chamber inside the plot at its boundary, the property connection sewer, the house connection chamber in the public right of way, sometimes a rider sewer, and a lateral sewer into the main sewer.

Table 48 The tertiary chain and its limits

Element	Where	Limit	Page
Property connection chamber	inside the property, at the boundary	—	G203-p17
Property connection sewer	chamber to house connection chamber	150 mm minimum (OD160 in G203 Table 6); at most 50 m long; slope 3 % to 10 %	G203-p18, p22
House connection chamber	usually 2.5 m from the property boundary, in the right of way	usually 1.2 m to 2.0 m deep; stubs or plugged ports for future connections	G203-p17, p19
Rider sewer	links house connection chambers, usually up to 3	OD160 minimum; slope 1 % to 10 %	G203-p17, p18, p22
Lateral sewer	house connection chamber or rider to the main sewer	OD200 minimum; at most 45 m; slope 1 % to 10 %	G203-p17, p18, p22

Source. G203 §3.2 to §3.4, pp 17–19; G203 Table 6, p22.

Ibri number. The test area holds 3,062 plots. The median distance from a plot's representative point to its nearest street pipe is 19.3 m, inside the 45 m a lateral may run. The plots far from any street pipe are the ones the tertiary layer will have to test one by one.

Where it lives. Not in the design engine yet: the current scope is street sewers only (Section 10.9).

10.4 The old tier values and the names used from now on

What it is for. To remove a clash of words. The design engine and NAMA's as-built data both say lateral for the street pipe, which the guideline calls a main sewer.

The rule. From now on the street pipe is a secondary main sewer, the collecting main of a district is a secondary header, and the trunk is primary. The word lateral is kept for the tertiary pipe alone. Outputs made before the change keep their old values as the record and are read through the mapping below.

Table 49 Tier values in the engine outputs and the names used from now on

Test-area design	Later engine	Guideline name	What it is	Pipes	km
trunk main	trunk	**Primary: trunk main**	the pipe to the plant, including the main pipe	92	6.04
sub main	sub main	**Secondary: header**	the collecting main of a district; the only pipe that joins the trunk	155	8.77
lateral	lateral	**Secondary: main sewer**	the street sewer every plot connects to	1,167	56.83
—	—	**Tertiary: rider, lateral**	house connection chamber to main sewer, at most 45 m	0	0

Source. _BRAIN/02_DESIGN_CRITERIA.md §2, the tier-names row (engineer, 2026-09-11); W14/docs/DESIGN_FLOWS_FOR_NETWORK.md §5. Guideline wording G203-p17, p21.

Ibri number. The test-area design is 71.64 km of pipe: 79.3 % main sewers, 12.2 % headers and 8.4 % trunk.

Where it lives. Field TIER in W8/shp/W8_pipes.shp (values trunk main, sub main, lateral) and in the later engine's shapefile and DXF exports (trunk, sub main, lateral). The renaming is applied in the engine copy from W13/tmp3 on.

10.5 The hierarchy learned from the built network

What it is for. To give the layout the shape a contractor builds. Matching the built network's average gradient, depth and chamber spacing shows the hydraulics are right. It shows nothing about whether the layout can be built, because the hierarchy is invisible in every one of those averages.

The rule. Properties feed the tertiary pipes, the tertiary pipes feed a main sewer, main sewers feed other main sewers, and only headers reach the trunk. A design in which every catchment finds its own way to the trunk is a set of branches no contractor would build.

The rule was read from NAMA's own manhole identifiers inside the test area, 2,101 pipes and 78.6 km. An identifier such as 5A-2-TM-MH185 is package 5A-2, trunk main, manhole 185; 5A-2-SM.2-MH391 is sub main 2; 5A-1-A49-MH3 is street-sewer zone A49. The designer's own upstream and downstream manhole fields then show where each zone drains.

Table 50 Where the 419 street-sewer zones of the built network drain

Drains into	Zones	Share
another street sewer	381	**91 %**
a sub main (header)	27	6 %
the trunk main	10	2 %

Sewage crosses a median of 11 street sewers before it reaches a header. All six headers drain to the trunk, so only about 16 things touch the trunk: 6 headers and 10 street sewers. An earlier design with no header tier had 30 things touching the trunk, 14 of them carrying under 100 properties and one carrying 3.

The number of joins was then swept on the test-area design to see where it stops working:

Table 51 Joins to the main pipe against pumping and dual-carriageway crossings, test area

Cap on joins	Joins built	Pumping stations	Deepest chamber, m	Dual crossings
none	31	0	9.12	1
20	19	0	10.33	1
16	14	1	11.79	1
12	11	3	11.97	2
8	8	3	11.97	2
6	6	2	11.90	15

Below about 14 joins the network starts buying pumping stations, and below 8 it starts crossing dual carriageways to consolidate. Around 20 is the tightest structure that still runs entirely on gravity without cutting across carriageways, close to the built network's 16.

Source. Project doctrine, _BRAIN/07_PROJECT_STATE.md §2 item 4 (learned from the as-built, 2026-08-23); measurements in W8/docs/LEARNING_FROM_ASBUILT.md (2026-08-21).

Ibri number. The test-area design kept 20 joins out of 55 candidates, from 55 streets facing the main pipe, with 0 pumping stations. In its pipe layer 7 of the pipes flagged as joins are headers and 12 are street main sewers. The rule adopted since then (Section 10.6) lets only headers join, so those street sewers will run into the nearest header instead.

Where it lives. W8/run/summary.json key trunk; field IS_JOIN in W8/shp/W8_pipes.shp; the as-built in W7/shp/EXISTING_SEWERLINE.shp (OP_STATUE = 1), fields US_MHID and DS_MHID.

10.6 How the layout is made

What it is for. To make a layout that is good, not only legal. The guideline says whether a pipe complies; these rules say how the network is arranged.

The rule. Twelve rules, agreed as the reasoning the design engine is built against:

1. **Sewage runs downhill.** Read the ground along every street and point each street downhill.
2. **Follow the arrows.** Where the falling streets stop is an outlet. A basin is not an outlet: a sub-network connects to the main pipe or the plant, never to a local sink.
3. **One spine per outlet.** From each outlet, the gravity path to the main pipe is the spine; the ground sets the number of joins, not a cap.
4. **Headers first.** The headers are the long, straight, low streets of each catchment, cut only at a crest. They earn their diameter from the houses they collect.
5. **Then hang the rest off them.** Every street drains to the header below it, every house to the street it fronts. One outlet per junction and no loops. A street that touches the main pipe does not join it; only headers join.
6. **A ridge is a boundary.** A street on a ridge drains whole to the lower side; a long one splits at its crest.
7. **Gradient follows the ground in three bands.** Flatter than the minimum: lay at the minimum. Between the minimum and the velocity limit: parallel to the ground at cover. Steeper than 3 m/s at peak: hold the gradient and take the rest as drops.
8. **Gradients in steps.** One tenth of the pipe's minimum gradient in G203-p29 Table 11: 0.5 mm/m at DN200, 0.27 mm/m at DN315, 0.075 mm/m at DN900. One gradient per run until the cover runs out. The diameter comes from the flow, never chosen to flatten a gradient.
9. **Lay from the heads.** That is the shallowest the route can ever be.
10. **Over 12 m: reroute first.** A different join, a helpful street, or the neighbouring catchment. Then cut for a pumping station where the pipe would pass 12 m, and restart at cover. Never dig past 12 m to avoid a pump; never pump to avoid a reroute.
11. **A closed hollow is a real pump.** So is a ridge between a pocket and every neighbour.
12. **Chambers by the guideline.** At every junction, head and change of gradient or diameter, spaced by G203-p30 Table 12, evenly divided and rounded, and clear of every plot.

The rules speak of sub-mains and laterals in the engine's old words; read them as headers and main sewers. The 12 m is a limit on depth, ground to invert, at every chamber and along every trench, with no exceptions: where a gravity sewer would pass it, a pumping station goes in before that point.

Rule 8 is the rule to design to. The design engine still lays every pipe on a fixed grid of 0.05 % (0.5 mm/m) whatever its size, which is how the test-area design was laid, and is to be aligned with the rule.

Source. Project rules, W13/docs/W13_DESIGN_LOGIC.md (agreed 2026-09-07, with dated amendments to 2026-09-08). 12 m: project doctrine of 2026-09-07 in the context of G203-p33 §4.6.3, which recommends a cover of about 10 to 12 m and makes excavation cost the trigger for pumping. Gradient steps: rule 8 of W13_DESIGN_LOGIC.md (2026-09-07); minimum gradients G203-p29 Table 11. Chamber spacing: G203-p30 Table 12.

Ibri number. The test-area design is the benchmark every change must still meet on its 5.51 km²: 71.6 km of sewer, 1,415 chambers and 0 pumping stations, deepest chamber 10.45 m. The network NAMA built there needs no pumping either.

Where it lives. W13/docs/W13_DESIGN_LOGIC.md; the benchmark run is W13/py/run_test_boundary.py (about 26 seconds), which reproduces W8/run/summary.json. The engine's gradient grid: SLOPE_STEP in W13/tmp3/py/sewnet/criteria.py.

10.7 The main pipe is an input

What it is for. To separate what is designed from what is given. The route of the main pipe to the plant is fixed by the engineer; the design connects the districts to it.

The rule. The main pipe is read from its drawn line and never derived. Sub-networks join it only through headers, at join points spaced apart; a street beside it runs into the nearest header. Its levels are set after the sub-networks are laid, because a join cannot sit below the pipe it feeds and the main pipe's profile depends on where the joins are.

Source. Project doctrine (2026-08-23; W13/docs/W13_DESIGN_LOGIC.md rules 3 and 5, 2026-09-07 and 2026-09-08). Its tier is primary by function (G203-p35); whether it also meets NWS's 800 mm condition is known only once it is sized.

Ibri number. On the test area the main pipe runs 6.15 km, 3.87 km of it inside the boundary, and both legs drain to their meeting point 792 m outside the boundary before continuing to the existing plant. The design found 55 candidate joins, kept 20, and reached them with connectors of median length 26.3 m.

Where it lives. Hydraulic/SHP/Main Pipe/Main Pipe.shp; W8/run/summary.json key trunk (source recorded there as "SHP/Main Pipe/Main Pipe.shp (given, not derived)").

10.8 Dual carriageways are excluded

What it is for. A dual carriageway cannot be dug up for a sewer, and a sewer under one cannot be maintained without closing it.

The rule. No pipe of any tier runs along a dual carriageway, the trunk included. A dual carriageway is crossed only by a short pipe at right angles, at an underpass or by trenchless work. The two carriageways are drawn as two parallel lines: they are excluded, not merged into one corridor. Where the road data carry the flag, dual = 1 marks a dual carriageway and dual = 2 a two-lane pair of which only one side is used. The street drawing used as the corridor source since 2026-09-07 already leaves the dual carriageways out. The design engine then adds its own short crossings at right angles between the two sides, which the route search may use at a cost where they save depth or a pumping station.

Source. Project rule (engineer, 2026-08-19), CLAUDE.md rule 7; W13/docs/W13_DESIGN_LOGIC.md, Inputs. Trenchless crossings and road settlement: G203-p21 §4.1, on NWS approval.

Ibri number. Of the 3,267 pipes NAMA built, 4 run within 4 m of a dual carriageway, 0.1 %: practice confirms the rule. On the test area 101 dual-carriageway segments were excluded, taking the street corridors from 97.4 km to 83.6 km. The engine then added 26 perpendicular dual-carriageway crossings on the test area.

Where it lives. Hydraulic/SHP/Road centerline 2 (field dual); Hydraulic/DWG/road network 03092026 eyeballed.dxf (the corridor source); W8/run/summary.json key road_treatment; the crossings added: W13/run/summary.json key road_treatment.dual_crossings_added, made by _dual_crossings in W13/py/sewnet/stages/road_treatment.py; the as-built check in W7/docs/CALIBRATION_vs_EXISTING.md.

10.9 Stub-outs and the tertiary layer

What it is for. A plot that is empty today will connect later. The network built now must carry its flow and leave it a way in, so the street is not opened again.

The rule. Chambers carry stubs or plugged ports for future connections. A stub-out is a capped connection at the frontage of a future plot, sized for that area's saturation flow; usually the minimum diameter, DN200, governs. The flow of a future plot is already inside Q_ULT, at its people times 171.3 l/d, so no separate allowance is added.

Source. G203-p19 §3.4 (stubs or plugged ports on chambers); project doctrine, _BRAIN/07_PROJECT_STATE.md §2 item 7. Future-plot rate: 0.85 × 164 + 0.54 × (0.22 + 0.14) × 164 l/d per person, G201-p60 Table 11 and G201-p71 Table 19, report R2 §15.4.

Ibri number. 14,203 of Ibri's 21,867 plots are empty today, and 60,509 of the study area's 77,265. On the test area 1,233 of 3,062 plots are empty. The street sewers already carry their saturation flow. The stub-outs, house connections and riders are not drawn yet: the test-area design recorded 0 stub-outs, and the engine's scope is street sewers only. The consequence to remember is that there is no house-by-house check yet: where a house sits below its road, the chamber that would have been deepened for it is not.

Where it lives. W8/run/summary.json key tertiary; scope in W13/docs/W13_DESIGN_LOGIC.md, section Scope at this stage; plot status Buiding_St = Future in W14/shp/PLOTS_load.shp.

10.10 Check it yourself

1. Open G203 at page 17 and read the last paragraph of §3.2; then G203 Table 5 on page 18 and G203 Table 6 on page 22. Confirm the 45 m, the 1 % and the OD200.
2. Load W8/shp/W8_pipes.shp in QGIS, style it by TIER and sum LEN_M per value. You should get 56.83 km of lateral, 8.77 km of sub main and 6.04 km of trunk main; read them as main sewer, header and primary.
3. Pick any street main sewer head in the same layer and follow it down to the main pipe. Count the street sewers it passes through before it reaches a header.
4. Load W7/shp/EXISTING_SEWERLINE.shp, filter OP_STATUE = 1, and read the manhole identifiers: TM, SM.n and the zone codes are the built network's tiers.
5. Filter Road centerline 2 on dual = 1, buffer it by 4 m and intersect it with the design's pipes. Only short crossings at right angles may remain.
6. Run python W13/py/run_test_boundary.py. It must still give about 71.6 km, 1,415 chambers and 0 pumping stations.

11 Gravity sewer hydraulics

This chapter turns a design flow into a pipe: its diameter, its gradient, its depth and the chambers along it. It carries every constraint the guideline sets on a gravity sewer, each with the page it comes from, and it ends with two Ibri pipes sized and checked rule by rule. Where this project has chosen a value the guideline does not give, the chapter says so in the same sentence.

Read the force words as the guideline writes them. "Shall" is a requirement. "Should" and "recommended" are advice. The difference decides what the designer may do when a rule cannot be met, so it is kept in every table below.

11.1 The names of the pipes, and the two flows every pipe carries

What it is for. A rule attaches to a kind of pipe, so the kind has to be named the way the guideline names it. The minimum gradient of a street sewer and of a lateral sewer differ by a factor of two, and using the wrong word applies the wrong one.

The rule. The guideline has three tiers. The primary network is the trunk mains, which carry the districts to the plant. The secondary network is the headers and main sewers laid under the streets: the main sewer is the street pipe every plot connects to, and the header collects main sewers across a district. The tertiary network is the rider sewers and lateral sewers, the short pipes from the house connection chamber to the main sewer, with a lateral no longer than 45 m. A trunk main is further defined as larger than 800 mm, more than 1,000 m long without connections, upstream of a plant or main pumping station.

Table 52 Pipe tiers in the guideline's words, with the pipes of the test-area design

Guideline tier	What it is	Pipes in the test area
Primary: trunk main	Carries the districts to the plant	92
Secondary: header	Collects main sewers across a district	155
Secondary: main sewer	The street sewer every plot connects to	1,167
Tertiary: rider and lateral sewers	House connection chamber to the main sewer, lateral at most 45 m	not modelled

The design outputs still carry the older engine's TIER values. The mapping from those values to the guideline's names is Table 49 in section 10.4, and the pipe counts above are read through it; it is not repeated here.

The rule. Every pipe is checked against two flows. Its size and capacity are set on the saturation flow Q_ULT of the plots upstream, 100 % connected, peaked and with infiltration added. Its self-cleansing is checked on the opening-year flow Q_2030 of the same plots multiplied by the connection ratio 0.61, peaked per pipe, with the peak formula chosen on the connected property count (recommended in the design-flow handoff, to be confirmed by the engineer). The two cases fail in opposite directions: an under-estimate surcharges the pipe, an over-estimate declares a silting pipe self-cleansing.

Source: G203-p17 and p21 (tiers), G203-p22 Table 6 (45 m), G203-p35 via _BRAIN/02 section 3 (trunk main definition); two flow cases: project rule (engineer, 2026-09-11), G1-p73 for 100 % coverage at the end of the period.

Worked Ibri number. Across the study area the sizing case is 60,099 m^3/d at saturation in 2070, and the self-cleansing case is 14,772 m^3/d. Multiplying saturation by 0.61 instead would give 36,660 m^3/d, 2.5 times the flow really expected in 2030, which is why that shortcut is not used.

Where it lives. Guideline mapping: _BRAIN/02_DESIGN_CRITERIA.md section 2, W14/docs/DESIGN_FLOWS_FOR_NETWORK.md section 5. Pipe counts: field TIER in W8/shp/W8_pipes.shp. Flows: fields Q_ULT and Q_2030 in W14/shp/PLOTS_load.shp; the ratio and the totals in W14/analysis/design_flows.json.

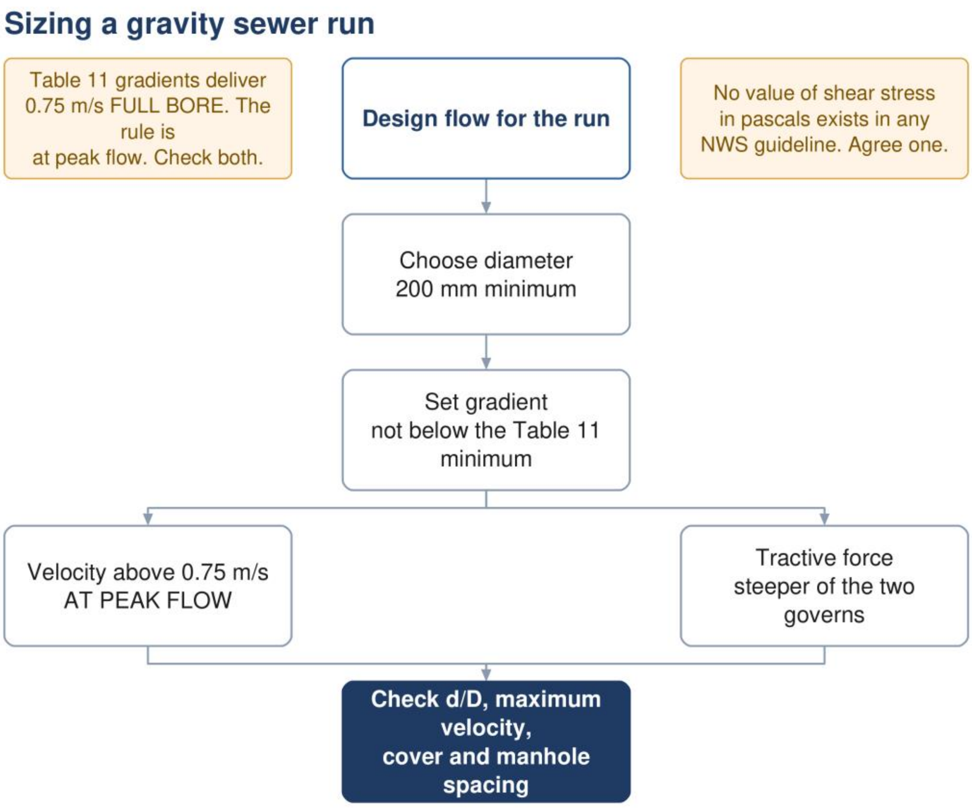

Figure 16 Sizing a gravity run. Both self-cleansing approaches apply and the steeper gradient governs; capacity and depth are checked last.

11.2 Colebrook-White, the design formula

What it is for. It gives the velocity and the capacity of a pipe at a given gradient. Every sizing check in this chapter, and the guideline's own minimum-gradient table, rests on it.

The rule. Gravity sewers shall be designed with a recognised formula, and the guideline names Colebrook-White and Manning. For Colebrook-White it fixes the roughness at ks = 1.5 mm for all pipe sizes and materials. The pipe-full velocity is

$$V = -2\sqrt{2gDS}\,\log_{10}\left[\frac{k_s}{3.7D} + \frac{2.51\nu}{D\sqrt{2gDS}}\right] \tag{47}$$

and the pipe-full flow is velocity times area:

$$Q_o = V \times A = V \times \frac{\pi D^2}{4} \tag{48}$$

Symbol	Meaning	Unit
V	pipe-full velocity	m/s
Q o	pipe-full flow	m³/s
g	acceleration due to gravity, 9.81	m/s²
D	pipe internal diameter	m
S	hydraulic gradient; equal to the pipe gradient in uniform flow	m/m
k s	roughness, **1.5 mm for all sizes and materials**	m
v	kinematic viscosity, **1.141 × 10⁻⁶ at 15 °C**	m²/s
A	pipe internal cross-sectional area	m²

For a part-full pipe the same law is written on the hydraulic radius, because the flow no longer fills the circle. With R = D/4 running full, equation (47) becomes

$$V = -2\sqrt{8gRS}\,\log_{10}\left[\frac{k_s}{14.8R} + \frac{0.6275\nu}{R\sqrt{8gRS}}\right] \tag{49}$$

with the geometry of the wetted segment at proportional depth y = d/D:

$$\theta = 2\arccos(1 - 2y), \quad A = \frac{D^2}{8}(\theta - \sin\theta), \quad P = \frac{D\theta}{2}, \quad R = \frac{A}{P} \tag{50}$$

Symbol	Meaning	Unit
R	hydraulic radius, flow area over wetted perimeter	m
y	proportional depth of flow d/D	—
θ	angle subtended at the centre by the water surface	rad
A, P	flow area and wetted perimeter at depth d	m², m

Viscosity comes from G203 Table 9, which gives three temperatures and asks for the 15 °C value in basic design as the conservative one:

Table 53 Kinematic viscosity (G203 Table 9, p25), and what it does to one pipe

Temperature	ν (m²/s × 10^{-6})	DN200 at 5.00 mm/m, pipe-full velocity
15 °C (design)	1.141	0.748 m/s
25 °C	0.897	0.750 m/s
35 °C	0.727	0.751 m/s

The velocity barely moves with temperature. At a roughness of 1.5 mm a sewer runs almost fully rough, the first term inside the bracket dominates and viscosity hardly matters. The roughness is the number that carries the design; it is not the roughness of new plastic but an allowance for slime, grit and age over the life of the sewer.

> **Three printing slips in G203 section 4.2.** The pipe-full flow on p24 is printed as velocity divided by area, which has no meaning as a flow; it is velocity times area, as in equation (48). The text on p27 sends the reader to G203 Figure 4 for the d/D against q/Q relation; that figure is G203 Figure 2 on p28, G203 Figure 4 being air flow. The two tractive-force constants on p27 differ by 2.3 % once the units are converted; section 11.7 says which one this tutorial uses. None changes a design; each will surface if a calculation is audited against the page.

Source: G203-p24 section 4.2.1 (formula, ks, licensed software), p25 Table 9 (viscosity), p28 section 4.2.4 (ks repeated).

Worked Ibri number. A DN200 PVC-U main sewer has a true bore of 188.2 mm. At the G203 Table 11 minimum of 5.00 mm/m, equation (47) gives V = 0.719 m/s and equation (48) Q o = 20.0 l/s running full. On the nominal 200 mm the same gradient gives 0.748 m/s and 23.5 l/s.

Where it lives. W13/py/sewnet/hydra.py, functions v_cw_R, v_full, q_full and solve_dod (the R-form with the segment geometry); constants KS and NU in W13/py/sewnet/criteria.py; the true bore in criteria.internal_diameter (PVC-U SN8 taken as SDR 34 up to 315 mm, GRP nominal above: an assumption until the pipe class is fixed). Results: fields VEL_MS, DOD and QPEAK_LS in W8/shp/W8_pipes.shp.

11.3 Manning, and why the project designs by Colebrook-White

What it is for. Manning is the other formula the guideline accepts, and the hydraulic model exports use it. The designer needs to know what roughness makes it agree with Colebrook-White.

The rule. Manning's formula, as printed, in its velocity, full-bore gradient and flow forms:

$$v = \frac{1}{n} R^{\frac{2}{3}} S^{\frac{1}{2}}, \quad Q = v\,A \tag{51}$$

$$S = \frac{v^2\, n^2\, (6.3448)}{D^{1.333}} \tag{52}$$

Symbol	Meaning	Unit
v	mean velocity	m/s
n	Manning roughness coefficient	—
R	hydraulic radius	m
S	slope of the energy grade line	m/m
D	pipe diameter (full-bore form only)	m
Q, A	flow and flow area	m^3/s, m^2

Equation (52) is equation (51) solved for S with R = D/4 running full. The constant is 4 to the power 4/3, which is 6.350; the printed 6.3448 is a rounding slip worth under 0.1 %.

G203 Table 8 gives n = 0.009 for plastic, 0.009 to 0.011 for PVC-U and fibreglass, 0.009 to 0.015 for polyethylene and 0.012 for cement-lined concrete. Those are clean-pipe values. Colebrook-White at ks = 1.5 mm is an aged-sewer value, and the guideline makes it mandatory for Colebrook-White. The two only agree if Manning is given the roughness of an aged sewer, not the roughness of the pipe as delivered.

Table 54 DN200 at 5.00 mm/m running full: Manning against Colebrook-White

Formula and roughness	Pipe-full velocity	Against Colebrook-White
Colebrook-White, ks 1.5 mm	0.748 m/s	—
Manning, n 0.009 (G203 Table 8, plastic)	1.066 m/s	+42 %
Manning, n 0.011	0.872 m/s	+17 %
Manning, n 0.013	0.738 m/s	−1.4 %

Source: G203-p24 and p25 (Manning's forms), p23 Table 8 (n), p24 and p28 (ks 1.5 mm for Colebrook-White).

Worked Ibri number. Manning with the plastic n of 0.009 overstates the DN200 velocity by 42 % against the mandated Colebrook-White basis. A model built that way would pass pipes that silt. The project designs by Colebrook-White and exports n = 0.013 to Manning-based software, which agrees within 1.4 % at this size.

Where it lives. W13/py/sewnet/criteria.py, MANNING_N_EXPORT = 0.013 (project choice, documented as the ks-equivalent for the SewerGEMS export); W13/py/sewnet/export_gems.py.

11.4 How full a pipe may run, and the part-full relation

What it is for. Capacity is held back at peak flow so that an unusual flow has somewhere to go and air can move over the water. This is the test that sets the diameter once flows grow.

The rule. At peak flow the proportional depth should not exceed the G203 Table 10 value. The guideline calls these recommended design criteria and defines d/D against the nominal diameter.

Table 55 Recommended depth of flow at peak flow (G203 Table 10, p27)

Sewer	Maximum d/D at peak flow
Pipe diameter up to 350 mm	0.65
Pipe diameter above 350 mm	0.50

The flow and velocity a pipe delivers at a given depth follow from equations (49) and (50). The guideline shows the relation as a curve (G203 Figure 2, p28). Recomputed here with Colebrook-White:

Table 56 Part-full ratios for DN200 PVC-U at 5.00 mm/m, Colebrook-White, ks 1.5 mm

d/D	q/Q o	v/V	Velocity (m/s)
0.10	0.020	0.387	0.278
0.20	0.087	0.609	0.438
0.30	0.195	0.774	0.556
0.40	0.337	0.902	0.648
0.50	0.500	1.000	0.719
0.60	0.672	1.072	0.771
0.65	0.756	1.099	0.790
0.70	0.837	1.120	0.805
0.80	0.977	1.139	0.819
0.90	1.066	1.124	0.808
0.94	1.076	1.103	0.793

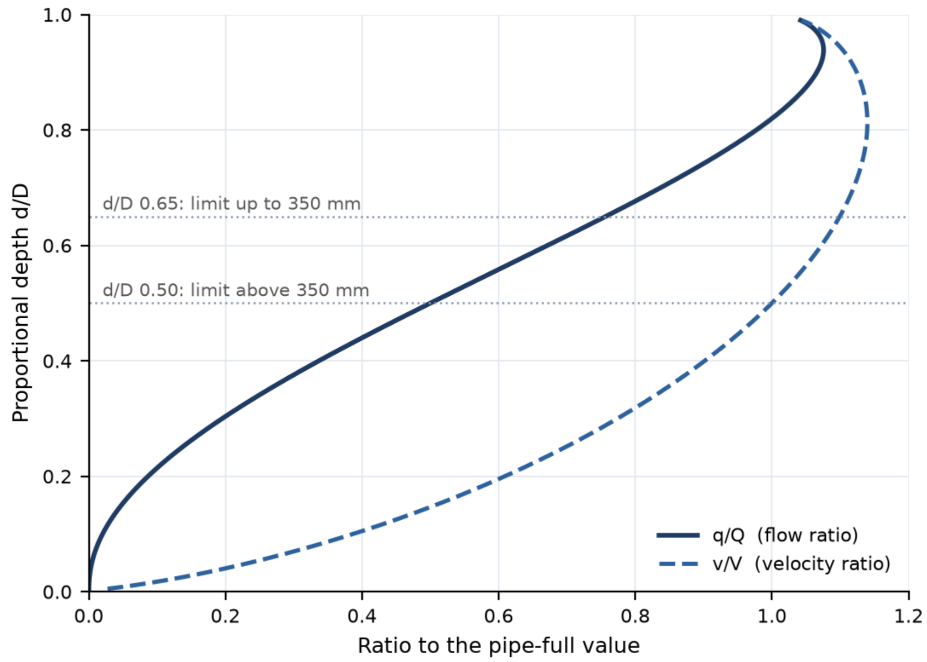

Figure 17 Part-full flow and velocity in a circular sewer, recomputed with Colebrook-White for DN200 PVC-U at 5.00 mm/m, with the G203 Table 10 limits.

Three things to take from the curve. At half depth the flow is exactly half the pipe-full flow and the velocity equals the pipe-full velocity. The velocity peaks near d/D = 0.8, about 1.14 times the full-bore value. The flow peaks near d/D = 0.94 at about 1.08 times, because the last part of the circle adds more wetted wall than area. The shape hardly depends on size: at DN600 and 1.25 mm/m the ratio at d/D 0.65 is 0.754 against 0.756 here.

Source: G203-p27 section 4.2.3 and Table 10; p28 Figure 2.

Worked Ibri number. At d/D 0.65 a DN200 PVC-U pipe at 5.00 mm/m carries 15.1 l/s. That is the largest peak it may take at its minimum gradient. In the test area the fullest pipe runs at d/D 0.634 at the flat-load design flow.

Where it lives. W13/py/sewnet/hydra.py, dod_limit and size_pipe (the smallest diameter whose d/D at the laid gradient is within the limit); DOD_MAX_SMALL, DOD_MAX_LARGE and DOD_DN_THRESHOLD in criteria.py. d/D is computed on the true bore, which is the stricter reading of a limit the guideline states on the nominal diameter.

11.5 Velocity: the self-cleansing minimum and the maximum

What it is for. A slow sewer lets solids settle and turn septic; a fast one erodes the pipe, releases hydrogen sulphide and makes chambers unsafe.

The rule. The minimum velocity shall be above 0.75 m/s at peak flow, with 0.90 m/s preferred. The maximum shall not exceed 3 m/s at the design depth of flow, and the maximum gradient is the one that respects it. The guideline does not say which year's peak the minimum applies to. The project tests it on the low case, the opening-year flow Q_2030 × 0.61, because that is when silting happens; the maximum is tested on the sizing flow Q_ULT, because that is when the pipe runs fastest.

Table 57 Velocity criteria

Criterion	Value	Condition	Force	Source
Self-cleansing velocity	above 0.75 m/s	at peak flow	shall	G203-p26
Preferred velocity	0.90 m/s	at peak flow	preferred	G203-p26
Maximum velocity	3 m/s	at the design depth of flow	shall	G203-p27
Maximum gradient	the one giving 3.0 m/s		should	G203-p29
Flow for the minimum	Q_2030 × 0.61, peaked	per pipe	project rule	engineer 2026-09-11
Flow for the maximum	Q_ULT, peaked, with infiltration	per pipe	project rule	engineer 2026-09-11

Two approaches to self-cleansing shall be used together, the velocity and the minimum tractive force, and the steeper gradient they give governs. At the head of a system, where 0.75 m/s cannot be reached at all, the tractive force alone sets the gradient. Section 11.7 gives the tractive-force rule. In the early years the actual flow is below the design flow; the guideline's answer is more frequent inspection and cleansing by the operator, and it sets no flow threshold, frequency or duration.

Source: G203-p25 to p27 section 4.2.2.1 (two approaches, steeper governs, head of system), p27 section 4.2.2.2 (maximum), p28 section 4.2.6 (early low flows), p29 section 4.3.2 (maximum gradient); flow cases: project rule (engineer, 2026-09-11).

Worked Ibri number. In the test-area design the fastest pipe runs at 1.09 m/s, a third of the maximum. On flat ground the maximum never binds; it binds on a steep hillside street, where the surplus fall is taken as a drop at a chamber instead (section 11.14).

Where it lives. V_SELF_CLEANSING, V_PREFERRED and V_MAX in W13/py/sewnet/criteria.py; hydra.smax_for (the gradient cap at the design flow). Low-case flows: W14/docs/DESIGN_FLOWS_FOR_NETWORK.md sections 2 to 4.

11.6 Minimum gradient from the self-cleansing velocity: G203 Table 11

What it is for. It gives each diameter the flattest gradient at which it may be laid. On Ibri's flat ground this table sets most of the network.

The rule. The gradient shall be enough to keep the design minimum of 0.75 m/s. G203 Table 11 turns that velocity into a gradient for each size using Colebrook-White. Sewers shall not be oversized to allow flatter slopes, and the slope must be uniform between successive manholes.

Table 58 Minimum sewer gradient (G203 Table 11, p29), with the Colebrook-White check

DN (mm)	mm/m	%	1 in	V full, nominal D (m/s)	V full, true bore (m/s)
200	5.00	0.500	200	0.748	0.719
250	3.75	0.375	267	0.751	0.721
315	2.70	0.270	370	0.741	0.712
400	2.05	0.205	488	0.754	0.754
500	1.55	0.155	645	0.756	0.756
600	1.25	0.125	800	0.763	0.763
700	1.00	0.100	1,000	0.753	0.753
800	0.85	0.085	1,176	0.755	0.755
900 and above	0.75	0.075	1,333	0.764	0.764

The mm/m column is the guideline's. The next two are the same number in other units. The last two are equation (47) evaluated at each gradient: on the nominal diameter every row returns 0.74 to 0.76 m/s, so the table is Colebrook-White at 0.75 m/s running full, on the nominal size.

G203 Table 11 is a pipe-full gradient, and that matters. The velocity rule is stated at peak flow, not running full. At the G203 Table 11 gradient a pipe reaches 0.82 m/s at d/D 0.65, 0.75 m/s at d/D 0.50 and 0.46 m/s at d/D 0.20 (DN200, nominal). A residential main sewer runs near d/D 0.2 for most of its life, so laying to G203 Table 11 is not compliance by itself; the velocity at the actual flow is what is tested. On the true bore of PVC-U the DN200 row gives only 0.72 m/s even running full.

The ban on oversizing closes the obvious escape on flat ground. A larger pipe has a flatter minimum, 1.25 mm/m at DN600 against 5.00 mm/m at DN200, so upsizing would keep a sewer shallow. The guideline forbids it: on flat ground the design accepts the depth and, when the depth runs out, pumps.

Source: G203-p29 section 4.3.1 and Table 11.

Worked Ibri number. 762 of the 1,414 pipes in the test-area design are laid at exactly 5.00 mm/m, the DN200 minimum, and the median laid gradient is 5.00 mm/m. The minimum is the design on this ground.

Where it lives. TABLE11 and TABLE11_FLOOR in W13/py/sewnet/criteria.py; hydra.smin_for takes the steeper of G203 Table 11 and the tractive minimum. The engine's gate reproduces G203 Table 11 from equation (47) within ±5 % before any design runs. Laid gradient: fields SLOPE_PMIL and SLOPE_PCT in W8/shp/W8_pipes.shp.

11.7 Minimum gradient from the tractive force

What it is for. Near the head of a network the flow is too small for any realistic gradient to reach 0.75 m/s. The tractive force asks a different question: does the water drag on the invert hard enough to move the sediment?

The rule. The tractive tension is the weight of the water along the slope divided by the wetted area it acts on:

$$\tau = \frac{W \sin\theta}{p\,L}, \quad W = \rho g a L \quad \Rightarrow \quad \tau = \rho g R \sin\theta \approx \rho g R S \qquad (53)$$

Mara, Sleigh and Taylor (2000), cited by the guideline, derived the minimum slope that moves particles, assuming d/D = 0.2 and n = 0.013:

$$S_{\min} = K\,\tau^{1.23}\,Q^{-0.461} \tag{54}$$

Symbol	Meaning	Unit
τ	tractive tension (boundary shear stress)	Pa
W	weight of the water in the length L	N
θ	angle of the pipe to the horizontal	rad
p, a	wetted perimeter and flow area	m, m²
L	length of pipe	m
ρ	density of the sewage	kg/m³
S min	minimum slope to move particles	m/m
K	2.33×10^{-4}, with Q in m³/s: the constant used for every tractive slope here	—
Q	peak flow in the pipe	m³/s

G203 prints the relation on p27 with two constants, 2.33×10^{-4} for Q in m³/s and 5.5×10^{-3} for Q in l/s. Converted to the same units they are not the same number: the l/s constant gives slopes 2.3 % flatter (4.56 against 4.67 mm/m at 1.5 l/s and 1 Pa). This tutorial uses $K = 2.33 \times 10^{-4}$ with Q in m³/s for every tractive slope, the steeper of the two. That form is recommended in the design-flow handoff, to be confirmed by the engineer.

The steeper of this gradient and the G203 Table 11 gradient is the minimum. Two numbers the method needs are not in the guideline. It gives no design value of τ anywhere in its 201 pages. And the low-flow threshold of 1.5 l/s that is often quoted with the method is not in G203 either; it is the minimum peak flow of the Mara simplified-sewerage literature, a single WC flush. The project carries τ as a parameter, runs at 1 Pa, and uses 1.5 l/s only as a tagged outside assumption until NWS confirms both.

On the low case every pipe takes one of four classes, tested in this order: velocity pass (0.75 m/s reached); early cleansing (peak flow below 1.5 l/s, too small for either test; the pipe goes on the flushing list and is not upsized, because DN200 is already the minimum); tractive pass (laid gradient at or above equation (54) at the real flow); and fails both with real flow, which is a design fault and the pipe is regraded. Testing the threshold before the tractive force keeps the relation from being extrapolated below the flows it was derived for.

Setting the design minimum gradient is a different use of the same relation. There the flow is floored at 1.5 l/s, because unfloored the formula demands an unbounded slope as the flow tends to zero. That floor is what makes the two methods meet at the head of a DN200 network.

Source: G203-p26 (tractive force, W = ρgaL), p27 (Mara relation, K, steeper governs, head of system); τ: GAP-9, no value in G203; 1.5 l/s: outside assumption (Mara, Sleigh and Taylor 2000), not in G203; classes: project rule (engineer, 2026-09-11); K in m³/s: recommended in W14/analysis/design_flows.json (rules, open_rulings_recommended), to be confirmed by the engineer.

Worked Ibri number. At Q = 1.5 l/s and τ = 1 Pa, equation (54) with $K = 2.33 \times 10^{-4}$ and Q in m³/s gives 4.67 mm/m: at the head of a network the two methods meet near the DN200 G203 Table 11 value of 5.00 mm/m. At τ = 2 Pa the same flow needs 10.95 mm/m, same K. τ is the single most sensitive number in the self-cleansing check, which is why it is reported as a parameter.

Where it lives. hydra.smin_tractive and smin_for in W13/py/sewnet/; TAU_PA = 1.0 and TRACTIVE_QMIN = 0.0015 m³/s in criteria.py (both listed in its ASSUMPTIONS register); classes: W14/docs/DESIGN_FLOWS_FOR_NETWORK.md section 4.

11.8 Tertiary slopes: G203 Table 5 is not G203 Table 11

What it is for. The short pipes from the property to the main sewer carry almost nothing and are never self-cleansing by velocity, so the guideline gives them their own, much steeper range.

The rule. G203 Table 5 sets typical slopes for the house connection, rider and lateral sewers.

Table 59 Typical pipe slopes for house connection, rider and lateral sewers (G203 Table 5, p18)

Category of sewer	Network	Minimum	Maximum
Property connection sewer	property connection	3 %	10 %
Rider sewer	tertiary	1 %	10 %
Lateral sewer	tertiary	1 %	10 %

The trap is the DN200 lateral sewer. It is the same size as a main sewer, and G203 Table 11 would let it lie at 0.5 %. It belongs to the tertiary network, so its minimum is 1 %, twice as steep. The property connection sewer is at least 150 mm, should not exceed 50 m, and needs 600 mm of cover.

Source: G203-p18 Table 4 (property connection sewer, 150 mm, 50 m) and Table 5 (slopes), p19 section 3.5 (600 mm cover).

Worked Ibri number. At concept stage the tertiary network is not laid: each plot is taken to connect to the main sewer in front of it. The rule enters the design as a depth check, since a main sewer must be deep enough for a 1 % lateral and a 3 % connection to reach it from the plot.

Where it lives. PCS_MIN_SLOPE = 0.03, RIDER_MIN_SLOPE = 0.01, PCS_MAX_LEN = 50 m, PCS_MIN_COVER = 0.60 m and LATERAL_MAX_LEN = 45 m in W13/py/sewnet/criteria.py; the connectability stage, W13/py/sewnet/stages/connectability.py.

11.9 Maximum gradient

What it is for. It stops a steep street from producing a pipe that runs faster than 3 m/s.

The rule. The maximum gradient should be the one at which the velocity at the design depth of flow reaches 3.0 m/s. Where the ground falls faster than that, the pipe is laid at or below the cap and the surplus fall is taken at a chamber as a drop.

Source: G203-p27 section 4.2.2.2, p29 section 4.3.2.

Worked Ibri number. For the secondary main sewer worked in section 11.16, carrying 2.12 l/s in DN200, the 3.0 m/s cap is not reached at any gradient below 100 %. For the header of section 11.17, 35.5 l/s in DN315, it is reached at a gradient of 10.6 %. Small flows in small pipes need very steep gradients to reach 3 m/s, so on Ibri's ground the cap matters mainly for the tertiary pipes, whose 10 % maximum in G203 Table 5 is far below it.

Where it lives. hydra.smax_for in W13/py/sewnet/hydra.py, which also reports when no gradient satisfies the cap and the pipe must be upsized.

11.10 Construction tolerance, and what it does at flat gradients

What it is for. A pipe is set out from its two ends. The design has to leave enough fall that the contractor's tolerance cannot turn it into a reverse gradient.

The rule. Line and level shall not deviate from the contract by more than 20 mm, and the deviations combined shall not create a reverse gradient. If both ends can be 20 mm out in opposite directions, a pipe needs more than 40 mm of fall to be sure of falling:

$$S \geq \frac{2\delta}{L}, \quad \delta = 0.020\,m \qquad (55)$$

Symbol	Meaning	Unit
S	laid gradient	m/m
δ	line and level tolerance, 20 mm	m
L	pipe length between chambers	m

Table 60 The 20 mm tolerance against the fall of a full-length pipe at its minimum gradient

DN (mm)	G203 Table 11 (mm/m)	Maximum spacing (m)	Fall over it (mm)	20 mm as share of fall (%)	Shortest pipe with 40 mm fall (m)
200	5.00	100	500	4.0	8.0
315	2.70	100	270	7.4	14.8
400	2.05	120	246	8.1	19.5
600	1.25	120	150	13.3	32.0
900	0.75	120	90	22.2	53.3

On a DN200 street pipe the tolerance is a small share of the fall. At trunk sizes it is not: a DN900 at 0.75 mm/m falls 90 mm in 120 m, and 20 mm is more than a fifth of it. Short pipes at flat trunk gradients carry the real risk, which is why flat trunk profiles must carry margin for the tolerance rather than sit exactly on G203 Table 11.

Source: G203-p29 section 4.3.1.

Worked Ibri number. The rule matters most at the joins onto the main pipe and on any future primary trunk main, where gradients are near 1 mm/m. In the local network it is almost never binding: a DN200 pipe at 5.00 mm/m has 40 mm of fall after 8 m.

Where it lives. FALL_TOLERANCE = 0.040 m (two times 20 mm) in W13/py/sewnet/criteria.py; stages/hydraulic.py steepens any pipe whose fall would be below it, and stages/audit.py checks every pipe.

11.11 Gradient steps: a tenth of the pipe's minimum

What it is for. The gradient on the drawing must be the gradient the invert levels came from, and one a contractor can set out. A computed 6.911 mm/m is neither.

The rule. Every pipe is laid at a whole number of gradient steps, and the step is a tenth of that pipe's own minimum gradient in G203 Table 11: 0.5 mm/m at DN200, 0.27 mm/m at DN315 and 0.075 mm/m at DN900. The required gradient is rounded up to the next step, since rounding down would breach the minimum. Along a street run one gradient is held until the cover is no longer enough; it changes only at a junction, or where holding it would breach the minimum cover or the maximum depth. The diameter is earned by the flow; it is never chosen to flatten the gradient. This is rule 8 of the layout rules in section 10.6.

$$S_{\text{laid}} = \left\lceil \frac{S_{\text{req}}}{\Delta S} \right\rceil \Delta S, \quad \Delta S = \frac{S_{\text{min,DN}}}{10} \tag{56}$$

Symbol	Meaning	Unit
S laid	gradient laid and drawn	m/m
S req	gradient required by the rules above	m/m
S min,DN	G203 Table 11 minimum gradient for the pipe's size	m/m
ΔS	gradient step, a tenth of S min,DN	m/m

Table 61 G203 Table 11 minima laid on rule 8 steps, and on the single 0.05 % grid of the test-area design

DN (mm)	G203 Table 11 (mm/m)	Rule 8 step (mm/m)	Laid minimum, rule 8 (mm/m)	Laid minimum, 0.05 % grid (mm/m)	Extra fall on the grid (%)
200	5.00	0.500	5.00	5.0	0
250	3.75	0.375	3.75	4.0	7
315	2.70	0.270	2.70	3.0	11
400	2.05	0.205	2.05	2.5	22
500	1.55	0.155	1.55	2.0	29
600	1.25	0.125	1.25	1.5	20
700	1.00	0.100	1.00	1.0	0
800	0.85	0.085	0.85	1.0	18
900 and above	0.75	0.075	0.75	1.0	33

Every G203 Table 11 minimum is itself a whole number of its own steps, so under this rule a pipe at its minimum is laid exactly on it at every size, and the step shrinks as the minimum flattens. A single step for all sizes cannot do that. The 0.05 % grid costs nothing at DN200, where 5.00 mm/m is already a round value, but it lays a DN315 at 3.0 instead of 2.70 mm/m and a DN900 at 1.0 instead of 0.75 mm/m, which sinks the DN900 250 mm deeper per kilometre.

The test-area design was laid on that single 0.05 % grid, and the design engine still lays it for every size; the engine is to be aligned with rule 8. At DN200, most of the network, the two agree. Above DN200 the worked examples of this chapter are recomputed on rule 8 steps, and each of their tables says which gradient it uses.

Source: project rule 8 of the network design logic (engineer, 2026-09-07), which replaced a single 0.05 % grid for all sizes (engineer, 2026-08-23); the guideline sets no rounding.

Worked Ibri number. The test-area design, laid on the 0.05 % grid, has 103 distinct gradients across 1,414 pipes, and every size above DN200 starts at the grid's rounded minimum rather than at G203 Table 11 (DN250 from 4.0 mm/m against 3.75 on rule 8; DN315 from 3.0 mm/m against 2.70 on rule 8; DN400 from 2.5 mm/m against 2.05 on rule 8; DN500 from 2.0 mm/m against 1.55 on rule 8). 100 of its 102 pipes above DN200 sit on that grid minimum; rule 8 lays each of them at its G203 Table 11 value. When the grid was introduced its cost was measured at 1.0 % more excavation, 0.12 m on the deepest chamber and no extra pumping station, with 448 distinct gradients reduced to 103.

Where it lives. Rule 8: W13/docs/W13_DESIGN_LOGIC.md, rule 8 (agreed 2026-09-07). The engine still lays the 0.05 % grid: SLOPE_STEP = 0.0005 in W13/tmp3/py/sewnet/criteria.py (the same in W13/py/sewnet/criteria.py), applied in stages/hydraulic.py, with the grid's measured cost in its ASSUMPTIONS register; it is to be aligned with rule 8. The rule 8 steps in this chapter are computed by step8 and round_up8 in TUTORIALS/T04/ch05_gravity.py. Fields SLOPE_PMIL and SLOPE_PCT on every pipe output carry the laid gradient.

11.12 Pipe sizes and materials

What it is for. It sets the smallest pipe each tier may use, and what it may be made of.

The rule. G203 Table 6 sets minimum sizes by outside diameter and the materials for open trench and trenchless work. The secondary network usually runs from 200 to 400 mm, and the guideline states that 400 mm is not a mandatory ceiling.

Table 62 Minimum sizes and materials (G203 Table 6, p22)

Category	Size	Open trench	Trenchless
Rider sewer, property connection sewer	OD 160 mm minimum	PVC-U, HDPE	GRP, HDPE, PVC-U
Lateral sewer, maximum length 45 m	OD 200 mm minimum	PVC-U, HDPE, GRP	GRP, HDPE, PVC-U
Main sewer	OD 200 mm minimum to 300 mm	PVC-U (up to 250 mm), HDPE, GRP	GRP, HDPE, PVC-U
Main sewer	350 mm and above	GRP, HDPE, GRP/PVC, lined RCC	GRP, HDPE

- **The 45 m limit is easy to miss.** G203 Table 6 puts the 45 m limit in the row heading of the lateral sewer; G203 section 3.2 on p17 attaches it to riders and laterals together. The project takes the conservative reading and applies 45 m to both.
- **The name is not the bore.** PVC-U is named by its outside diameter, GRP by its inside. A DN200 PVC-U pipe passes water through about 188 mm. Sizing on the name overstates the capacity of every plastic pipe: 23.5 against 20.0 l/s running full at 5.00 mm/m.
- **Trenchless is the exception.** Trenchless work is used only where a trench is not feasible, with NWS approval; at major road crossings possible settlement can decide the method.

Source: G203-p17 section 3.2, p21 section 4.1, p22 Table 6, p23 (secondary range).

Worked Ibri number. In the test-area design 1,312 of 1,414 pipes are DN200: in a residential network the size is set by the minimum, not by the flow.

Where it lives. DN_SERIES, DN_MIN_MAIN = 200, DN_TERTIARY = 160, internal_diameter and material in W13/py/sewnet/criteria.py; field MAT on every pipe.

11.13 Cover and the maximum depth

What it is for. Minimum cover protects the pipe and keeps it clear of other services. Maximum depth is what ends a gravity system: past it, the sewage is pumped.

The rule. The minimum depth shall be 1.3 m to the crown. Where circumstances force less, concrete protection is required and the cover over pipe and protection together shall be at least 0.5 m; the 0.5 m is a conditional exception, not a floor. A 3 m horizontal clearance from other utilities is required. For the maximum, the guideline recommends approximately 10 to 12 m of cover, requires deeper pipes to be investigated with the manufacturer, and tells the Engineer to incorporate pumping stations where excavation cost becomes prohibitive.

This project holds 12 m as a hard limit with no exceptions. It applies to every chamber and to the trench between chambers, and it is measured from ground to invert, which is stricter than cover by the pipe's outside diameter. Where a gravity sewer would pass 12 m, a pumping station goes in before that point: the sewage is lifted and the pipe restarts at normal cover. Pumps are not removed by digging deeper, and no area is exempted by calling it a pumping pocket. The guideline's real trigger is cost, and until the options are costed the project does not claim the licence the cost test would give. The rising main from each station is sized on the pump duty, not on the arriving gravity flow: at least 0.75 m/s at design minimum flow (1.0 m/s for intermittent, start-stop pumping; 1.2 m/s in a vertical main) and at most 2.5 m/s (G203-p50 section 8.1). The 3.0 m/s maximum of section 11.5 belongs to the gravity sewer, not to the rising main.

Depth accumulates wherever the ground falls more slowly than the pipe must. Starting at minimum depth, the distance a sewer can travel before it reaches the limit is

$$L_{\max} = \frac{h_{\max} - h_0}{S - i_g} \tag{57}$$

Symbol	Meaning	Unit
L max	distance to the depth limit	m
h max	depth limit, ground to invert: 12 m	m
h 0	starting depth: 1.3 m cover + outside diameter + bedding	m
S	laid gradient	m/m
i g	ground fall along the route (zero on flat ground, negative uphill)	m/m

Source: G203-p33 section 4.6.3 (1.3 m, 0.5 m with protection, 3 m, approximately 10 to 12 m, manufacturer, pumping on cost); 12 m hard limit: project doctrine (PROJECT_STATE section 2 item 3, settled 2026-08-19; standing decision 2026-09-07 until costing exists); rising main velocity G203-p50 section 8.1.

Worked Ibri number. A DN200 main sewer starting at 1.55 m to invert and laid at 5.00 mm/m across flat ground reaches 12 m after 2,090 m. On a rising street it gets there sooner, by the rise. The test-area design stays inside the limit with its deepest chamber at 10.45 m and 0 pumping stations, over 71.6 km and 1,415 chambers.

Where it lives. MIN_COVER_CROWN = 1.3, MAX_DEPTH = 12.0, WALL_ALLOW = 0.05 and invert_depth_min in W13/py/sewnet/criteria.py; the hard limit in stages/hydraulic.py and the check of every chamber and every trench in stages/audit.py; fields DEPTH and IS_PUMP in W8/shp/W8_manholes.shp.

11.14 Manholes: where, how far apart, drops and the inlet angle

What it is for. Chambers give access for cleaning and control every change in the flow. Their spacing, their drops and the angle at which pipes enter them decide whether the network can be maintained and whether solids drop out at the junctions.

The rule. Manholes shall be provided at every change of gradient, every change of diameter, every junction of two or more pipes, at regular spacing on straight lines, and at the end of each lateral sewer. G203 Table 12 gives the recommended maximum spacing, and any alteration needs NWS pre-approval.

Table 63 Maximum spacing between manholes (G203 Table 12, p30)

Pipe diameter (mm)	Maximum spacing (m)
200 to 315	100
350 to 900	120
1 000 to 1 400	150
More than 1 400	200

The rule. Where a pipe arrives more than 600 mm above the outgoing invert, a backdrop is required and shall be built outside the manhole. Internal backdrops are allowed only for new connections to existing manholes where an external one is not practicable, and never in a manhole under 1.5 m in diameter. A backdrop should be no higher than 2 m; beyond that a vortex drop shaft should be used. At a property connection, falls over 600 mm are not permitted and a backdrop goes outside the manhole.

Table 64 Drops at a chamber (G203 p30 and p19)

Invert difference	What is required	Force
Up to 600 mm	within the chamber, by benching and channels	—
More than 600 mm	a backdrop, built outside the manhole	shall
More than 2 m	a vortex drop shaft	should

The rule. Benching and channels shall give smooth transitions, and no inlet pipe shall have an angle less than 90° to the direction of flow. The project reads the angle between the incoming and the outgoing pipe at the chamber: 180° is straight through, 90° a square junction, and less than 90° means the incoming flow is turned back on itself, where solids drop out. Where a street meets at a bad angle, a bend chamber goes a few metres short of the junction so the turn is made in two halves; where there is no room (2 m clear of plots, 3 m between chambers) the junction is flagged for a purpose-made chamber with a curved channel.

Source: G203-p29 and p30 section 4.4 (locations, drops, inlet angle, Table 12); p19 section 3.6 (property connections); practice at bad angles: PROJECT_STATE section 2 item 6.

Worked Ibri number. In the test-area design the longest pipe is 100.0 m at DN200, within G203 Table 12. 163 chambers receive a pipe more than 600 mm above the outgoing invert and so need an external backdrop; 73 of those drops exceed 2 m and need a vortex drop shaft. 74 chambers are flagged for a curved-channel chamber because an inlet arrives under 85°.

> **The inlet angle is a stated deviation in the engine.** The guideline says 90°. The engine accepts 85°, because enforcing 90° exactly added roughly 200 chambers purely to turn the flow, with no construction benefit; anything sharper is flagged for a curved-channel chamber, never fixed by adding a chamber. The deviation has to be carried to the deliverable as a deviation, not presented as compliance.

Where it lives. mh_max_spacing, MH_SPLIT_LEN = 100 m, DROP_TRIGGER = 0.60 m, BACKDROP_MAX = 2.0 m and INLET_MIN_DEG = 85 (stated deviation, engineer 2026-08-20, recorded in the ASSUMPTIONS register) in W13/py/sewnet/criteria.py; fields N_DROPS and VORTEX in W8/shp/W8_manholes.shp, DROP_DN on each pipe, and the flagged chambers in W8/shp/W8_sharp_inlets.shp.

11.15 What NAMA's built network confirms

What it is for. A rule set can be self-consistent and still produce a network no designer would draw. The as-built network inside the test area is the check against real practice.

The rule. Compare the design with the as-built on the measures the rules control, and read the result for what it can and cannot show.

Table 65 Design against the as-built network, test area

Measure	NAMA as-built	Design run compared	Reading
Median laid gradient	4.98 mm/m	5.00 mm/m	matches
Median depth	1.92 m	1.75 m	matches
90th percentile depth	4.58 m	5.02 m	matches
Share deeper than 6 m	1.4 %	6.1 %	the design digs deeper more often
Median manhole spacing	29.8 m	43.4 m	the design is wider

The gradients and depths say the minimum-gradient and cover rules behave the way a real designer's did. Two further lessons came out of the comparison. Tighter manhole spacing was tested as a way to keep trenches shallower and it does not: the depth is set by the gradient, not by the chambers. And the averages are blind to the layout: in the built network most street sewers drain into another street sewer and only about sixteen pipes touch the trunk, a hierarchy none of these statistics can see. Matching the hydraulics says nothing about whether the layout is buildable.

Source: W7/docs/CALIBRATION_vs_EXISTING.md (3,267 as-built pipes with levels); W8/docs/LEARNING_FROM_ASBUILT.md (hierarchy).

Worked Ibri number. The live test-area design has a median laid gradient of 5.00 mm/m against 4.98 mm/m as built.

Where it lives. The calibration and the hierarchy study are the two documents above; the design figures in the table are the run they record, and the median gradient is read from W8/shp/W8_pipes.shp.

11.16 Worked example 1: a secondary main sewer

What it is for. To take one real street pipe through every rule in order, from the plots it serves to its class on the self-cleansing test. It replaces the earlier worked pipe, which was loaded at a flat five people per property and peaked by holding Merrimack below 100 properties; both of those have been superseded.

The rule. Sum the plots upstream; peak by Merrimack above 100 properties and by Peltier at or below; add infiltration by the length of sewer upstream; choose the smallest diameter whose d/D at the laid gradient is within G203 Table 10, never below that size's G203 Table 11 minimum on rule 8 steps; check the maximum velocity; then take the low case through the velocity and tractive tests and give the pipe its class.

$$PF = 1.5 + \frac{1}{\sqrt{Q_{\mathrm{m}}}} \tag{58}$$

$$Q_{\mathrm{pdf}} = 2.65\, Q_{\mathrm{adf}}{}^{0.879} \tag{59}$$

$$Q_{\mathrm{inf}} = i\, L \tag{60}$$

Symbol	Meaning	Unit
PF	Peltier peak factor, 100 properties or fewer	—
Q m	average daily flow	l/s
Q pdf, Q adf	peak and average daily flow, over 100 properties	Ml/d
i	infiltration allowance, 720	l/d per km
L	length of sewer upstream	km

Source: Peaks G1-p71 (Merrimack) and G1-p72 (Peltier, ceiling 5.0); infiltration G1-p72; sizing G203-p27 to p30; flow cases, classes and the property count per year: W14/docs/DESIGN_FLOWS_FOR_NETWORK.md sections 2 to 4; the connected count for the low-case formula and K in m³/s: recommended in W14/analysis/design_flows.json, to be confirmed by the engineer; gradient steps: rule 8 (engineer, 2026-09-07).

The pipe is P-0042, a DN200 secondary main sewer 14.60 m long. It is laid at 5.00 mm/m, a whole number of rule 8 steps and also a value on the 0.05 % grid the test-area design was laid on, so the two agree here. It is the last pipe of its catchment, 9.8 ha and 903 m of sewer, so everything the catchment produces passes through it into chamber MH-1385 on the main pipe.

Table 66 P-0042: the plots it serves

Quantity	Value	How it is obtained
Plots	55	plot centre inside the catchment
Domestic properties 2024	25	sum of G_DOM
Properties at saturation	57.5	G_DOM + (POP_ULT − POP) / OR_S, summed
Properties in 2030	29.9	G_DOM + (POP_2030 − POP) / OR_S, summed
Connected properties in 2030	18.2	properties in 2030 × 0.61
Q_ULT, average	69.7 m³/d	sum of Q_ULT
Q_2030, average	41.0 m³/d	sum of Q_2030
Low case, average	25.0 m³/d	Q_2030 × 0.61
Sewer upstream	0.903 km	all pipe in the catchment

Table 67 P-0042: the two design flows

Step	Working	Result
Sizing case: method	57.5 properties at or under 100	Peltier
Sizing case: peak	Peltier: Q m = 69.7 / 86.4 = 0.807 l/s, PF = 1.5 + 1/√0.807 = 2.61, peak = 2.11 l/s	2.11 l/s
Infiltration	720 l/d/km × 0.903 km = 650 l/d	0.008 l/s
Design flow	peak + infiltration	**2.12 l/s**
Low case: method	18.2 connected properties in 2030 at or under 100	Peltier
Low case: peak	Peltier: Q m = 25.0 / 86.4 = 0.289 l/s, PF = 1.5 + 1/√0.289 = 3.36, peak = 0.97 l/s	**0.97 l/s**

Both peak factors are below the recommended ceiling of 5.0. The low case carries no infiltration: the flow handoff defines it as the connected opening-year sewage, peaked, and adding the 0.008 l/s allowance would only flatter the self-cleansing test, which is the direction to avoid. The layout itself was drawn with a flat five people per property, which gave this catchment 77.1 m³/d; on the counted plot loads it is 69.7 m³/d.

The low-case formula is chosen on the connected count, 18.2 properties (29.9 standing in 2030 × 0.61), because the flow it peaks is the connected flow. That count is recommended in the design-flow handoff, to be confirmed by the engineer. Counted on the properties standing, the formula here would still be Peltier.

Table 68 P-0042: choosing the diameter for 2.12 l/s, gradients on rule 8 steps

Size	G203 Table 11 (mm/m)	Laid at, rule 8 (mm/m)	d/D	Limit	v (m/s)	Result
DN200	5.00	5.00	0.220	0.65	0.46	**smallest that passes**

The minimum size carries the flow with a large margin. The diameter is set by G203 Table 6, not by the flow, which is the normal case for a residential street sewer.

Table 69 P-0042: every check, at the rule 8 gradient; depths and drop as laid in the test-area design

Check	Value	Limit	Result	Source
d/D at design flow	0.220	≤ 0.65	pass	G203-p27 Tab 10
Velocity at design flow	0.46 m/s	≤ 3.0 m/s	pass	G203-p27
Laid gradient against G203 Table 11	5.00 mm/m	≥ 5.00 mm/m	pass	G203-p29
Gradient on a rule 8 step	5.00 mm/m	multiple of 0.500 mm/m	pass	project rule 8
Fall against tolerance	73 mm	> 40 mm	pass	G203-p29
Length against G203 Table 12	14.60 m	≤ 100 m	pass	G203-p30
Depth at the downstream chamber	7.41 m	≤ 12 m	pass	project, G203-p33
Deepest chamber in the catchment	4.07 m	≤ 12 m	pass	project, G203-p33
Drop into the downstream chamber	3.28 m	0.6 m backdrop; 2 m vortex	vortex drop shaft	G203-p30
Low case: velocity	0.37 m/s at d/D 0.151	≥ 0.75 m/s	not reached	G203-p26
Low case: tractive, τ = 1 Pa, K 2.33 × 10^{-4} (Q in m³/s)	S min 5.70 mm/m	laid 5.00 mm/m	not met	G203-p27
Class	**early cleansing**			project rule

Worked Ibri number. The pipe is sized by the minimum and carries 2.12 l/s at saturation, d/D 0.22. In the opening year it carries 0.97 l/s at peak and 0.37 m/s. Neither test is met, and the flow is below the 1.5 l/s outside threshold, so it is an early-cleansing pipe: it goes on the flushing list for the early years (G203-p28) and is not upsized, since DN200 is already the smallest main sewer. Nor is it regraded. Equation (54), with $K = 2.33 \times 10^{-4}$ and Q in m^3/s, evaluated at this flow asks for 5.70 mm/m, but that is the relation extrapolated below the flow range it was derived for; what is missing is flow, not gradient. The drop of 3.28 m into the main-pipe chamber exceeds 2 m, so the join needs a vortex drop shaft rather than a backdrop.

Where it lives. Computed in this chapter (TUTORIALS/T04/ch05_gravity.py, function example) from W14/shp/PLOTS_load.shp (Q_ULT, Q_2030, G_DOM, POP, POP_2030, POP_ULT, OR_S), W14/analysis/design_flows.json (connection ratio, infiltration) and the W8 design: catchment polygon and PIPE_M in W8/shp/W8_catchments.shp, pipe P-0042 in W8_pipes.shp, chamber MH-1385 in W8_manholes.shp. The engine itself still loads the flat per-plot figure until it reads PLOTS_load.shp (DESIGN_FLOWS_FOR_NETWORK.md section 6). The rule 8 gradient is computed by step8 and round_up8 in the same script.

11.17 Worked example 2: a header

What it is for. To size a pipe where the flow, not the minimum, sets the diameter, and to show the self-cleansing test passing on the tractive force when the velocity is not reached.

The rule. Sum the plots upstream; peak by Merrimack above 100 properties and by Peltier at or below; add infiltration by the length of sewer upstream; choose the smallest diameter whose d/D at the laid gradient is within G203 Table 10, never below that size's G203 Table 11 minimum on rule 8 steps; check the maximum velocity; then take the low case through the velocity and tractive tests and give the pipe its class.

Source: Peaks G1-p71 (Merrimack) and G1-p72 (Peltier, ceiling 5.0); infiltration G1-p72; sizing G203-p27 to p30; flow cases, classes and the property count per year: W14/docs/DESIGN_FLOWS_FOR_NETWORK.md sections 2 to 4; the connected count for the low-case formula and K in m³/s: recommended in W14/analysis/design_flows.json, to be confirmed by the engineer; gradient steps: rule 8 (engineer, 2026-09-07).

The pipe is P-0741, a DN315 secondary header 51.33 m long. The test-area design laid it at 3.00 mm/m, its minimum rounded up to the 0.05 % grid; on rule 8 steps the same minimum is 2.70 mm/m, and the example is worked at 2.70 mm/m. It is the last pipe of its catchment, 174.9 ha and 19,494 m of sewer, so everything the catchment produces passes through it into chamber MH-0652 on the main pipe.

Table 70 P-0741: the plots it serves

Quantity	Value	How it is obtained
Plots	998	plot centre inside the catchment
Domestic properties 2024	814	sum of G_DOM
Properties at saturation	1,137.1	G_DOM + (POP_ULT − POP) / OR_S, summed
Properties in 2030	862.5	G_DOM + (POP_2030 − POP) / OR_S, summed
Connected properties in 2030	526.1	properties in 2030 × 0.61
Q_ULT, average	1,175.6 m³/d	sum of Q_ULT
Q_2030, average	889.9 m³/d	sum of Q_2030
Low case, average	542.9 m³/d	Q_2030 × 0.61
Sewer upstream	19.494 km	all pipe in the catchment

Table 71 P-0741: the two design flows

Step	Working	Result
Sizing case: method	1,137.1 properties over 100	Merrimack
Sizing case: peak	Merrimack: Q adf = 1.1756 Ml/d, Q pdf = 2.65 × 1.1756^0.879 = 3.0549 Ml/d, PF = 2.60, peak = 35.36 l/s	35.36 l/s
Infiltration	720 l/d/km × 19.494 km = 14,036 l/d	0.162 l/s
Design flow	peak + infiltration	**35.52 l/s**
Low case: method	526.1 connected properties in 2030 over 100	Merrimack
Low case: peak	Merrimack: Q adf = 0.5429 Ml/d, Q pdf = 2.65 × 0.5429^0.879 = 1.5490 Ml/d, PF = 2.85, peak = 17.93 l/s	**17.93 l/s**

Both peak factors are below the recommended ceiling of 5.0. The low case carries no infiltration: the flow handoff defines it as the connected opening-year sewage, peaked, and adding the 0.162 l/s allowance would only flatter the self-cleansing test, which is the direction to avoid. The layout itself was drawn with a flat five people per property, which gave this catchment 1,174.3 m³/d; on the counted plot loads it is 1,175.6 m³/d.

The low-case formula is chosen on the connected count, 526.1 properties (862.5 standing in 2030 × 0.61), because the flow it peaks is the connected flow. That count is recommended in the design-flow handoff, to be confirmed by the engineer. Counted on the properties standing, the formula here would still be Merrimack.

Section 9.10 works the same header with 1,004 plots and a design flow of 35.92 l/s, against 998 plots and 35.52 l/s here. The difference is the allocation of plots, not the method. Chapter 9 gives each plot to its nearest street pipe and sums down the tree, which draws in a net 6 more plots whose centres lie just outside this catchment's polygon; here a plot counts only if its centre lies inside the catchment. The sewer upstream is the same, 19.494 km, so the infiltration is the same. At 35.92 l/s the same DN315 at 2.70 mm/m runs at d/D 0.635, also within the limit, so either allocation gives the same pipe.

Table 72 P-0741: choosing the diameter for 35.52 l/s, gradients on rule 8 steps

Size	G203 Table 11 (mm/m)	Laid at, rule 8 (mm/m)	d/D	Limit	v (m/s)	Result
DN200	5.00	5.00	above 0.95	0.65	—	cannot carry the flow
DN250	3.75	3.75	above 0.95	0.65	—	cannot carry the flow
DN315	2.70	2.70	0.631	0.65	0.77	**smallest that passes**

A smaller pipe laid steeper does not rescue it: at its own minimum gradient, a whole number of rule 8 steps, it either cannot carry the flow or runs too full. The size is set by the flow, and laying the chosen size at its flatter minimum is exactly what G203 Table 11 allows. Choosing a larger size to lay it flatter still would be the oversizing the guideline forbids.

Table 73 P-0741: every check, at the rule 8 gradient; depths and drop as laid in the test-area design

Check	Value	Limit	Result	Source
d/D at design flow	0.631	≤ 0.65	pass	G203-p27 Tab 10
Velocity at design flow	0.77 m/s	≤ 3.0 m/s	pass	G203-p27
Laid gradient against G203 Table 11	2.70 mm/m	≥ 2.70 mm/m	pass	G203-p29
Gradient on a rule 8 step	2.70 mm/m	multiple of 0.270 mm/m	pass	project rule 8
Fall against tolerance	139 mm	> 40 mm	pass	G203-p29
Length against G203 Table 12	51.33 m	≤ 100 m	pass	G203-p30
Depth at the downstream chamber	9.56 m	≤ 12 m	pass	project, G203-p33
Deepest chamber in the catchment	10.45 m	≤ 12 m	pass	project, G203-p33
Drop into the downstream chamber	0.00 m	0.6 m backdrop; 2 m vortex	none needed	G203-p30
Low case: velocity	0.66 m/s at d/D 0.418	≥ 0.75 m/s	not reached	G203-p26
Low case: tractive, τ = 1 Pa, K 2.33×10^{-4} (Q in m^3/s)	S min 1.49 mm/m	laid 2.70 mm/m	pass	G203-p27
Class	**tractive pass**			project rule

Worked Ibri number. The header runs at d/D 0.63 at saturation, inside the 0.65 limit, at 0.77 m/s. In the opening year it carries 17.93 l/s at peak and reaches only 0.66 m/s, so the velocity test is not met; the tractive minimum at that flow, equation (54) with $K = 2.33 \times 10^{-4}$ and Q in m^3/s, is 1.49 mm/m against 2.70 mm/m laid, so it is a tractive pass. The steeper of the two approaches governs, and here it is G203 Table 11: 2.70 mm/m for DN315, which is already a whole number of rule 8 steps (0.27 mm/m at this size), so the pipe is laid at 2.70 mm/m. The single 0.05 % grid of the test-area design laid it at 3.00 mm/m.

The class of this pipe depends on a number the guideline does not give. At τ = 2 Pa the tractive minimum at the same flow, same K, rises to 3.49 mm/m, above the 2.70 mm/m laid, and the class becomes "fails both, regrade". A header carrying real flow would then have to be steepened, and the extra fall goes into depth downstream, where the join chamber already sits at 9.56 m. The design value of τ has to be agreed with NWS before the network is finalised.

Where it lives. Computed in this chapter (TUTORIALS/T04/ch05_gravity.py, function example) from W14/shp/PLOTS_load.shp (Q_ULT, Q_2030, G_DOM, POP, POP_2030, POP_ULT, OR_S), W14/analysis/design_flows.json (connection ratio, infiltration) and the W8 design: catchment polygon and PIPE_M in W8/shp/W8_catchments.shp, pipe P-0741 in W8_pipes.shp, chamber MH-0652 in W8_manholes.shp. The engine itself still loads the flat per-plot figure until it reads PLOTS_load.shp (DESIGN_FLOWS_FOR_NETWORK.md section 6). The rule 8 gradient is computed by step8 and round_up8 in the same script; the nearest-pipe allocation of section 9.10 is _net() in TUTORIALS/T04/ch04_peak_tiers.py.

11.18 Every constraint in one place

Requirements of the guidelines, with the page each comes from and the force of its wording. Project decisions follow in a separate table so the two can never be confused.

Table 74 Guideline constraints on a gravity sewer

Constraint	Value	Force	Source
Design formula	Colebrook-White or Manning; licensed software	shall	G203-p24

Constraint	Value	Force	Source
Colebrook-White roughness	ks 1.5 mm, all sizes and materials	shall	G203-p24, p28
Kinematic viscosity	1.141×10^{-6} m²/s at 15 °C	should	G203-p25 Tab 9
Manning n	0.009 to 0.015 by material	typical	G203-p23 Tab 8
Self-cleansing velocity	above 0.75 m/s at peak flow	shall	G203-p26
Preferred velocity	0.90 m/s	preferred	G203-p26
Two approaches	velocity and tractive force; steeper governs	shall	G203-p25 to p27
Head of system	tractive force alone	shall	G203-p27
Tractive relation	S min = K τ^1.23 Q^−0.461; K 2.33×10^{-4} (Q in m³/s) or 5.5×10^{-3} (Q in l/s), 2.3 % apart	method	G203-p27
Maximum velocity	3 m/s at the design depth of flow	shall	G203-p27
Maximum gradient	the one giving 3.0 m/s	should	G203-p29
Depth of flow at peak	d/D 0.65 up to 350 mm, 0.50 above	recommended	G203-p27 Tab 10
Part-full relation	d/D against q/Q	figure	G203-p28 Fig 2
Early low flows	more frequent inspection and cleansing	should	G203-p28
Minimum gradient by size	G203 Table 11, 5.00 to 0.75 mm/m	shall	G203-p29 Tab 11
No oversizing for flatter slopes	prohibited	shall	G203-p29
Uniform slope between manholes	required	must	G203-p29
Line and level tolerance	20 mm, never a reverse gradient	shall	G203-p29
Tertiary slopes	connection 3 to 10 %; rider, lateral 1 to 10 %	typical	G203-p18 Tab 5
Property connection sewer	150 mm minimum, 50 m maximum	should	G203-p18 Tab 4
Property connection cover	600 mm minimum	required	G203-p19
Minimum sizes	OD 160 rider and connection; OD 200 lateral and main	table	G203-p22 Tab 6
Lateral sewer length	45 m maximum	table	G203-p17, p22 Tab 6
Secondary network size	200 to 400 mm, 400 not mandatory	usual	G203-p23
Minimum cover	1.3 m to crown	shall	G203-p33
Shallower than 1.3 m	concrete protection, 0.5 m over pipe and protection	shall	G203-p33
Horizontal clearance	3 m	required	G203-p33
Maximum cover	approximately 10 to 12 m	recommended	G203-p33
Beyond it	investigate with manufacturer; pump where excavation cost is prohibitive	shall	G203-p33
Manhole locations	gradient, diameter, junction, regular spacing, end of lateral	shall	G203-p29, p30
Manhole spacing	G203 Table 12: 100, 120, 150, 200 m	recommended	G203-p30 Tab 12
Altering the spacing	NWS pre-approval	has to	G203-p30
Backdrop	above 600 mm, outside the manhole	shall	G203-p30
Backdrop height	2 m, then vortex drop shaft	should	G203-p30
Internal backdrops	not in manholes under 1.5 m diameter	not permitted	G203-p30
Inlet angle	not less than 90° to the direction of flow	shall	G203-p19, p30
Wadis and washout areas	pipes and chambers avoided	must, shall	G203-p30, p33
Access	24 hours, 7 days to pipes and chambers	shall	G203-p33
Corridor width	2.00 m for 200 to 500 mm; 2.80 m for 600 to 900 mm	indicative	G203-p32, p33 Tab 13
Peak factor, over 100 properties	Merrimack, Q pdf = 2.65 Q adf^0.879 (Ml/d)	is to be used	G1-p71
Peak factor, alternative	Peltier, 1.5 + 1/√Q m (l/s)	method	G1-p72
Peak factor ceiling	5.0	recommended	G1-p72
Infiltration, new networks	720 l/d per km of sewer	should	G1-p72
Coverage at end of period	100 %	assumed	G1-p73

Table 75 Project decisions the guideline does not make

Question	Guideline gives	Project decision	Source
Which flow sizes the pipe	nothing	Q_ULT, saturation, 100 % connected	engineer 2026-09-11
Which flow tests self-cleansing	"at peak flow"	Q_2030 × 0.61, peaked per pipe	engineer 2026-09-11
Peak factor at 100 properties or fewer	no formula stated for that range	Peltier	engineer 2026-09-11
Property count that picks the low-case formula	nothing	connected: properties of 2030 × 0.61	recommended in the design-flow handoff, to be confirmed by the engineer
Tractive constant	two, 2.3 % apart	$K = 2.33 \times 10^{-4}$ with Q in m^3/s	recommended in the design-flow handoff, to be confirmed by the engineer
Tractive tension τ	method and constant only	1 Pa, a parameter, reported	GAP-9
Low-flow threshold	nothing	1.5 l/s, outside assumption (Mara)	tagged
Maximum depth	approximately 10 to 12 m cover, recommended	12 m ground to invert, hard, no exceptions	PROJECT_STATE 2 item 3
Gradient steps	nothing	a tenth of the size's G203 Table 11 minimum; the test-area design and the engine are still on a 0.05 % grid, to be aligned	rule 8, engineer 2026-09-07
Inlet angle in the engine	90°	85°, stated deviation, sharper flagged	engineer 2026-08-20
Rider length	45 m on the lateral row	45 m on riders as well	02 section 6
True bore of PVC-U	OD-designated sizes	SDR 34 (SN8) until the pipe class is fixed	assumption

11.19 Check it yourself

1. Evaluate equation (47) at D = 0.200 m and S = 0.005 with ks 1.5 mm and ν 1.141×10^{-6}. You should get 0.75 m/s: that is G203 Table 11's first row. Repeat for DN900 at 0.75 mm/m.
2. Open W8/shp/W8_pipes.shp. Check that no pipe has SLOPE_PMIL below the G203 Table 11 value for its DN_MM, that every value is a multiple of 0.5 (the 0.05 % grid the file was laid on; rule 8 has the same step at DN200 and finer steps above it), and that LEN_M never exceeds the G203 Table 12 spacing for its size.
3. Open W8/shp/W8_manholes.shp. Check that no DEPTH exceeds 12 m and that IS_PUMP sums to zero. Then look at a deep chamber and confirm the pipes either side stay under 12 m along their whole length, not only at the ends.
4. Select the plots of W14/shp/PLOTS_load.shp whose centre lies inside catchment 23 of W8/shp/W8_catchments.shp. Sum Q_ULT and Q_2030 and compare with section 11.17; then compare the plot count with the nearest-pipe allocation of section 9.10.
5. Recompute the header's d/D by hand: take its design peak, divide by the pipe-full flow of DN315 at 2.70 mm/m, its rule 8 gradient, on the true bore, and read d/D from the part-full table in section 11.4.
6. Change τ from 1 to 2 Pa and redo the tractive test of both worked examples. Note which class changes, and why the parameter is reported rather than fixed.
7. In W8/shp/W8_pipes.shp, list every pipe with DROP_DN above 0.6 m and above 2 m. Each needs an external backdrop or a vortex drop shaft at its downstream chamber.

8. Read G203 pages 17 to 33 and G201 pages 71 to 73 against the constraint table of section 11.18, row by row. Every value should be found on the page given, with the force word shown.

12 Self-cleansing and the early years

A sewer is sized for the flow it carries when its catchment is full, which in this study is 2070. It opens around 2030 carrying a fraction of that. Solids that settle in the opening years stay where they are unless the flow, or a maintenance crew, moves them. This chapter tests every pipe on the flow it will actually carry when it opens, puts it in one of four classes, and says what the design and the operator do about each class.

> **Here over-estimating the flow is the unsafe error.** Everywhere else in the method a generous flow is the conservative choice. In this check it is the opposite: too much early flow declares a pipe self-cleansing while it silts, and the operator is told it needs no washing. Every choice in this chapter leans towards the smaller flow.

12.1 What the guideline requires

What it is for. Before any number is computed, the reader needs to know which tests are mandatory, which one governs, and where the guideline lets one of them go.

The rule. The guideline states it in four sentences, quoted here exactly.

> *“Sanitary sewers must be designed so that sediment does not accumulate during periods of low flow without providing some period with enough flow to clean out the pipes. To ensure the suspended sediment carrying capacity of the sewer, two approaches shall be used: 1. Self-cleansing velocity 2. Minimum Tractive force”* G203 §4.2.2.1, p25–26

> *“Steeper gradient calculated based on self-cleansing velocity and minimum tractive force methodology shall be adopted as minimum pipe gradient.”* G203 §4.2.2.1, p27

> *“At the head of the sewerage systems, the flow velocity based on the minimum self-cleansing may not be attainable. In these circumstances, the minimum pipe gradient for the sewer shall be calculated based on the hydraulic design approach of minimum tractive force.”* G203 §4.2.2.1, p27

Three things follow. Both tests are mandatory: the verb is shall. Where both can be met, the steeper of the two gradients is the minimum. Where the velocity cannot be reached, which the guideline places at the head of the system, the tractive test alone sets the gradient. The guideline does not define the head of the system by flow or by number of properties; this method reads it as any pipe that cannot reach the velocity at its opening-year peak flow. G203 section 4.3 (pp 28–29) repeats the two approaches and turns the velocity approach into G203 Table 11, a minimum gradient for each diameter, reproduced in Chapter 11 as the table of minimum sewer gradients.

The two tests are used in two roles. In the design, each pipe is laid at no less than the steeper of its G203 Table 11 gradient and the tractive slope at its peak flow. In the audit, the pipe as laid is tested again on the opening-year flow of Section 12.4, which is smaller than the design flow and so demands a steeper tractive slope. A pipe that passed in the design can therefore fail the audit, and that is the purpose of the audit.

Source. G203 §4.2.2.1 p25–27 and §4.3 p28–29, read from the PDF.

Ibri number. A street pipe in Ibri must drain about 909 average plots before its opening-year peak flow can reach 0.75 m/s on a DN200 at its G203 Table 11 gradient (Section 12.2). In a branching network most of the pipe length sits near the heads and drains far fewer, so on most of the network the tractive test is the one that decides.

Where it lives. W13/tmp3/py/sewnet/hydra.py, smin_for (the steeper of G203 Table 11 and the tractive slope at the pipe's peak flow); stages/audit.py, _self_cleansing.

12.2 The self-cleansing velocity

What it is for. The first test. A pipe that reaches the velocity at its opening-year peak flow moves its own solids and needs nothing more.

The rule. The velocity at the peak flow must reach 0.75 m/s. The preferred 0.90 m/s is reported for each pipe but is not the pass mark.

> *"To provide a self-cleansing regime within gravity sewers, the minimum velocity in the pipe shall be above 0.75 m/s at peak flow, with preferred velocity at 0.90m/s."* G203 §4.2.2.1, p26

The velocity of a pipe running part full is the Colebrook-White equation of G203 p24 with the diameter replaced by four times the hydraulic radius of the wetted section. At full bore 4R equals D and the guideline's own form returns.

$$v = -2\sqrt{8gRS} \cdot \log_{10}\left[\frac{k_s}{14.8R} + \frac{2.51\nu}{4R\sqrt{8gRS}}\right] \qquad (61)$$

Symbol	Meaning	Unit
v	mean velocity at the depth of flow	m/s
R	hydraulic radius of the wetted section, area over wetted perimeter	m
S	gradient of the pipe	m/m
g	acceleration due to gravity, 9.81	m/s²
ks	roughness, 1.5 mm for all sizes and materials (G203 p24, p28)	m
ν	kinematic viscosity, 1.141 × 10⁻⁶ at 15 °C (G203 p25 Tab 9)	m²/s

Source. G203 p24 (Colebrook-White, ks), p25 Table 9 (ν), p26 (0.75 and 0.90 m/s), p29 Table 11. The part-full form is a project calculation of the same equation; the guideline requires the design itself to be run in NWS-approved software (p24).

Ibri number. A DN200 at its G203 Table 11 gradient of 5.00 mm/m, running full, moves at 0.75 m/s and carries 23.5 l/s. G203 Table 11 is therefore the velocity approach evaluated at full bore. The table below shows what happens below full bore, for the DN200 and for a DN315 at its own minimum of 2.70 mm/m.

Table 76 Velocity and flow at the G203 Table 11 gradient, by proportional depth (Colebrook-White, ks 1.5 mm, 15 °C)

d/D	DN200, 5.00 mm/m: v (m/s)	DN200: q (l/s)	DN315, 2.70 mm/m: v (m/s)	DN315: q (l/s)	0.75 m/s
0.65	0.82	17.8	0.81	43.6	passes
0.50	0.75	11.8	0.74	28.9	**no margin**
0.30	0.58	4.6	0.58	11.3	fails
0.20	0.46	2.0	0.46	5.1	fails
0.10	0.29	0.5	0.29	1.2	fails

A pipe at its minimum gradient reaches 0.75 m/s only when it runs just over half full: d/D 0.502, a flow of 11.9 l/s in the DN200. At exactly half depth, the row d/D 0.50 of the table, it carries 11.8 l/s at a velocity a fraction under 0.75 m/s, which is why that row says no margin. In the opening year that takes about 909 average Ibri plots upstream. Laying steeper than the minimum buys an earlier self-cleansing velocity, and that is a real trade: a little more excavation now against fewer years of washing. The values agree with the part-full table of the concept methodology, which was typed from a Manning ratio; they differ only at d/D 0.10, by 0.02 m/s.

Where it lives. W13/tmp3/py/sewnet/hydra.py, pipe_state(dn, slope, q_peak) returns depth and velocity; criteria.py, V_SELF_CLEANSING = 0.75 and V_PREFERRED = 0.90. The engine uses the true internal bore of the pipe it lays; this chapter uses the nominal diameter, which is what reproduces G203 Table 11, so engine velocities differ slightly.

12.3 The minimum tractive force

What it is for. The second test, and the only one a small flow can meet. It asks whether the flow pulls hard enough on the pipe wall to move a particle, whatever its velocity.

The rule. The force balance of G203 p26 (weight of the fluid along the pipe equal to the friction on its wetted perimeter, with W = ρgaL) gives the tractive tension on the wall:

$$\tau = \rho g R S \tag{62}$$

Mara, Sleigh and Taylor solved it for the slope, for a pipe running 20 % deep:

$$S_{\min} = K \cdot \tau^{1.23} \cdot Q^{-0.461} \tag{63}$$

Symbol	Meaning	Unit
τ	tractive tension on the wall; no design value in G203 (GAP-9)	Pa
ρ	density of the fluid, 1,000	kg/m³
g, R, S	gravity, hydraulic radius, gradient, as in the velocity equation	m/s², m, m/m
Smin	minimum slope to move particles	m/m
Q	peak flow in the pipe	m³/s
K	2.33 × 10⁻⁴ with Q in m³/s (G203 p27), used for every slope in this chapter	—

> *"Mara, Sleigh, and Taylor (2000) developed the following relationship for minimum slope based on the assumption of d/D = 0.2 and n = 0.013"* G203 §4.2.2.1, p27

The equation is printed in the guideline as an image, which is how an earlier copy of the criteria lost the τ term. Its origin can be checked. Put Manning with n = 0.013 at d/D = 0.2 (area 0.1118 D², hydraulic radius 0.1206 D) into equation (62), and eliminate D. The exponents come out as 16/13 = 1.231 and 6/13 = 0.462, and K as 2.33 × 10⁻⁴ with Q in m³/s, the guideline's value. So the formula is simply the gradient at which a pipe running 20 % deep puts τ on its wall. Two consequences matter for design. The diameter has dropped out: the tractive slope depends on the flow and on τ only, so a larger pipe never helps this test. And the formula says nothing about a pipe running at another depth; it is a rule for small flows.

> **One constant, in m³/s.** G203 p27 also prints a second constant, 5.5 × 10⁻³ with Q in l/s. It is not the same equation in other units: converting 2.33 × 10⁻⁴ to litres gives 2.33 × 10⁻⁴ × 1,000^0.461 = 5.63 × 10⁻³, so the l/s constant gives slopes 2.3 % flatter. The m³/s form gives the steeper slope, which is the safe direction for this test, and it is the one the engine uses. Every slope in this chapter is computed with K = 2.33 × 10⁻⁴ and Q in m³/s, and every table names it: recommended in the design-flow handoff, to be confirmed by the engineer. The difference is far smaller than the uncertainty in τ.

The guideline defines τ, prints the equation and its constant, and gives no design value for τ anywhere in the document. It is therefore a parameter, recorded as GAP-9. Every class count is reported at τ = 1, 1.5 and 2 Pa, and no single value is presented as the guideline's. The engine runs at 1 Pa as a working value tagged GAP-9 until NWS confirms one. The table shows how much the choice moves the answer.

Table 77 Tractive slope Smin (mm/m) against τ and peak flow, K = 2.33 × 10⁻⁴ with Q in m³/s (flows shown in l/s); bold where steeper than the DN200 G203 Table 11 gradient of 5.00 mm/m

τ (Pa)	Q = 1 l/s	Q = 2 l/s	Q = 5 l/s	Flow above which DN200 at 5.00 mm/m passes (l/s)
1.0	**5.63**	4.09	2.68	1.3
1.5	**9.27**	**6.73**	4.41	3.8
2.0	**13.20**	**9.59**	**6.29**	8.2

The slope rises as τ^1.23: 1.65 times at 1.5 Pa and 2.35 times at 2 Pa. At 1 Pa a DN200 laid at its G203 Table 11 gradient passes the tractive test for any peak flow above 1.3 l/s; at 2 Pa it needs 8.2 l/s. The choice of τ decides whether most street pipes pass or fail, which is why it must come from NWS and not from this study.

Source. G203 §4.2.2.1 p26 (force balance, τ in Pa), p27 (the formula, K, the assumptions d/D 0.2 and n 0.013); _BRAIN/05_GAPS.md GAP-9 (no τ in G203); the check of K is a project calculation.

Ibri number. Pipe B of Section 12.6, a DN200 at 5.00 mm/m draining 200 average Ibri plots, carries 2.23 l/s at its opening-year peak. Its tractive slope is 3.89 mm/m at 1 Pa, so it passes; at 1.5 Pa it is 6.41 mm/m and the same pipe fails.

Where it lives. W13/tmp3/py/sewnet/hydra.py, smin_tractive(Q); criteria.py, TRACTIVE_K = 2.33e-4 (Q in m³/s), TAU_PA = 1.0, TRACTIVE_QMIN = 0.0015; stages/audit.py, selfclean_stats reports how many pipes would fail at 2 Pa.

12.4 The two flow cases

What it is for. Sizing and self-cleansing fail in opposite directions, so they cannot share a flow. Each takes its own.

The rule. Size every pipe on the saturation flow of the plots upstream, peaked, plus infiltration. Test self-cleansing on the plots' own 2030 flow times the connection ratio, peaked per pipe on the connected property count, and feed that one flow to both the velocity and the tractive test.

$$Q_{\text{size}} = f_{\text{pk}}\left(\textstyle\sum_i Q_{\text{ULT,i}}\right) + q_{\text{inf}}L \qquad (64)$$

$$Q_{\text{sc}} = f_{\text{pk}}\left(0.61 \times \textstyle\sum_i Q_{2030,\text{i}}\right) \qquad (65)$$

Symbol	Meaning	Unit
Q size	peak flow that sizes the pipe	l/s
Q sc	peak flow for both self-cleansing tests	l/s
Q ULT,i	average dry-weather flow of plot i at saturation (2070), field Q_ULT	m³/d
Q 2030,i	average dry-weather flow of plot i in 2030, field Q_2030	m³/d
0.61	connection ratio reached by 2028 (Inception R0)	—
f pk	peak per pipe: Merrimack over 100 properties, Peltier at 100 or fewer; in the low case the properties counted are the connected ones	—
q inf	infiltration, 720 l/d per km of new sewer (G201-p72)	l/d/km
L	length of sewer upstream of the pipe	km

The peak is taken per pipe on the properties upstream, counted for each plot as its domestic properties today plus its added people over the settlement's occupancy. The sizing flow counts the properties at saturation, all connected. In the low case the count that picks the formula is the connected one, the 2030 properties × 0.61: recommended in the design-flow handoff, to be confirmed by the engineer. The smaller count picks Peltier more often, and at the same flow Peltier gives the lower peak, which is the safe direction for this test. Over 100 properties Merrimack applies, Qpdf = 2.65 Qadf^0.879 with both flows in Ml/d (G201-p71). At 100 or fewer Peltier applies, PF = 1.5 + 1/√Qm with Qm in l/s (G201-p72). The guideline recommends that the hourly peak factor not exceed 5.0; it is a recommendation, reported where exceeded and never applied as a silent cap (G201-p72). Infiltration is added to the sizing flow only. Left out of the self-cleansing flow it errs on the safe side, and it is small: 0.0083 l/s per kilometre of pipe.

Why these two. Sizing fails on an under-estimate, so it takes the largest flow the pipe will ever carry, at full connection, which G201-p73 sets for the end of the planning period. Self-cleansing fails on an over-estimate, so it takes the smallest flow that is still real: the opening year the terms of reference name (construction year or 2030), at the connection ratio actually forecast for it. Using each plot's own 2030 flow matters because growth is not even. A district still empty in 2030 carries almost nothing in 2030, whatever it will carry at saturation.

The tempting shortcut, saturation flow times the connection ratio, is never used. It mixes 2070 people with a 2028 connection share and gives every new district 61 % of its saturation flow in 2030. The tables show the size of the error.

Table 78 The two flow cases over the whole study area, average dry-weather flow

Flow	m^3/d	Used for
2030, all properties connected	24,216	no test; the base of the low case
2030 × 0.61, the low case	**14,772**	**self-cleansing, both tests**
Saturation, 2070	**60,099**	**sizing, with peak and infiltration**
Saturation × 0.61	36,660	never: 2.5 times the real 2030 flow

Table 79 Why the shortcut fails: the low case against saturation × the connection ratio, by settlement, m^3/d

Settlement	Q 2030 × 0.61	Q saturation × 0.61	Overstated by	Full in
Ibri	8,160.0	14,669.8	1.8	2056
At Tayyib	545.0	4,110.6	7.5	2068
Shalashil	18.5	1,238.3	66.9	2061

In Ibri, largely built today, the shortcut overstates the opening-year flow 1.8 times. In Shalashil, almost empty today, it overstates it 67 times, and every pipe there would be declared self-cleansing while carrying almost nothing.

Source. Project rule, the two flow cases (engineer, 2026-09-11), _BRAIN/02_DESIGN_CRITERIA.md §1; G201-p71 and p72 (peaks, infiltration, PF), G201-p73 (coverage 100 % by the end of the period); connection ratio from the Inception R0. The connected property count of the low case, the Mara constant in m^3/s, leaving infiltration out of the low case and the order of the tests are open rulings, recommended in the design-flow handoff, to be confirmed by the engineer (design_flows.json, rules.open_rulings_recommended; W14/docs/DESIGN_FLOWS_FOR_NETWORK.md §8).

Ibri number. Whole area in 2030: 24,216 m^3/d × 0.61 = 14,772 m^3/d for self-cleansing; 60,099 m^3/d at saturation for sizing. Properties: 26,255 standing in 2030, 16,016 of them connected in the low case; 70,328 at saturation.

Where it lives. W14/shp/PLOTS_load.shp, fields Q_2030, Q_ULT, G_DOM, POP, POP_2030, POP_ULT, OR_S; W14/analysis/design_flows.json, keys rules, totals and low_case_2030_m3d, written by W14/py/make_design_flows.py; handoff W14/docs/DESIGN_FLOWS_FOR_NETWORK.md §2–3.

12.5 The four classes

What it is for. One label per pipe that tells the designer what to fix and the operator what to wash.

The rule. Every pipe of every tier (primary, secondary header, secondary main sewer) is tested on its low-case peak Q sc, at the diameter and gradient as laid, in this order:

1. The velocity at Q sc reaches 0.75 m/s: velocity pass. Nothing to do.
2. Otherwise, Q sc is below 1.5 l/s: early cleansing. The flow is too small for either test to mean anything. The pipe goes on the flushing list; it is not upsized and not regraded.
3. Otherwise, the gradient is at least the tractive slope Smin at Q sc and the chosen τ: tractive pass. Nothing to do; the count is reported at each τ.
4. Otherwise: fails both. The pipe carries a real flow and is too flat for it. This is a design fault: steepen the pipe to at least Smin and run the design again.

The order matters. Velocity comes first because a steep pipe scours at any flow, however small. The low-flow test comes before the tractive test because at a very small flow the formula returns a gradient no street can take: at 0.5 l/s and 1 Pa it asks for 7.75 mm/m. That case is the one the guideline hands to the operator in G203 §4.2.6 (Section 12.7), not to the designer.

Table 80 The four self-cleansing classes, tested on the low-case peak flow

Class	Test at Q sc	Action
Velocity pass	v ≥ 0.75 m/s	none
Early cleansing	v < 0.75 m/s and Q sc < 1.5 l/s (outside assumption)	flushing list (G203 §4.2.6); not upsized, DN200 is the minimum
Tractive pass	v < 0.75 m/s, Q sc ≥ 1.5 l/s, S ≥ Smin(τ, Q sc)	none; counts reported at τ = 1, 1.5 and 2 Pa
Fails both: regrade	v < 0.75 m/s, Q sc ≥ 1.5 l/s, S < Smin(τ, Q sc)	design fault: steepen to at least Smin

Source. G203 §4.2.2.1 p25–27 and §4.2.6 p28; project rule, the four classes (engineer, 2026-09-11), _BRAIN/02_DESIGN_CRITERIA.md §1; the 1.5 l/s threshold is an outside assumption (Section 12.8).

Ibri number. The four worked pipes of Section 12.6, one in each class.

Where it lives. Not yet in the engine. W14/docs/PROMPT_W13_SELFCLEANSING_AUDIT.md specifies it for W13/tmp3: the loads wired from PLOTS_load.shp, a field CLEANSE on every pipe, counts and lengths by class and tier, and a QGIS map coloured by class. Today stages/audit.py _self_cleansing tests the design flow, not the low case, and gives a pass or a fail, not four classes.

12.6 Four worked pipes

Each pipe below is constructed from the real per-plot averages of its settlement: every plot counted, built or empty, so a pipe through a mixed district gets a mixed district's flow. The diameter is the smallest that carries the sizing flow within the depth limits of G203 Table 10 (p27); the gradient is the G203 Table 11 minimum for that diameter unless the ground is steeper. Every gradient is laid in steps of a tenth of that diameter's minimum, rounded up: 0.5 mm/m at DN200 and 0.27 mm/m at DN315, so each G203 Table 11 minimum is itself ten steps. The low-case formula is picked on the connected property count (Section 12.4). Infiltration on the sizing flow takes 23.7 m of sewer per plot served, the ratio of the test-boundary layout.

Table 81 Per-plot averages behind the worked pipes (PLOTS_load.shp, every plot)

Settlement	Plots	Q_2030 (m³/d)	× 0.61 (m³/d)	Q_ULT (m³/d)	Properties 2030	× 0.61, connected	Properties at saturation
Ibri	21,867	0.6117	0.3732	1.0998	0.588	0.359	1.057
At Tayyib	10,871	0.0822	0.0501	0.6199	0.089	0.054	0.870

Pipe A: Ibri secondary header, 1,500 plots, DN315 at 8.10 mm/m. Sized on 47.9 l/s at saturation (Merrimack, infiltration included), it runs at d/D 0.49. The ground falls at 8.00 mm/m, so it is laid at 8.10 mm/m, the next DN315 step, rather than its G203 Table 11 minimum of 2.70. On the low case it carries 18.42 l/s (Merrimack, 538 connected properties) at 0.98 m/s. Class: velocity pass, at any τ.

Pipe B: Ibri secondary main sewer, 200 plots, DN200 at 5.00 mm/m. A street pipe at the DN200 minimum. Low case 2.23 l/s (Peltier, 72 connected properties), velocity 0.47 m/s: fails the velocity test. Tractive slope 3.89 mm/m at 1 Pa, under its 5.00 mm/m: tractive pass at 1 Pa. At 1.5 Pa the slope is 6.41 mm/m and the same pipe fails both.

Pipe C: Ibri secondary main sewer, head run, 30 plots, DN200 at 5.00 mm/m. The first run of a street. Low case 0.55 l/s (Peltier, 11 connected properties, PF 4.28), velocity 0.31 m/s, below 1.5 l/s: early cleansing. Its sizing flow is 1.20 l/s, far inside a DN200, so DN200 is set by the minimum diameter, not by the flow.

Pipe D: At Tayyib secondary header, 1,500 plots, DN315 at 2.70 mm/m. At Tayyib is 7.5 times larger at saturation than in 2030, so this header is sized on 29.1 l/s (Merrimack) and laid at its G203 Table 11 minimum of 2.70 mm/m. That gradient is on the DN315 grid: 10 steps of 0.27 mm/m. In 2030 it carries 2.24 l/s (Peltier, 82 connected properties) at 0.36 m/s. The tractive slope at 1 Pa is 3.88 mm/m, steeper than the 2.70 mm/m laid: fails both, regrade.

Table 82 The four worked pipes: sizing flow, low-case flow and class at τ = 1 Pa; Smin with K = 2.33 × 10^{-4} and Q in m^3/s

Pipe	DN, mm/m	Q size (l/s) / d/D	Q sc (l/s)	v at Q sc (m/s)	Smin, 1 Pa (mm/m)	Class
A	DN315, 8.10	47.9 / 0.49	18.42 (Merrimack)	0.98	1.47	**velocity pass**
B	DN200, 5.00	8.1 / 0.41	2.23 (Peltier)	0.47	3.89	**tractive pass**
C	DN200, 5.00	1.2 / 0.15	0.55 (Peltier)	0.31	7.39	**early cleansing**
D	DN315, 2.70	29.1 / 0.50	2.24 (Peltier)	0.36	3.88	**fails both: regrade**

Table 83 The same pipes at the three values of τ, Smin with K = 2.33 × 10^{-4} and Q in m^3/s

Pipe	τ = 1 Pa (Smin mm/m: class)	τ = 1.5 Pa	τ = 2 Pa
A	1.47: velocity pass	2.42: velocity pass	3.45: velocity pass
B	3.89: tractive pass	6.41: fails both: regrade	9.13: fails both: regrade
C	7.39: early cleansing	12.16: early cleansing	17.33: early cleansing
D	3.88: fails both: regrade	6.39: fails both: regrade	9.11: fails both: regrade

Two lessons. Pipes A and C do not move with τ: A passes on velocity and C sits below the low-flow threshold. Pipes B and D do, and pipe B is the common case, a DN200 street pipe at its minimum gradient; whether the network has a handful of regrades or hundreds depends on τ. That is why the counts are always reported at all three values.

The property count changes the formula for pipe B and pipe D. Pipe B has 72 connected properties and takes Peltier, 2.23 l/s; on its 118 standing properties it would take Merrimack, 3.13 l/s. Pipe D has 82 connected properties and takes Peltier, 2.24 l/s; on its 134 standing properties it would take Merrimack, 3.15 l/s. The standing count would declare more early flow than the pipe will carry, the unsafe error of this chapter.

Pipe D is cheap to put right at 1 Pa. Its tractive slope of 3.88 mm/m rounds up to 4.05 mm/m, the next DN315 step, and steepening from 2.70 costs 0.135 m of extra fall per 100 m. At 2 Pa the tractive slope is 9.11 mm/m and the pipe is laid at 9.18 mm/m, 0.65 m more per 100 m, and on flat ground that fall is taken from the depth budget of the whole run.

In Ibri a pipe needs about 121 average plots upstream before its low-case peak reaches 1.5 l/s, and about 909 before a DN200 at its minimum gradient runs at 0.75 m/s. A branching network has most of its pipes near the heads, below the first figure, so in the opening years the flushing list, not the regrade list, is likely to be the long one; the audit will give the real count.

Where it lives. This chapter's module computes the four pipes (TUTORIALS/T04/ch06_selfclean.py, calc()); the per-plot averages are read from W14/shp/PLOTS_load.shp. The same numbers per real pipe will come from the engine audit of Section 12.5. The gradient steps are W13/docs/W13_DESIGN_LOGIC.md rule 8 (agreed 2026-09-07). The engine still rounds every gradient up to a flat 0.05 % grid (SLOPE_STEP = 0.0005 in W13/tmp3/py/sewnet/criteria.py, applied in stages/hydraulic.py), which is the grid W8 laid; it is to be aligned with rule 8.

12.7 What the guideline says about the early years

What it is for. It is the guideline's only word on low early flows, and the basis for handing the early-cleansing class to the operator.

The rule. Quoted exactly:

> *"During early development phases, the actual flow will usually be below the flow of design inducing risk of clogging due to low velocity. The operator should proceed to more frequent inspections and cleansings during this period."* G203 §4.2.6, p28

It recognises the problem and gives it to the operator. It sets no flow threshold, no inspection frequency and no duration, and its verb is should, a recommendation. The threshold of the early-cleansing class, the washing frequency and the length of the washing period are therefore not guideline values: the first is an outside assumption (Section 12.8), the other two are for NWS as operator to set.

Source. G203 §4.2.6 p28, read from the PDF.

Ibri number. Pipe C of Section 12.6, 0.55 l/s at 0.31 m/s in 2030, is exactly the pipe this clause describes.

Where it lives. The flushing list the audit will write (Section 12.9).

12.8 The 1.5 l/s low-flow figure

What it is for. It is the line between a pipe judged by the tractive test and a pipe put on the flushing list.

The rule. A pipe whose low-case peak is below 1.5 l/s, and which does not reach 0.75 m/s, is early cleansing. The figure is tagged as an outside assumption wherever it is used, in the audit output and in the map legend, and is to be confirmed with NWS.

Source. Outside assumption, tagged. The figure is not in PAM-GUD-203: the whole document, 201 pages, was searched and 1.5 l/s occurs nowhere; the word flush appears only for pumping stations, force mains, the treatment works and odour control. It comes from the simplified-sewerage literature of Mara, where it is the minimum peak flow, the peak of one WC flush. G203 cites Mara, Sleigh and Taylor only for the tractive formula (p27).

In that literature the figure is used the other way round, as a floor: a flow below 1.5 l/s is raised to 1.5 l/s before the slope is computed. Used that way at 1 Pa it gives 4.67 mm/m. That is flatter than the DN200 G203 Table 11 gradient of 5.00 mm/m, so every DN200 head pipe would pass on paper, on the strength of a flush that a street with a few occupied houses in 2030 may not deliver often. This method uses the figure as a threshold instead, before the formula, and puts those pipes on the flushing list.

> **The engine still applies the floor.** smin_tractive in W13/tmp3/py/sewnet/hydra.py raises any flow below TRACTIVE_QMIN = 1.5 l/s to 1.5 l/s. That is correct for the design gradient, where it keeps the formula finite. The four-class audit must test the threshold first, as in Section 12.5; otherwise every early-cleansing pipe is reported as a tractive pass.

Ibri number. About 121 average Ibri plots produce a low-case peak of 1.5 l/s. The threshold therefore decides the length of the flushing list, and that is why it needs NWS confirmation before the list is costed.

Where it lives. criteria.py TRACTIVE_QMIN (the floor); the class threshold goes into the audit of W14/docs/PROMPT_W13_SELFCLEANSING_AUDIT.md, step 5; the search is recorded in _BRAIN/02_DESIGN_CRITERIA.md §1, row dated 2026-09-11.

12.9 The output: the flushing list and the washing cost

What it is for. It turns the classes into work: a list the operator can wash from, and a cost the appraisal can carry.

The rule. The flushing list holds every early-cleansing pipe: identifier, tier, diameter, gradient, length, low-case peak, velocity, and the tractive slope at each τ. Regrade pipes do not go on it; they are steepened before the design is issued. The washing effort is the listed length times the washing frequency, and it enters the option appraisal as an operating cost.

An early-cleansing pipe is not upsized, for three reasons. It is already a DN200, the minimum main sewer (G203 p22 Table 6). A larger pipe makes things worse: pipe C in a DN250 at the same gradient runs at 0.30 m/s against 0.31 m/s, and the tractive slope does not depend on the diameter at all. And the guideline forbids the related move of oversizing to flatten a gradient:

> *"Sewers shall not be oversized to facilitate flatter slopes."* G203 §4.3.1, p29

The only real lever is gradient, and for pipe C it is out of reach: 0.75 m/s at 0.55 l/s needs 69 mm/m, 6.4 m more fall per 100 m than the 5.00 mm/m laid. So the pipe is washed until its flow grows, which is exactly what G203 §4.2.6 asks of the operator.

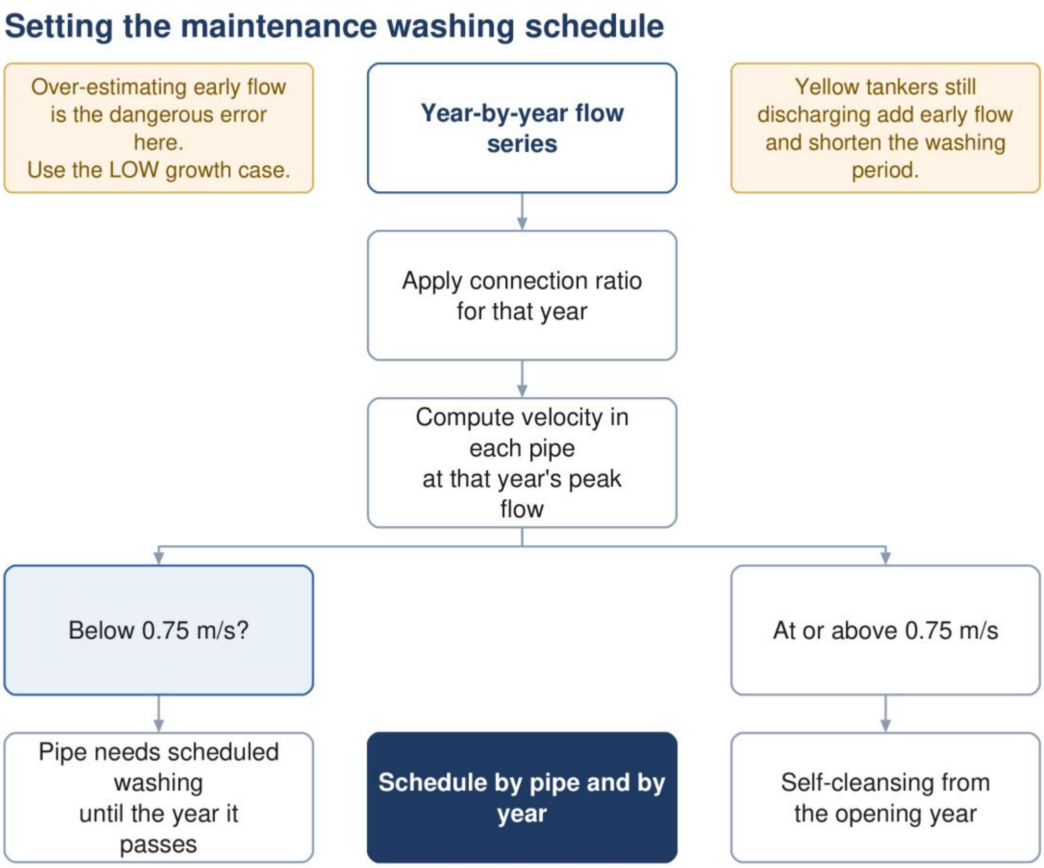

Figure 18 Setting the washing schedule, from the concept methodology. It shows the velocity test only; Sections 12.3 and 12.5 add the tractive test and the four classes. Its tanker note is not applied: no tanker flow is in any design flow until the delivery records are received.

The washing belongs in the appraisal because it changes the ranking. A network laid at minimum gradients is cheaper to build and dearer to run in its first years; one laid steeper costs more excavation and less washing. Unless the washing appears as an operating cost, built up from the listed length, the frequency and a rate per kilometre, the comparison between options is wrong. The appraisal carries it in net present value and life-cycle cost at 5 % over 25 years.

How long a pipe stays on the list is not yet fixed. The class is set on the 2030 low case. The plots also carry the 2055 flow, and a listed pipe that passes on it leaves the list by then at the latest, but the connection ratio after 2028 is not given by the Inception R0, so the year in between cannot be read yet.

Source. G203 §4.2.6 p28 (operator cleansing), p22 Table 6 (DN200 minimum), p29 §4.3.1 (no oversizing); project rule, financial method: NPV and life-cycle cost at 5 % over 25 years, OPEX built bottom-up from duty (_BRAIN/07_PROJECT_STATE.md §2 item 1f, 2026-09-01).

Ibri number. Pipe C: 0.55 l/s in 2030, DN200 at 5.00 mm/m, on the list; pipe D: regraded to 4.05 mm/m at 1 Pa, off it.

Where it lives. The audit field CLEANSE and the class table of W14/docs/PROMPT_W13_SELFCLEANSING_AUDIT.md, step 7; the operating-cost line in the appraisal method (W9/py/make_appraisal_figure.py, report §21.3 and §35).

12.10 What changed from the earlier tutorials

- **Self-cleansing.** The concept methodology tested velocity only, on an undefined low growth case. Both tests are now applied, on the plots' own 2030 flow times 0.61, with four classes.
- **Tractive force.** The gravity-sewer tutorial gave the formula and warned that τ has no value. τ is now carried at three values, and every slope uses one constant, 2.33×10^{-4} with Q in m^3/s.
- **Velocity table.** The part-full velocities are now computed by Colebrook-White with ks 1.5 mm, not typed.

12.11 Check it yourself

1. In design_flows.json, multiply totals 2030 qadf_m3d by connection_ratio_2030: 24,216 × 0.61 should give low_case_2030_m3d, 14,772.
2. Pick one pipe in the engine output. Select the plots upstream in PLOTS_load.shp, sum Q_2030 × 0.61 and the 2030 properties × 0.61, pick Merrimack or Peltier on that connected count, and compare with the audit's low-case flow.
3. Compute Smin by hand at 2 l/s and 1 Pa: $2.33 \times 10^{-4} \times 0.002$^−0.461 = 4.09 mm/m, the value the engine's own test expects.
4. Check that the audit reports its class counts at τ = 1, 1.5 and 2 Pa, and that no output presents one τ as the guideline's.
5. Check that every early-cleansing pipe is a DN200 and that none was upsized.
6. Check that every fails-both pipe is steeper in the next run, and that the layout gate still holds after the regrade.
7. Search the G203 PDF for 1.5 l/s and for flush, and confirm the threshold is tagged as an outside assumption in the output and on the map legend.

13 Lifting stations and force mains

A lifting station takes sewage that gravity can no longer carry, raises it, and pushes it through a pressure pipe, the rising main or force main, to a chamber from which gravity takes over again. Every station is land to acquire, a power supply, a running cost for the life of the scheme and one more way for the network to fail. This chapter sets out when a station goes in, how many there are, and how the pumps, the wet well and the rising main are sized.

The guideline's position. Pumping stations and pressure mains shall be avoided whenever gravity sewer designs are feasible and cost effective (G203 p181, §11.5.3.1(a)). A network with no station is the outcome the guideline asks for, not an exception to defend.

13.1 When a station goes in

What it is for. To decide, chamber by chamber, whether the sewer may carry on by gravity or must be lifted.

The rule. No chamber, and no point of the trench between two chambers, may lie more than 12 m below the ground. Where a gravity sewer would pass that depth, a station goes in before that point: the sewage is lifted and the sewer restarts at normal cover, so the next stretch runs by gravity again. Before a station is accepted, the route is searched again for a shallower way, and only the pumping that survives that search is real. A station is never removed by digging past 12 m, and depth past 12 m is never excused by calling the area a pumping pocket.

The guideline states the depth as a recommendation, not a limit. It recommends a maximum cover of approximately 10 to 12 m, sends deeper cover to the pipe manufacturers, and requires pumping stations where the cost of excavation becomes prohibitive. Its trigger for a station is cost. The project holds 12 m as a hard limit because the network has not been costed yet, and the cost test is the only thing that could justify going deeper. Once costs exist, each excursion past 12 m becomes an economic question of its own, never a general relaxation. The earlier concept methodology taught the guideline's wording, that the trigger is cost and not depth; until costs exist the depth check decides.

Source. *G203 p33 §4.6.3 (recommended maximum cover, manufacturer check, pumping where excavation cost is prohibitive); G203 p181 §11.5.3.1(a). Project rule: 12 m with no exceptions (engineer, 2026-08-19), a hard limit until the cost analysis exists (engineer, 2026-09-07).*

Worked Ibri number. On the 5.51 km² test boundary the design lays 1,415 chambers over 71.6 km of sewer. The deepest chamber is 10.45 m, 1.55 m inside the limit, and the network needs no lifting station. The network NWS built on the same ground has no station either.

Where it lives. W13/py/sewnet/criteria.py, MAX_DEPTH = 12.0. The depth test on every chamber and on the trench between chambers is in W13/py/sewnet/stages/audit.py; a chamber past the limit is gathered into a pocket by _pockets in stages/hydraulic.py. The gate figures are keys n_nodes, net_km, max_depth_m and stations of W13/run/summary.json, written by W13/py/run_test_boundary.py.

13.2 One station per pocket, and cascades

What it is for. To keep the number of stations as low as the ground allows. The count, more than the size, is what costs land, power connections and operators.

The rule. Four steps, in this order.

1. Chambers that fail the 12 m test and touch one another form one non-gravity pocket. A pocket gets one station, at its lowest chamber, never one per failing branch.

2. A pocket serving fewer than 50 plots is not given a station at concept stage. It stays in the concept design with its 12 m failure flagged, and detail design resolves it by re-routing, by regrading within 12 m, or by a station, never by going past 12 m. It is listed, never dropped.
3. Two stations within about 1.5 km of each other are examined together: whether the upstream one can discharge into the gravity sewer that feeds the downstream one, so the two work in cascade on one outlet, or whether a single station placed lower can serve both pockets. The arrangement with fewer stations is preferred, because the count drives land acquisition.
4. Every station site is approved by NWS in advance, at concept or preliminary stage. The pump floor, transformer and generator sit above the maximum flood level, with floors at least 300 mm above the 1:50-year flood, and sites under overhead power lines are avoided.

Source. *Project rule: one station per contiguous non-gravity pocket, cascade within about 1.5 km, pockets under 50 plots to detail design (CLAUDE.md rule 9; doctrine §2 item 3). Siting and approval: G203 p38 §7.2. Land: G203 p43 Table 21.*

Worked Ibri number. On the test boundary the station report is empty: 0 stations, 0 pockets small enough to absorb and 0 pairs close enough to cascade. Each station avoided saves a site of 50 to 100 m² for the smallest type and at least 900 m² for the largest (G203 p43 Table 21), plus its access and its power supply.

Where it lives. W13/py/sewnet/stages/hydraulic.py, _pockets: contiguous failing chambers, site at the lowest chamber, flagged absorb below SLS_MIN_PLOTS = 50 (the code counts properties where the rule says plots). W13/py/sewnet/pipeline.py, _station_report: pairs within SLS_CASCADE_M = 1500 m. Output: summary.json key stations, fields small_enough_to_absorb and close_enough_to_cascade.

13.3 How many pumps, and of what type

What it is for. To fix the flow the station must deliver and the number of pumps it carries.

The rule. The pumps shall be capable of handling the design peak flow, and there shall be enough of them that the design peak can still be delivered with any one pump out of service. A station serving a small area has at least two identical units, duty and standby, either one able to carry the design flow. The station type follows the design flow (table below). At site level the guideline asks for a modular N + 1: N elements to meet the average day, the extra one to meet the peak. Pumps rotate duty after each start and stop so running hours even out.

The design peak is taken at saturation, on the plots' Q_ULT, because that is the flow the civil works must pass. Pumps are mechanical plant with a 15-year service rating, so the pump sets themselves are phased: the first set is chosen for the flows of its own life, while the wet well, the rising main and the site are built for saturation.

Table 84 Pumping station types (G203 p40 Table 17; land from p43 Table 21)

Type	Design flow	Duty pumps, minimum	Standby	Land, minimum
Type 1	up to 100 l/s	1	1	50 to 100 m²
Type 2	over 100 to 300 l/s	2	1	200 to 400 m²
Type 3	over 300 l/s	3	1	900 m² or more

Two tables disagree on Type 3. G203 Table 17 on p40 gives a Type 3 station three duty pumps and one standby. The dry-arrangement table on p42 gives it two duty and one standby. Types 1 and 2 agree in both. The three-pump reading is the conservative one and is used here.

Source. *G203 p39 §7.3 (identical duty and standby units; peak with any one pump out), p40 §7.4 and Table 17 (types, 15-year pump rating, duty rotation), p42 (dry arrangement), p38 §7.1 (20-year life for non-structural mechanical installations); G201 p33 (N + 1 at site level). Project rule: pumps phased, buried civil works at saturation (doctrine §2 item 1).*

Worked Ibri number. Take the flows of Hijar as a station-sized example. No station is proposed there; its flows are simply a realistic size for one. At saturation Hijar produces 461.8 m³/d on average, 5.34 l/s. It serves about 674 properties, well over 100, so the peak is Merrimack's: 2.65 × 0.462^0.879 = 1.344 Ml/d, 15.6 l/s, a peak factor of 2.91, under the 5.0 the guideline recommends as a ceiling for the hourly factor. Each kilometre of sewer upstream adds 720 l/d of infiltration, 0.0083 l/s, which moves the duty only in the third figure. The duty is therefore 15.6 l/s: a Type 1 station with one duty and one standby pump, each delivering 15.6 l/s. One pump covers both the average day and the peak, so N = 1 and N + 1 = 2, the same two pumps.

Where it lives. Not yet in the pipe-laying code, whose station report gives the count, the properties served, the lift and the rising main length (stages/hydraulic.py, fields n_props, lift_m, rising_main_m). Duty and type are set by hand from the pocket's upstream Q_ULT in W14/shp/PLOTS_load.shp. Example flows: W14/analysis/design_flows.json, settlement HIJAR (q_ult, q_2030_low, pop_ult).

13.4 The wet well

What it is for. To give the pump enough volume between its start and stop levels that the motor is not started more often than it can stand, and enough suction head to run without cavitating.

The rule. The minimum live volume, between the start and stop levels of a single constant-speed pump, is

$$V = 0.25\,Q\,T,\ \ T = \frac{3600}{n} \tag{66}$$

Symbol	Meaning	Unit
V	live volume between pump start and stop levels	m³
Q	capacity of a single pump	m³/s
T	shortest allowed on-off cycle	s
n	starts per hour allowed by the motor maker; at least 10 up to 30 kW	1/h

The factor 0.25 comes from the pump cycle. With an inflow q into a well of live volume V and a pump of capacity Q, one cycle lasts V/q to fill and V/(Q − q) to empty. That sum is shortest when the inflow is half the pump capacity:

$$t_c = \frac{V}{q} + \frac{V}{Q-q} \geq \frac{4V}{Q}\ \ at\ \ q = \frac{Q}{2} \tag{67}$$

Requiring that shortest cycle to be no shorter than T gives V = QT/4. The formula is for a single constant-speed pump; the guideline gives nothing for variable-speed or duty-assist operation. Successive start and stop levels are 200 to 300 mm apart, starts are delayed 5 to 10 s, and computational and physical modelling is to be considered for stations of 0.5 m³/s and more. Storage for the emergency overflow sits above the start level of the last duty pump. Larger motors take their starts per hour from the manufacturer and NEMA MG 1. The pumps also need suction head, with a margin of at least 1 m over what the pump requires:

$$\mathrm{NPSH_a} = H_\mathrm{a} - H_\mathrm{vpa} - H_\mathrm{st} - H_\mathrm{f} \tag{68}$$

Symbol	Meaning	Unit
H a	absolute pressure on the liquid surface	m
H vpa	vapour pressure	m
H st	static head	m
H f	friction head	m

The guideline prints this without units or a sign convention. As printed it is a suction lift; with the flooded suction the guideline prefers, the static term adds.

Source. *G203 p48 §7.8 (live volume, starts, level spacing, modelling); p47 §7.5 (storage above the last duty start) and §7.6 (NPSH, 1 m margin); p38 §7.2 (flooded suction preferred). The cycle derivation is standard pump-sump theory, not in the guideline.*

Worked Ibri number. For the example duty of 15.6 l/s and 10 starts an hour, T = 360 s and V = 0.25 × 0.0156 × 360 = 1.40 m³ of live volume. That is the minimum between two levels; the well is deeper by the pump submergence and the overflow storage.

Where it lives. Not in the code; a hand calculation at concept stage, repeated in the pump maker's selection at preliminary design.

13.5 Sizing the rising main

What it is for. To choose the bore of the pressure pipe so that solids keep moving at the lowest delivery and the pipe is neither scoured nor surged at the highest.

The rule. The rising main is sized on the pump duty, not on the flow arriving by gravity. A pump does not run at the arrival rate: it fills the well and empties it at its own rate, so the main carries the duty flow whenever it carries anything. The duty can never be less than the design peak arriving, or the well would never empty.

- **Lowest velocity:** at the design minimum flow, meaning the pump's delivery at maximum static head, at least 0.75 m/s for raw sewage; at least 1.0 m/s where the flow is intermittent, as it is with fixed-speed pumps that start and stop; at least 1.2 m/s in a vertical main.
- **Highest velocity:** not greater than 2.5 m/s in the worst case.
- **Smallest bore:** 75 mm internal for non-clog pumps, 50 mm with grinder pumps.
- **Choice:** where several bores satisfy the velocities, a cost comparison shall find the size with the lowest whole-life cost of main and pumping together. Where early flows are far below later ones, two or more rising mains may be warranted.

$$v = \frac{4\,Q_\mathrm{d}}{\pi\,D^2} \tag{69}$$

Symbol	Meaning	Unit
v	mean velocity in the main while the pump runs	m/s
Q d	pump duty, the delivery at the operating point	m³/s
D	internal diameter of the main	m

The minimum flow of G203 Table 16 (p40) is the other half of the check. The initial minimum flow is the average flow times a factor: 0.25 at 50 l/s, 0.35 at 500, 0.45 at 2,500 and 0.50 at 5,000 l/s. The guideline names it as the flow for sizing the main against deposition. With fixed-speed pumps the main still sees the full duty whenever a pump runs, and what the minimum flow changes is how long sewage waits between cycles. With variable-speed pumps the delivery turns down toward the inflow, and then the minimum flow itself must still meet the velocity floor. That split is our reading of p40 against p50; the guideline does not spell it out. The time sewage spends in the main is

$$t_\mathrm{r} = \frac{\pi\,D^2\,L}{4\,\bar{q}} \tag{70}$$

Symbol	Meaning	Unit
t r	mean retention time in the main	s
L	length of the main	m
q̄	average inflow to the station	m³/s

The guideline asks for a main short enough to hold sewage no longer than half an hour, and admits in the same sentence that this is very rarely achieved. Retention is the link to the next chapter.

3.0 m/s is the gravity sewer's limit, not the rising main's. An earlier project note and the pipe-laying code both give the rising main a window of 0.75 to 3.0 m/s and cite G203 p50. That page says 2.5 m/s. The 3.0 m/s figure belongs to the gravity sewer at its design depth of flow (G203 p27 §4.2.2.2; p29 §4.3.2). The code also takes 0.75 m/s as the floor where start-stop pumping requires 1.0 m/s, and it bores the main as a gravity plastic pipe from

OD160. For a rising main fed by fixed-speed pumps use 1.0 to 2.5 m/s; 0.75 m/s is the floor only where delivery at the design minimum flow is continuous, and a vertical main needs 1.2 m/s.

Source. *G203 p50 §8 and §8.1 (velocities, smallest bore, cost comparison, two mains); p50 §8.2.1 (half-hour retention); p40 §7.4 and Table 16 (minimum flow); p41 Table 17 (station pipework 2.5 m/s at maximum flow, 0.6 m/s at minimum, 0.5 m/s with grinder pumps). Project rule: the main is sized on the pump duty (doctrine §2 item 3).*

Worked Ibri number. For the example duty of 15.6 l/s, four internal bores give the velocities in the table below. Bores of 100 and 125 mm pass the 1.0 to 2.5 m/s window. The retention columns assume a main of 1,500 m, a length taken from the cascade distance and not from any route; the saturation average is 5.34 l/s and the 2030 low-case average 1.98 l/s.

Table 85 Rising main candidates for the example station, duty 15.6 l/s, assumed length 1,500 m

Internal bore, mm	Velocity at duty, m/s	1.0 to 2.5 m/s	Volume, m³	Retention at saturation, min	Retention 2030 low case, min
100	1.98	**pass**	11.8	37	99
125	1.27	**pass**	18.4	57	155
150	0.88	fail	26.5	83	223
200	0.50	fail	47.1	147	396

Every candidate holds sewage well past the half hour, and longest in the opening years. G203 Table 16 starts at 50 l/s, so below it the lowest factor, 0.25, is taken (our reading; the table gives no value there): a minimum inflow of 0.50 l/s, at which the 1.40 m³ well takes 47 minutes to fill once. The smallest passing bore, 100 mm, keeps retention lowest and is carried into the next chapter as the example main; the final choice waits for the whole-life cost comparison.

Where it lives. W13/py/sewnet/stages/hydraulic.py, _size_rising_mains: takes the first DN of DN_SERIES from OD160 whose bore passes the arriving peak at V_MAX = 3.0 m/s and sets the duty to at least 0.75 m/s times the bore area. It must be corrected to the G203 p50 window (at least 1.0 m/s for start-stop pumping, at most 2.5 m/s) and a pressure-pipe bore before any station is designed with it. Rising main length per station: summary.json, stations.rising_main_m.

13.6 Profile, valves and termination

What it is for. To lay the main so that it stays full, can be isolated, emptied and vented, and hands its sewage to the gravity system without releasing gas.

The rule. The main follows these requirements.

- **Profile:** laid at predetermined gradients, recommended minimum 1:500 rising and 1:300 falling, never flatter than 1:750, ideally rising all the way to the discharge; air valves at high points, washouts at low points. It is designed to run full and remain full at all times.
- **Layout:** in the public right of way, in straight lines, with pre-formed and anchored bends; on a highway in the carriageway, at least 1 m from the kerb line.
- **Access and valves:** an access point every 500 m; in-line isolation valves at about 500 m and never more than 800 m apart; washouts sized to empty a section in 3 to 4 hours, 100 mm for a main up to 400 mm, 150 mm to 800 mm, 200 mm to 1,200 mm and 300 mm above; double-orifice air valves at high points.
- **Cover:** 1.3 m to the crown without protection, 0.5 m with protection, 1.5 m at a wadi crossing, the same as for a gravity sewer.
- **Material:** ductile iron and HDPE for the pressure main.
- **Termination:** at a manhole, entering no more than 300 mm above the receiving flow line; a water seal and forced venting through odour control where the discharge is turbulent; a lined or corrosion-resistant receiving manhole; hydrogen sulphide monitoring there to control dosing at the start of the main; the main kept primed, ideally with a vertical bell-mouth.

- **Twin mains:** only with a dedicated hydraulic study that keeps 0.75 m/s in every operating mode, each pipe restrained independently.

Source. *G203 p50 §8 (access every 500 m) and §8.2.1 (profile); p51 §8.2.1 (run full) and §8.2.2 (layout); p52 §8.2.3 (twin mains) and §8.2.4 (cover); p53 §8.3 (material); pp 53 to 54 §8.4 (valves); p55 §8.5 (termination).*

Worked Ibri number. The example main of 1,500 m needs 2 intermediate access points and 2 intermediate isolation valves at the 500 m spacing, and, being under 400 mm, 100 mm washouts. At 1:500 rising over its whole length it would climb 3.0 m; the profile has to fit that inside the ground and the cover rules.

Where it lives. Not in the code at concept stage. The route and profile of a rising main are drawn by hand along the street network and checked against these rules.

13.7 Surge

What it is for. To check that a pump trip or start does not burst the main or pull a vacuum in it.

The rule. The pressure change from a sudden change of velocity is

$$\Delta H = \frac{c}{g}\,\Delta v \tag{71}$$

Symbol	Meaning	Unit
ΔH	change in pressure head	m
c	wave propagation speed in the pipe	m/s
g	acceleration due to gravity, 9.81	m/s²
Δv	change in flow velocity	m/s

The equation is for checking; the analysis itself is done in NWS-approved licensed software. The number of simultaneous pump starts and stops modelled follows N + 1, N being the pumps at the station, so that waves reflected from earlier events can combine into the true worst case. The model is run first with roughness near zero for the largest swing, then with realistic roughness. No vapour cavities and no column separation are allowed; the minimum pressure may not fall below the pipe maker's limit or 0.2 bar below atmospheric, whichever is higher; the maximum may not exceed the test pressure or the weakest component's rating. On a wastewater force main, air admitted at an air valve can be acceptable if the protection is designed for it. The wave speed is described only as depending on the pipe and the liquid; no formula is given, so it comes from the pipe maker's data or standard theory.

Source. *G201 pp 145 to 147 (Appendix III, Joukowsky, N + 1, criteria, force-main concession); scope p25 (final SewerGEMS model including surge).*

Worked Ibri number. A pump trip on the example main stops a flow moving at 1.98 m/s. Each 100 m/s of wave speed gives ΔH = 100 / 9.81 × 1.98 = 20.2 m of head swing. The swing is set almost wholly by the wave speed, and the wave speed by the pipe material and wall chosen, so the material decision and the surge protection are one decision.

Where it lives. Not modelled yet; one SewerGEMS transient run per station at preliminary design.

13.8 Emergency overflow

What it is for. To decide in advance where sewage goes when a station fails, so that it does not go into houses.

The rule. Every station shall have an emergency overflow, approved by the Environmental Authority, and only extreme events should use it. The overflow is never placed at an upstream manhole, where closing the inlet penstock for maintenance could spill by accident; the wet-well overflow carries a dip tube or baffle board. A method to prevent or minimise overflows is submitted for approval, chosen on least cost and least operational complexity from four: power from two independent sources or a generator; storage in trunk sewers and basins that drains back; emergency storage; and emergency pumping. For the network, retention basins are sized for 24 hours of nominal flow and emptied by tanker, and relief sewers for 1.5 times the nominal flow.

> **No station storage duration is given in the station chapter.** G203 section 7.5 names emergency storage and stops. The 24-hour figure comes from the network overflow section and is applied to a station here by analogy; the 48 to 72-hour figure elsewhere belongs to a lagoon at a treatment plant and is not transplanted. The design proposes, NWS approves.

Source. *G203 pp 46 to 47 §7.5; p191 §12.1.1 (retention basin 24 h, relief sewer 1.5 times).*

Worked Ibri number. Twenty-four hours of nominal flow is one day's average flow: for the example station at saturation, 462 m^3, against a live volume of 1.40 m^3. The emergency store is a separate structure, not a deeper wet well. A relief sewer at 1.5 times the nominal flow would carry 8.0 l/s.

Where it lives. Not in the code; a site-layout item for each station.

13.9 Check it yourself

1. After a gate run, open W13/run/summary.json: stations.count is 0 and max_depth_m is under 12.0. If a station appears, check it sits at the lowest chamber of its pocket and that the route was searched again before it was accepted.
2. For any station, sum Q_ULT of the plots upstream in W14/shp/PLOTS_load.shp, peak it by Merrimack when more than 100 properties are served, and confirm the duty is at least that peak and the type follows G203 Table 17 (Table 84 here, the pumping station types).
3. Recompute $V = 0.25\,Q\,T$ with the duty in m^3/s and $T = 360$ s.
4. For the chosen bore, confirm $1.0\ m/s \le 4Q/\pi D^2 \le 2.5\ m/s$ at the pump's delivery with fixed-speed, start-stop pumps; 0.75 m/s is the floor only for continuous delivery at the design minimum flow, and 1.2 m/s in a vertical main (G203 p50 §8.1). A bore justified against 3.0 m/s has been checked against the wrong limit.
5. Compute $\pi D^2 L/4\bar{q}$ on the 2030 low-case average and carry the retention to the odour assessment.
6. List every pair of stations within 1,500 m and confirm each was examined for a cascade or a merge.

14 Septicity and odour

Sewage that stands without oxygen turns septic. Sulphate-reducing bacteria in the slime on the pipe wall produce sulphide. Where the flow is turbulent, it leaves the water as hydrogen sulphide gas, which is toxic, smells at very low concentrations and oxidises on the pipe crown to sulphuric acid that attacks concrete and corrodes metal. Heat speeds every step, so in Ibri this is a design matter, not an operating one.

14.1 What the designer owes

What it is for. To know what the guideline requires on odour and where in the network the risk sits.

The rule. The designer of the sewers, stations and rising mains shall give NWS a dedicated hydrogen sulphide management evaluation, covering prevention (sizing, chemical injection) and correction (monitoring, flushing, coatings, odour treatment). The network is to be designed to minimise the conditions that generate odour: short retention, little turbulence, and no over-sized trunk sewers, which let solids settle. The risk concentrates in three places: rising mains, where sewage sits without air; very flat gravity sewers; and every drop, where turbulence strips the gas out of the water.

Source. *G203 p162 §11.1 (the evaluation); p166 §11.3.2 and p168 §11.4.2 (retention, turbulence, no over-sizing of trunk sewers); p185 (very flat sewers carry the greatest risk).*

Worked Ibri number. Some over-sizing is built into the design basis by necessity: pipes are sized on saturation and cleansed on the opening year. Ibri produces 24,049 m³/d at saturation and 8,160 m³/d in the 2030 low case, 34 % of it. A pipe sized for saturation runs at about 34 % of its design average in its first years, which is when deposits and septic slime form. That is why the early-cleansing list of the self-cleansing check exists.

Where it lives. W14/analysis/design_flows.json, settlements, fields q_ult and q_2030_low. The evaluation is a report deliverable, not yet written.

14.2 The three levers

What it is for. To know what the designer can actually change.

The rule. Three things decide how much sulphide forms and how much escapes.

- **Retention.** long retention lets sulphide build. A force main should ideally hold sewage no more than half an hour. Stations and mains are designed to minimise detention, and twin mains are considered to keep low flows moving.
- **Velocity.** low velocity means deposition and long retention together. Gravity sewers at very low slopes carry the greatest risk.
- **Turbulence.** every drop, jump and free discharge strips dissolved sulphide into the air, which is where the smell and the corrosion happen. Discharges are submerged where possible, and a backdrop (required where the invert drops more than 600 mm, external, up to 2 m high, a vortex shaft beyond) is a turbulence point.

Source. *G203 p50 §8.2.1 (half hour); pp 181 to 182 §11.5.3.1 (b), (c), (d), (f); p185 (flat sewers); p30 §4.4 (backdrops).*

Worked Ibri number. The example rising main of the last chapter, 100 mm over 1,500 m, holds sewage for 99 minutes in the 2030 low case, 3.3 times the half-hour ideal, before any wait in the wet well. On the test boundary the gravity design has 177 drops of more than 600 mm, entering 163 chambers that need an external backdrop. The test-boundary gate run counts 78 drops of more than 2 m, beyond a backdrop, that need a vortex shaft. Each is a place where dissolved sulphide leaves the water.

Where it lives. Drops and backdrop chambers: field N_DROPS of W8/shp/W8_manholes.shp, summed, and counted where above zero (the gate run's W13/run/summary.json key drops gives 177). Vortex drops: summary.json key vortex_sites, the drops of type vortex (W13/py/run_test_boundary.py; W13/py/sewnet/stages/hydraulic.py).

14.3 Screening by the Fayoux score

What it is for. A quick qualitative ranking of where hydrogen sulphide will form.

The rule. Score four factors from the table and add the scores. As printed: 0 to 5 no risk, 5 to 10 low risk, 10 to 30 significant risk, and 20 to 30 certain risk. The last two bands overlap in the guideline and the scale stops at 30. The inputs are the maximum possible temperature, and the residence time and minimum average velocity at night-time flow. Where a value falls between columns, the less favourable column is taken; that is our reading, as is treating a sum above 30 as certain.

Table 86 Fayoux qualitative risk of hydrogen sulphide formation (G203 p185 Table 99)

Factor					
Temperature	5 °C	10 °C	15 °C	20 °C	> 20 °C
score	0	2	4	10	20
Residence time	1 h	3 h	6 h	12 h	24 h
score	0	1	4	6	15
Average velocity over 24 h	1 m/s	0.8 m/s	0.6 m/s	0.4 m/s	0.2 m/s
score, instantaneous 0.6 m/s	–	–	–	10	15
score, instantaneous 1.0 m/s	0	1	2	6	10
score, instantaneous 1.5 m/s	0	0	0	2	6
Redox potential Eh	+200 mV	+100 mV	0 mV	−100 mV	−200 mV
score	0	3	15	30	> 30

Source. *G203 p185 Table 99 (after Fayoux, 1988). Temperature: inland climate, maximum yearly average 35 °C (G201 p78 Table 24); the guidelines give no sewage temperature.*

Worked Ibri number. For the example main, with the 2030 low-case daily average standing in for the night-time flow (night flow is lower, so the result is a lower bound): temperature above 20 °C scores 20; a residence of 1.7 h scores 1; a 24-hour average velocity of 0.25 m/s falls in the 0.2 m/s column and, with 1.98 m/s while pumping, in the 1.5 m/s row, scoring 6. No redox record is held, so that factor is not scored. The sum is at least 27: significant to certain risk. The lesson is the temperature term. Every sewer in Ibri starts at 20, inside the significant band before anything else is counted, so the method cannot tell a good option from a bad one here. Computed retention and velocity do that.

Where it lives. A hand calculation, with inputs from the rising main sizing in 13.5.

14.4 The Pomeroy–Parkhurst equations

What it is for. A quantitative estimate of sulphide build-up in gravity sewers and rising mains.

The rule. For a gravity sewer the rate of change of sulphide is a generation term less a loss term:

$$\frac{dS}{dt} = 3.23\,M'\,[\mathrm{EBOD}]\,r^{-1} - 2.1\,N\,(s\,v)^{0.375}\,[S]\,d_{\mathrm{m}}{}^{-1} \tag{72}$$

Symbol	Meaning	Unit
[S]	sulphide concentration	mg/l
t	retention time	not stated
M′	effective sulphide flux in gravity sewers	not stated
[EBOD]	effective BOD	mg/l
r	hydraulic radius	not stated
N	empirical factor for loss of sulphide	not stated
s	slope of the energy grade	not stated
v	mean velocity	not stated
d m	mean hydraulic depth, area over top width	not stated

The effective BOD corrects the 20 °C BOD to the sewage temperature:

$$(\mathrm{EBOD}) = \mathrm{BOD}_5\,(1.07)^{T-20} \tag{73}$$

Symbol	Meaning	Unit
EBOD	effective BOD	mg/l
BOD5	five-day BOD at 20 °C	mg/l
T	sewage temperature	°C

For rising mains the guideline gives a modified form:

$$S = K_{\mathrm{s}}\,\mathrm{BOD}_5\,t\,f(T) \tag{74}$$

Symbol	Meaning	Unit
K s	sulphide generation rate constant	not stated
t	retention time	not stated
f(T)	temperature function	form not stated

The equations cannot be evaluated from the guideline. No value is given for M′ or N, and most symbols carry no units. The rising main form gives K s neither a value nor a unit and f(T) no form. Both establish that sulphide rises with retention and temperature; neither yields a number without going to the original Pomeroy and Parkhurst work, and every coefficient taken from there is an outside assumption, tagged as such.

Source. *G203 pp 185 to 186 §11.5.3 (Pomeroy–Parkhurst, EBOD correction, rising-main form).*

Worked Ibri number. The temperature factor 1.07^(T − 20) is 1.40 at 25 °C, 1.97 at 30 °C and 2.76 at 35 °C, the inland maximum yearly average. Whatever the coefficients turn out to be, the generation term in Ibri runs at two to three times its rate at 20 °C, and it scales straight into sulphide per minute of retention.

Where it lives. Not in the code; used only if the evaluation of 14.1 needs a number the design values of 14.5 do not give.

14.5 Design concentrations at the end of a rising main

What it is for. To size monitoring and management where no field data exists.

The rule. Where hydrogen sulphide generation is expected (long retention, septic conditions) and no field data is available, management and monitoring at the termination of the pressure main, at the treatment plant or in the network, are designed for an average concentration of 50 to 100 ppm and a peak of no more than 200 ppm. The termination manhole is studied for a monitoring system that controls chemical injection at the start of the main, and is sealed, force-vented through odour control where the discharge is turbulent, and lined or built of corrosion-resistant material.

Source. *G203 p47 §7.7; p55 §8.5.*

Worked Ibri number. The example main scores significant to certain risk and holds sewage for about 99 minutes in its first years, so the 50 to 100 ppm and 200 ppm values apply at its termination. The test-boundary design has 0 m of rising main and no termination to treat. The existing network has 10.0 km of built force main behind 1 station, whose termination is the same design case if it is kept.

Where it lives. W14/report/data_facts.py, WASTEWATER, for the existing force main and station; summary.json stations.rising_main_m for the design.

14.6 Design measures, in the order the guideline prefers them

What it is for. To act on the risk in order of permanence and cost, treatment last.

The rule. Work down the list; go to the next measure only when the one before is exhausted.

1. Avoid pumping wherever gravity is feasible and cost-effective.
2. Where pumping stays, minimise detention: twin or dual mains to keep low flows moving, the main draining back to the well between cycles where it can, no partly full sections.
3. Discharge submerged to absorb energy. Inverted siphons shall not be allowed.
4. Bring gravity sewers into stations with the least free fall and turbulence.
5. Keep air moving: wet-well levels that do not block air in the sewer, and vents no smaller than 150 mm and 6 m above ground, with a UV-resistant cap.
6. Do not over-size trunk sewers.
7. Only then treat. Vapour-phase odour treatment comes first and liquid-phase dosing (metal salts, oxidants and the like) supplements it. Every odour control system is N + 1, and an integrated system removes 99.95 % of hydrogen sulphide and 85 % of odour. The guideline gives no dose rates, so any dosing proposal is the designer's to substantiate.

Source. *G203 pp 181 to 182 §11.5.3.1 (a) to (j) and §11.5.3.2; p183 (dosing chemicals); p32 §4.5 (vents); pp 166, 168 (no over-sizing); p175 (N + 1, 99.95 % and 85 %).*

Worked Ibri number. Measure 1 is met on the test boundary: 0 stations and 0 m of rising main. The measures that remain there are 5 and 6: ventilation at the 163 backdrop chambers, and pipes sized on saturation and nothing larger.

Where it lives. summary.json keys stations and vortex_sites; W8/shp/W8_manholes.shp, field N_DROPS.

14.7 Check it yourself

1. For each rising main compute $\pi D^2 L/4\bar{q}$ on the 2030 low-case average and flag any retention over 30 minutes.
2. Score each main and each very flat gravity reach with G203 Table 99, the Fayoux table (Table 86 here), at night-time flow; record the band and which factor drove it.
3. Recompute $1.07^{(T-20)}$ at the temperature used, and write down where that temperature came from.
4. Check every rising main enters its receiving manhole no more than 300 mm above the flow line and carries a monitoring point.
5. Count the backdrop chambers (N_DROPS above zero in W8/shp/W8_manholes.shp) and the vortex drops (vortex_sites in summary.json) and confirm each is ventilated.

15 Utilities and crossings

A sewer shares its corridor with water mains, power cables, telecom ducts and, in places, fuel lines and aflaj. A clash found on site costs a redesign under traffic; found at concept stage it is a routing choice. This chapter covers what we hold, how the clashes are found, the separations the guideline sets, and the three kinds of crossing: roads, wadis and aflaj.

15.1 What we hold, and what we do not

What it is for. To know which clashes can be checked now and which wait for the owners' records.

The rule. Only a record of an asset's route, and ideally its depth, can be used for a clash check. A customer point locates a customer, not the cable that feeds it.

Source. *Received-data register of the Concept Design Report, Part B.*

Worked Ibri number. The electricity file holds 33,971 meters: points with a tariff and a coordinate, no cable route and no depth. The potable water network is the PAEW dataset, 647.8 km of mains inside the study area, the one utility whose routes are held. Electricity, telecom and gas service records: to be requested from the respective owners.

Where it lives. W14/report/data_facts.py, lists OTHER, WATER and REGISTER; facts_w14.meter_counts().

15.2 Finding the clashes

What it is for. To find every clash before the alignment is fixed.

The rule. Six steps.

1. Obtain service drawings from every owner with assets in the area: electricity distribution, telecom operators, gas and fuel operators, the municipality, the roads authority and the falaj owners.
2. Superimpose the proposed alignment on each set.
3. Choose the trial-pit sites: road intersections, crossings of major services, and the routes of primary sewers and rising mains.
4. Agree the pit programme with NWS, obtain the municipal excavation permit and notify the owners.
5. Where a clash cannot be avoided, price moving our asset against moving theirs, and obtain the owner's agreement for the cheaper one.
6. Show every crossing with another underground utility on the sewer's hydraulic profile; the scope requires the profile to highlight them.

The scope provides fifty trial pits at critical locations the consultant proposes. The formal utility survey belongs to the preliminary stage, but the concept design must already show that each route can be built, so the desk study and the critical pits start now. Detection covers oil, gas, water and effluent pipelines, cables and irrigation aflaj, by trial pits, probing, ground radar and electro-location.

Source. *Scope p7 items 7 and 13 (trial pits, fifty); p11 item 2 (profile highlighting crossings); p18 §4.1.2.2 (utility surveys, re-routing); G203 p198 §13.3 (underground services detection).*

Worked Ibri number. On the 5.51 km² test boundary alone the design crosses dual carriageways at 26 places, each a trial-pit candidate. Over a 531.4 km² study area, fifty pits go to the crossings that decide a route, not to confirm what the records already show.

Where it lives. W13/run/summary.json, road_treatment.dual_crossings_added (W13/py/sewnet/stages/road_treatment.py).

15.3 Clearances

What it is for. To keep enough distance from other services to build, maintain and repair each without disturbing the other.

The rule. The separations below apply; beyond them, the owning authority's own requirement governs.

Table 87 Separation from other services

Situation	Requirement	Source
Gravity sewer to other utilities	**3 m** minimum horizontal clearance	G203 p33 §4.6.3
Another utility in the same trench	on a separate bench on undisturbed soil	G203 p33
Shallow sewer beneath a major road or highway	a proper design check	G203 p33
Force main to water main, horizontal	**3.0 m** minimum	G203 p51 §8.2.2
Force main crossing a water main	**under** it, **450 mm** outside to outside; one full water pipe length centred on the crossing	G203 p51
Force main on a highway	in the carriageway, at least 1 m from the kerb line; manholes at least 0.5 m from it	G203 p51
Sewer in a private area	at least 3 m between buildings and sewer; manholes accessible 24 hours a day	G203 p51
Service corridor, DN200 to DN500	2.0 m reservation	G203 p32 Table 13

Source. *As in the table.*

Worked Ibri number. The 647.8 km of PAEW water main inside the study area is the one set against which these separations can be checked now. Every rising main must keep 3.0 m from it and pass under it with 450 mm clear.

Where it lives. W14/report/data_facts.py, WATER; the PAEW layer in the RECEIVED DATA group of the project QGIS file.

15.4 Roads and dual carriageways

What it is for. To keep sewers out of roads that cannot be dug up, and to cross roads with the least disruption.

The rule. No sewer of any kind runs along a dual carriageway, primary sewer included, because it cannot be dug up for maintenance. A dual carriageway is crossed only by a short pipe at right angles to it. Dual carriageways are read from the dual column of the road centrelines: 1 is a dual carriageway and is excluded; 2 is a two-lane pair of which only one side is used. Existing roads are crossed by trenchless methods wherever possible; where a road is cut, it is reinstated to the Oman Highway Design Manual, and pressure mains get valves on both sides of a major road crossing.

Source. *Project rule (engineer, 2026-08-19; CLAUDE.md rule 7); G201 p85 §9.1 (trenchless crossing, reinstatement to the Oman Highway Design Manual) and §9.2 (valves either side of a major road crossing); G203 p21 and p35 (trenchless, subject to NWS approval).*

Worked Ibri number. On the 5.51 km² test boundary, 101 of the 1,725 road segments are dual carriageway and carry no sewer; the design crosses them at 26 places. The network built in 2006 agrees with the rule in practice: 4 of its 3,267 pipes, 0.1 %, run within 4 m of a dual carriageway.

Where it lives. W13/py/sewnet/stages/road_treatment.py; summary.json keys s1.dual_1, road_treatment.dual_excluded and road_treatment.dual_crossings_added. The as-built count: W7/docs/CALIBRATION_vs_EXISTING.md.

15.5 Wadis and aflaj

What it is for. To keep pipes out of ground that floods and scours, and to cross it safely where it cannot be avoided.

The rule. Pipes and chambers are not laid in wadis or in ground subject to washout: the guideline says this must be avoided and shall be avoided. The design avoids it; where it finds no other way, the presence is a recorded and justified exception carried into the deliverable, never a silent one. Crossing a wadi is a separate matter with its own rules.

- **Cover:** 1.5 m to the crown at a wadi crossing, gravity sewer and force main alike; 2.0 m in soft soil.
- **Pipe:** ductile iron over the crossing and 15 m either side, with mechanical or detachable joints; protection to NWS drawing PAM-STD-404, checked against flotation of the empty pipe.
- **Valves:** isolation and air valves on both sides of an active or major crossing and a washout at the low point on one side; no chamber or marker post in the bed or on the banks, and every valve reachable while the wadi is in flood.
- **Data and approval:** bed profiles, flood frequency (1 in 20, 50 and 100 years), bed material and bed-level change from the agencies, and MoAFWR approval.
- **Aflaj:** buffer zones around each falaj, minimum safe distances for excavation, and protection written into the contractor's environmental plan.

Source. *G203 p30 §4.4.1(a) and p33 (avoid wadis); G203 p52 §8.2.4 (1.5 m); G201 pp 85 to 86 §9.3 (wadi crossings, 2.0 m in soft soil) and §9.4 (aflaj). Project rule: recorded exception (engineer, 2026-09-07). Wadi ground is taken as classes 4 to 6 of the 50-year flood-hazard grid, a project assumption pending a scour check.*

Worked Ibri number. A DN200 sewer, the smallest street pipe, has its invert at least 1.5 + 0.2 = 1.7 m under the wadi bed at a crossing, and 2.2 m in soft soil, before any allowance for scour. On flat ground that extra depth has to be found upstream and recovered downstream, so a wadi crossing can deepen the chambers on either side.

Where it lives. W13/py/sewnet/criteria.py, HAZARD_WADI_CLASSES = (4, 5, 6); the flood grid is RunConfig.hazard in pipeline.py.

15.6 Check it yourself

1. List every utility owner in the area and the date each set of service drawings was requested and received.
2. Overlay the rising mains on the PAEW mains and measure the closest approach; anything under 3.0 m is a clash.
3. Filter the road centrelines on dual = 1 and confirm no design pipe runs along one; every pipe touching one must cross it at right angles.
4. Overlay the network on flood-hazard classes 4 to 6; every chamber inside is a recorded exception with its reason.
5. At each wadi crossing check the crown cover is 1.5 m, or 2.0 m in soft soil, on the long section.

16 Hydraulic modelling

The model is not a check run after the design; it is a contract deliverable in its own right, submitted in native editable format and updated after construction. On this project it has a second job: it is the independent referee of a network that the project's own code lays and sizes.

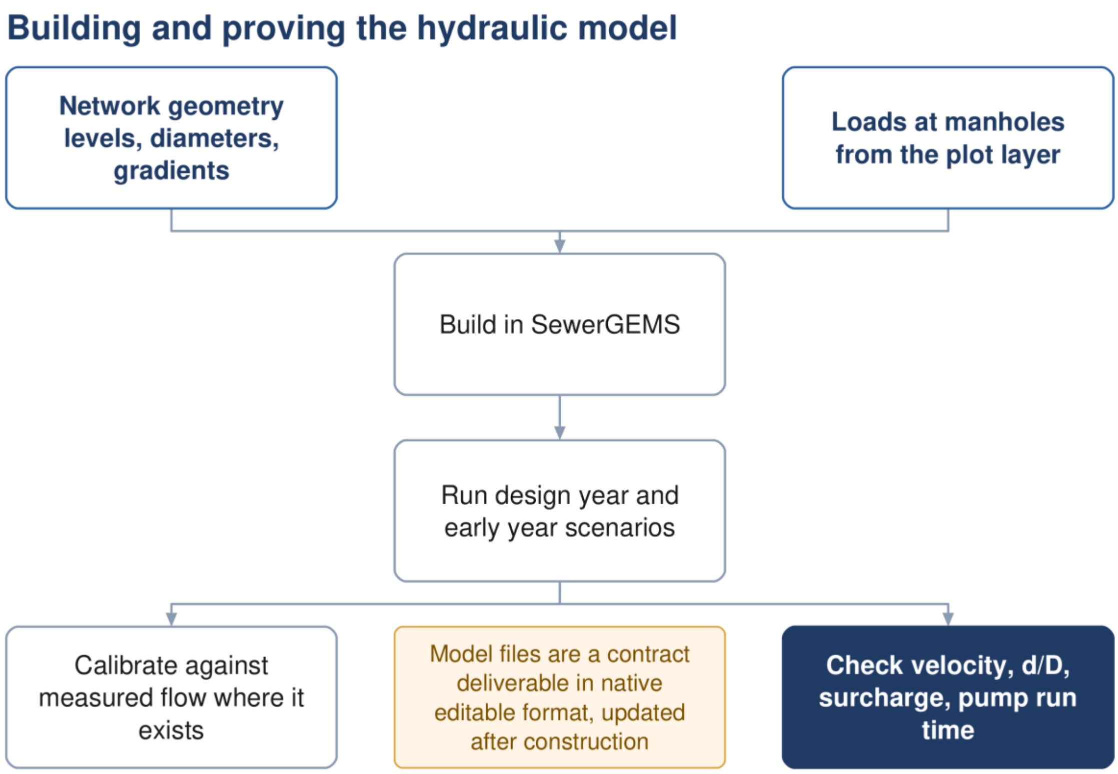

Figure 19 Building and proving the model. The early-year run matters as much as the design-year run.

16.1 The model is a contract deliverable

What it is for. To know what model is owed, in what software, and when.

The rule. The concept hydraulic calculations for the wastewater network are prepared on SewerGEMS, and those for the treated effluent network on WaterGEMS. Models are submitted for the design years start (to be agreed), 2030, 2055 and ultimate, and the final models include surge. Static, extended-period and surge models go to NWS after each design phase, and again after construction. The method follows the WaPUG and CIWEM codes of practice and the US EPA SWMM manuals, and the calculations run in licensed software approved by NWS.

> **The tender names two packages.** The scope names SewerGEMS and WaterGEMS, or other software agreed during the project. The staffing schedule asks the network design engineer for proficiency in Mike Urban. The modelling software is one of the matters put to NWS for confirmation.

Source. *Scope p8 item 19, p14, p17 and p25; tender p123 (staffing); G201 p109 §13.4.2 and p144 (submissions, method); G203 p24 §4.2.1 (licensed software).*

Worked Ibri number. The flows the model years carry, whole study area, are in the table below.

Table 88 Model-year flows, average dry weather, whole study area

Year	Role	People	Qadf, m³/d
2024	today, from the meters	119,893	20,852
2030	opening year	139,535	24,216
2055	model year	244,914	42,266
2070	saturation, the sizing case	349,029	60,099

Where it lives. W14/analysis/design_flows.json, totals; generated by W14/py/make_design_flows.py.

16.2 SewerGEMS as the referee

What it is for. To prove the design with software nobody on the project wrote.

The rule. The network is laid and sized by the project's own code. SewerGEMS then re-solves the same network, and two gates stand between the code and a design called verified.

1. **Solver gate.** The code's partly-full Colebrook-White solver reproduces the minimum gradients of G203 Table 11 within 5 %.
2. **Referee gate.** Every pipe's discharge, velocity and d/D from SewerGEMS agrees with the code within 5 %. A pipe outside that is investigated before the design is accepted. Small differences are expected from junction losses, which the code ignores at concept stage, and from SewerGEMS's gradually-varied flow against the code's normal depth.

Three import traps have cost time before. Levels go in as elevations, never depths. After the build, every conduit's Set Invert to Start Node and Set Invert to Stop Node must be set to False, or the mapped inverts are overwritten and every drop manhole is lost. A loads import replaces each manhole's whole load collection, so the full table is always imported, never a part. The exported base flow is an average: the model must apply the same peak as the code, Merrimack or Peltier per pipe, or the comparison compares two different flows. Check it on three pipes of different size before comparing the rest.

Source. *Project rule: the verification regime (engineer, 2026-08-18; W4/docs/PLAN.md §3b); G203 p29 Table 11; G203 p24 §4.2.1.*

Worked Ibri number. The test-boundary package carries 1,415 manholes, 1,414 conduits and one outfall, with the code's discharge, velocity and d/D beside empty SewerGEMS columns for every conduit.

Where it lives. W13/py/sewnet/export_gems.py writes MANHOLES, CONDUITS and OUTFALL shapefiles, LOADS.xlsx and REFEREE_pipes.csv; the import walk-through is W8/sewergems/IMPORT_PROCEDURE.md. The exported loads are still the old flat figure per property (criteria.PLOT_QADF_LS) under a Fixed pattern; they are replaced by the plots' Q_ULT before the referee run.

16.3 What is loaded, and where

What it is for. To put the right flow on every pipe.

The rule. Each plot carries its average dry-weather flow for 2024, 2030, 2055 and saturation, and nothing else. The model adds the rest, per pipe.

1. Sum the flow of the plots upstream of the pipe, in m³/d, and count the properties upstream.
2. Peak it. Over 100 properties, Merrimack, with both flows in Ml/d; 100 or fewer, Peltier, with the average in l/s. The hourly peak factor should not exceed 5.0; that is a recommendation, not a cap to apply silently. In the self-cleansing case the count that chooses the formula is the connected one, the 2030 properties × 0.61, as recommended in the design-flow handoff, to be confirmed by the engineer.
3. Add infiltration of 720 l/d per km of new sewer, by each pipe's own length, accumulated downstream. It is added unpeaked; the guideline does not state the order, so that is our reading, to be confirmed. Storm water is not considered.

4. Load each plot at the manhole it drains to, from the plot layer, never spread evenly over the network.

$$Q_{\text{pdf}} = 2.65\, Q_{\text{adf}}^{0.879} \tag{75}$$

$$\text{PF} = 1.5 + \frac{1}{\sqrt{Q_{\text{m}}}} \tag{76}$$

$$Q_{\text{inf}} = 720\, L \tag{77}$$

Symbol	Meaning	Unit
Q pdf	peak dry-weather flow, Merrimack	Ml/d
Q adf	average dry-weather flow upstream	Ml/d
PF	peak factor, Peltier	—
Q m	average flow upstream, Peltier	l/s
Q inf	infiltration	l/d
L	length of sewer upstream	km

Source. *G201 p71 §7.4.2 (Merrimack, over 100 properties), p72 (Peltier, 5.0 recommendation, §7.4.3 infiltration), p73 (100 % coverage by the end of the period). Project rule: size on Q_ULT, infiltration per pipe and never per plot (engineer, 2026-09-11).*

Worked Ibri number. Ibri at saturation: Qadf = 24,048.8 m³/d, 24.049 Ml/d, so Qpdf = 2.65 × 24.049^0.879 = 43.37 Ml/d, 502 l/s, a peak factor of 1.80. At the head of a system, a pipe serving 60 future properties at Ibri's occupancy of 6.07 carries 364 people × 171.3 l/d = 0.72 l/s on average; Peltier gives PF = 1.5 + 1/√0.72 = 2.68 and a peak of 1.93 l/s. The 71.64 km of the test-boundary network takes 720 × 71.64 = 51,581 l/d of infiltration, 51.6 m³/d or 0.60 l/s at its outfall.

Where it lives. W14/shp/PLOTS_load.shp, fields QADF, Q_2030, Q_2055, Q_ULT, G_DOM, POP, POP_ULT and OR_S; W14/docs/DESIGN_FLOWS_FOR_NETWORK.md; W13/py/sewnet/stages/loads.py (INFILT_L_D_KM = 720 on upstream length, unpeaked). The code still reads the flat load until it is joined to PLOTS_load.

16.4 Which runs

What it is for. To run each check on the flow that can make it fail.

The rule. Accuracy is directional. Capacity fails on an under-estimate; the self-cleansing check fails on an over-estimate, because a pipe credited with more early flow than it gets is declared self-cleansing while it silts. So capacity is run on the high case and cleansing on the low one, and saturation × 0.61 is never used: it mixes 2070 people with a 2028 connection share.

Table 89 The model runs

Run	Load	What it decides
Capacity	Q_ULT upstream, peaked per pipe, plus infiltration	pipe size; depth of flow and 3.0 m/s maximum (G203 p27)
Self-cleansing	Q_2030 × 0.61 upstream, peaked per pipe	velocity test (0.75 m/s at peak) and tractive-slope test; the four classes
Model years	Q_2030 and Q_2055, peaked	pump staging, surcharge in between
Plant	settlement totals by year, plus network infiltration and 10 %	plant capacity and phasing
Surge	pump trips and starts, N + 1 events	rising main protection

Source. *Project rule: two flow cases (engineer, 2026-09-11); accuracy is directional (doctrine §2 item 1d). G201 p73 (coverage 100 % by the end of the period, plant +10 %); G203 pp 25 to 27 §4.2.2.1; scope p14 (design years).*

Worked Ibri number. Whole area: 60,099 m³/d at saturation; 24,216 m³/d in 2030, × 0.61 = 14,772 m³/d. Saturation × 0.61 would give 36,660 m³/d, 2.48 times the flow expected in 2030, and would pass as self-cleansing pipes that silt. Ibri alone peaks at 194 l/s in the 2030 low case against 502 l/s at saturation. The plant case is 66,109 m³/d with the 10 % margin.

Where it lives. W14/analysis/design_flows.json: rules, totals, low_case_2030_m3d, stp_ultimate_with_margin_m3d.

16.5 Calibration

What it is for. To make the model reproduce what the network actually does before it is trusted to predict.

The rule. Where measured flow exists, the model is calibrated against it to the acceptance below. The existing network enters the model only after it has been verified by survey, because NWS itself declares its records inaccurate.

Table 90 Wastewater model calibration acceptance (G201 p145 Table 32)

Parameter	Typical acceptance
Peak flow	± 10 to 15 %
Volume	± 15 %
Timing	correct peak arrival
Pump runtime	± 10 %

Source. *G201 p145 Table 32; tender p201 (existing layout inaccurate, as-built in scope).*

Worked Ibri number. No measured flow is held. Plant inflow and tanker records: requested. The concept model is therefore uncalibrated and says so. The existing plant's nominal capacity on record is 1,800 m^3/d, against 20,852 m^3/d the study area generates today on the network-accounted basis.

Where it lives. W14/report/data_facts.py, REGISTER and WASTEWATER.

16.6 Check it yourself

1. Run the solver gate: the code's minimum gradients against G203 Table 11, every row within 5 %.
2. Import the package, set both Set Invert flags to False, run steady state and paste discharge, velocity and d/D into REFEREE_pipes.csv; list every pipe outside 5 %.
3. Pick three pipes of different size and check the model's peak equals Merrimack or Peltier on the upstream flow.
4. Sum the manhole loads and compare with the sum of Q_ULT over the plots inside the boundary; they must match before infiltration.
5. Confirm the self-cleansing run uses Q_2030 × 0.61 per plot, not saturation × 0.61.

17 The existing network

Ibri already has sewers. The concept design has to say what happens to them: what is kept, what is upgraded, what is replaced, and how the new network joins the old. That is a priced deliverable in its own right, not a preface to the new design.

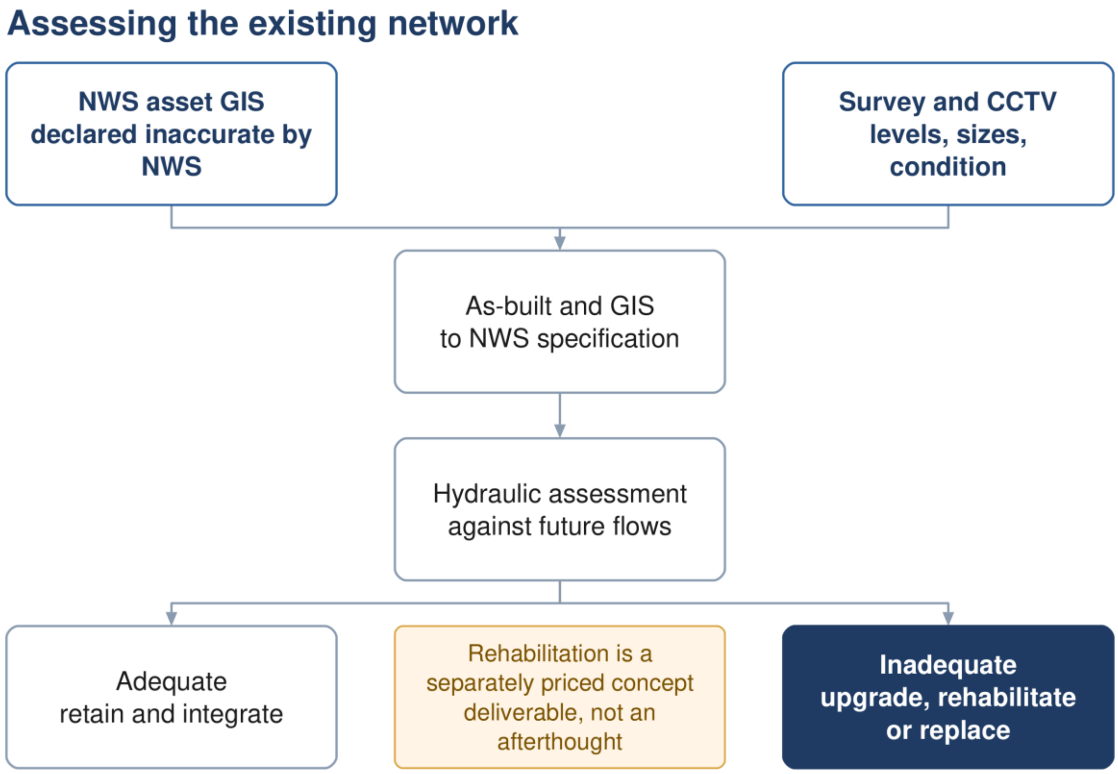

Figure 20 Assessing the existing network. Rehabilitation is a priced deliverable in its own right.

17.1 Two networks in one dataset

What it is for. To know what is actually in the ground before measuring anything.

The rule. The wastewater asset dataset holds a built network and a proposed one, told apart by the operational status field OP_STATUE and confirmed by four more fields that agree with it on every record: installation date, source, project code and remark. Built assets carry 1 January 2006, a drawing or CCTV source and project codes 5A-1 to 5A-5. Proposed ones carry no date, an asset-planning source, the project code SUREKHA and a remark naming them as large-urban-area networks. Filter on status first, then measure, and quote the two separately wherever either is quoted.

Table 91 The wastewater dataset inside the study area, split on status

Asset	Built, features	Built, length or size	Proposed, features	Proposed, length or size
Gravity sewer	3,267	**111.6 km**	129	199.3 km
Force main	1	**10.0 km**	8	23.2 km
Treated effluent main	none	**none**	8	45.7 km
Pumping station	1	**1**	—	—
Treatment plant	1	**1,800 m³/d**	1	29,038 m³/d, shown as design

> **Never quote a combined length.** Adding the two gravity figures gives 310.9 km, a figure that has been quoted for this dataset before and is wrong: 199.3 km of it is a plan, not a pipe. A proposed alignment has no capacity, no condition and no connection. State built and proposed separately, every time.

Source. *Measured in the project GIS against the 531.4 km² boundary, EPSG:32640 (2026-08-30); tender p201.*

Worked Ibri number. Built: 111.6 km of gravity sewer in 3,267 segments, 10.0 km of force main, 1 pumping station, a treatment plant of 1,800 m³/d, and no treated effluent main. Proposed: 199.3 km of gravity sewer, 23.2 km of pumping main and 45.7 km of treated effluent main.

Where it lives. W14/report/data_facts.py, WASTEWATER and CONSTRUCTED; field OP_STATUE in the received asset dataset.

17.2 What the built records carry

What it is for. To know what may be taken from the records and what must be surveyed.

The rule. Nothing in the built records is used as a design value. The owner's own remark on every built record is that the data is not reliable and must be used only for reference. Levels, sizes and condition come from the survey and the CCTV.

Source. *Attribute count on the built gravity layer; remark field of the dataset.*

Worked Ibri number. Of the 3,267 built gravity segments (111.6 km), the nominal diameter field is empty on every one. Upstream invert and ground levels are filled on 2,144 (66 %), an outside diameter on 2,142 (OD160 on 1,997 and OD200 on 145), and a material (PVC on 2,142). The segments join 3,268 manholes by ID. The partly filled fields are leads for the survey, not data: they may be design values rather than measurements, and the remark covers them too.

> **Say it in those words.** The received-data register reads that levels and diameters are not recorded. That is exact for the nominal diameter and loose for the rest: level and outside-diameter fields are filled on about two-thirds of the segments. The conclusion stands, because the owner disowns every value, but the reason is the remark, not an empty table.

Where it lives. W7/shp/EXISTING_SEWERLINE.shp, OP_STATUE = 1: fields N_DIAMETER, OUT_DIAMET, IN_DIAMETE, US_INVERT_, DS_INVERT_, US_GROUND_, MATERIAL, REMARKS, US_MHID and DS_MHID. The file also holds 55 proposed segments (OP_STATUE = 0); filter before any count.

17.3 What the built network is still good for

What it is for. Calibrating the layout, not the hydraulics.

The rule. Unreliable levels still carry a pattern. Read in bulk, the built network shows how the owner's own contractors laid a sewer on this ground, and a new design that departs far from that pattern should be able to say why. The manhole IDs encode the tier: 5A-2-TM-MH185 is a manhole on a trunk main and 5A-2-SM.2-MH391 is on sub main 2. The as-built uses the older engine's three words, trunk main, sub main and lateral, and is read through the same mapping, Table 49 in section 10.4. Matching averages of gradient and depth says the hydraulics are plausible and nothing about whether the layout can be built; the hierarchy does.

Source. *Manhole IDs and US_MHID/DS_MHID traced inside the test boundary; tier words G203 p21; the tier mapping of section 10.4.*

Worked Ibri number. Inside the test boundary, 381 of the 419 street-sewer zones, 91 %, drain into another street sewer, and only about 16 connections reach the trunk main: 6 sub mains and 10 street sewers. A design with thirty things joining the trunk is laid unlike anything a contractor has built here.

Check the basis before quoting an as-built comparison. The earlier gradient calibration (median 4.98 mm/m as built against 5.00 mm/m designed) was measured on an extract of 3,322 segments and 188.6 km, which includes the 55 proposed segments. Repeat it on OP_STATUE = 1 before quoting it.

Where it lives. W8/docs/LEARNING_FROM_ASBUILT.md (hierarchy); W7/docs/CALIBRATION_vs_EXISTING.md (gradient, dual carriageway).

17.4 Method

What it is for. The order of work for the existing network.

The rule. Seven steps.

1. Separate built from proposed on the status field and confirm the split on the date, source and project code before any quantity is taken.
2. Survey cover and invert levels, diameters, materials, gradients, house connections, riders, lifting and pumping stations and rising mains.
3. Inspect condition by CCTV to NF EN 13508-2, early in design rather than at detailed design.
4. Build the as-built and GIS to NWS specification and upload it, subject to NWS acceptance.
5. Model the verified network against the design flows to find where capacity runs out, and when.
6. Classify each asset: adequate and kept; adequate after rehabilitation; or replaced.
7. Design the integration points between old and new networks, and between the old plant and the new one.

Source. *Tender p201 (layout inaccurate; as-built and GIS in scope; integrate, upgrade and rehabilitate); scope p12 and p15 (as-built content, upload); scope p14 (integrity of existing networks against the identified flows); G203 p197 (CCTV to NF EN 13508-2, early).*

Worked Ibri number. The survey starts from 3,268 manholes on 111.6 km of built gravity sewer, plus 10.0 km of force main and its station, for which the register reads: one station located; equipment and duty details being surveyed.

Where it lives. W14/report/data_facts.py, REGISTER; W7/shp/EXISTING_SEWERLINE.shp.

17.5 What the assessment must produce

What it is for. To carry the existing network into the options on equal terms with the new works.

The rule. A hydraulic verdict on every existing asset against the design flows; a rehabilitation and replacement schedule with quantities; the integration design; and the cost of all of it, carried into the options appraisal beside the new works. An option that reuses more of the old network is cheaper to build and may cost more to run, and only the whole-life comparison over 25 years at 5 % settles it. A costed do-nothing option is the baseline.

Source. *Scope p14 and p25 (integration, hydraulic assessment and rehabilitation of the existing systems); G201 p57 and pp 95 to 96 (25 years, 5 %). Project rule: financial method (engineer, 2026-09-01).*

Worked Ibri number. None of the verdicts can be computed yet: with no reliable diameter or level, the capacity of the 111.6 km is unknown until the survey reports. The scale of the question is known. The existing plant's nominal 1,800 m³/d is 8.6 % of the 20,852 m³/d the study area generates today, and the study area reaches 60,099 m³/d at saturation in 2070.

Where it lives. W14/report/data_facts.py, WASTEWATER; facts_w14.totals().

17.6 Check it yourself

1. Filter the dataset on OP_STATUE and confirm every built record carries 1/1/2006, a dwg or cctv source, a 5A project code and the reference-only remark.
2. Measure built and proposed lengths separately in EPSG:32640 inside the boundary and compare them with the table in 17.1.
3. Count the filled diameter and level fields on the built segments.
4. Before quoting any comparison with the as-built, confirm it was measured on OP_STATUE = 1 alone.
5. Trace a few TM and SM manhole IDs through US_MHID and DS_MHID and draw the hierarchy they encode.

18 Treatment plant flows and loads

The network is sized pipe by pipe on the saturation flow and laid once. The plant is sized differently. It takes the flow of the whole catchment year by year, with a margin, and the mass of organic matter and solids that flow carries, and it is built in steps that follow the series. This chapter turns the settlement totals into the plant flows, the loads, the size category and a phasing, and says where each rule comes from.

18.1 The incoming flow

What it is for. The incoming flow is the average flow the plant must treat. The peak flows, the loads per cubic metre and the capacity of every phase are built on it.

The rule. The average sewage flow of the served catchment, plus infiltration, plus a ten per cent design margin on the sum.

$$Q_{\text{in}} = (1 + m) \times (Q_{\text{adf}} + Q_{\text{inf}}) \quad (78)$$

$$Q_{\text{inf}} = 0.72 \times L_{\text{new}} + 0.10 \times Q_{\text{adf,existing}} \quad (79)$$

Symbol	Meaning	Unit
Q in	incoming flow, the plant's design average flow	m³/d
Q adf	average sewage flow of the served catchment in the year, 100 % connected	m³/d
Q inf	infiltration	m³/d
m	design margin, 0.10	—
L new	length of new sewer draining to the plant (720 l/d per km = 0.72 m³/d per km)	km
Q adf,existing	the part of Q adf that drains through the existing network	m³/d

Tanker deliveries join the sum when their records are received (Section 18.4). The margin is applied to the sewage and the infiltration together, because the guideline says it covers fluctuations in population, consumption and infiltration alike. It is applied on top of duty and standby equipment and is never netted against them.

The margin carries different force in the two guidelines. The general guideline says a ten per cent margin should be applied when designing new plants. The wastewater guideline, in G203 Table 29, says incoming flows shall be calculated with it. The stronger wording governs, so the margin is always applied.

Infiltration also has two rates. New sewers take 720 l/d per kilometre. An existing network inland, outside the influence of groundwater, takes ten per cent of its wastewater flow. How much of the catchment drains through the existing network will be known from the as-built survey.

Source. G203-p65 §10.2.2.1 Table 29 (incoming flows, "shall"); G201-p73 §7.4.5 (margin "should be applied when designing new STPs", "over and above any redundancies"); G201-p72 §7.4.3 (720 l/d/km new, 10 % existing inland); G203-p65 (capacity on the per-capita consumption of the Integrated Master Plan, which is the 164 l/d of G201-p60 Table 11).

Worked for Ibri. The 25 settlements generate 60,099 m³/d at saturation in 2070. With the margin, the design average flow is 60,099 × 1.10 = 66,109 m³/d, which is the figure the design-flow file carries (66,108.9 m³/d). Infiltration is not yet in it, because the network for the whole study area has not been laid. Its scale can be judged now. Every 100 km of new sewer adds 72 m³/d, and it takes 835 km of new sewer before infiltration reaches one per cent of the saturation flow.

Where it lives. Annual flow per settlement: W14/py/growth_by_settlement.py writes W14/analysis/W14_growth_by_settlement.xlsx, sheets "Qadf by year m3d" and "Five-year Qadf", read by facts_w14.totals(). Margin: W14/py/make_design_flows.py, constant MARGIN, key stp_ultimate_with_margin_m3d in W14/analysis/design_flows.json. Infiltration: added per pipe by length in the network engine (JSON rule infiltration_applies), summed at the plant.

Tutorial T01 carried an ultimate flow of about 49,700 m³/d from the Inception Report. That figure rested on the connected population and a single occupancy rate, and the counted saturation flow above replaces it.

18.2 The plant flows and what each one sizes

What it is for. A plant is not sized on one flow. Its parts respond to different time scales, and the guideline names the flow for each.

Table 92 The design flows of a treatment plant

Flow	Definition	What it sizes	Page
Average annual flow (AAF)	the average of the daily volumes over a continuous 12 months	biological treatment, with the design inlet load	G203-p65, p66
Maximum daily flow (MDF)	the largest volume in a continuous 24 hours	no rule attached, and no factor given in either guideline	G203-p65
Peak hourly flow (PHF), also written QPDF	the largest volume in one hour	hydraulic pass-through process structures	G203-p65, p66
Design peak instantaneous flow	the instantaneous maximum flow rate	defined only; no method or factor given	G203-p66

The rule. Except where the specification or NWS says otherwise, pass-through structures are sized on the peak hourly flow and biological treatment on the average annual flow and the design inlet load. Both the hydraulic and the load design include the recycled liquors and the received tanker volumes. The guideline gives the peak hourly flow the symbol QPDF, which is the peak flow of the general guideline. The whole catchment holds far more than 100 properties, so the Merrimack formula is the one to use.

$$Q_{\mathrm{pdf}} = 2.65 \times Q_{\mathrm{adf}}{}^{0.879} \qquad P_{\mathrm{f}} = \frac{Q_{\mathrm{pdf}}}{Q_{\mathrm{adf}}} \tag{80}$$

Symbol	Meaning	Unit
Q pdf	peak flow, taken as the plant's peak hourly flow	Ml/d
Q adf	average daily flow of the catchment	Ml/d
P f	peak factor	—

The margin applies to the peak as it does to the average, since both come from the same accumulated flow. Infiltration is added after peaking, without a daily pattern. The guideline does not state that order; it is the reading used here and is to be confirmed with NWS.

The maximum daily flow cannot be computed from the guidelines, because neither gives a factor for it. It is read from a year of daily inflow records at the existing works. Until those arrive, it is not quoted.

Source. G203-p65 Table 29 (the four definitions, "PHF or QPDF"); G203-p66 §10.2.2.1 ("Hydraulic pass-through process structures are to be sized based on peak hourly flow", "Biological treatment systems ... based on the AAF and the design inlet load", recycled liquors and tanker volumes "shall" be included); G201-p71 §7.4.2 (Merrimack, "over 100 properties"); G201-p72 (Peltier alternative, Qm in l/s; hourly factor "recommended" not above 5.0). Order of infiltration and peaking: _BRAIN/02_DESIGN_CRITERIA.md §11.4, GAP-23.

Worked for Ibri. At saturation Q adf = 60.099 Ml/d, so Q pdf = 2.65 × 60.099^0.879 = 97.02 Ml/d and the peak factor is 1.614. With the margin, the design peak hourly flow is 106,724 m³/d. At opening in 2030 the factor is higher, 1.802, because the formula peaks a smaller flow harder. The Peltier alternative gives 1.538 at saturation (Qm = 695.6 l/s), within 5 per cent of Merrimack. At plant scale the choice of formula barely matters, and the 5.0 ceiling never comes into play.

Where it lives. The same formulae run per pipe in the network engine (design_flows.json rules peak_over_100_properties and peak_100_properties_or_fewer). The plant peak is computed in this chapter from facts_w14.totals(); no pipeline script writes it yet.

18.3 The plant flows by year

What it is for. The series says when capacity is needed. The network is laid once for saturation; the plant is built in step with this table.

The rule. Each year's average flow is the sum of the 25 settlements, every metered property counted as connected. The design average and the design peak each carry the margin. The plant, like the pipes, is sized on the full load, because under-estimating the total undersizes the plant. The connection ratio is used only where over-estimating is the risk.

Source. G201-p73 §7.4.4 (coverage "100% by the end of the planning period"); project rule, Concept Design Report R2 §10.1 row "Connection to the sewer" (W14/report/rpt_cd.py) and the directional-accuracy rule in _BRAIN/02_DESIGN_CRITERIA.md §11.1.

Table 93 Plant flows at five-year intervals to saturation, 100 % connected

Year	People	AAF, m³/d	AAF + 10 %, m³/d	Peak factor	PHF + 10 %, m³/d
2024	119,893	20,852	22,937	1.83	42,089
2025	122,766	21,344	23,478	1.83	42,961
2030	139,535	24,216	26,638	1.80	48,003
2035	157,710	27,329	30,062	1.78	53,387
2040	178,781	30,938	34,032	1.75	59,537
2045	197,153	34,085	37,494	1.73	64,829
2050	218,926	37,815	41,596	1.71	71,023
2055	244,914	42,266	46,492	1.68	78,322
2060	275,608	47,523	52,276	1.66	86,824
2065	310,307	53,467	58,813	1.64	96,299
2070	349,029	60,099	66,109	1.61	106,724

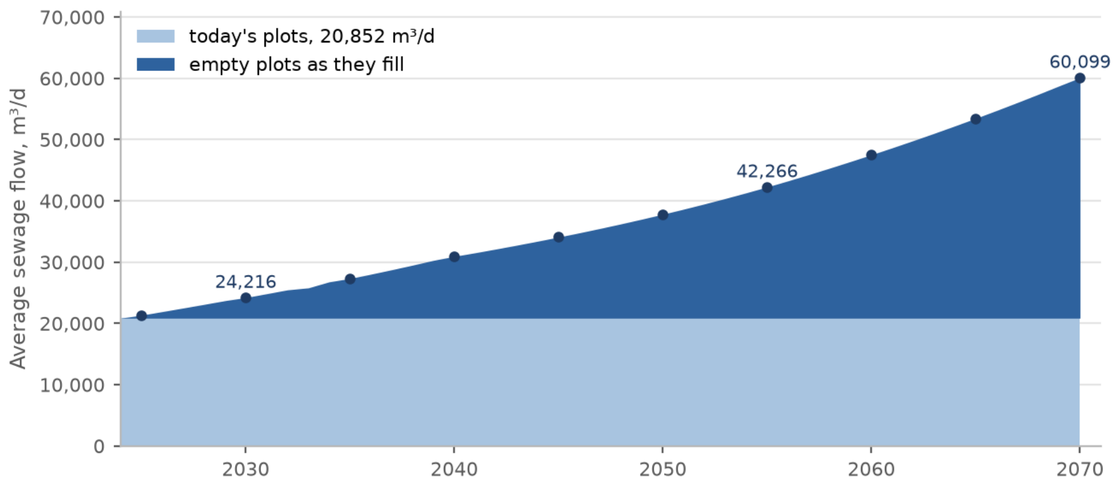

Figure 21 Average sewage flow of the study area by year, from today's plots and from the empty plots as they fill. The plant is phased on this curve.

Worked for Ibri. The design average rises from 26,638 m³/d at opening in 2030 to 66,109 m³/d at saturation in 2070, a factor of 2.48. If connection stood at the Inception Report's 0.61 at opening, the network would deliver 14,772 m³/d in 2030. That is not a sizing figure. It is the load the first phase must treat stably, and the flow for the network's early self-cleansing check.

Where it lives. facts_w14.totals() and facts_w14.five_year_rows("q"); low case: design_flows.json key low_case_2030_m3d (W14/py/make_design_flows.py, CONNECT_2030). Chart: W14/report/charts_w14.py, C09_flow.

18.4 Tankers and septage reception

What it is for. Sewage delivered by road bypasses the network and arrives at the plant. The guideline requires it in both the hydraulic and the load design, and it is much stronger than network sewage.

The rule. Received tanker volumes are included in the flows and loads. Septage and leachate are accepted only through four provisions:

- **Reception.** a dedicated tanker discharge station with screening and oil and grease removal;
- **Equalisation.** a flow equalisation tank for tanker discharge alone, so the main process sees no shock load;
- **Lagoon.** an emergency lagoon of 48 to 72 hours of plant capacity, only in specific cases and under NWS requirement and approval, pumped back to the inlet when capacity allows;
- **Operation.** sampling before acceptance, operating procedures, records for traceability, and contracts with tanker operators that fix acceptable characteristics.

Where the COD to BOD ratio of an influent is above 2.2, the guideline calls it not effectively treatable by biological means, and a separate line for high-strength wastewater is required if the volumes are significant. The guideline does not say what significant means.

Table 94 Tankered sewage against network sewage, design values in mg/l

Parameter	Tanker, average to maximum (G203 Table 31)	Network sewage (G203 Table 30)
BOD_5	350 – 1,050	350 – 400
COD	1,350 – 5,000	700 – 900
Total suspended solids	900 – 4,300	400 – 500
Total Kjeldahl nitrogen as N	115 – 265	60 – 80
Ammonia nitrogen as N	70 – 125	40 – 50
Total phosphorus	16 – 35	10 – 15
Fat, oil and grease	22 – 200	50 – 100

G203 Table 31 is measured at Qurayyat plant from 2021 to 2024 and is a reference, not a design value for Ibri. Nor is the company-wide observation that yellow tankers brought about 17 per cent of the flow reaching NWS plants in 2024. It is a network average, and it says nothing about Ibri.

Source. G203-p66 (tanker volumes "shall" be included); G203-p73 §10.3.1 (the four provisions, "must be accounted for"); G203-p74 (COD/BOD above 2.2, separate line "if volumes are significant"); G203-p67 Table 30, p68 Table 31; G201-p73 §7.4.4 (17 %, "pollution loads of tankers are higher").

Worked for Ibri. At the G203 Table 31 averages (Table 94 here) the COD to BOD ratio of tankered sewage is 1,350 / 350 = 3.86, above the 2.2 limit. Every 1,000 m^3/d of it brings 350 kg/d of BOD, 1,350 kg/d of COD and 900 kg/d of suspended solids. The same volume of Ibri network sewage at saturation carries about 348 kg of BOD, 627 to 767 kg of COD and 465 kg of suspended solids (Section 18.5). An average tanker matches the network in BOD. It is the COD and the solids that it adds out of proportion, and the G203 Table 31 maxima run from 3 to nearly 5 times the averages. No tanker volume is in any flow or load in this tutorial. The delivery records at the existing works (volume, origin and strength) are requested, and the flow and load are added when they arrive. Known sources outside the network include the Madayn industrial city and the power-station camp.

Where it lives. design_flows.json rule tankers ("pending: ... not in any flow"); Concept Design Report R2 §10.1, §16 and §26.1 (W14/report/rpt_cd.py, rpt_ef.py); _BRAIN/05_GAPS.md GAP-19 (camps tankered to the works) and GAP-20 (Ibri's own tanker share).

Tutorial T01 wrote a tanker term into the plant inflow. It stays in the rule, but it carries no number until the delivery records are held.

18.5 Organic loads

What it is for. The biology is sized on the mass of organic matter and solids, not on the volume. A plant with the right flow and the wrong load will not meet the effluent standard.

The rule. The load is the population times a per-person contribution. The concentration follows by dividing by the flow. The second relation is a mass balance that the guideline does not print.

$$L = \frac{P \times l}{1000} \qquad C = \frac{1000 \times L}{Q} \tag{81}$$

Symbol	Meaning	Unit
L	load of the determinand	kg/d
P	resident population served	persons
l	per-person contribution: at least 60 for BOD_5, 80 for suspended solids	g/person/d
C	concentration	mg/l
Q	average annual flow	m^3/d

The per-person figures are floors, not typical values; the guideline's words are "at least". For a new plant, loads are projected from actual waste load data where it exists and compared with the per-person figures. For an existing works that is upgraded or extended, the organic design is based on the measured strength from the Client's laboratory, with an increment for growth. Ibri has an existing works on the same catchment, so its laboratory records govern once they are received. The 60 g and 80 g are then the check, not the basis.

The margin of Section 18.1 is stated for flow. Whether it also applies to load is not stated, so the loads here carry none. The organic peak factors of Section 18.6 are the guideline's allowance on load.

Source. G203-p74 §10.3.1 ("at least 60 g of BOD5 per capita per day and 80 g of suspended solids"; existing works "shall be based upon the actual strength ... from the Client's laboratory"; LIMS); G203-p73 (new works: projections from actual data, compared with per-capita contribution); G203-p67 Table 30 (network sewage ranges, used only after agreement with NWS).

Table 95 Organic and solids loads at 60 g BOD and 80 g suspended solids per person

Year	People	AAF, m^3/d	BOD, kg/d	TSS, kg/d	BOD, mg/l	TSS, mg/l
2024	119,893	20,852	7,194	9,591	345	460
2030	139,535	24,216	8,372	11,163	346	461
2055	244,914	42,266	14,695	19,593	348	464
2070	349,029	60,099	20,942	27,922	348	465

Worked for Ibri. At saturation 349,029 people × 60 g = 20,942 kg/d of BOD, and × 80 g = 27,922 kg/d of suspended solids. Over 60,099 m^3/d that is 348 mg/l of BOD and 465 mg/l of solids, at the lower edge of the G203 Table 30 ranges of 350 to 400 and 400 to 500 mg/l (Table 94 here). It sits low for a reason. The flow includes the non-domestic, governmental and industrial sewage, while the per-person load counts residents only. The concentration is the cross-check on the whole chain. A figure near 200 mg/l would mean the flow per person had been overstated.

Where it lives. Computed in this chapter from facts_w14.totals() ("pop" and "q"); no pipeline script carries plant loads yet. The laboratory (LIMS) records of the existing works are a data request.

18.6 COD, nitrogen and the organic peak factors

What it is for. COD sets the oxygen demand and shows whether the influent is biologically treatable. The peak loads set the aeration capacity.

The rule. Domestic COD lies between 1.8 and 2.2 times BOD. The peak load is the average load times a peak factor that depends on the plant size.

$$L_{\mathrm{COD}} = r \times L_{\mathrm{BOD}} \quad L_{\mathrm{peak}} = f_{\mathrm{L}} \times L \tag{82}$$

Symbol	Meaning	Unit
L COD, L BOD	average COD and BOD loads	kg/d
r	COD to BOD ratio, 1.8 to 2.2 for domestic sewage	—
L peak	peak load for aeration design	kg/d
f L	organic peak factor: BOD and COD 1.5 small, 1.2 medium to large; TKN 2.0 small, 1.5 medium to large	—

The guideline gives no per-person figure for nitrogen. The G203 Table 30 range of 60 to 80 mg/l of total Kjeldahl nitrogen (Table 94 here), applied to the flow, is the only guideline basis. It may be used only after agreement with NWS. Where surge loads are critical, equalisation of flow and load is considered.

Source. G203-p74 §10.3.1 (ratio 1.8 - 2.2; peak factors "For small STPs = 1.5; For medium-large STPs = 1.2", TKN "2" and "1.5"; equalisation); G203-p67 Table 30 (TKN 60 - 80 mg/l).

Worked for Ibri. The plant is large (Section 18.7), so the medium-to-large factors apply. At saturation, COD is 1.8 to 2.2 × 20,942 = 37,695 to 46,072 kg/d. The peak BOD load is 1.2 × 20,942 = 25,130 kg/d, and the peak COD load is 45,234 to 55,286 kg/d. Nitrogen on G203 Table 30 is 3,606 to 4,808 kg/d of TKN, and at the factor of 1.5 the peak is 5,409 to 7,212 kg/d. Nitrogen, not BOD, drives the process choice (Section 19.4).

Where it lives. Computed in this chapter from facts_w14.totals(); to be replaced by the measured strength when the laboratory records are received.

18.7 The size category, and why phasing is the pivotal decision

What it is for. The category decides which rules apply to the plant. On this project it also decides whose scope the plant is.

Table 96 Treatment plant size categories

Category	Capacity
Small	below 500 m³/d
Medium	500 m³/d up to 20,000 m³/d
Large	**20,000 m³/d and above**

The rule. A large plant is held to four rules that a medium plant is not:

- **Modelling.** computational fluid dynamics modelling is carried out unless the Terms of Reference say otherwise;
- **Disinfection.** the chlorine contact tank is replaced by a treated effluent storage tank that does its job (Section 19.5);
- **Buffer.** the residential buffer is set by odour dispersion modelling, between 300 and 1,000 m to the 5 odour-unit contour, instead of a flat 500 m;
- **Peak factors.** the organic peak factors are 1.2 for BOD and COD and 1.5 for nitrogen (Section 18.6).

The Terms of Reference add a fifth rule, and it is the one that matters commercially. The preliminary design and the construction tender of the new plant are in the consultant's scope only if the design capacity of its Phase I is below 20,000 m³/d. Above that, they are not part of the scope unless NWS instructs otherwise.

Source. G203-p65 §10.2.1 (categories); G203-p66 (CFD for large plants); G203-p135 §10.6.7.4 (TSE storage tank for large plants); G201-p43-44 Table 8 (buffer); G203-p74 (peak factors); Terms of Reference, Data/scope.pdf, PDF pages 3, 4, 5, 6, 16, 22, 27 and 30 (Phase I below 20,000 m³/d).

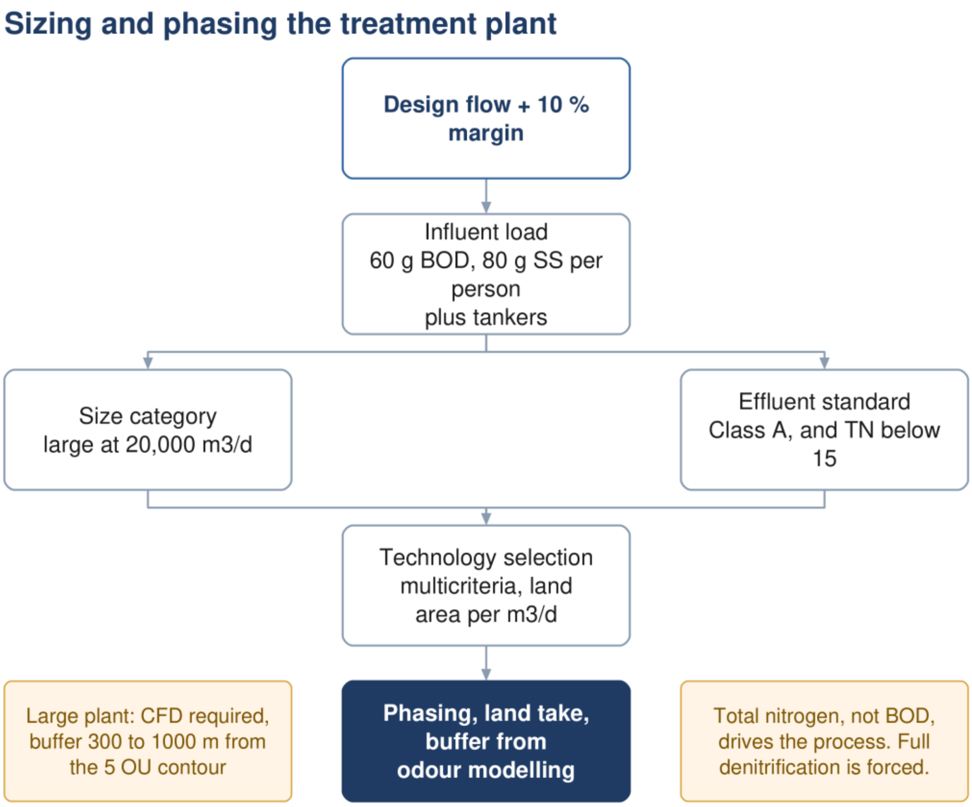

Figure 22 Sizing and phasing the plant: the design flow and the loads set the category and the process, and the category sets the buffer and the modelling.

Worked for Ibri. The study area already generates 20,852 m³/d in 2024, above the large-plant threshold before any margin. At opening in 2030 the full design average is 26,638 m³/d. For Phase I of a new plant to stay below 20,000 m³/d at opening, the existing works would have to go on treating at least 6,638 m³/d. With a fifteen-year horizon to 2045, the design average is 37,494 m³/d and the existing works would have to keep 17,494 m³/d. The capacity of the existing works is not yet known. Now take the Inception connection ratio instead: 14,772 × 1.10 = 16,249 m³/d, below the threshold. The category of Phase I therefore flips on two things nobody has measured yet: the connection path and the capacity left in the existing works. That is why phasing, not process, is the pivotal plant decision. The position adopted sizes on the full load; the choice of Phase I is NWS's.

Where it lives. facts_w14.totals(); _BRAIN/05_GAPS.md GAP-7 (existing works capacity and inlet level, requested from NWS); _BRAIN/07_PROJECT_STATE.md, project summary (phasing as the pivotal decision).

18.8 Phasing, worked

What it is for. Capacity follows demand. A plant built for saturation on opening day would run at a fraction of its load for decades, with the capital spent before the sewage arrives and a biological process that is hard to keep stable.

The rule. The plant is designed for a horizon of at least fifteen years. The planning life of twenty-five years is the ultimate design capacity and the period of the cost comparison. The plant is built as several process lines that add capacity for later horizons or phases. The number of lines in any year follows from the design average flow and the capacity of one line.

$$N(y) = \left\lceil \frac{(1 + m) \times Q_{\mathrm{adf}}(y)}{C_{\mathrm{line}}} \right\rceil \qquad (83)$$

Symbol	Meaning	Unit
N(y)	number of duty process lines needed in year y	—
Q adf(y)	average sewage flow in year y, 100 % connected	m³/d
m	design margin, 0.10	—
C line	treatment capacity of one line	m³/d

A modular plant adds one element on top: N elements meet the average day and N + 1 meet the peak. The margin sits on top of that standby element and is never counted as part of it. The guideline does not say whether a single phase may be shorter than fifteen years when the plant as a whole meets the horizon. It is read here as applying to the plant, with phases as the lines within it. This is to be confirmed with NWS.

Source. G203-p65 §10.2.1 (horizon "at least 15 years"); G201-p57 §7.1 (planning life 25 years, "ultimate design capacity" and NPV period); G201-p53 §6.6.2 ("shall be designed using several process lines"); G201-p33 §4.3 (N + 1, "should"); G201-p73 §7.4.5 (margin over and above redundancy).

Table 97 Illustration: 6 equal lines of 11,018 m³/d, and the year each is first needed

Line	Capacity with this line, m³/d	First needed	AAF that year, m³/d	AAF + 10 % that year, m³/d
1	11,018	already by 2024	20,852	22,937
2	22,036	already by 2024	20,852	22,937
3	33,054	already by 2024	20,852	22,937
4	44,073	2039	30,287	33,315
5	55,091	2053	40,390	44,429
6	66,109	2063	51,004	56,105

Worked for Ibri. Divide the saturation design average of 66,109 m³/d into 6 equal lines of 11,018 m³/d. A line is first needed in the year the design average passes the lines already built. Three lines are needed already in 2024, because 22,937 m³/d exceeds two lines. At opening, 3 lines (33,054 m³/d) run at 73 per cent of their capacity on the full-connection flow, and at 45 per cent on the Inception connection ratio. The second figure is the turndown the process must survive. A Phase I to the fifteen-year horizon of 2045 needs 4 lines. A second phase to the planning life of 2055 needs 5. Saturation needs all 6. The years in the table are when capacity must be in service, and each contract starts one construction period earlier. The standby line of the N + 1 rule is added to every count.

This is an illustration, not the phasing. The number and size of the lines, the part the existing works plays, the connection path and the tanker volume are all decided in the plant options. The method does not change: read the year each line is needed from the annual series, check the opening turndown on the low case, and check Phase I against the 20,000 m³/d threshold of Section 18.7.

Where it lives. Annual series: W14/analysis/W14_growth_by_settlement.xlsx sheet "Qadf by year m3d" (every year to 2100), through facts_w14.totals()["q"]; the line years are computed in this chapter.

18.9 Check it yourself

1. Sum the 2070 column of sheet "Qadf by year m3d" in W14_growth_by_settlement.xlsx. It gives 60,099 m³/d. Multiply by 1.10 and compare with stp_ultimate_with_margin_m3d in design_flows.json.
2. Recompute the plant peak: 2.65 × 60.099^0.879 in Ml/d. Divide by 60.099 and check the factor of 1.614.
3. Recompute the saturation BOD: 349,029 × 60 / 1000 kg/d, then × 1000 / 60,099. The concentration should sit near the G203 Table 30 range (Table 94 here). If it does not, the flow per person is wrong.

4. Find the first year in the annual sheet whose flow × 1.10 exceeds three lines (33,054 m^3/d). It should match the table for line 4.
5. When the laboratory and tanker records arrive, replace the 60 g and 80 g with the measured loads, add the tanker volume and load, and check the combined COD to BOD ratio against 2.2.

19 Treated effluent

The plant's output is also a resource. This chapter sets how much treated effluent is produced and delivered, how the network that carries it is sized, which customers are studied one by one, and the quality and disinfection it must reach.

19.1 How much is produced and how much reaches customers

What it is for. The volume of effluent available sets how many customers the treated effluent network can serve, and how much surplus has to be stored or discharged.

The rule. Ninety-five per cent of the plant inflow becomes treated effluent. A further ten per cent of what is produced is lost in the distribution network.

$$Q_{\mathrm{TSE}} = 0.95 \times Q_{\mathrm{inflow}} \tag{84}$$

$$Q_{\mathrm{delivered}} = 0.90 \times Q_{\mathrm{TSE}} = 0.855 \times Q_{\mathrm{inflow}} \tag{85}$$

Symbol	Meaning	Unit
Q inflow	sewage actually arriving at the plant, without the design margin	m³/d
Q TSE	treated effluent produced	m³/d
Q delivered	treated effluent reaching the customers	m³/d

The missing five per cent is evaporation, sludge dewatering, process reuse and the plant's own landscaping and washing. The two ratios carry different force. The general guideline says the 95 per cent can be assumed, but the 10 per cent network loss shall be assumed. The margin is capacity, not sewage, so it is not in the inflow. Tanker volumes join the inflow when their records are received.

Source. G201-p73 §7.4.6.1 (ratio "Set at 95%"); G201-p75 §7.4.6.3(a) ("it can be assumed that 95 percent ... will be available as TSE"); G201-p76 §7.4.6.3(b) ("a system loss of 10 percent of all produced TSE shall be assumed").

Table 98 Treated effluent produced and delivered

Case	Inflow, m³/d	Produced, m³/d	Delivered, m³/d
2030, at the Inception connection ratio of 0.61	14,772	14,033	12,630
2030, fully connected	24,216	23,006	20,705
2055	42,266	40,153	36,137
2070, saturation	60,099	57,094	51,385

Worked for Ibri. At saturation the plant receives 60,099 m³/d, produces 57,094 m³/d and delivers 51,385 m³/d. At opening on the Inception connection ratio it delivers only 12,630 m³/d. The safe direction reverses here. The treated effluent pipes are sized on the high case. Customer contracts must be written against the low case, because effluent promised and not produced is the failure a customer sees.

Where it lives. Computed in this chapter from facts_w14.totals() and design_flows.json low_case_2030_m3d. The report carries the delivered equation in R2 §17 (W14/report/rpt_cd.py). No treated effluent demand script exists yet.

19.2 Summer-peak sizing and the seasonal factors

What it is for. Irrigation demand swings with the season, while the plant's output is nearly flat. The network is sized on the worst month, and the difference between supply and demand has to go somewhere.

The rule. The treated effluent system is sized for the summer peak demand. Monthly demand is a percentage of the summer demand:

Table 99 Annual variation of treated effluent demand, per cent of summer demand

Months	Per cent of summer demand
December, January, February	50
March, April, May	75
June, July, August	**100**
September, October, November	75

Within the day, demand bunches into the irrigation hours. It is modelled with daily patterns obtained from the authority or the customers; the guideline's own patterns are for reference only.

Source. G201-p76 §7.4.6.4(a) Table 23 and (b) ("The TSE system shall be sized to accommodate the peak demand experienced during the summer months"); G201-p76 (c) and G201-p77 (f) (daily patterns).

Worked for Ibri. Averaged over the year, the G201 Table 23 profile (Table 99 here) is 75 per cent of summer demand. Suppose customers are signed so that their summer demand equals the 51,385 m³/d delivered at saturation. In December to February they take half of it, and 25,692 m³/d is surplus. Over the year the surplus averages 12,846 m³/d. Suppose instead demand is matched to the annual average. Then summer demand is 68,513 m³/d and summer falls short by 17,128 m³/d. Either way the gap is storage, or discharge to a wadi at Class A. That is why the Terms of Reference ask for treated effluent discharge to wadis and emergency lagoons as well as the network.

Where it lives. Computed in this chapter. Terms of Reference, Data/scope.pdf PDF page 22, item 8 (d) and (e).

19.3 Customers and the large consumers

What it is for. The demand the network is sized on is the sum of its customers. The largest of them set the hydraulics on their own.

The rule. Customers are public (landscaping of highways, secondary roads, interchanges and roundabouts, and public parks) or private (community parks, golf courses, private gardens, and nurseries and farms). Demand is built from summer planting rates. Any consumer averaging more than 500 m³/d is studied individually, with an irrigation timing pattern that reflects its actual settings. A consumer with large storage has its demand spread evenly over the hours its storage covers.

Table 100 Summer irrigation demand for concept design

Planting	Summer demand	Planting	Summer demand
Shrubs	20 – 40 l/plant/d	Ground cover	10 l/m²/d
Palm trees	120 – 165 l/plant/d	Seasonal flowers	10 l/m²/d
Other trees	40 – 80 l/plant/d	Grass	12 l/m²/d
Hedges	10 l/m/d	Roads and junctions, mixed	10 l/m²/d

Planting densities for concept design are 15 m spacing for trees and 3 m for shrubs. The road rate applies where the vegetation is unknown and needs the municipality's approval.

Source. G201-p74 Table 20 (consumers); G201-p75 Tables 21 and 22 and item (j); G201-p76 §7.4.6.4 (d) ("larger than 500 m3/day is to be studied individually") and (e) (storage).

Worked for Ibri. A consumer of 500 m³/d is 1.0 per cent of the effluent delivered at saturation. On the G201 Table 21 rates (Table 100 here), 500 m³/d is 4.2 ha of grass in summer, or 3,030 to 4,167 palm trees. A golf course, a large farm or the municipality's road landscaping each crosses that line, so each is modelled on its own. The customer list is itself a deliverable of the Terms of Reference and is not yet held.

Where it lives. Terms of Reference, Data/scope.pdf PDF page 22, item 10 (list of treated effluent customers and their demands). No customer dataset is in W14.

19.4 Quality: Class A and total nitrogen

What it is for. The effluent standard decides the treatment process. At Ibri the nitrogen limit, not the BOD limit, is the one that decides.

The rule. The process must meet Class A or B of Ministerial Decision 145/1993, and any effluent discharged to a wadi must meet at least Class A, with the approval of the Environmental Authority and APSR. Separately, the technology selected must meet a combined total nitrogen below 15 mg/l as N, where total nitrogen is organic nitrogen plus ammonia, ammonium, nitrite and nitrate.

Table 101 Class A limits that shape the process, mg/l unless stated

Parameter	Class A	Parameter	Class A
BOD_5	15	Ammoniacal nitrogen as N	5
COD	150	Organic (Kjeldahl) nitrogen as N	5
Suspended solids	15	Nitrate as NO_3	50
Faecal coliforms	200 per 100 ml	Total phosphorus as P	30
Viable nematode ova	fewer than 1 per l	Total nitrogen (NWS)	**below 15 as N**

Source. G203-p69 §10.2.4.1 and Table 34 (BOD, COD, SS), G203-p70 (nitrogen, phosphorus), G203-p71 (coliforms, ova; "Total Nitrogen limit < 15 mg/L as N"); G203-p73 §10.2.4.3 (wadi discharge at least Class A, EA and APSR approval).

Worked for Ibri. Add the three Class A nitrogen limits on one basis. Nitrate at 50 mg/l as NO_3 is 50 × 14 / 62 = 11.3 mg/l as N. Ammoniacal 5 plus organic 5 plus nitrate 11.3 gives 21.3 mg/l, which Class A would allow. NWS allows 15. The plant therefore needs full nitrification and denitrification, and that narrows the technologies before BOD or solids enter the choice. The tighter limits of Decision 159/2005 apply only where the wadi reaches the sea at a reasonable distance. Ibri's wadis are inland, so applying them would over-specify the plant.

Where it lives. Concept Design Report R2 §13 treatment plant criteria table and its footnote (W14/report/rpt_cd.py); _BRAIN/02_DESIGN_CRITERIA.md §9 (wadi row).

19.5 Chlorine residual: two requirements, one dose

What it is for. The effluent must stay disinfected all the way to the furthest customer. Chlorine decays along the network, so one number at the plant is not enough.

The rule. There are two requirements at two places, and the dose must satisfy both.

$$0.3 \le C_{\text{discharge}} \le 1.0 \quad \text{(up to 3.0 permitted for decay in the network)} \tag{86}$$

$$0.3 < C_{\text{consumer}} < 1.0 \tag{87}$$

Symbol	Meaning	Unit
C discharge	total chlorine residual at the plant's point of discharge	mg/l
C consumer	total chlorine residual at every consumer end	mg/l

The second requirement governs the first. The allowable maximum at the discharge depends on the length and character of the treated effluent network. The discharge set-point therefore comes out of the network model's water age, not out of the plant design. For a large plant, disinfection takes place in a treated effluent storage tank that works as the contact tank. It gives 30 minutes at peak flow, a residual of 0.5 mg/l and a contact value of at least 15 mg·min/l, with a baffling factor of 0.7 or more.

Source. G203-p71 §10.2.4.1 ("at least between 0.3 mg/L and 1.0 mg/L" at the STP point of discharge, "up to 3.0 mg/L are permitted"); G203-p130 §10.6.7 ("greater than 0.3 mg/L and less than 1.0 mg/L at the consumer end"; maximum at discharge "depends on the length and characteristics of the TSE distribution network"); G203-p135 §10.6.7.4 (large plants: TSE storage tank as contact tank; 30 min, 0.5 mg/l, CT 15 mg/L.min, T10/T 0.7).

Worked for Ibri. The three contact criteria are one condition. A residual of 0.5 mg/l held for 30 minutes gives 0.5 × 30 = 15 mg·min/l, exactly the minimum, so neither may be relaxed without raising the other. Take the design peak hourly flow at saturation, 106,724 m^3/d, through the tank without the five per cent process loss, which is the safe side. The effective contact volume is 106,724 × 30 / 1,440 = 2,223 m^3. The storage the tank must also provide for the customers comes on top of that.

Where it lives. Computed in this chapter from facts_w14.totals(). The discharge set-point waits for the treated effluent network model.

19.6 Check it yourself

1. Multiply the saturation flow of 60,099 m^3/d by 0.95 and then by 0.90, and compare with the table in Section 19.1.
2. Average the twelve monthly percentages of G201 Table 23, reproduced here as Table 99, "Annual variation of treated effluent demand". The result is 75, and the ratio of summer demand to the annual average is 1 / 0.75.
3. Convert the three Class A nitrogen limits to mg/l as N and add them. The sum exceeds 15, which is why the plant must denitrify.
4. When the network model is built, read the water age at the furthest customer and confirm that the discharge residual still leaves more than 0.3 mg/l there.

20 Sludge

Sludge handling and disposal are an integral part of every treatment plant, and their plans and specifications are part of its design. This chapter sets the quantity, where the sludge goes, and what the disposal route forces on the dewatering and drying.

Source. G203-p135 §10.6.8 ("must be considered as an integral part of any complete treatment system").

20.1 How much

What it is for. The sludge quantity sizes the thickening, dewatering, drying and storage, and the vehicle traffic to the outlet.

The rule. For planning, sludge is taken as a fixed yield per cubic metre of inflow.

$$M_s = 0.25 \times Q_{\text{inflow}} \tag{88}$$

Symbol	Meaning	Unit
M s	sludge produced, read as dry solids	kg/d
Q inflow	plant inflow	m³/d

The figure is a master-plan baseline and a starting point for the sludge strategy, not a design value. The guideline says the quantity depends heavily on the influent quality and the process, and it does not say whether the 0.25 is dry solids or wet cake. It is read here as dry solids. The process design replaces it with a yield worked from the loads of Section 18.5.

Source. G201-p78 §7.4.7 ("a baseline value of 0.25 kg/m³ is used as a general guideline"). Dry-solids reading: interpretation, to be confirmed with NWS.

Table 102 Sludge at 0.25 kg per cubic metre of inflow, 100 % connected

Year	Inflow, m³/d	Sludge, kg/d dry solids	Sludge, t/year dry solids
2030	24,216	6,054	2,210
2055	42,266	10,566	3,857
2070	60,099	15,025	5,484

Worked for Ibri. At saturation, 0.25 × 60,099 = 15,025 kg/d, about 15.0 t/d of dry solids or 5,484 t a year. Against the 20,942 kg/d of BOD arriving (Section 18.5), that is 0.72 kg of sludge per kg of BOD. The guideline gives no yield per kilogram of BOD to check this against, which is why the process model has to produce the design figure.

Where it lives. Computed in this chapter from facts_w14.totals(); no pipeline script carries sludge yet.

Tutorial T01 called the 0.25 kg/m³ an Inception Report planning rate. It is the general guideline's own master-plan baseline.

20.2 Where it goes

What it is for. The outlet fixes the dryness the sludge must reach, and so the treatment line.

The rule. Sludge is reused unless no form of reuse is possible. Its quality must meet the heavy-metal limits of Table 2 of Ministerial Decision 145/93, which the guideline cites but does not reproduce. Sludge above those limits goes to landfill only with the Ministry's prior approval, and on the landfill operator's terms.

Ibri is the governorate's sludge treatment centre. The national sludge plan, as the wastewater guideline records it, names composting at Ibri plant as the sludge treatment centre for Adh Dhahirah. The plant receives sludge from the governorate as well as producing its own, and that has to be carried into the land area, the odour buffer and the vehicle access. The quantity it will receive from other plants is not yet known.

Table 103 Landfill acceptance of sludge

Criterion	Requirement
Solids content	**at least 80 %, dried before disposal**
pH	neutral, for stability
Temperature	ambient, no heat generation
Daily quantity	not more than 60 t/d, depending on the landfill and its other waste
Standing	the operator may stop receiving sludge at any time

Source. G203-p136 Table 67 ("Adh Dhahirah: STC - composting in Ibri STP"), §10.6.8.1 (MD 145/93 reuse and heavy metals; landfill with prior Ministry approval; landfill criteria); G203-p137 (temperature, 60 tons/day, right to stop).

20.3 Dewatering and drying

What it is for. Dewatering turns liquid sludge into cake. Its capacity and the dryness it reaches decide whether the cake can leave the site.

The rule. Dewatering is sized on the peak weekly average flow at 11 hours a day, 7 days a week. It has a standby unit, or else storage for at least four days of production. Solids recovery is at least 95 per cent. Cake exceeds 22 per cent dry solids, or 18 per cent for surplus activated sludge alone. Thermal dryers take cake of 20 to 25 per cent and produce more than 90 per cent.

$$M_{\mathrm{h}} = \frac{M_{\mathrm{s,week}}}{11} \qquad M_{\mathrm{cake}} = \frac{M_{\mathrm{s}}}{DS} \tag{89}$$

Symbol	Meaning	Unit
M h	dewatering throughput per operating hour	kg DS/h
M s,week	daily sludge in the peak week	kg DS/d
M cake	cake leaving the dewatering, wet mass	kg/d
DS	dry-solids fraction of the cake	—

Source. G203-p151 §10.6.8.6 (peak weekly average flow, 11 h/d, 7 d/week, standby, 4 days storage, 22 %DS and 18 %DS, recovery 95 %); G203-p155 (thermal dryers fed at 20 - 25 %DS, more than 90 %DS out).

Worked for Ibri. The peak-week factor is not in the guideline and will come from the existing works' records, so take the average week. At saturation, dewatering handles 15,025 / 11 = 1,366 kg DS per operating hour. At the 22 per cent minimum dryness the cake is 15,025 / 0.22 = 68.3 t/d, above the landfill's 60 t/d on its own. Dried to the landfill's 80 per cent it is 18.8 t/d. Mechanical dewatering alone cannot reach the landfill criterion. Drying or composting is forced whichever outlet is used, and composting is what the national plan already assigns to Ibri.

Where it lives. Computed in this chapter from facts_w14.totals(). The peak-week factor and the sludge the works already produces are requests to NWS, with the existing works data (_BRAIN/05_GAPS.md GAP-7).

20.4 Check it yourself

1. Multiply the saturation flow of 60,099 m³/d by 0.25 and compare with the table in Section 20.1.
2. Divide the dry solids by 0.22 and by 0.80 and set each against the landfill's 60 t/d.
3. When the existing works' records arrive, compare their measured sludge per cubic metre with 0.25, and replace the baseline in the process model.

21 Cost estimation

Three options per system are compared on cost, and the comparison decides which one is built. An estimate that leaves out a whole category of work does not merely understate the total. It distorts the ranking, because the omission usually falls harder on one option than on another. This chapter sets out the basis of the estimate, how a capital cost is built up, what it must contain and the lines that are usually missed. Chapter 22 turns the estimate into a decision.

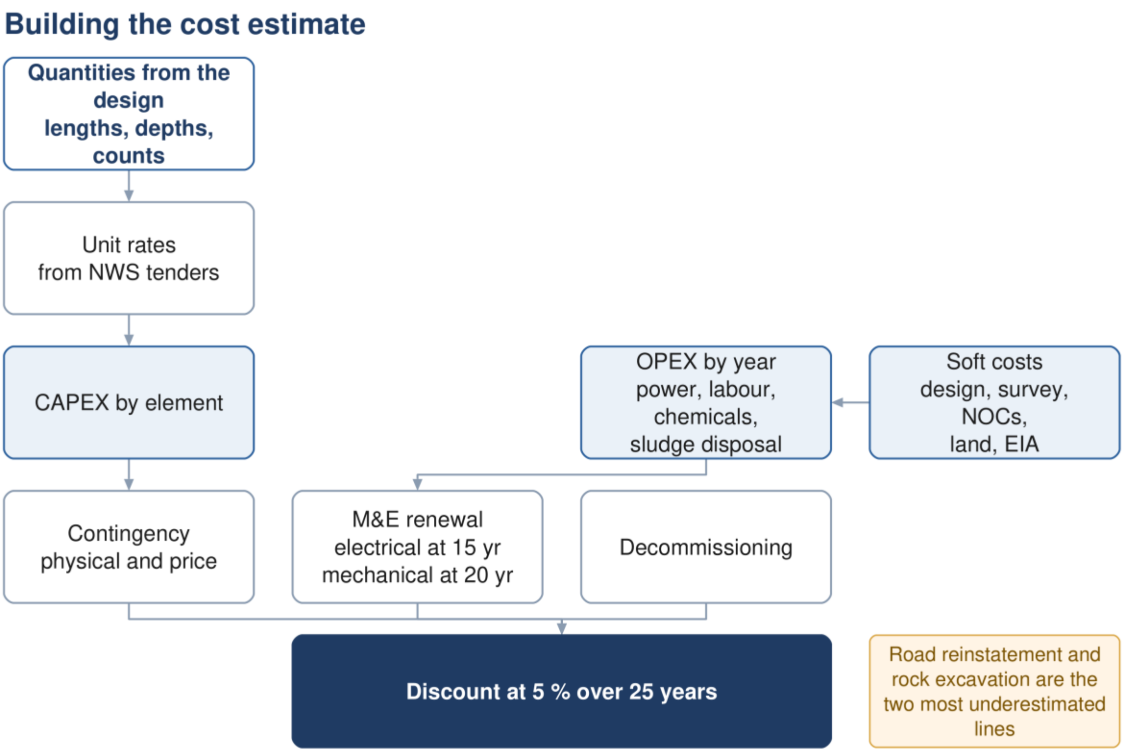

Figure 23 Building the estimate. Renewal and decommissioning sit inside the twenty-five year horizon, not beyond it.

21.1 The basis of the estimate

What it is for. The basis fixes the accuracy, the price date, the rates and the presentation before a single rate is written, so that every option is priced on the same footing.

The rule. The estimate is built to the requirements in the table.

Table 104 Basis of the cost estimate

Item	Requirement	Source
Accuracy	**± 20 %** at this stage, stated in the report; ± 30 % at feasibility and ± 10 % at detailed design	G201 p17–18
Contingencies	the stage accuracy is reached by reducing contingencies and uncertainty, so the contingencies are stated lines, not hidden in the rates	G201 p18
Unit rates	the most recent market rates, preferably from NWS tender or contract documents	G201 p95
Price basis	capital and operating cost from the start of construction, at current base prices	G201 p96
Format and method	NWS advises the required format and methodology on request, before the design phase	G201 p96
Presentation	by asset category at feasibility; **by system element, as a cost curve**, at preliminary design; a bill of quantities at detailed design	G201 Table 2, p21–22
Method of measurement	none is named in the three guidelines; CESMM3 is proposed, to be confirmed when NWS advises the format	project proposal

Source: PAM-GUD-201 §1.6 pp 17–18, Table 2 pp 19–22, §12.2 pp 95–96. The guideline names three stages: feasibility, preliminary and detailed. It places the comparison and selection of options by multi-criteria analysis in the preliminary design (p18), which is the work of this concept stage, so ± 20 % applies. That reading is ours. T03 listed CESMM3 as a requirement; a search of all three guidelines finds no method of measurement, so it is carried here as a proposal.

Worked Ibri number. A cost curve for the plant must span the capacities its phases will take. With the plant margin of 10 % (G201 p73) that is at least 26,638 m³/d, the 24,216 m³/d of 2030 with the margin, up to 66,109 m³/d, the 60,099 m³/d of saturation in 2070 with the margin. Both ends are a Large plant (20,000 m³/d or more, G203 p65), so the curve is priced on large-plant rates throughout.

Where it lives. No cost model exists yet. The basis is stated in report R2 §35.1 (W14/report/rpt_gh.py). The accuracy row is in _BRAIN/02_DESIGN_CRITERIA.md, "Cost/schedule accuracy by phase". The flows and the margin: W14/analysis/design_flows.json, keys totals and stp_margin.

21.2 How a capital cost is built up

What it is for. One structure for every option and every system, so that two estimates differ only where the designs differ.

The rule. Each item is priced as a unit cost times a quantity, the items are summed, and preliminaries and the two contingencies are added as stated fractions of that sum.

$$C_{\text{cap}} = \left(1 + f_{\text{pre}} + f_{\text{phys}} + f_{\text{price}}\right) \times \sum_k c_k q_k \tag{90}$$

Symbol	Meaning	Unit
C cap	capital cost of the option	OMR
c k	unit cost of item k: per metre of sewer by diameter and depth band, per chamber by depth band, per connection, per m³/d of plant capacity, per station by duty	OMR per unit
q k	quantity of item k, taken from the design of that option	m, no., m³/d
f pre	preliminaries, as a fraction of the sum	—
f phys	physical contingency: quantity and scope the design does not yet know; it shrinks as the design develops	—
f price	price contingency: escalation from the base date to the middle of construction	—

Two pricing rules follow from how a sewer network spends its money. A gravity sewer is priced by diameter and by depth band, because in a deep trench the excavation, the support and the dewatering cost more than the pipe; a flat rate per diameter prices a 9 m trench like a 2 m one. And every option is costed as it is designed, with its own pumping stations, its own storage and the energy that follows from them, never as the preferred option with one quantity changed.

> **Contingency is not a single number.** Keep the physical allowance, which covers what the design does not yet know, apart from the price allowance, which covers inflation between the estimate and construction. Merged, they hide which one drives the total, and only the first should shrink as the design develops.

Source: the structure is adopted from the two NWS pre-investment appraisals, which price unit rate times quantity by diameter and material, then add preliminaries and contingency as percentages of the sub-total (one used 10 % and 20 %, so its total is 1.30 times the sub-total). The guidelines give no unit rates, no uplifts and no build-up. Project rule (engineer, 2026-09-01), which also sets depth-band pricing and costing each option as designed.

Worked Ibri number. The connection line shows why every quantity needs its year. The saturation network serves 70,328 domestic properties. 22,559 exist in 2024 and 26,255 by 2030; the other 44,073 are connected as the empty plots fill, after the network opens. They are a cost in the years they are made, and Chapter 22 discounts them there. Put in year zero they would be charged at full value instead of their present value.

> **The only NWS-sourced rates held are water-supply rates.** The rates in the two NWS pre-investment appraisals are for water mains in polyethylene, ductile iron and steel, and for a concrete reservoir. They carry no sewer, chamber or treatment rate, and the 2019 and 2023 sets are identical to the digit although both claim recent tender prices. Use them only as a cross-check, escalated to a stated base date. Priced bills of quantities from completed schemes have been requested; when they arrive they become the primary basis.

Where it lives. No script prices anything yet. The structure and the rate provenance: W9/analysis/W9_PIAD_financial_review.md §2.1, §6 and §6b; report R2 §35.1–35.2 (W14/report/rpt_gh.py). Connections by year: W14/analysis/design_flows.json, key properties, written by W14/py/make_design_flows.py as G_DOM + (POP_year − POP) / OR_S summed over the plots.

21.3 What the estimate must contain

What it is for. The largest single risk in a sewerage estimate is that the pipe is priced carefully and everything around it is not. In urban work the trench, the reinstatement and the traffic usually cost more than the pipe. The tables below are the checklist against that.

The rule. Every group below is priced for every option, or is stated as not applicable to it with the reason.

Table 105 Sewer network items

Group	Items
Earthworks	Excavation by depth band; rock excavation priced separately; dewatering where the water table is met; trench support by depth; disposal of surplus and unsuitable material
Pipework	Pipe supply and lay by diameter and material; bedding and surround; selected backfill and imported fill; jointing; specials and fittings
Chambers	Manholes by depth band and diameter; backdrops where inverts differ by more than 600 mm (G203 p30); drop shafts; benching and channels; covers by loading class
Tertiary sewers and connections	Property connection sewers and connection chambers; rider and lateral sewers (a lateral no longer than 45 m, G203 Table 6); connections to the existing network; stubs and end caps for plots not yet built
Surface	**Road reinstatement by surface type**: asphalt, interlock, kerb, footpath, unpaved; line marking; **traffic management and diversions**
Crossings	Road, wadi and utility crossings; every dual-carriageway crossing, which the layout allows only as a short perpendicular pipe; trenchless drives where open cut is not permitted; carrier and sleeve pipes; thrust and reception pits
Enabling	Trial pits; utility diversions and protection; temporary works; site establishment
Proving	Water and air tightness testing; CCTV survey; compaction testing; as-built survey

Table 106 Lifting station items

Group	Items
Land and site	Land acquisition and Krooki fees; clearance and levelling; access road and turning area; boundary wall and gates; landscaping
Civil	Wet well by depth; dry well or valve chamber; base slab and dewatering during construction; protective lining; superstructure or cover slab; emergency storage or overflow structure
Mechanical	Duty and standby pumps; screens or macerators; station pipework, valves and non-return valves; lifting davit or gantry; sump pump; ventilation
Electrical and control	Incoming supply and connection charges; transformer; motor control centre; standby generator and fuel storage; instrumentation, SCADA and telemetry; small power and lighting
Odour and safety	Odour control unit and ducting; gas detection; fire detection where required; welfare facilities
Commissioning	Testing, commissioning, operator training, spares

Table 107 Force main items

Group	Items
Pipework	Pipe supply and lay by diameter and material; excavation, bedding and backfill as for gravity; thrust and anchor blocks; restrained joints
Appurtenances	Air valves and their chambers; washouts at low points; in-line isolation valves at about 500 m and never more than 800 m apart (G203 p53–54); access every 500 m (G203 p50); marker posts and warning tape
Termination	Receiving manhole; vent and odour control at the discharge
Proving	Pressure testing; pigging or flushing provision; surge protection identified by the transient study

Table 108 Treatment plant items

Group	Items
Land and site	Land acquisition; site preparation and earthworks; **protection against the 25-year and 100-year flood levels, the plant to stay operational in a flood** (G203 p63, Table 27); internal roads; fencing; buffer planting; solar area if provided
Inlet works	Coarse and fine screens; screenings handling; grit removal; fat, oil and grease removal; **tanker discharge facility** with its screening, sampling, flow measurement and equalisation; flow equalisation
Biological treatment	Reactor civils; aeration equipment and blowers; mixers; internal recycle pumps; membranes or media where used
Separation and tertiary	Clarifiers with scrapers; return and surplus sludge pumping; tertiary filtration; disinfection; treated effluent storage acting as contact tank
Sludge line	Thickening; digestion where provided; dewatering with a standby unit; drying or composting; storage; **reception of sludge from other plants**, because Ibri is the governorate's designated sludge treatment centre (G203 Table 67, p136)
Chemical systems	Coagulant, polymer, pH correction, carbon source and disinfectant dosing; bulk storage and bunding
Odour	Covers and enclosures; extraction ducting; treatment units; continuous monitoring
Electrical and control	Incoming supply and substation; motor control centres; standby generation; solar photovoltaic; instrumentation, control and SCADA
Buildings	Administration, laboratory, workshop, stores, welfare; blower and dewatering buildings; guard house
Emergency and disposal	Emergency lagoon of 48 to 72 hours, only where NWS requires and approves it (G203 p73); raw sewage diversion; wadi outfall and its approvals; treated effluent filling station
Commissioning	Wet and dry commissioning; process proving to Class A; operator training; initial spares

Table 109 Treated effluent network items

Group	Items
Distribution	Pipework by diameter; excavation and reinstatement; valves, air valves and washouts; crossings
Storage and boosting	Reservoirs; booster stations; chlorine top-up where the network is long
Customers	Customer connections and meters; filling stations with their identification, metering and control systems

Table 110 Costs outside the works

Group	Items
Professional	Design and supervision fees; topographic and utility survey; geotechnical investigation; hydraulic modelling
Consents	Environmental impact assessment; NOCs and permits; land and Krooki fees; municipality and roads authority approvals
Provisions	**Physical contingency**; **price contingency** to the middle of construction; risk allowance carried from the risk register

Source: the stages of the life-cycle cost, from planning to disposal, G201 §12.4 p96; the item values as cited in the tables. The lists themselves are the designer's checklist, not a guideline table.

Worked Ibri number. At saturation the study area produces 60,099 m³/d and Ibri town alone 24,049 m³/d. Either is a Large plant (G203 p65), and a Large plant takes a residential buffer of 300 to 1,000 m set by odour modelling, the 5 OU contour, rather than a fixed distance (G201 Table 8, p43). The land line of the estimate therefore cannot be priced until the odour contour is drawn, and it will differ between a process with covered and treated air and one without.

Where it lives. Report R2 §35.2 (W14/report/rpt_gh.py) states the scope in client terms. The flows by settlement: W14/analysis/design_flows.json, key settlements.

21.4 Five lines that get missed

What it is for. Each of these is routinely omitted, and each is large enough to move a ranking.

- **Rock excavation:** priced as if the ground were soft. In this terrain it rarely is, and the difference is several times the rate.
- **Road reinstatement:** often the largest single line in urban sewerage, and sensitive to surface type in a way an average rate hides.
- **Renewal inside the horizon:** instrumentation and control lasts 15 years, mechanical plant 20 and electrical plant 15 to 50 (G201 Table 10, p57). A 25-year appraisal therefore contains at least one replacement of the first two. Leaving it out flatters every option heavy in plant, which is the decentralised one.
- **Sludge haulage:** not marginal here, because the plant is the governorate's designated sludge treatment centre and receives sludge from beyond its own catchment (G203 Table 67, p136).
- **Decommissioning:** named in the guideline's own life-cycle scope (G201 p96) and almost always left out.

Where it lives. Report R2 §35.3 (W14/report/rpt_gh.py) carries renewal as a cost in the year it occurs.

21.5 Check it yourself

1. Add the printed line items of any estimate and compare the sum with the printed total. In one of the NWS appraisals the printed sub-total, preliminaries and contingency sum to 32,637,940 OMR against a printed total of 33,066,420 OMR: the total had been refreshed and the lines had not (W9/analysis/W9_PIAD_financial_review.md, finding F7).
2. Check that every rate carries its date and its escalation to the stated base date. A rate unchanged across two studies four years apart has been copied, not priced.
3. Check that gravity sewer rates change with depth band as well as with diameter.
4. Check that no option shares a structure, a station or a reservoir with another option, and that each carries the energy of its own pumping.
5. Check the connection quantity against the property count by year: 26,255 by 2030 and 70,328 at saturation (design_flows.json, key properties).
6. Check that the accuracy is stated as ± 20 % and that the physical and the price contingency are separate lines.

22 Financial appraisal

The estimate says what each option costs to build. The appraisal says what each costs to own for twenty-five years, in money of one date, so that options of different shapes can be compared. A scheme that is cheap to build and expensive to run can lose to one that is the reverse, and the appraisal is where that becomes visible.

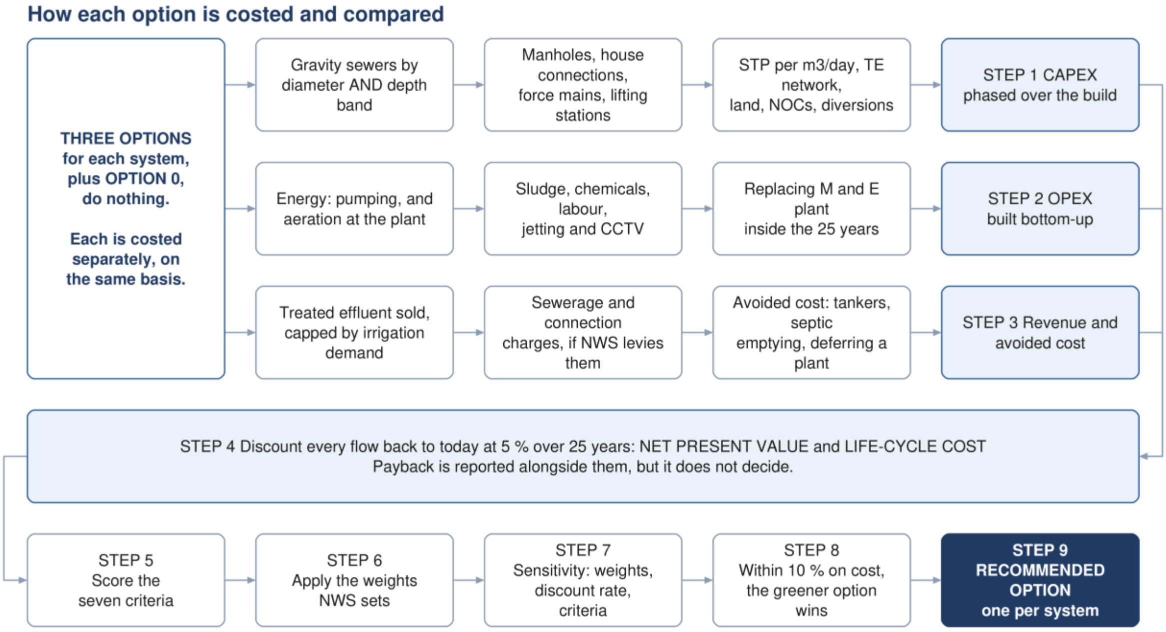

Figure 24 The appraisal end to end. Each option is costed on three streams, the streams are discounted together, and only then are the options scored.

None of the formulae in this chapter comes from NWS. Net present value, life-cycle cost and carbon accounting are required by the general guideline, and none of them is written as an equation anywhere in it. Cite ISO 15686-5 for life-cycle cost and net present value, and ISO 14064 with the GHG Protocol for carbon (G201 p99). What NWS does fix is the discount rate, the period, the cost components and the decision rules.

22.1 The decision rule

What it is for. One measure decides between options; the others inform. Saying which is which before any number exists stops the choice drifting toward whichever measure favours a preferred option.

The rule. Net present value and total life-cycle cost, at 5 % over 25 years, decide. Payback is reported beside them and does not decide. Total life-cycle cost is also the quantity the 10 % tie-break of Chapter 23 is measured on.

Source: G201 §7.1 p57: the 25-year planning life is "the period over which the NPV will be calculated for the purpose of comparing schemes for total lifetime cost analysis". G201 §12.4 p96: NPV is applied to find the lowest total lifetime cost, at 5 % unless NWS instructs otherwise, calculated from the start of construction at current base prices. Payback reported but not deciding: project rule (engineer, 2026-09-01).

Worked Ibri number. Even counted from the opening year 2030, the 25-year window closes in 2055, when the study area generates 42,266 m³/d. The pipes are sized on the 60,099 m³/d of saturation in 2070, so inside the appraisal the network never carries more than 70.3 % of its design flow. Counted from the start of construction, as the guideline says, the window closes earlier still. Capacity built for 2070 is partly idle for the whole appraisal, which is why phasing moves the net present value.

Where it lives. The rule: _BRAIN/07_PROJECT_STATE.md §2 item 1f; report R2 §35.4 (W14/report/rpt_gh.py). The flow by year: W14/analysis/W14_growth_by_settlement.xlsx, sheet "Qadf by year m3d".

22.2 Net present value and life-cycle cost

What it is for. Capital, operating cost, renewal and benefit arise in different years. Discounting brings them to one date so they can be added.

The rule. The net present value of an option is the sum of its net cash flows, each divided by the discount factor of its year:

$$\mathrm{NPV} \ = \ \sum_{t=0}^{n} \frac{C_t}{(1+r)^t} \qquad (91)$$

Symbol	Meaning	Unit
NPV	net present value of the option	OMR
C t	net cash flow in year t: treated effluent income, charges and avoided cost, less capital, operating and renewal cost; residual value enters the final year as a credit	OMR
r	discount rate, 0.05 unless NWS instructs otherwise	—
t	year, counted from the start of construction	year
n	appraisal period, 25 years	year

Total life-cycle cost is the same sum taken over the cost streams alone: capital, operating, renewal and decommissioning, less residual value. It carries no benefit, so it can be compared between options whose benefits are the same, which is the case for options that meet the same functional requirement.

A constant amount paid every year from year 1 to year n has a present value equal to the amount times the annuity factor:

$$a_n \ = \ \frac{1-(1+r)^{-n}}{r} \qquad (92)$$

Symbol	Meaning	Unit
a n	annuity factor: present value of one unit a year for n years	year

Source: neither equation is in PAM-GUD-201, -202 or -203; both are standard discounting as in ISO 15686-5. The rate and the period are G201 p96 and p57.

Worked Ibri number. At r = 0.05, one rial spent in year 25 is worth 0.295 today, and a_25 = 14.09. A flat operating cost of X a year from year 1 is therefore worth 14.09 X, not 25 X; multiplying by 25 overstates it 1.77 times. If the stream starts in year 4, after a construction period, its present value is 11.37 X and 25 X overstates it 2.2 times. And the Ibri stream is not flat: the average flow grows from 24,216 m³/d in 2030 to 42,266 m³/d in 2055, 1.75 times, so every cost that follows flow is built year by year, never as one annual figure.

Where it lives. No appraisal model exists yet. The equation in client terms: report R2 §35.4 (W14/report/rpt_gh.py). The discounting arithmetic above is computed in this module from RATE and YEARS.

22.3 The cash flow to be built

What it is for. The discounting is only as good as the cash flow it is applied to.

The rule. For each option, a year-by-year cash flow across the appraisal period with the streams below.

Table 111 Streams of the cash flow

Stream	Timing	Note
Capital	in the years each phase is built	phased, never lumped at year zero
Connections	in the year each plot connects	follows the property count by year
Operating	every year from commissioning	follows the connected flow and grows with it
Renewal	instrumentation and control at 15 years, mechanical at 20, electrical 15 to 50 (G201 Table 10, p57)	at least one cycle falls inside the period
Residual value	final year, as a credit	the life left beyond year 25 in civil works and pipework (50 years, G201 Table 10); ISO 15686-5 practice, to agree with NWS
Decommissioning	end of life	named in the life-cycle scope (G201 p96)
Benefits	every year from commissioning	treated effluent income, charges, avoided cost (22.5)

Phasing is what makes the comparison honest. A central plant built in two phases and a set of local plants built as districts develop have very different cash-flow shapes, and discounting is sensitive to shape. Lumping the capital of both at year zero erases the difference the appraisal exists to reveal.

Source: G201 §7.1 p57, Table 10 (the lives "are used for financial asset depreciation calculations"); G201 §12.4 p96 for the stages the life-cycle cost covers. Residual value is not named in the guideline.

Worked Ibri number. In the opening year 2030 the fully connected flow would be 24,216 m³/d, but the early-year case carries the connection ratio of 0.61 and gives 14,772 m³/d. The operating cost and the benefits of the first years follow the connected flow, not the full one.

Where it lives. The early-year flow: W14/analysis/design_flows.json, key low_case_2030_m3d, and rules.connection_ratio_2030.

22.4 Operating cost, built from duty

What it is for. Operating cost is where options that look alike on capital come apart, so it has to be built from what each asset does.

The rule. Each component is quantified from its driver. Operating cost is never taken as a percentage of capital: for a wastewater system energy and sludge dominate, and neither follows capital value.

Table 112 Operating cost components and their drivers

Component	Driven by	Basis and source
Power	kWh a year: pump duty flow and head; aeration from oxygen demand	**latest APSR tariff** (G201 p96)
Labour and staffing	posts and shift pattern	per hour, day, month or year (G201 p96)
Vehicles and equipment	fleet	per hour, day, month or year (G201 p96)
Spare parts and consumables	installed plant	annual provision (G201 p96)
Chemicals	dose rate times flow treated	where applicable (G201 p96)
Maintenance and repairs	asset type and condition	annual provision (G201 p96)
Sludge handling and disposal	tonnes of dry solids a year	treatment, haulage, gate fees (ours)
Network cleansing	pipes on the early-cleansing list	jetting and CCTV by year (ours; Chapter 12)
Monitoring	sampling regime	laboratory, odour, effluent compliance (ours)

Power deserves attention twice. More than 80 % of NWS operational emissions come from electricity (G201 p99), so the energy line drives the operating cost and the carbon score together, and the two move in the same direction rather than trading off.

Source: G201 §12.3 p96 lists the six components marked; the guideline says "shall include, but not be limited to". Duty-based, never a percentage of capital: project rule (engineer, 2026-09-01).

Worked Ibri number. At the indicative 0.25 kg of sludge per m³ of inflow (G201 p78) the plant makes 6.1 t a day at the 2030 flow and 10.6 t a day at the 2055 flow. The sludge line of the operating cost grows by the same 1.75 times as the flow inside the window, and the haulage is longer because the plant also receives other plants' sludge.

Where it lives. Report R2 §35.3 (W14/report/rpt_gh.py). The early-cleansing list comes from the self-cleansing classes of the network run (Chapter 12; W14/docs/DESIGN_FLOWS_FOR_NETWORK.md §4).

22.5 Benefits: avoided cost comes first

What it is for. A net present value needs the benefit side, and for a sewerage scheme that side is mostly money not spent.

The rule. Three streams, each measured against a costed Option 0, do nothing:

1. connection and sewerage charges, if NWS applies any, at the rate it sets;
2. treated effluent sales: volume, offtaker and price set by NWS, capped by irrigation demand and not by what the plant produces;
3. avoided cost: tanker haulage of sewage to the plant, septic-tank emptying, and the deferral of a new plant.

For a sewerage scheme the third stream usually dominates, the opposite of a water-supply scheme where volumetric billing carries the case. Name each stream for what it is. A tariff margin or an avoided cost called revenue misleads every reader of the result. Option 0 is costed in full: continued tanker haulage to the existing plant, septic emptying, and the environmental and public-health consequences of no collection. Without it, an avoided cost has no baseline.

Source: the guidelines carry no tariff, no revenue treatment and no avoided-cost rule. Treated effluent produced at 95 % of the inlet flow, G201 p73. The three streams and Option 0: project rule (engineer, 2026-09-01).

Worked Ibri number. At 95 % of the inflow the plant produces 23,006 m³/d of treated effluent at the 2030 flow and 57,094 m³/d at saturation. That is a ceiling on effluent income, not a forecast of it; the sales volume comes from the irrigation demand of Chapter 19. The avoided tanker haulage cannot be put in figures yet: no filling-station or delivery record is held, and no tanker flow is carried in any design flow.

Where it lives. The tanker status: design_flows.json, rules.tankers. The revenue problem for a sewerage scheme: W9/analysis/W9_PIAD_financial_review.md, last section.

22.6 Payback: reported, not deciding

What it is for. NWS reads an investment case by its payback, so it is reported.

The rule. Report the discounted payback: the first year in which the cumulative discounted net cash flow reaches zero. Do not rank options on it. It ignores everything after that year, and on a sewerage scheme a payback built on revenue alone leaves out most of the benefit.

Source: project rule (engineer, 2026-09-01). Payback is not mentioned in G201 §12.

Worked example from an NWS appraisal. The published paybacks of one water-supply appraisal were reproduced to the month by its own formula, yet it billed a peak-day volume that included losses, with the domestic and non-domestic split inverted: annual income was overstated 2.09 times, and the recommended option's 17 months became about 36 once corrected. A payback is exactly as good as the volume it is billed on.

Where it lives. W9/analysis/W9_PIAD_financial_review.md, findings F1 and F2.

22.7 Sensitivity

What it is for. A ranking that flips under a plausible change of input is not a recommendation, and sensitivity is how that is found out.

The rule. The multi-criteria analysis is rerun varying exactly three things:

1. the weighting between the categories under evaluation;
2. the discount rate applied to calculate the net present value;
3. the input design criteria, for example a reduction in demand.

Source: G201 §12.8 pp 105–106.

Worked Ibri number. The third is the one that bites here. The early-year flow rests on a connection ratio (14,772 m^3/d in 2030 against 24,216 fully connected), the saturation year rests on the capacity of the empty plots (229,136 people) rather than on a forecast, and no tanker or private-well water is in any flow. A sensitivity run that does not move the flow is not testing the input most likely to be wrong.

Where it lives. The inputs: design_flows.json (low_case_2030_m3d, rules.connection_ratio_2030); the capacity: W14/analysis/settlements_today.csv, column FUT_CAP_POP.

22.8 Risk

The designer identifies the risks, such as regulatory change, climate impact and technology obsolescence (G201 §12.8 p105), and carries a register from concept onward. For the appraisal each risk needs a likelihood, an impact in money or time, and an owner; the risk-adjusted cost then sits beside the base estimate. The NWS risk model scores likelihood and impact on a five-by-five matrix, separately for investment and delivery risk. Score every option, the recommended one included: in one NWS appraisal the recommended option scored zero because it was never assessed. Risks that fall on one option and not another are the ones that change the ranking.

Where it lives. W9/analysis/W9_PIAD_financial_review.md, finding F6 and §6; report R2 §36 (W14/report/rpt_gh.py).

22.9 How appraisals of this kind go wrong

Two NWS pre-investment appraisals were read line by line against their own workbooks. The faults found are not unusual, and each is worth checking for deliberately.

Table 113 Faults found in two NWS appraisals, and what to do instead

Fault	What to do instead
Revenue billed on a peak-day volume that includes losses	Bill on the average day and on billable volume only; losses are non-revenue by definition
The split between customer categories taken from the wrong place, so the high tariff falls on most of the volume	Take the split from the land-use allocation that sized the network, and check it against the demand table
Alternatives that differ from the preferred option in one quantity, sharing its structures and omitting their own pumping	Cost each option as designed, with its own plant, storage and energy
Operating cost as a percentage of capital, with energy and labour at zero	Build operating cost from duty; for wastewater, energy and sludge dominate
Annual operating cost multiplied by the period and called a lifetime cost	Discount it; at 5 % the plain multiplication overstates a 25-year stream by a factor between about 1.8 and 2.2, depending on when it starts
A tariff margin or an avoided cost described as revenue	Name each stream, and state the baseline it is measured against
Unit rates carried forward unchanged from an earlier study	Escalate to a stated base date, and say which
The recommended option left out of the risk assessment	Score every option, the recommended one included

None of these changed the ranking in the appraisals reviewed: rebuilt on the guideline basis, the recommended option still led on life-cycle cost by 27.8 % against its nearest rival. They would change it in a close case, and they change every absolute figure that a budget, a tariff submission or a board paper depends on.

Where it lives. W9/analysis/W9_PIAD_financial_review.md §3 and §4; _BRAIN/07_PROJECT_STATE.md §2 item 1g.

22.10 What the appraisal must produce

- capital cost by option and by phase
- operating cost by option and by year
- net present value and total life-cycle cost of each option at 5 % over 25 years
- discounted payback, reported as an indicator
- carbon in tonnes of CO_2 equivalent a year and per cubic metre, against the NWS benchmark (G201 p99)
- in-country value, assessed at this stage in outline
- the sensitivity results on all three variables
- a risk register with costed impacts, every option scored
- the input to the NWS pre-investment appraisal document

22.11 Check it yourself

1. Recompute a_25 at 5 %: $(1 - 1.05^{-25}) / 0.05 = 14.09$. Any lifetime operating cost equal to 25 times the annual figure has not been discounted.
2. Check that the cash flow starts at the start of construction, as G201 p96 requires, and that capital is spread over the construction years.
3. Check that at least one replacement of mechanical plant and of instrumentation and control falls inside the period.
4. Check that every benefit is named as income, charge or avoided cost, and that each avoided cost has its Option 0 line.

5. Check that the sensitivity moves the flow, not only the weights and the rate.
6. Check the operating cost of the first years against the early-year flow, 14,772 m³/d in 2030, not against the fully connected 24,216 m³/d.

23 Options and the recommendation

The guideline requires the designer to present options and choose between them systematically. This chapter covers how many options, of what kind, the criteria they are scored on, who sets the weights, and the rule that decides between two options that cost nearly the same.

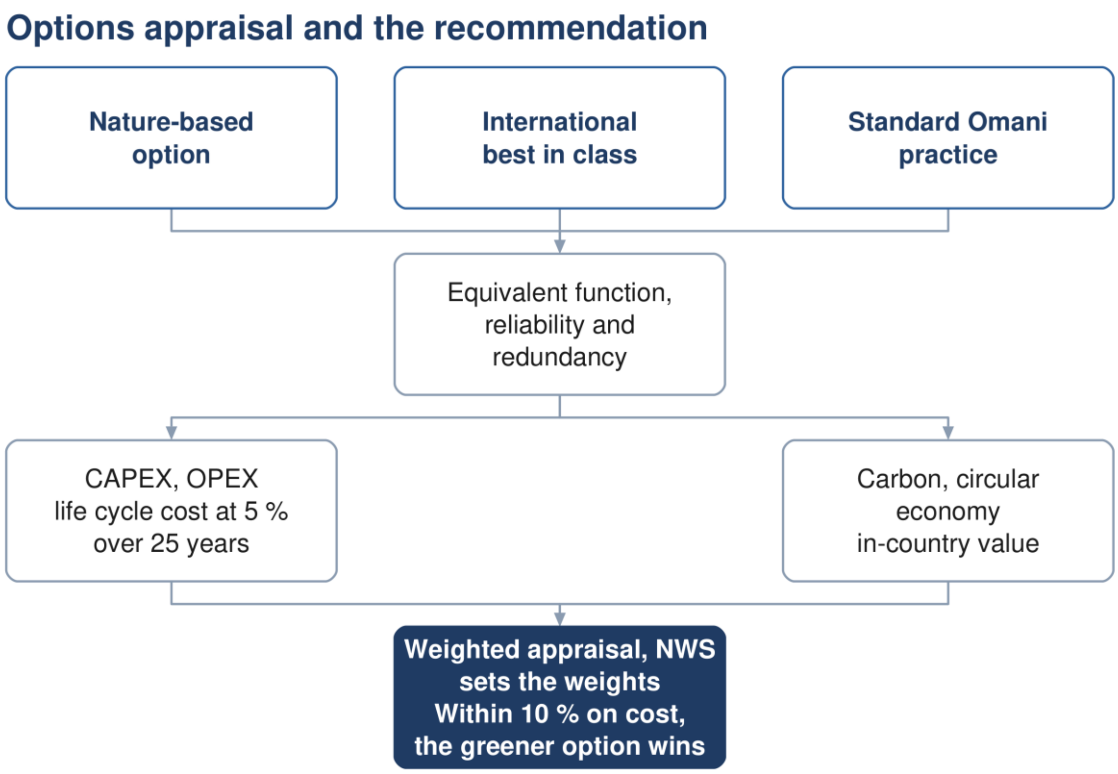

Figure 25 Options appraisal. Where total lifetime costs sit within ten per cent, the greener option is adopted.

23.1 How many options

What it is for. Enough genuinely different options that the recommendation is a choice, not a confirmation.

The rule. At least three options for each system: the sewer network, the treated effluent network and the treatment plant, and three for each plant if more than one plant is proposed. Lifting stations are not a separate set; their number follows from the network layout, and they are compared inside it. Every option meets the same functional requirements and specification, with similar reliability and redundancy, so that the comparison is not biased by scope.

Source: G201 §12 and §12.1, p95: "a minimum of three (3) design options" and "equivalent functional requirements and specifications with similar levels of reliability and redundancy". Three per system and per plant, and lifting stations inside the network options: project rule (engineer, 2026-08-31).

Worked Ibri number. At saturation the study area produces 60,099 m³/d and Ibri town alone 24,049 m³/d, both at or above the 20,000 m³/d of a Large plant (G203 p65). Splitting the load between plants therefore still leaves at least one Large plant, and every plant proposed brings its own three options into the comparison.

Where it lives. _BRAIN/07_PROJECT_STATE.md §2 item 1e; report R2 §21.1 (W14/report/rpt_ef.py); flows by settlement in design_flows.json.

23.2 Three archetypes

What it is for. Three variants of one scheme are a sensitivity test, not an options appraisal. The archetypes force the options apart.

The rule. The guideline names the kind of option to be included, as guidance:

1. an option that pushes the environmental sustainability ambitions of NWS and the Sultanate, including nature-based solutions;
2. an international best-in-class, state-of-the-art solution;
3. a standard solution based on current practice and technology established in Oman.

The guideline names them and stops. What distinguishes them in design terms is ours, set out below and offered to NWS for comment; it is never presented as a guideline requirement.

Table 114 Proposed meaning of the three archetypes in design terms

	Sustainability-led	International best practice	Established local practice
Network	Gravity maximised, accepting deeper excavation to avoid pumping	Layout optimised by model, pumping where it lowers whole-life cost	Conventional layout, pumping where the ground requires
Treatment	Lower-energy process; nature-based polishing where the scale permits	Highest-performing process, smallest footprint	Process established in Oman, simple to run and maintain
Energy	Generation on site, with a self-sufficiency target	High-efficiency plant and advanced process control	Grid supply with standby generation
Reuse	Reuse maximised, wadi discharge minimised	High reuse at high effluent quality	Reuse to the demand identified, surplus discharged
Materials	Low-carbon and locally sourced where performance permits	Specified for performance and durability	Standard specification
Consequence	Higher capital, lowest carbon and operating cost	Highest capital, least land, best effluent	Lowest capital, most land, highest operating cost

One limit binds the first archetype. Constructed wetlands are recommended only for small plants in rural areas, approximately up to 500 m³/d or 4,000 population equivalent (G203 p101). Nature-based treatment therefore serves outlying settlements or polishing, never the main plant.

Source: G201 §12.1 p95 for the archetypes; G203 §10.5 p101 for the wetland limit. The design-terms table: report R2 §21.2, our interpretation.

Worked Ibri number. 7 of the 25 settlements stay at or under 500 m³/d even at saturation: Hijar (462), Al Qali (135), Al Akheedar (110), Miayrid (108), Satwah (67), Usaybuq (38), Ash Shiab (16) m³/d. Together they house 5,464 people in 2070, 1.6 % of the study area, and each is under 4,000 people. That is the whole field on which a wetland can treat a settlement's sewage outright; everywhere else nature-based treatment can only polish.

Where it lives. design_flows.json, key settlements (q_ult, pop_ult); report R2 §21.2 (W14/report/rpt_ef.py).

23.3 Seven criteria, and who weights them

What it is for. Cost alone does not choose. The criteria bring in what the money does not show.

The rule. Each option is assessed on the parameters of G201 §12 (capital and operating cost, life-cycle cost, carbon over the lifetime, resource efficiency under a circular economy including decommissioning, in-country value, and the degree of nature-based solution), over 25 years. A weighted multi-criteria analysis then compares the options on seven criteria:

1. total lifetime cost

2. sustainability: carbon footprint, circular economy, nature-based solutions
3. social development and in-country value
4. adaptability and resilience
5. operability
6. constructability
7. environmental impact

NWS sets the weights. The guideline specifies no scoring scale, no normalisation and no rule for combining the scores, so the method is proposed and agreed before any option is scored. The simplest form, and the one proposed, is a weighted sum:

$$S_j = \sum_{k=1}^{7} w_k s_{\mathrm{jk}}, \quad \sum_{k=1}^{7} w_k = 1 \tag{93}$$

Symbol	Meaning	Unit
S j	weighted score of option j	—
w k	weight of criterion k, set by NWS	—
s jk	score of option j on criterion k, on a scale agreed with NWS	—

Carbon is expressed in tonnes of CO_2 equivalent a year and per cubic metre, with all Scope 1 and Scope 2 emissions and Scope 3 for the significant elements (G201 p99). Nature-based solutions are measured by the carbon of the conventional materials they avoid (G201 p104).

Source: G201 §12 p95 (parameters, the 25 years, NWS weights), §12.6 pp 104–105 (the seven criteria), p99 (carbon). The weighted-sum form is not in the guidelines; it is ours, to be agreed with NWS.

Worked Ibri number. The benchmark each option's carbon is set against: at 1.17×10^{-3} t CO_2e per m³ of treated effluent (G201 p99), the 23,006 m³/d produced at the 2030 flow corresponds to 9,825 t CO_2e a year, and the 57,094 m³/d at saturation to 24,382 t CO_2e a year. An option that beats those figures is better than the NWS average; one that does not has to say why.

Where it lives. Report R2 §21.3–21.4 (W14/report/rpt_ef.py) and §34 (W14/report/rpt_gh.py). No scoring sheet exists yet.

23.4 Sensitivity of the ranking

The ranking is rerun under each of the three sensitivity variables of Section 22.7: the weights, the discount rate and the design inputs. An option that wins under one set of weights only is a fragile recommendation, and the report says so. The weights are NWS's to set, so the sensitivity on weights is also the evidence NWS needs to set them knowingly.

23.5 The tie-break

What it is for. Where two options cost nearly the same, the guideline decides for the greener one, which makes the sustainability work decisive rather than decorative.

The rule. If equivalent options are within 10 % on total lifetime cost, the more sustainable option is adopted. The sentence leaves three things open, and each is fixed here as a proposal to NWS:

- **Base:** the gap is measured against the lowest total lifetime cost, which gives the larger percentage and so the stricter test;
- **More sustainable:** the option with the better sustainability score of the multi-criteria analysis (carbon, circular economy, nature-based solutions);
- **Order:** inside the band the tie-break overrides the weighted score; outside it the weighted score decides.

$$\frac{\mathrm{LCC}_j - \mathrm{LCC}_{\min}}{\mathrm{LCC}_{\min}} \leq 0.10 \tag{94}$$

Symbol	Meaning	Unit
LCC j	total lifetime cost of option j, present value	OMR
LCC min	the lowest total lifetime cost among the equivalent options	OMR

Source: G201 §12.9 p106: "If equivalent options are within 10% Total Lifetime Cost, the more sustainable option will be adopted." The base, the meaning of more sustainable and the order: ours, to agree with NWS.

The base matters only in a close case, and a close case is exactly where the rule applies. Two options whose costs differ by a factor of 1.10 are 10.0 % apart measured on the cheaper one and 9.1 % apart measured on the dearer one. Between those two figures the verdict depends on the base, so the report states which base it used.

Worked example from an NWS appraisal, rebuilt on the guideline basis: the winner's total lifetime cost was 34,047,919 OMR against 47,135,167 OMR for the nearest rival. That is 38.4 % above the winner, or 27.8 % below the rival; outside the band on either base, so cost decided. Three options costed over 25 years often land far closer than that, and then the sustainability scoring decides the outcome.

Where it lives. Report R2 §21.4 (W14/report/rpt_ef.py); _BRAIN/07_PROJECT_STATE.md §2 item 1e; the worked example: W9/analysis/W9_PIAD_financial_review.md §4.

23.6 Value engineering

A project estimated under OMR 5 million takes a general value engineering review by the design team at concept. At OMR 5 million or more, a formal study by an independent certified consultant is required at concept and at preliminary design. The table's footnote lowers that threshold for any sewage treatment plant or pumping station worth more than OMR 2 million (G201 Table 27, p93).

> **Formal value engineering falls inside the concept programme.** A Large plant of the order of 60,099 m^3/d is expected to clear the OMR 2 million footnote by a wide margin, so an independent certified value engineering consultant is appointed within the concept stage, and again at preliminary design. At concept the study reviews the alternatives and builds agreement on the direction (G201 §11.3.2 p93), so it belongs before the options are fixed, not after.

Where it lives. Report R2 §37 (W14/report/rpt_gh.py).

23.7 The recommendation and the roadmap

The recommendation is the optimum solution on the whole analysis, with the tie-break applied (G201 §12.9 p106). For the recommended option the designer then prepares an implementation roadmap: the scope of the later design stages, the procurement strategy, in particular whether the works are designed by the designer or by an EPC contractor, the phasing, and a framework for monitoring performance.

Where it lives. Report R2 §38–39 (W14/report/rpt_gh.py); the appraisal figure: W9/py/make_appraisal_figure.py.

23.8 Check it yourself

1. Count the options per system and per plant. Fewer than three anywhere is non-compliant with G201 p95.
2. Put the options side by side and check that their functional requirement, effluent standard and redundancy are the same. A cheaper option that delivers less is not an option.
3. Check that no option is the preferred option with one quantity changed; each carries its own structures and pumping.

4. Check that the weights came from NWS in writing, and that the scoring scale was agreed before the scoring.
5. Recompute the gap between the two cheapest options against the cheaper one. If it is at or under 10 %, check that the greener option was adopted and that the report says why.
6. Check that any wetland option serves a settlement under 500 m³/d at saturation (design_flows.json, q_ult).

24 Where the guidelines cannot be relied on

A reference that presents its source as flawless is not usable. Everything in this chapter was checked at the page. Where the guideline is wrong, this tutorial reproduces what it prints and says what is wrong with it; it never silently corrects the client's own standard. The chapter then sets out where this project departs from the guidelines, what is still waiting for data, and how the numbers compare with the Inception Report.

24.1 Formulae that are not in the guidelines

Each of these is commonly assumed to be in the guidelines and is not. Citing NWS for any of them would be a fabricated reference.

Table 115 Formulae to cite elsewhere

Formula	Cite instead	What NWS does supply
Net present value	ISO 15686-5	5 % discount rate, 25-year period (G201 p96, p57)
Life-cycle cost	ISO 15686-5	the list of cost stages (G201 p96)
Carbon footprint	ISO 14064, GHG Protocol	units and benchmark values (G201 p99)
Multi-criteria score	designer's own, agreed with NWS	the criteria and who weights them (G201 p95, p105)
Payback	standard finance	nothing
Total dynamic head	standard hydraulics	the component list, in prose
Pump power	standard hydraulics	nothing; no efficiency assumption
Darcy-Weisbach	standard form	roughness values only (G202 p104)
Hazen-Williams	standard form	C values only
Wave celerity	Korteweg	a qualitative description
Mass load from flow and concentration	standard	loads and concentrations separately
Oxygen demand, sludge age, clarifier area	Metcalf and Eddy, or DWA	ranges to land inside (G203 p75, p102)
Surge vessel volume	none	a 20 % minimum allowance (G201 p147)
Emergency storage at a pumping station	none	nothing; propose and obtain approval
Roughness of a raw sewage force main	none	potable and treated effluent values only
Minimum low flow for self-cleansing	Mara, Sleigh and Taylor	nothing (24.4)
Aquifer recharge criteria	none	absent from all three documents

Where it lives. TUTORIALS/T03_R01/EQUATION_REGISTER.md Part 2, updated; Appendix A.2 points back to this table rather than repeating it.

24.2 A mandatory method that cannot be completed: tractive force

What it is for. At the head of a system the flow is too small to reach the self-cleansing velocity, and the tractive-force gradient is the only defence against silting.

The rule. Two approaches shall be used for the sediment-carrying capacity of a sewer: the self-cleansing velocity and the minimum tractive force. The steeper of the two gradients is adopted as the minimum. At the head of a system, where the velocity cannot be reached, the gradient is set by the tractive force alone:

$$S_{\min} = K\tau^{1.23}Q^{-0.461} \quad (95)$$

Symbol	Meaning	Unit
S min	minimum gradient	m/m
K	coefficient, 2.33×10^{-4} with Q in m^3/s (G203 p27); the guideline's second constant, for Q in l/s, is discussed below	—
τ	tractive tension the flow must exert on the wall	Pa
Q	peak flow in the pipe	m^3/s

The value of τ is not in the guidelines. The check is mandatory and the equation is given, but no value of tractive tension in pascals appears anywhere in the wastewater guideline's 201 pages or the general guideline's 152. Any value used comes from the literature and must be agreed with NWS in writing. This is the largest hole in the gravity design chain.

Source: G203 §4.2.2.1 pp 25–27; the equation and K at p27, printed only as an image. No τ in G203 or G201: open item GAP-9. Interim handling: τ is a parameter; 1 Pa (Mara literature) is tagged [GAP-9] wherever it is used, never quoted as an NWS value.

Worked Ibri number. At τ = 1 Pa [GAP-9] and the 1.5 l/s low flow [outside assumption, 24.4], entered as 0.0015 m^3/s with K = 2.33×10^{-4}, S min = 4.67 mm/m: flatter than the 5.00 mm/m minimum for a DN200 sewer in G203 Table 11 (p29). At that τ a DN200 laid at its G203 Table 11 minimum already meets the tractive test; at a higher τ it may not. The exponent makes τ decisive: doubling it multiplies S min by 2^1.23 = 2.35, to 11.0 mm/m. The count of pipes in each self-cleansing class swings with a number nobody has fixed, which is why the classes are always reported against the τ used.

Two cautions on the formula itself. G203 p27 prints two coefficients, 2.33×10^{-4} for Q in m^3/s and 5.5×10^{-3} for Q in l/s, and they agree only to 2.3 %: restated for Q in l/s, 2.33×10^{-4} becomes 5.63×10^{-3}, not 5.5×10^{-3}. The l/s constant therefore gives gradients 2.3 % flatter; in the example above it gives 4.56 mm/m instead of 4.67. This tutorial uses K = 2.33×10^{-4} with Q in m^3/s everywhere, which gives the steeper gradient and so errs on the safe side. That choice is recommended in the design-flow handoff, to be confirmed by the engineer. And the formula assumes a flow depth of 0.2 D and n = 0.013; outside those it is an approximation.

Where it lives. _BRAIN/05_GAPS.md, GAP-9; _BRAIN/02_DESIGN_CRITERIA.md, "Min tractive force method"; design_flows.json, rules.tractive_tension and rules.open_rulings_recommended.mara_constant; W14/docs/DESIGN_FLOWS_FOR_NETWORK.md §4 (the four classes) and §8 (the constant).

24.3 Errors of substance

The defects below change a number or make a clause unusable. The full register of twenty-eight is in Appendix A.

Table 116 Errors of substance in the guidelines

Where	What is wrong	How to handle it
G203 p24	full-bore flow printed as velocity divided by area, which is dimensionally impossible	use velocity times area; note the defect
G201 p72	Peltier printed with 1 in the numerator; the published form carries 2.5	reproduce as printed; query NWS
G201 p62	Adh Dhahirah tanker ratio of 333 % cannot be true; the volumes are sound	do not use the ratio; query NWS
G201 p144	pipeline emptying time evaluates to $m^{1.5} \cdot s^2$, not seconds	the guideline calls it a quick check; use a transient model
G203 p25	Manning constant 6.3448 against an exact 6.3496	quote the guideline value; footnote the exact one
G203 p40 / p42	two tables disagree on duty pumps for the largest station type	take the conservative reading; flag
G203 p146	aerobic digestion text requires 45 days; its own table gives 10 to 15	the text governs; note the conflict
G203 p124	clarifier surface loading and overflow rate do not reconcile, both at average flow	reproduce both; do not pick one silently
G203 p185	hydrogen sulphide risk bands overlap and leave a gap	reproduce verbatim; flag
G203 p71	chlorine residual worded as "at least between"	0.3 to 1.0 at the consumer; up to 3.0 at the plant

24.4 The 1.5 l/s low-flow threshold

What it is for. A pipe whose flow is too small for either self-cleansing test is neither regraded nor upsized; it goes on the flushing list. Some threshold has to say when the flow is too small.

The rule. Below a peak flow of 1.5 l/s on the early-year case, a pipe that passes neither test is classed early cleansing and flagged for flushing, not upsized; the DN200 minimum governs its size.

Source: not in PAM-GUD-203: the whole document was searched on 2026-09-11. It is the minimum peak flow of the Mara simplified-sewerage literature, a WC flush, which G203 cites only for the tractive-slope formula (p27). G203 §4.2.6 p28 asks for more frequent inspection and cleansing of pipes that cannot self-cleanse and gives no threshold. Tagged outside assumption; to be confirmed with NWS.

Worked Ibri number. One Ibri dwelling at the settlement's occupancy of 6.07 returns 0.85 × 164 × 6.07 = 846 l/d, or 0.0098 l/s. Unpeaked and fully connected, a pipe needs about 153 such dwellings upstream before its average flow alone reaches 1.5 l/s; peaking lowers that count, the connection ratio of 0.61 raises it. Either way, the threshold decides how many pipe heads land on the flushing list.

Where it lives. design_flows.json, rules.low_flow_1_5_l_s; _BRAIN/02_DESIGN_CRITERIA.md, "1.5 L/s is NOT in PAM-GUD-203".

24.5 Cross-references that do not resolve

- the Terms of Reference cite item 2.1 of Section 05 of the wastewater manual, which no longer exists after the March 2026 renumbering; the content is now §10.1
- a package-plant peak factor of 3.0 is attributed to the general guideline, which nowhere prints 3.0 (G203 p97)
- two internal references to the tanker discharge section point to the wrong subsection; it is §10.6.2 (G203 p73, p83)
- the general guideline is cited as AM-GUD rather than PAM-GUD (G203 p51, p65)
- value engineering timing is concept in one table and detailed design only in another (G201 p93 against p20)

- environmental impact assessment timing is scoping at preliminary in one clause and a full assessment at concept in another (G201 p19 against p44)

24.6 Departures from the guidelines

What it is for. Where the data held cannot support the guideline's method, the method used instead is declared, with the reason, so that NWS can confirm it rather than discover it.

The rule. Every departure is stated in the report with the guideline position, the position adopted and the reason, and confirmation is requested.

Table 117 Departures and the positions adopted

Subject	Guideline position	Position adopted	Reason
Occupancy rate (G201 p58–59)	Population divided by housing units, both from NCSI	Settlement population divided by the domestic electricity meters counted in the settlement, with a floor and a cap	Housing units are not published at settlement level
Non-domestic and governmental demand (G201 p60–61)	Unit rates per pupil, bed, employee and floor area where detailed land use is available (G201 Table 12)	The published governorate ratios	The quantities G201 Table 12 needs are not recorded in any dataset held
Allocation of non-domestic demand (G201 p59)	Distributed across the served population	Placed on the plots whose meters generate it	A residential street generates no commercial flow; the total is unchanged
Industrial estates (G201 p61)	Determined case by case with the developer	An assumed workforce at the dry-industry unit rate	No workforce or discharge record supplied; to be replaced by the record
Water supplied by tanker (G201 p62, p73)	Assessed from the tanker filling-station records	Tanker-supplied households carry the domestic rate like any metered household; no separate tanker term	No filling-station record held; requested
Other water sources (G201 p70)	Private wells and other non-network abstraction assessed	Not assessed; every flow is on the network-accounted basis	No record of private abstraction held; requested
Connection to the sewer (G201 p73)	Coverage rising to the whole served area by the end of the period	Every metered property connected from 2024 for sizing; the early-year self-cleansing case takes the Inception Report connection ratio	Sizing on the full load is the safe side; the ratio is carried where over-estimating is the risk
Design-flow standard (G201 p71)	Where detailed land use exists, design flows to a stated international standard such as BS EN 752	The planning ratios and return rates of the guideline, plot by plot	The quantities a detailed method needs are not held
Growth beyond the forecast (G201 p58)	Not extrapolated more than ten years beyond the available forecast	The Inception Report series to 2100, used for its growth rates only	The saturation year rests on the capacity of the cadastre; the horizon to 2100 was instructed by NWS

Source: report R2 §10.1, Table "Departures from the design guidelines". Dated project rules behind the rows: load basis locked (engineer, 2026-08-30); occupancy per settlement with floor and cap (engineer, 2026-09-10); the estates (engineer, 2026-09-09 and -10); the two flow cases (engineer, 2026-09-11).

Worked Ibri number. The occupancy rows in figures: over the study area 119,893 people on 22,559 domestic properties is 5.31 persons per property; Ibri derives 6.07; the floor is 4.0 and the cap 6.12, and 13 settlements under a thousand people take 4.0 outright. The estates carry 6,300 workers at 93 l/d (G201 Table 12, p61), 316 m³/d of sewage. The connection row matters in the direction of the error: saturation times the connection ratio would give 36,660 m³/d for the early years, against the 14,772 m³/d really expected in 2030, and would declare pipes self-cleansing while they silt.

Where it lives. W14/report/rpt_cd.py, part_c; W14/analysis/occupancy_by_settlement.csv; design_flows.json (low_case_2030_m3d, rules).

24.7 Items still waiting for data

Each item below is either a missing value or a missing record. None is filled by a guess: each is a parameter, a tagged assumption or a flow left out and said to be left out.

Table 118 Pending items and what closes them

Item	Status	What closes it
Tractive tension τ (GAP-9)	no value in G203 or G201; a parameter; 1 Pa tagged where used	NWS confirms a value in writing
1.5 l/s low-flow threshold	not in G203; Mara, tagged	NWS confirms or sets another
Tanker supply and sewage tankers (GAP-19, GAP-20)	not in any flow; the governorate tanker ratio is unusable (24.3)	filling-station and delivery records
Private wells and other sources (GAP-11)	not assessed	abstraction records, field survey
G201 Table 12 quantities (GAP-10, GAP-15)	not supplied; ratios used	pupils, beds, employees, floor areas
Existing plant capacity and inlet (GAP-7)	unknown	NWS operating data
Infiltration basis, peaking formula, tanker catchment (GAP-23)	guideline values used; differ from the Inception Report (24.8)	NWS ruling
Meters far from any plot	126 meters more than 15 m from a plot carry no load	open with the engineer

Where it lives. _BRAIN/05_GAPS.md; design_flows.json, keys rules and free_meters.

24.8 Reconciliation with the Inception Report

What it is for. The Inception Report R0 carries its own demand workbook, and NWS has read its numbers. Every difference is explained here, so no figure in the concept report looks like a silent change.

Two positions of the earlier flow tutorial have changed: occupancy is now derived per settlement rather than taken as a single 5, and Merrimack is the formula for any pipe serving more than 100 properties, with Peltier below that, rather than Peltier throughout.

Table 119 Reconciliation register against the Inception Report R0

Item	Guideline	Inception Report	This method	Status
Domestic consumption	164 l/c/d (G201 Table 11, p60)	163.5, computed from billing	164 l/c/d	consistent
Occupancy	population ÷ housing units, NCSI (G201 p58)	a single rate	per settlement, floor 4.0, cap 6.12; Ibri 6.07; 5.31 over the area	departure (24.6)
Population 2024, 25 settlements	NCSI series (G201 p58)	116,452	119,893: the floor adds 688 in 2 settlements; the under-1,000 rule adds 2,955 in 8 and removes 203 in 5	explained
Non-domestic and governmental	22 % and 14 % (G201 Table 11, p60), or G201 Table 12 (p61)	ratios	ratios for the volume, placed on the meters	departure (24.6)
Return to sewer	85 % and 54 % (G201 Table 19, p71)	same	85 % and 54 %	consistent
Infiltration	720 l/d/km, new network (G201 p72)	10 % of the flow	720 l/d/km per pipe, never on the plot	confirm with NWS
Peak	Merrimack over 100 properties, Peltier alternative; hourly factor ≤ 5.0 recommended (G201 p71–72)	Peltier plus a 20 % weekly peak	Merrimack or Peltier per pipe; no weekly peak	confirm: peaking twice
Tanker flow	coverage as for water (G201 p73)	settlements within 25 km	not in any flow; records requested	open

Item	Guideline	Inception Report	This method	Status
Connection	100 % by the end of the period (G201 p73)	0.51 in 2024 rising to 0.61 by 2028	sizing at 100 %; early-year case Q 2030 × 0.61 = 14,772 m³/d	adopted for the low case
Horizon	no more than ten years beyond the forecast (G201 p58)	series to 2100	rates only; saturation 2070 on the capacity of the cadastre	departure (24.6)
Ultimate average flow	none	about 49,700 m³/d	60,099 m³/d in 2070; 66,109 with the plant margin	superseded
Plant margin and effluent	+10 %, 95 % (G201 p73)	same	same	consistent
Sludge	0.25 kg/m³, indicative (G201 p78)	0.25 kg/m³	same	consistent; also in G201

Source: the guideline column from _BRAIN/02_DESIGN_CRITERIA.md with pages; the Inception Report column from its demand workbook as decoded in TUTORIALS/T01_Sewage_Flow_and_Load_Calculation.md, and its ultimate flow as footnoted in report R2 §15.7. T01 called the sludge rate an Inception-only value; it is also G201 p78.

Where it lives. The census figure: the demand workbook _CLIENT/Ibri Sewer Demand R0 2026 08 03.xlsx, sheet "Project Pop Settlements", read by facts_w14.settlement_table (workbook_2024). This method's figures: facts_w14.totals and design_flows.json.

24.9 What to do about all this

Three rules keep the report defensible. Quote the guideline as it is printed, and say separately what is wrong with it; never publish a silently corrected version of the client's own standard. Where a formula is not NWS's, attribute it to whoever it belongs to. And where the guideline is silent on something it makes mandatory, propose a value, say where it came from, and obtain agreement in writing before the design depends on it.

24.10 Check it yourself

1. Pick any guideline value in the report and open the page it cites. If the value is not on that page, the citation is wrong, whatever the value.
2. Search the report for τ. Every use must carry its value and the tag [GAP-9].
3. Search the report for 1.5 l/s. It must never be attributed to PAM-GUD-203.
4. Recompute the early-year flow two ways: 60,099 × 0.61 = 36,660 m³/d is wrong; the sum of Q_2030 × 0.61 over the plots, 14,772 m³/d, is right.
5. Add the settlement column WB_2024 of Settlements_merged.shp (116,452) and compare it with POP_2024 (119,893); the difference is the floor and the small-settlement rule, nothing else.
6. For each departure, check that the report states the guideline position, the position adopted and the reason, and asks for confirmation.

Appendix A Equations, fields and the rerun order

The reference half of the tutorial: every equation with its source, the defects in the guidelines, the fields of the two layers that carry the load, and the order in which the scripts are rerun after a change.

A.1 Equations in the guidelines

Guidelines PAM-GUD-201 (152 pages), -202 (177) and -203 (201), all Revision 01, March 2026; the file names say v1.0. The printed page equals the PDF page in all three. Every equation was re-rendered from the PDF and read visually: text extraction mangles fraction bars, radicals and exponents, and three equations exist only as images.

Table 120 Equations that exist in the guidelines

No.	Equation	Source	Note
1	Population = plots × properties per plot × occupancy rate	G201 p58	one of three permitted approaches
2	Occupancy rate = population ÷ housing units	G201 p58	from NCSI data
3	LPCD = domestic water accounted ÷ (OR × active domestic accounts)	G201 p60	the derivation behind G201 Table 11
4	Q pdf = 2.65 Q adf^0.879 (Merrimack)	G201 p71	Ml/d both sides; over 100 properties
5	Pf = Q pdf ÷ Q adf	G201 p71	dimensionless
6	Q pdf = Q adf × Pf WW	G201 p72	m^3/d, one page from eq. 4
7	Pf WW = 1.5 + 1 ÷ √Q m (Peltier)	G201 p72	Q m in l/s; defect D1
8	P HF = Q max hour ÷ Q avg	G201 p63	water side
9	P DF = Q max day ÷ Q avg	G201 p63	water side
10	V = −2 √(2gDS) $\log_{10}$[k s ÷ 3.7D + 2.51ν ÷ (D √(2gDS))] (Colebrook-White)	G203 p24	k s = 1.5 mm; ν = 1.141 × 10^{-6} m^2/s
11	Q o = V × A	G203 p24	printed V ÷ A; defect D2
12	Q p = Q ÷ Q o	G203 p24	proportional discharge
13	v = (1/n) R^(2/3) S^(1/2) (Manning)	G203 p25	permitted, not required
14	S = v^2 n^2 × 6.3448 ÷ D^1.333	G203 p25	full bore; defect D3
15	Q = (1/n) R^(2/3) S^(1/2) A	G203 p25	
16	τ = W sin θ ÷ (p L) (tractive tension)	G203 p26	image only
17	W = ρ g a L	G203 p26	companion to 16
18	S min = K τ^1.23 Q^−0.461	G203 p27	image only; K printed as 2.33 × 10^{-4} (Q in m^3/s) and 5.5 × 10^{-3} (Q in l/s), 2.3 % apart; this method uses 2.33 × 10^{-4} with Q in m^3/s (Section 24.2); no τ, defect D4
19	H L = β (w/b)^(4/3) h sin θ (Kirschmer)	G203 p104	exponent printed as 1.33
20	V = 0.25 Q T (wet well live volume)	G203 p48	T = 3600 ÷ starts per hour; 10 starts/h up to 30 kW
21	NPSHa = Ha − Hvpa − Hst − Hf	G203 p47	margin ≥ 1 m
22	ΔH = (c/g) Δv (Joukowsky)	G201 p146	not in G202 or G203
23	Pipeline emptying time	G201 p144	dimensionally inconsistent; defect D5
24	dS/dt = 3.23 M′ [EBOD] r^{-1} − 2.1 N (s v)^0.375 [S] d m^{-1} (Pomeroy-Parkhurst)	G203 p186	two of nine symbols carry units
25	EBOD = BOD × 1.07^(T−20)	G203 p186	
26	S = K s BOD_5 t f(T) (force mains)	G203 p186	unusable: K s and f(T) undefined
27	Q TSE = 0.95 × Q STP inlet	G201 p73, p75	
28	TSE network loss = 10 % of TSE produced	G201 p76	"shall"
29	Sludge = 0.25 kg/m^3 × inlet volume	G201 p78	indicative
30	Aerated biomass = total biomass × aerated fraction of the cycle	G203 p94	
31	Total N = Org-N + NH_3 + NH_4 + NO_2-N + NO_3-N	G203 p71	

A.2 Equations of this method

The equations the load chain, the design flows and the appraisal add to the guideline set. Each is either built from guideline values, whose pages are given, or is a dated project rule.

Table 121 Equations added by this method

No.	Equation	Source	Where it lives
M1	$OR_s = P_{s,2024} \div N_{dom,s}$; floor 4.0, cap 6.12; 4.0 outright under 1,000 people	G201 p58 form; project rule 2026-09-10	plot_class_v2_apply.py; field OR_S
M2	$Capacity_s$ = empty home-shaped plots × home share × properties per home plot × OR_s	project rule 2026-09-10	plot_class_v2_apply.py; FUT_CAP, CAP_POP
M3	$Q_{plot} = 0.85 \times 164 \times OR \times N_{dom} + 0.54 \times (U_{nd} N_{nd} + U_{gov} N_{gov} + 93 N_w)$, l/d	G201 Tables 11, 12, 19 (p60, p61, p71); project rule 2026-09-10	plot_class_v2_apply.py; QADF
M4	$U_{nd} = 0.22 \times 164 \times P_s \div N_{nd,s}$ (meters outside the estates)	G201 Table 11 p60; placement, project rule	U_NDOM
M5	$U_{gov} = 0.14 \times 164 \times P_s \div N_{gov,s}$ (meters outside the estates)	G201 Table 11 p60; placement, project rule	U_GOV
M6	$Q_{future\ plot}$ = people × 171.3 l/d, where 171.3 = 164 × (0.85 + 0.22 × 0.54 + 0.14 × 0.54)	G201 Tables 11, 19	growth_by_settlement.py; Q_2030, Q_2055, Q_ULT
M7	Properties in year y = G_DOM + (POP_y − POP) ÷ OR_S	project rule	make_design_flows.py; design_flows.json properties
M8	$Q_{low} = Q_{2030} \times 0.61$, summed per pipe, then peaked on the connected count, properties of 2030 × 0.61 (Merrimack over 100, Peltier at or under)	project rule 2026-09-11; ratio from the Inception Report; the connected count recommended in the design-flow handoff, to be confirmed by the engineer	design_flows.json low_case_2030_m3d
M9	Infiltration = 720 l/d/km × pipe length, added per pipe	G201 p72	the network engine, never the plot
M10	Plant capacity = (settlement total + network infiltration + tankers when known) × 1.10	G201 p73	design_flows.json stp_ultimate_with_margin_m3d (before infiltration)
M11	$C_{cap} = (1 + f_{pre} + f_{phys} + f_{price}) \times \Sigma c_k q_k$	NWS appraisal structure; project rule 2026-09-01	Section 21.2
M12	$NPV = \Sigma C_t \div (1 + r)^t$, t = 0 … n; r = 0.05, n = 25	ISO 15686-5; r and n G201 p96, p57	Section 22.2
M13	$a_n = [1 - (1 + r)^{-n}] \div r$	standard discounting	Section 22.2
M14	$S_j = \Sigma w_k s_{jk}$, $\Sigma w_k = 1$	ours, to agree with NWS	Section 23.3
M15	$(LCC_j - LCC_{min}) \div LCC_{min} \le 0.10$	G201 p106; the base is ours	Section 23.5

The formulae that are not in the guidelines, and what to cite instead, are the table of Section 24.1.

A.3 Register of defects in the guidelines

Each was verified at the page. Reproduce the guideline's value where it is quoted, and flag it; never repair it silently.

Table 122 Defects in PAM-GUD-201, -202 and -203

No.	Defect	Page	Handling
D1	Peltier numerator printed as 1, not the classical 2.5	G201 p72	print as printed; query NWS
D2	$Q_o = V \div A$, dimensionally impossible	G203 p24	use V × A; note the typo
D3	Manning constant 6.3448 against exact $4^{(4/3)} = 6.3496$	G203 p25	quote 6.3448; footnote
D4	no value of τ in pascals in G203 or G201	G203 p26–27	state the gap; GAP-9
D5	pipeline emptying time evaluates to $m^{1.5} \cdot s^2$	G201 p144	quote with the caveat
D6	Adh Dhahirah tanker ratio of 333 % impossible	G201 p62	volumes sound, ratio not; query
D7	G203 Tables 17 and 19 disagree on Type 3 duty pumps (3 against 2)	G203 p40, p42	G203 Table 17, the conservative one

No.	Defect	Page	Handling
D8	Fayoux H_2S risk bands overlap; nothing above 30	G203 p185	reproduce; flag
D9	aerobic digestion: text 45 days, G203 Table 77 10–15 days	G203 p146	text governs
D10	clarifier loading 1.0–1.4 m/h against overflow 16–28 $m^3/m^2 \cdot d$	G203 p124	reproduce both; flag
D11	chlorine residual "at least between 0.3 and 1.0"	G203 p71, p130	0.3–1.0 at consumer; to 3.0 at plant
D12	OSEC free chlorine "> 0.06 mg/L", orders of magnitude out	G203 p132	report, do not use
D13	air valve size bands overlap and leave a gap	G203 p53	reproduce as printed
D14	washout size bands overlap and leave gaps	G203 p54	reproduce as printed
D15	package plant peak factor 3.0 attributed to G201, not there	G203 p97	query NWS
D16	trunk main "length above 1,000 mm without connexions"	G203 p35	quote with [sic]; metres meant
D17	motor service factor "1.15%"	G202 p96	1.15 intended
D18	"AM-GUD-201" and "AM-GUD-202" for PAM-GUD	G203 p65, p51	typos
D19	G203 Table 63 preceded by an orphan caption "Table 58"	G203 p124	numbering defect
D20	tanker discharge references point to §10.6.4 and §10.6.3; it is §10.6.2	G203 p73, p83	two wrong references
D21	primary settler detention "1.2" with no unit	G203 p119	read as 1.2 h; state the inference
D22	G201 Table 2 header row collapsed to zero height	G201 p19	recovered by character extraction
D23	G201 Table 2 field investigations: detailed cell blank	G201 p19	reproduce as blank
D24	value engineering timing: G201 Table 27 concept, G201 Table 2 detailed only	G201 p93, p20	G201 §11.3.2 supports G201 Table 27
D25	EIA timing: scoping at preliminary against full EIA at concept	G201 p19, p44	query NWS
D26	two undefined asterisks: G201 Table 12 header and "*Q_adf"	G201 p61, p71	no footnote exists
D27	design life 20 years against pump service rating 15 years	G203 p38, p40	different things; state both
D28	"nominal flow" undefined for retention basin and relief sewer	G203 p191	query NWS

A.4 Fields of the plot layer

W14/shp/PLOTS_load.shp carries, for each of the 77,265 plots, the fields below. It is the load table of the design: the network reads the flow of each plot from it, and every settlement figure is a sum over it.

Table 123 Fields of the plot layer

Field	Content
Name	cadastral identifier, as received
Moh_Classi, Classes	class code and class name of the cadastre, as received
Buiding_St	built or future, as received
SETTLE	the settlement the plot belongs to
AREA_M2	plot area, m^2
N_ACC	meters on the plot, all tariffs
N_DOM, N_DOMADD	meters on the primary and subsidised tariffs; on the additional-account tariff
N_COM, N_GOV, N_AGR, N_CRT, N_IND	meters on the commercial tariff (with fisheries and tourism), the government tariff (with defence), and the agricultural, large-consumer and industrial tariffs
G_DOM	domestic meters, one property each (primary, subsidised and additional tariffs)
G_NDOM, G_GOV, G_SPEC, G_AGR	non-domestic, governmental, special and agricultural meters, the large consumers placed by their identified use
DERIVED	use of the plot
WHYC	the rule that set the use
ESTATE	the industrial estate the plot lies in, if any
WORKERS	workers assigned to an estate plot, by area
HOMESHAPE	1 if the plot is home-shaped: 200 to 1,000 m^2, compact, not a strip
COMPACT, ASPECT	area against the smallest enclosing rectangle; that rectangle's aspect ratio
NDVI_MEAN, NDVI_SHARE, GREEN_M2	the satellite vegetation test: mean index, share of green pixels, green area in m^2
OR_S	occupancy adopted for the settlement, persons per property
PPP_S	properties per home plot adopted for the settlement
HOMESH_S	home share adopted for the settlement

Field	Content
U_NDOM, U_GOV	non-domestic water per shop meter and governmental water per government meter in the settlement, l/d
POP	people in 2024: G_DOM × OR_S
W_DOM, W_NDOM, W_GOV, W_SPEC, W_TOT	water by stream and in total, m³/d, 2024
S_DOM, S_NDOM, S_GOV, S_SPEC	sewage by stream, m³/d: 0.85 of the domestic water, 0.54 of the rest
QADF	average dry-weather sewage flow of the plot in 2024, m³/d: the sum of the four streams
FUT_CAP	1 if an empty plot counts as a future home
FUT_PROPS	properties expected per counted plot: the settlement's ratio times its home share
SPREAD_W	weight for placing the settlement's growth: the plot's area capped at 1,000 m² on an empty plot of up to 2,000 m² that is not a grove, industrial or heritage; zero on every other plot
POP_2030, POP_2055, POP_ULT	people on the plot in 2030, 2055 and the saturation year
Q_2030, Q_2055, Q_ULT	average sewage flow in 2030, 2055 and the saturation year, m³/d; a future plot at the future-plot rate per person (M6)
SAT_YEAR	the year the plot's settlement is full
ULT_YEAR	the saturation year of the study area

A.5 Fields of the settlement layer

W14/shp/Settlements_merged.shp carries the same at settlement level for the 25 settlements, with the rates and the five-year steps. The plots sum to it in every flow and population column.

Table 124 Fields of the settlement layer

Field	Content
SETTLE, AREA_KM2	name; area in km²
WB_2024, WB_2100	population of the Inception Report series in 2024 and 2100
PROPS	domestic properties (meters) in 2024
OR_RAW, OR_S	occupancy derived and adopted
PPP_RAW, PPP_S	properties per home plot derived and adopted
HOMESH_RAW, HOMESH_S	home share derived and adopted
SMALL	1 if under 1,000 people in 2024: the small-settlement rule applies
BUILT_PL, EMPTY_PL, HOME_MEAS	built plots; empty plots; home plots the ratio was measured on
CAP_PLOTS, CAP_POP	empty plots counted as future homes; the people they hold at saturation
G_NDOM, NDOM_POOL, G_GOV, GOV_POOL, G_AGR, G_SPEC	meters by category; of the shop and government meters, those outside the estates that share the pool
WORKERS	workers of the industrial estates in the settlement
U_NDOM, U_GOV	non-domestic water per shop meter, governmental water per government meter, l/d
POP_2024	people in 2024
W_DOM, W_NDOM, W_GOV, W_SPEC, W_TOT	water by stream and in total, m³/d, 2024
S_DOM, S_NDOM, S_GOV, S_SPEC, Q_2024	sewage by stream and in total, m³/d, 2024
POP_2025 … POP_2070, Q_2025 … Q_2070	people and average sewage flow at five-year steps to the saturation year
POP_ULT, Q_ULT	people and flow in the saturation year of the study area
SAT_YEAR, SAT_POP, SAT_Q	the year the settlement is full, and its people and flow then
SAT_OWN	the year it would fill on its own growth alone; 0 if not before 2100
IN_PEOPLE, OUT_PEOPLE, MAIN_TO, INSHARE	people received from and sent to other settlements by the saturation year; the main receiver; the share of capacity taken by overflow
ULT_YEAR, UNHOUSED	the saturation year of the study area; growth of the series to 2100 with nowhere to go

Where it lives. Both field lists are the ones delivered with report R2, Appendix A4 (W14/report/rpt_app.py).

A.6 The rerun order

Every number in this tutorial and in the concept report comes out of five steps, run in this order after any change to the data or a rule. Each step reads what the one before wrote; running one out of order leaves the layers disagreeing with each other.

Table 125 The rerun order and what each step writes

Step	Script, and where it runs	Reads	Writes
1	W14/py/plots_meters_load.py, inside QGIS with the project open	the plots, the electricity meters, the identified large-consumer accounts and projects, the towns and the boundary, as layers of the QGIS project	shp/ELE_meters_on_plots.shp (every meter, its plot and category); shp/PLOTS_load.shp (meters and today's load); shp/Settlements_merged.shp; analysis/plots_load_today.csv; analysis/settlements_today.csv
2	W14/py/plot_class_v2_apply.py, plain Python	PLOTS_load.shp; analysis/ndvi_plots.csv and vegfrac_plots.npy; the Inception demand workbook	PLOTS_load.shp (use, OR_S, U_NDOM, U_GOV, loads, FUT_CAP, SPREAD_W); analysis/plot_class_audit.csv; settlements_today.csv; occupancy_by_settlement.csv
3	W14/py/growth_by_settlement.py, plain Python	settlements_today.csv; PLOTS_load.shp; Settlements_merged.shp; the Inception demand workbook	analysis/W14_growth_by_settlement.xlsx (people and flow by settlement and year, the five-year tables, the overflow routes, the rules); PLOTS_load.shp (POP_ and Q_ fields, SAT_YEAR, ULT_YEAR); Settlements_merged.shp (five-year, saturation and overflow fields)
4	W14/py/make_design_flows.py, plain Python	the outputs of steps 2 and 3, through facts_w14	docs/DESIGN_FLOWS_FOR_NETWORK.md; analysis/design_flows.json
5	W14/report/build.py --pdf, plain Python with Word	facts_w14 and the figures in W14/report/img	report/R2/Ibri_Concept_Design_Report_R2.docx and .pdf (the REV constant selects the revision folder)

Never stop after step 1. The loader still carries the single occupancy rate of an earlier stage among its constants. The occupancy per settlement, the plot use and the loads are applied by step 2, which rewrites the plot layer. A plot layer read between steps 1 and 2 carries the old rate.

Two inputs of step 2 are made once and rerun only when their source changes: the satellite vegetation index per plot (W14/py/ndvi_plots.py, writing analysis/ndvi_plots.csv) and the vegetation fraction per plot (W14/py/vegfrac_plots.py, writing analysis/vegfrac_plots.npy). After step 5 this tutorial is rebuilt with TUTORIALS/T04/build.py, which reads the same facts, so its numbers follow.

A.7 Check it yourself

1. After step 3, sum QADF over the plot layer and compare it with Q_2024 summed over the settlement layer: both should give 20,852 m³/d.
2. After step 3, sum Q_ULT over the plots: 60,099 m³/d in 2070.
3. After step 4, open design_flows.json and check that its totals match the workbook sheet "Qadf by year m3d" for 2024, 2030, 2055 and the saturation year.
4. After step 5, read the exported report figures as images, not only the file list: a layer with its categories switched off still exports a complete-looking legend.
5. Check the future-plot rate by hand: 164 × (0.85 + 0.22 × 0.54 + 0.14 × 0.54) = 171.3 l/d per person.